

**Dubai Metro, UAE
Green Line – Al-Qusais Depot**

**33KV Transformer Shed
(Building 26)**

**Structural Design Calculations
(DCP4 Submission)**

Nov 2007

Client: JT Metro JV	Contract No. (if any):
Project Title: Dubai Metro, UAE	Project No.: JRD4561
Document No.: DM001-D-GBD-WTS-DR-DCC-545029	Controlled Copy No.:
Document Title: Green Line – Al-Qusais Depot 33KV Transformer Shed (Building 26) Structural Design Calculations (DCP4 Submission)	
Covering Letter / Transmittal Ref. No.:	Date of Issue: 25 Nov 2007

Revision, Review and Approval/Authorization Records

Revision	Description	Prepared by / date	Reviewed by / date	App. or Auth. by / date
A3	ISSUED FOR APPROVAL DCP4	D. Shah Nov 07	M. Lake Nov 07	J. Hill Nov 07
A2	DCP3 Re-Submission	D. Shah Nov 07	M. Lake Nov 07	J. Hill Nov 07
A1	DCP3 Submission	D. Shah Sep 07	P. Scott Sep 07	J. Hill Sep 07

Distribution (if insufficient space, please use separate paper)

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3	Atkins (Paul Shepherd-Smith)
4	Atkins (Dipak Shah)

Note: App. and Auth. mean "Approved" and "Authorized" respectively.

Document No.: DM001-D-GBD-WTS-DR-DCC-545029	Date: 25 Nov 2007
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1 Introduction

The Dubai Metro Project is under construction by the Dubai Rapid Link Consortium (DURL). The Civil Works are contracted to the Japan Turkey Metro Joint Venture (JTMJV), and the Rail Systems Works are contracted to Mitsubishi Heavy Industries (MHI). The rolling stock will be provided by Kinki Sharyo.

The 33KV Transformer Shed is situated in a building 19m long, 17m wide, and 8.41m high.

This submission for 33KV Transformer Shed (Building 26) in Al-Qusais Depot is made by the Designers for the JTMJV, and as such concentrates on the scope of the Civil Works Contract, which includes the Architecture, Structure and M&E Design.

2 Building Form and Structural System

33KV Transformer Shed is 19m long, 17m wide and 8.41m high. This facility will contain various rooms to house transformer, AC Switchgear & AHU.

For a structurally efficient solution together with fulfilling functional requirement, a 3x3 bay RCC frame spaced 6.5m, 6m & 6.5m respectively along grid A, B, C & D and spaced at 5.04m, 6.85m & 5.31m respectively along grid 1, 2, 3 & 4 is proposed. This RCC structure is covered with insulated metal panelled roof which is supported by steel frame spaced at 6.5m, 6m & 6.5m respectively along grid 1, 2, 3 & 4. This frame made up of I beam supported by 2 column between span. The building has been designed for 3 hour fire rating provisions. The building is to be clad with insulated standing seam aluminium sheeting, fixed to cold formed Z purlins which span on to structural frame.

A 225mm thick suspended ground level slab monolithic with RC Beams is proposed to resist various forces from equipments.

3 Drawings

To complement the purpose of this re-submission, the following Structural drawings are submitted along with this report.

	<u>Drawing Title</u>	<u>Drawing No.</u>
Structural Drawings		
1	Al-Qusais-Depot - General Notes And Typical Details Sheet 1 of 3	DM001-D-GDP-CDP-DD-DCC-551990
2	Al-Qusais-Depot - General Notes And Typical Details Sheet 2 of 3	DM001-D-GDP-CDP-DD-DCC-551991

	<u>Drawing Title</u>	<u>Drawing No.</u>
3	Al-Qusais-Depot - General Notes And Typical Details Sheet 3 of 3	DM001-D-GDP-CDP-DD-DCC-551992
4	Al-Qusais Depot, Building - 26 - 33 KV Transformer Shed - Foundation Layout Plan	DM001-D-GBD-WTS-DD-DCC-551800
5	Al-Qusais Depot, Building - 26 - 33 KV Transformer Shed - Ground Slab Layout Plan	DM001-D-GBD-WTS-DD-DCC-551801
6	Al-Qusais Depot, Building - 26 - 33 KV Transformer Shed - RC Roof Layout Plan	DM001-D-GBD-WTS-DD-DCC-551802
7	Al-Qusais Depot, Building - 26 - 33 KV Transformer Shed - Steel Roof Layout Plan	DM001-D-GBD-WTS-DD-DCC-551803
8	Al-Qusais Depot, Building - 26 - 33 KV Transformer Shed - Section A-A	DM001-D-GBD-WTS-DD-DCC-551804
9	Al-Qusais Depot, Building - 26 - 33 KV Transformer Shed - Section B-B	DM001-D-GBD-WTS-DD-DCC-551805
10	Al-Qusais Depot, Building - 26 - 33 KV Transformer Shed - Elevations	DM001-D-GBD-WTS-DD-DCC-551807
11	Al-Qusais Depot, Building - 26 - 33 KV Transformer Shed - Footing Details	DM001-D-GBD-WTS-DD-DCC-551809
12	Al-Qusais Depot, Building - 26 - 33 KV Transformer Shed - Ground Slab Reinforcement Details	DM001-D-GBD-WTS-DD-DCC-551812
13	Al-Qusais Depot, Building - 26 - 33 KV Transformer Shed - Roof Slab Reinforcement Details	DM001-D-GBD-WTS-DD-DCC-551813
14	Al-Qusais Depot, Building - 26 - 33 KV Transformer Shed - Pit Reinforcement Details	DM001-D-GBD-WTS-DD-DCC-551815
15	Al-Qusais Depot, Building - 26 - 33 KV Transformer Shed - Ground Slab Level Beam Schedule – Sheet 1	DM001-D-GBD-WTS-DD-DCC-551817
16	Al-Qusais Depot, Building - 26 - 33 KV Transformer Shed - Ground Slab Level Beam Schedule – Sheet 2	DM001-D-GBD-WTS-DD-DCC-551818
17	Al-Qusais Depot, Building - 26 - 33 KV Transformer Shed - RC Roof Level Beam Schedule	DM001-D-GBD-WTS-DD-DCC-551820
18	Al-Qusais Depot, Building - 26 - 33 KV Transformer Shed - Roof Setting Out - Plan & Section	DM001-D-GBD-WTS-DD-DCC-551825

4 3D- Frame Analysis and Design

Roof loading :

Dead load

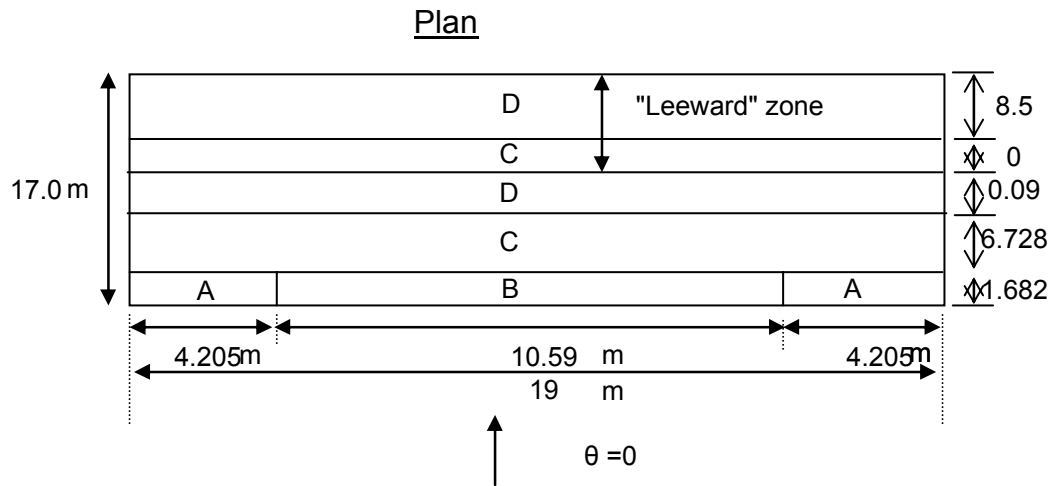
W_{DL} :

Sheeting	0.1 kPa	
Purlins	0.1 kPa	
Insulation	0.3 kPa	
M&E Finishes	0.3 kPa	
	<u>0.8 kPa</u>	
		$\times 1 = 0.8 \text{ kN/m}$

Live load

		0.6 kPa	$\times 1 = 0.6 \text{ kN/m}$
$P1_{LL}$:	A/C Equipment S/W		
	= 5 kN		(point load at mid span)

Average C_{pe} for roof (Wind direction normal to eave)

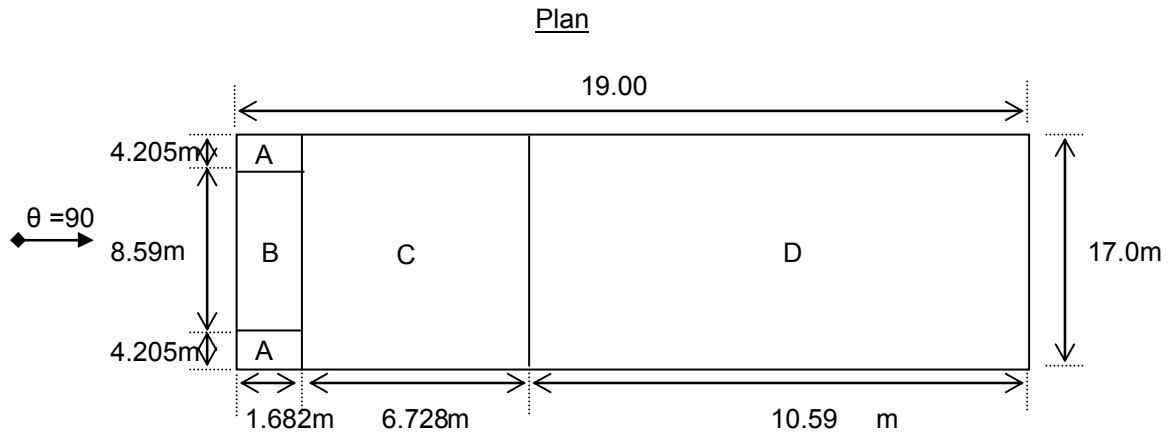


Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 19.00 \text{ m} \\
 H &= 8.41 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 2 \times 8.41 \\
 &= 16.82 \text{ m}
 \end{aligned}$$

Zone	Windward		Leeward		
	Area (m^2)	C_{pe}	Area	C_{pe1}	C_{pe2}
A	14.1	-2			
B	17.8	-1.4			
C	127.8	-0.7	0.0	-0.7	-0.07
D	1.7	-0.2	161.5	-0.2	0.2
Total	161.5		161.5		
Average		-0.78		-0.20	0.20

Average C_{pe} for roof (Wind direction normal to gable end)



Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 17 \text{ m} \\
 H &= 8.41 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 2 \times 8.41 \\
 &= 16.82 \text{ m}
 \end{aligned}$$

For max^m uplift, assume zone C coefficient=>

$$C_{pe} = -0.7$$

For max^m downward force, assume zone D coefficient=>

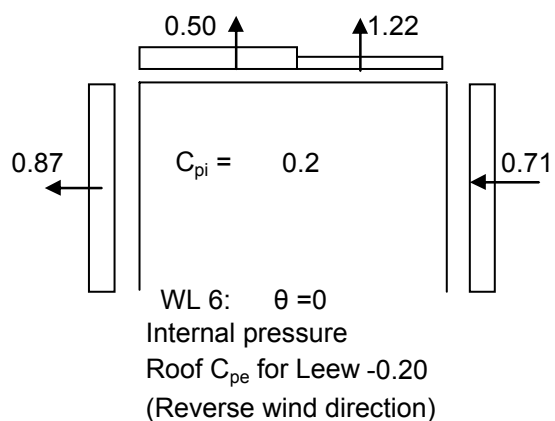
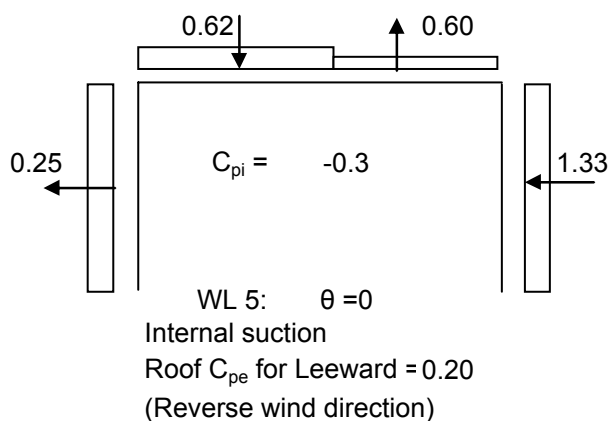
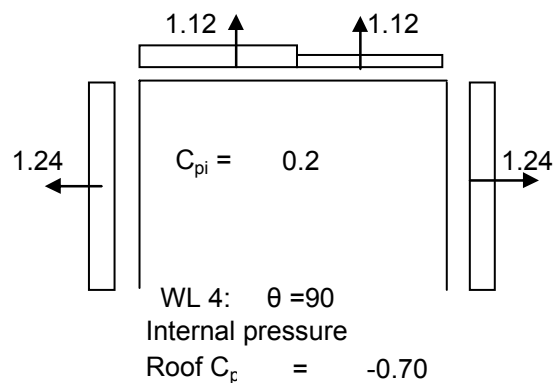
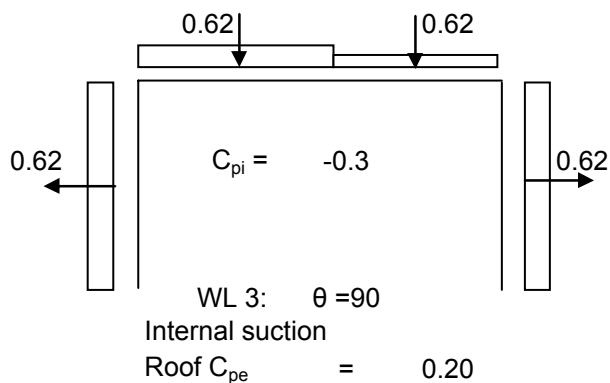
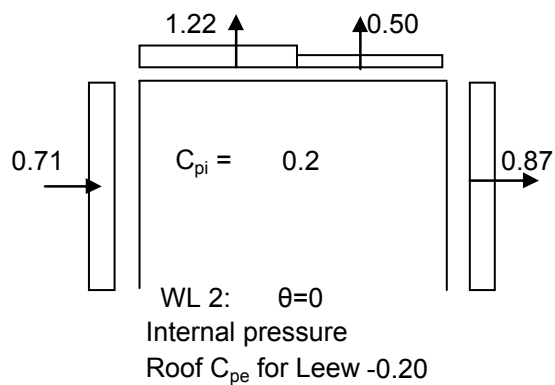
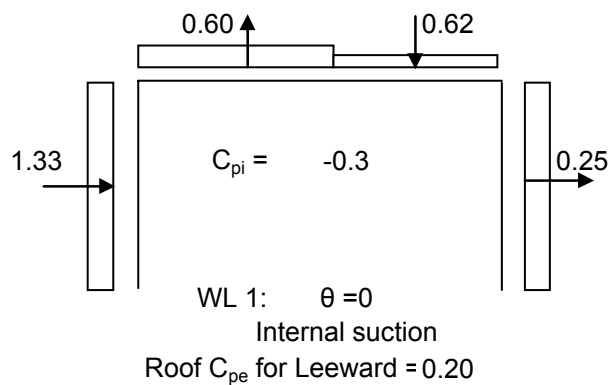
$$C_{pe} = 0.2$$

Wind Load Computation

$$\begin{aligned}
 D &= 17.00 \text{ m} \\
 H &= 8.41 \text{ m} \\
 D/H &= 2.02
 \end{aligned}$$

Element	Effective Wind Speed V_e m/s	Dynamic Wind Pr. q kN/m ²	External Pressure Coefficient, C_{pe}			
			$\theta = 0$ (normal to eave)		$\theta = 90$ (normal to gable end)	
			Windward	Leeward		
Roof	45	1.24	-0.78	-0.2 0.2	-0.7 0.2	(Zone C) (Zone D)
Wall	45	1.24	0.77	-0.5 -0.5	-0.8 -0.8	(Zone B) (Zone B)

Wind Load = $1 \times q \times (C_{pe} - C_{pi})$ kN/m (For unit width)



Parameters taken In STAAD For EQ Calculations

Design Information:

1) Seismic Zone	=	2A	(from Design Brief)
2) Occupancy Category	=	1	(Tanle-16-k, Conservatively adopt 1)
3) Lateral Force procedure	=	Static Procedure	(Section 1629.8 seismic zone 2A, structures not more than 5 stories)
4)Importance Factor	I	= 1	(Table 16-K)
5)Numeric. Coef: Strength & Ductility	R	= 4.5	(Table 16-N for OMRF - steel only)
	R	= 3.5	(Table 16-N for OMRF - Concrete only)
6)Soil Type	=	S _D	(from Design Brief)
7)C _t (as per clause 1630.2.2)	=	0.0853	(for steel moment resisting frame)
	=	0.0731	(for RC moment resisting frame)
8) Near Source Factor	N _a	= 1	(From Table 16-S)
9)Near Source Factor	N _v	= 1	(From Table 16-T)

In STAAD :

Z	=	0.15	(For Zone 2A)
I	=	1	
RWX =RWZ	=	4.5	(for steel structure)
	=	3.5	(for concrete structure)
STYP	=	4	(Type of Soil S _D)
C _t	=	0.0853	(for steel moment resisting frame)
	=	0.0731	(for RC moment resisting frame)
P _x	=	0.381	(for steel structure)
	=	0.48	(for concrete structure)
P _z	=	0.381	(for steel structure)
	=	0.48	(for concrete structure)
N _a	=	1	
N _v	=	1	

3 hour Fire rating considerations as per BS 8110-1:1997

(Table 3.4 Page 24, Fig. 3.2 Page 25)

		Code Requirement	Provided	Units
Concrete Beams →	Width	240	300/400	mm
	Cover	40	40	mm
Concrete Columns →	Width	400	500/550	mm
	Cover	25	50	mm
Concrete Slabs →	Thickness	150	175/225	mm
	Cover	35	35	mm

Load Cases For R.C. Structure

Basic Load Cases

Number	Name
21	EQX (EARTHQUAKE LOAD)
22	EQZ (EARTHQUAKE LOAD)
1	SELFWEIGHT
2	DL
3	LL
4	PH(X-DIR)
5	PH(Z-DIR)
11	WL1 (0 DEG ,INTERNAL SUCTION -0.3)
12	WL2 (0 DEG ,INTERNAL PRESSURE +0.2)
13	WL3 (90 DEG ,INTERNAL SUCTION -0.3, WW&LW +0.2)
14	WL4 (90 DEG ,INTERNAL PRESSURE +0.2, WW&LW -0.7)
15	WL5 (0 DEG ,INTERNAL SUCTION -0.3, LEE WARD -0.7,
16	WL6 (0 DEG ,INTERNAL PRESSURE +0.2, LEE WARD -0.2,

Combination Load Cases

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
101	1.0DL+1.0LL	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	1.00
102	1.0DL+0.5LL	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.50
103	0.8DL+0.8LL+0.8WL1	1	SELFWEIGHT	0.80
		2	DL	0.80
		3	LL	0.80
		11	WL1 (0 DEG ,INTERNAL SUCTION -0.3)	0.80
104	1.0DL+1.0WL1	1	SELFWEIGHT	1.00
		2	DL	1.00
		11	WL1 (0 DEG ,INTERNAL SUCTION -0.3)	1.00
105	0.8DL+0.8LL+0.8WL2	1	SELFWEIGHT	0.80
		2	DL	0.80
		3	LL	0.80
		12	WL2 (0 DEG ,INTERNAL PRESSURE	0.80
106	1.0DL+1.0WL2	1	SELFWEIGHT	1.00
		2	DL	1.00
		12	WL2 (0 DEG ,INTERNAL PRESSURE	1.00
107	0.8DL+0.8LL+0.8WL3	1	SELFWEIGHT	0.80
		2	DL	0.80
		3	LL	0.80
		13	WL3 (90 DEG ,INTERNAL SUCTION -0.3,	0.80
108	1.0DL+1.0WL3	1	SELFWEIGHT	1.00
		2	DL	1.00
		13	WL3 (90 DEG ,INTERNAL SUCTION -0.3,	1.00
109	0.8DL+0.8LL+0.8WL4	1	SELFWEIGHT	0.80
		2	DL	0.80
		3	LL	0.80
		14	WL4 (90 DEG ,INTERNAL PRESSURE	0.80
110	1.0DL+1.0WL4	1	SELFWEIGHT	1.00
		2	DL	1.00

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		14	WL4 (90 DEG ,INTERNAL PRESSURE	1.00
111	0.8DL+0.8LL+0.8WL5	1	SELFWEIGHT	0.80
		2	DL	0.80
		3	LL	0.80
		15	WL5 (0 DEG ,INTERNAL SUCTION -0.3,	0.80
112	1.0DL+1.0WL4	1	SELFWEIGHT	1.00
		2	DL	1.00
		15	WL5 (0 DEG ,INTERNAL SUCTION -0.3,	1.00
113	0.8DL+0.8LL+0.8WL6	1	SELFWEIGHT	0.80
		2	DL	0.80
		3	LL	0.80
		16	WL6 (0 DEG ,INTERNAL PRESSURE	0.80
114	1.0DL+1.0WL6	1	SELFWEIGHT	1.00
		2	DL	1.00
		16	WL6 (0 DEG ,INTERNAL PRESSURE	1.00
115	DL+0.72EQX+0.22EQZ	1	SELFWEIGHT	1.00
		2	DL	1.00
		21	EQX (EARTHQUAKE LOAD)	0.72
		22	EQZ (EARTHQUAKE LOAD)	0.22
116	DL+0.72EQX-0.22EQZ	1	SELFWEIGHT	1.00
		2	DL	1.00
		21	EQX (EARTHQUAKE LOAD)	0.72
		22	EQZ (EARTHQUAKE LOAD)	-0.22
117	DL-0.72EQX+0.22EQZ	1	SELFWEIGHT	1.00
		2	DL	1.00
		21	EQX (EARTHQUAKE LOAD)	-0.72
		22	EQZ (EARTHQUAKE LOAD)	0.22
118	DL-0.72EQX+0.22EQZ	1	SELFWEIGHT	1.00
		2	DL	1.00
		21	EQX (EARTHQUAKE LOAD)	-0.72
		22	EQZ (EARTHQUAKE LOAD)	-0.22
119	DL+0.72EQZ+0.22EQX	1	SELFWEIGHT	1.00
		2	DL	1.00
		22	EQZ (EARTHQUAKE LOAD)	0.72
		21	EQX (EARTHQUAKE LOAD)	0.22
120	DL+0.72EQZ-0.22EQX	1	SELFWEIGHT	1.00
		2	DL	1.00
		22	EQZ (EARTHQUAKE LOAD)	0.72
		21	EQX (EARTHQUAKE LOAD)	-0.22
121	DL-0.72EQZ+0.22EQX	1	SELFWEIGHT	1.00
		2	DL	1.00
		22	EQZ (EARTHQUAKE LOAD)	-0.72
		21	EQX (EARTHQUAKE LOAD)	0.22
122	DL-0.72EQZ-0.22EQX	1	SELFWEIGHT	1.00
		2	DL	1.00
		22	EQZ (EARTHQUAKE LOAD)	-0.72
		21	EQX (EARTHQUAKE LOAD)	-0.22
123	0.9DL+0.72EQX+0.22EQZ	1	SELFWEIGHT	0.90
		2	DL	0.90
		21	EQX (EARTHQUAKE LOAD)	0.72
		22	EQZ (EARTHQUAKE LOAD)	0.22
124	0.9DL+0.72EQX-0.22EQZ	1	SELFWEIGHT	0.90
		2	DL	0.90
		21	EQX (EARTHQUAKE LOAD)	0.72
		22	EQZ (EARTHQUAKE LOAD)	-0.22
125	0.9DL-0.72EQX+0.22EQZ	1	SELFWEIGHT	0.90
		2	DL	0.90
		21	EQX (EARTHQUAKE LOAD)	-0.72
		22	EQZ (EARTHQUAKE LOAD)	0.22
126	0.9DL-0.72EQX+0.22EQZ	1	SELFWEIGHT	0.90
		2	DL	0.90
		21	EQX (EARTHQUAKE LOAD)	-0.72
		22	EQZ (EARTHQUAKE LOAD)	-0.22

127	0.9DL+0.72EQZ+0.22EQX	1	SELFWEIGHT	0.90
		2	DL	0.90
		22	EQZ (EARTHQUAKE LOAD)	0.72
		21	EQX (EARTHQUAKE LOAD)	0.22
128	0.9DL+0.72EQZ-0.22EQX	1	SELFWEIGHT	0.90
		2	DL	0.90
		22	EQZ (EARTHQUAKE LOAD)	0.72
		21	EQX (EARTHQUAKE LOAD)	-0.22
129	0.9DL-0.72EQZ+0.22EQX	1	SELFWEIGHT	0.90
		2	DL	0.90
		22	EQZ (EARTHQUAKE LOAD)	-0.72
		21	EQX (EARTHQUAKE LOAD)	0.22
130	0.9DL-0.72EQZ-0.22EQX	1	SELFWEIGHT	0.90
		2	DL	0.90
		22	EQZ (EARTHQUAKE LOAD)	-0.72
		21	EQX (EARTHQUAKE LOAD)	-0.22
131	1.0DL+0.75LL+0.54EQX+0.17	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		21	EQX (EARTHQUAKE LOAD)	0.54
		22	EQZ (EARTHQUAKE LOAD)	0.17
132	1.0DL+0.75LL+0.54EQX-0.17	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		21	EQX (EARTHQUAKE LOAD)	0.54
		22	EQZ (EARTHQUAKE LOAD)	-0.17
133	1.0DL+0.75LL-0.54EQX+0.17	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		21	EQX (EARTHQUAKE LOAD)	-0.54
		22	EQZ (EARTHQUAKE LOAD)	0.17
134	1.0DL+0.75LL-0.54EQX-0.17 EQZ	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		21	EQX (EARTHQUAKE LOAD)	-0.54
		22	EQZ (EARTHQUAKE LOAD)	-0.17
135	1.0DL+0.75LL+0.54EQZ+0.17	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		22	EQZ (EARTHQUAKE LOAD)	0.54
		21	EQX (EARTHQUAKE LOAD)	0.17
136	1.0DL+0.75LL+0.54EQZ-0.17	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		22	EQZ (EARTHQUAKE LOAD)	0.54
		21	EQX (EARTHQUAKE LOAD)	-0.17
137	1.0DL+0.75LL-0.54EQZ+0.17	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		22	EQZ (EARTHQUAKE LOAD)	-0.54
		21	EQX (EARTHQUAKE LOAD)	0.17
138	1.0DL+0.75LL-0.54EQZ-0.17 EQX	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		22	EQZ (EARTHQUAKE LOAD)	-0.54
		21	EQX (EARTHQUAKE LOAD)	-0.17
200	1.4DL+1.6LL	1	SELFWEIGHT	1.40
		2	DL	1.40
		3	LL	1.60
201	1.2DL+1.2LL+1.2WL1	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	1.20
		11	WL1 (0 DEG ,INTERNAL SUCTION -0.3)	1.20

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202	1.4DL+1.4WL1	1	SELFWEIGHT	1.40
		2	DL	1.40
		11	WL1 (0 DEG ,INTERNAL SUCTION -0.3)	1.40
203	1DL+1.4WL1	1	SELFWEIGHT	1.00
		2	DL	1.00
		11	WL1 (0 DEG ,INTERNAL SUCTION -0.3)	1.40
204	1.2DL+1.2LL+1.2WL2	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	1.20
		12	WL2 (0 DEG ,INTERNAL PRESSURE	1.20
205	1.4DL+1.4WL2	1	SELFWEIGHT	1.40
		2	DL	1.40
		12	WL2 (0 DEG ,INTERNAL PRESSURE	1.40
206	1DL+1.4WL2	1	SELFWEIGHT	1.00
		2	DL	1.00
		12	WL2 (0 DEG ,INTERNAL PRESSURE	1.40
207	1.2DL+1.2LL+1.2WL3	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	1.20
		13	WL3 (90 DEG ,INTERNAL SUCTION -0.3,	1.20
208	1.4DL+1.4WL3	1	SELFWEIGHT	1.40
		2	DL	1.40
		13	WL3 (90 DEG ,INTERNAL SUCTION -0.3,	1.40
209	1DL+1.4WL3	1	SELFWEIGHT	1.00
		2	DL	1.00
		13	WL3 (90 DEG ,INTERNAL SUCTION -0.3,	1.40
210	1.2DL+1.2LL+1.2WL4	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	1.20
		14	WL4 (90 DEG ,INTERNAL PRESSURE	1.20
211	1.4DL+1.4WL4	1	SELFWEIGHT	1.40
		2	DL	1.40
		14	WL4 (90 DEG ,INTERNAL PRESSURE	1.40
212	1DL+1.4WL4	1	SELFWEIGHT	1.00
		2	DL	1.00
		14	WL4 (90 DEG ,INTERNAL PRESSURE	1.40
213	1.2DL+1.2LL+1.2WL5	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	1.20
		15	WL5 (0 DEG ,INTERNAL SUCTION -0.3,	1.20
214	1.4DL+1.4WL5	1	SELFWEIGHT	1.40
		2	DL	1.40
		15	WL5 (0 DEG ,INTERNAL SUCTION -0.3,	1.40
215	1DL+1.4WL5	1	SELFWEIGHT	1.00
		2	DL	1.00
		15	WL5 (0 DEG ,INTERNAL SUCTION -0.3,	1.40
216	1.2DL+1.2LL+1.2WL6	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	1.20
		16	WL6 (0 DEG ,INTERNAL PRESSURE	1.20
217	1.4DL+1.4WL6	1	SELFWEIGHT	1.40
		2	DL	1.40
		16	WL6 (0 DEG ,INTERNAL PRESSURE	1.40
218	1DL+1.4WL6	1	SELFWEIGHT	1.00
		2	DL	1.00
		16	WL6 (0 DEG ,INTERNAL PRESSURE	1.40
219	1.32DL+1.1LL+1.1EQX+0.33 EQZ	1	SELFWEIGHT	1.32
		2	DL	1.32
		3	LL	1.10
		21	EQX (EARTHQUAKE LOAD)	1.10
		22	EQZ (EARTHQUAKE LOAD)	0.33
220	1.32DL+1.1LL+1.1EQX-0.33 EQZ	1	SELFWEIGHT	1.32
		2	DL	1.32

		3	LL	1.10
		21	EQX (EARTHQUAKE LOAD)	1.10
		22	EQZ (EARTHQUAKE LOAD)	-0.33
221	1.32DL+1.1LL-1.1EQX+0.33 EQZ	1	SELFWEIGHT	1.32
		2	DL	1.32
		3	LL	1.10
		21	EQX (EARTHQUAKE LOAD)	-1.10
		22	EQZ (EARTHQUAKE LOAD)	0.33
222	1.32DL+1.1LL-1.1EQX-0.33 EQZ	1	SELFWEIGHT	1.32
		2	DL	1.32
		3	LL	1.10
		21	EQX (EARTHQUAKE LOAD)	-1.10
		22	EQZ (EARTHQUAKE LOAD)	-0.33
223	1.32DL+1.1LL+1.1EQZ+0.33 EQX	1	SELFWEIGHT	1.32
		2	DL	1.32
		3	LL	1.10
		22	EQZ (EARTHQUAKE LOAD)	1.10
		21	EQX (EARTHQUAKE LOAD)	0.33
224	1.32DL+1.1LL+1.1EQZ-0.33 EQX	1	SELFWEIGHT	1.32
		2	DL	1.32
		3	LL	1.10
		22	EQZ (EARTHQUAKE LOAD)	1.10
		21	EQX (EARTHQUAKE LOAD)	-0.33
225	1.32DL+1.1LL-1.1EQZ+0.33 EQX	1	SELFWEIGHT	1.32
		2	DL	1.32
		3	LL	1.10
		22	EQZ (EARTHQUAKE LOAD)	-1.10
		21	EQX (EARTHQUAKE LOAD)	0.33
226	1.32DL+1.1LL-1.1EQZ-0.33 EQX	1	SELFWEIGHT	1.32
		2	DL	1.32
		3	LL	1.10
		22	EQZ (EARTHQUAKE LOAD)	-1.10
		21	EQX (EARTHQUAKE LOAD)	-0.33
227	0.99DL+1.1EQX+0.33EQZ	1	SELFWEIGHT	0.99
		2	DL	0.99
		21	EQX (EARTHQUAKE LOAD)	1.10
		22	EQZ (EARTHQUAKE LOAD)	0.33
228	0.99DL+1.1EQX-0.33EQZ	1	SELFWEIGHT	0.99
		2	DL	0.99
		21	EQX (EARTHQUAKE LOAD)	1.10
		22	EQZ (EARTHQUAKE LOAD)	-0.33
229	0.99DL-1.1EQX+0.33EQZ	1	SELFWEIGHT	0.99
		2	DL	0.99
		21	EQX (EARTHQUAKE LOAD)	-1.10
		22	EQZ (EARTHQUAKE LOAD)	0.33
230	0.99DL-1.1EQX-0.33EQZ	1	SELFWEIGHT	0.99
		2	DL	0.99
		21	EQX (EARTHQUAKE LOAD)	-1.10
		22	EQZ (EARTHQUAKE LOAD)	-0.33
231	0.99DL+1.1EQZ+0.33EQX	1	SELFWEIGHT	0.99
		2	DL	0.99
		22	EQZ (EARTHQUAKE LOAD)	1.10
		21	EQX (EARTHQUAKE LOAD)	0.33
232	0.99DL+1.1EQZ-0.33EQX	1	SELFWEIGHT	0.99
		2	DL	0.99
		22	EQZ (EARTHQUAKE LOAD)	1.10
		21	EQX (EARTHQUAKE LOAD)	-0.33
233	0.99DL-1.1EQZ+0.33EQX	1	SELFWEIGHT	0.99
		2	DL	0.99
		22	EQZ (EARTHQUAKE LOAD)	-1.10
		21	EQX (EARTHQUAKE LOAD)	0.33
234	0.99DL-1.1EQZ-0.33EQX	1	SELFWEIGHT	0.99
		2	DL	0.99

		22	EQZ (EARTHQUAKE LOAD)	-1.10
		21	EQX (EARTHQUAKE LOAD)	-0.33
235	1.0DL+1.0LL+1.0PHX	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	1.00
		4	PH(X-DIR)	1.00
236	1.0DL+1.0LL+1.0PHZ	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	1.00
		5	PH(Z-DIR)	1.00
301	SELF+DL	1	SELFWEIGHT	1.00
		2	DL	1.00
302	LL	3	LL	1.00
303	0.8(LL+WL1)	3	LL	0.80
		11	WL1 (0 DEG ,INTERNAL SUCTION -0.3)	0.80
304	0.8(LL+WL2)	3	LL	0.80
		12	WL2 (0 DEG ,INTERNAL PRESSURE	0.80
305	0.8(LL+WL3)	3	LL	0.80
		13	WL3 (90 DEG ,INTERNAL SUCTION -0.3,	0.80
306	0.8(LL+WL4)	3	LL	0.80
		14	WL4 (90 DEG ,INTERNAL PRESSURE	0.80
307	0.8(LL+WL5)	3	LL	0.80
		15	WL5 (0 DEG ,INTERNAL SUCTION -0.3,	0.80
308	0.8(LL+WL6)	3	LL	0.80
		16	WL6 (0 DEG ,INTERNAL PRESSURE	0.80
309	0.8(LL+EQX/1.4+0.3EQZ/1.4)	3	LL	0.80
		21	EQX (EARTHQUAKE LOAD)	0.57
		22	EQZ (EARTHQUAKE LOAD)	0.17
310	0.8(LL+EQX/1.4-0.3EQZ/1.4)	3	LL	0.80
		21	EQX (EARTHQUAKE LOAD)	0.57
		22	EQZ (EARTHQUAKE LOAD)	-0.17
311	0.8(LL-EQX/1.4+0.3EQZ/1.4)	3	LL	0.80
		21	EQX (EARTHQUAKE LOAD)	-0.57
		22	EQZ (EARTHQUAKE LOAD)	0.17
312	0.8(LL-EQX/1.4-0.3EQZ/1.4)	3	LL	0.80
		21	EQX (EARTHQUAKE LOAD)	-0.57
		22	EQZ (EARTHQUAKE LOAD)	-0.17
313	0.8(LL+EQZ/1.4+0.3EQX/1.4)	3	LL	0.80
		22	EQZ (EARTHQUAKE LOAD)	0.57
		21	EQX (EARTHQUAKE LOAD)	0.17
314	0.8(LL+EQZ/1.4-0.3EQX/1.4)	3	LL	0.80
		22	EQZ (EARTHQUAKE LOAD)	0.57
		21	EQX (EARTHQUAKE LOAD)	-0.17
315	0.8(LL-EQZ/1.4+0.3EQX/1.4)	3	LL	0.80
		21	EQX (EARTHQUAKE LOAD)	-0.57
		22	EQZ (EARTHQUAKE LOAD)	0.17
316	0.8(LL-EQZ/1.4-0.3EQX/1.4)	3	LL	0.80
		21	EQX (EARTHQUAKE LOAD)	-0.57
		22	EQZ (EARTHQUAKE LOAD)	-0.17

Load Cases For Steel Structure

Basic Load Cases

Number	Name
21	EQX (EARTHQUAKE LOAD)
22	EQZ (EARTHQUAKE LOAD)
1	SELFWEIGHT
2	DL
3	LL
4	PH(X-DIR)
5	PH(Z-DIR)
11	WL1 (0 DEG,INTERNAL SUCTION (CPI=-0.3)
12	WL2 (0 DEG ,INTERNAL PRESSURE CPI=+0.2)
13	WL3 (90 DEG ,INTERNAL SUCTION CPI=-0.3, WW&LW CPE=+0.2)
14	WL4 (90 DEG ,INTERNAL PRESSURE CPI=+0.2, WW&LW CPE=-0.7)
15	WL5 (0 DEG ,INTERNAL SUCTION CPI=-0.3, LEE WARD -0.7, REVERSE)
16	WL6 (0 DEG ,INTERNAL PRESSURE CPI=+0.2, LEE WARD -0.2, REVERSE)

Combination Load Cases

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
101	1.0DL+1.0LL	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	1.00
102	1.0DL+0.5LL	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.50
103	0.8DL+0.8LL+0.8WL1	1	SELFWEIGHT	0.80
		2	DL	0.80
		3	LL	0.80
		11	WL1 (0 DEG,INTERNAL SUCTION (CPI=-	0.80
104	1.0DL+1.0WL1	1	SELFWEIGHT	1.00
		2	DL	1.00
		11	WL1 (0 DEG,INTERNAL SUCTION (CPI=-	1.00
105	0.8DL+0.8LL+0.8WL2	1	SELFWEIGHT	0.80
		2	DL	0.80
		3	LL	0.80
		12	WL2 (0 DEG ,INTERNAL PRESSURE	0.80
106	1.0DL+1.0WL2	1	SELFWEIGHT	1.00
		2	DL	1.00
		12	WL2 (0 DEG ,INTERNAL PRESSURE	1.00
107	0.8DL+0.8LL+0.8WL3	1	SELFWEIGHT	0.80
		2	DL	0.80
		3	LL	0.80
		13	WL3 (90 DEG ,INTERNAL SUCTION CPI=-	0.80
108	1.0DL+1.0WL3	1	SELFWEIGHT	1.00
		2	DL	1.00
		13	WL3 (90 DEG ,INTERNAL SUCTION CPI=-	1.00
109	0.8DL+0.8LL+0.8WL4	1	SELFWEIGHT	0.80
		2	DL	0.80
		3	LL	0.80
		14	WL4 (90 DEG ,INTERNAL PRESSURE	0.80
110	1.0DL+1.0WL4	1	SELFWEIGHT	1.00
		2	DL	1.00

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		14	WL4 (90 DEG ,INTERNAL PRESSURE	1.00
111	0.8DL+0.8LL+0.8WL5	1	SELFWEIGHT	0.80
		2	DL	0.80
		3	LL	0.80
		15	WL5 (0 DEG ,INTERNAL SUCTION CPI=-	0.80
112	1.0DL+1.0WL4	1	SELFWEIGHT	1.00
		2	DL	1.00
		15	WL5 (0 DEG ,INTERNAL SUCTION CPI=-	1.00
113	0.8DL+0.8LL+0.8WL6	1	SELFWEIGHT	0.80
		2	DL	0.80
		3	LL	0.80
		16	WL6 (0 DEG ,INTERNAL PRESSURE	0.80
114	1.0DL+1.0WL6	1	SELFWEIGHT	1.00
		2	DL	1.00
		16	WL6 (0 DEG ,INTERNAL PRESSURE	1.00
115	DL+0.72EQX+0.22EQZ	1	SELFWEIGHT	1.00
		2	DL	1.00
		21	EQX (EARTHQUAKE LOAD)	0.72
		22	EQZ (EARTHQUAKE LOAD)	0.22
116	DL+0.72EQX-0.22EQZ	1	SELFWEIGHT	1.00
		2	DL	1.00
		21	EQX (EARTHQUAKE LOAD)	0.72
		22	EQZ (EARTHQUAKE LOAD)	-0.22
117	DL-0.72EQX+0.22EQZ	1	SELFWEIGHT	1.00
		2	DL	1.00
		21	EQX (EARTHQUAKE LOAD)	-0.72
		22	EQZ (EARTHQUAKE LOAD)	0.22
118	DL-0.72EQX+0.22EQZ	1	SELFWEIGHT	1.00
		2	DL	1.00
		21	EQX (EARTHQUAKE LOAD)	-0.72
		22	EQZ (EARTHQUAKE LOAD)	-0.22
119	DL+0.72EQZ+0.22EQX	1	SELFWEIGHT	1.00
		2	DL	1.00
		22	EQZ (EARTHQUAKE LOAD)	0.72
		21	EQX (EARTHQUAKE LOAD)	0.22
120	DL+0.72EQZ-0.22EQX	1	SELFWEIGHT	1.00
		2	DL	1.00
		22	EQZ (EARTHQUAKE LOAD)	0.72
		21	EQX (EARTHQUAKE LOAD)	-0.22
121	DL-0.72EQZ+0.22EQX	1	SELFWEIGHT	1.00
		2	DL	1.00
		22	EQZ (EARTHQUAKE LOAD)	-0.72
		21	EQX (EARTHQUAKE LOAD)	0.22
122	DL-0.72EQZ-0.22EQX	1	SELFWEIGHT	1.00
		2	DL	1.00
		22	EQZ (EARTHQUAKE LOAD)	-0.72
		21	EQX (EARTHQUAKE LOAD)	-0.22
123	0.9DL+0.72EQX+0.22EQZ	1	SELFWEIGHT	0.90
		2	DL	0.90
		21	EQX (EARTHQUAKE LOAD)	0.72
		22	EQZ (EARTHQUAKE LOAD)	0.22
124	0.9DL+0.72EQX-0.22EQZ	1	SELFWEIGHT	0.90
		2	DL	0.90
		21	EQX (EARTHQUAKE LOAD)	0.72
		22	EQZ (EARTHQUAKE LOAD)	-0.22
125	0.9DL-0.72EQX+0.22EQZ	1	SELFWEIGHT	0.90
		2	DL	0.90
		21	EQX (EARTHQUAKE LOAD)	-0.72
		22	EQZ (EARTHQUAKE LOAD)	0.22
126	0.9DL-0.72EQX+0.22EQZ	1	SELFWEIGHT	0.90
		2	DL	0.90
		21	EQX (EARTHQUAKE LOAD)	-0.72
		22	EQZ (EARTHQUAKE LOAD)	-0.22

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127	0.9DL+0.72EQZ+0.22EQX	1	SELFWEIGHT	0.90
		2	DL	0.90
		22	EQZ (EARTHQUAKE LOAD)	0.72
		21	EQX (EARTHQUAKE LOAD)	0.22
128	0.9DL+0.72EQZ-0.22EQX	1	SELFWEIGHT	0.90
		2	DL	0.90
		22	EQZ (EARTHQUAKE LOAD)	0.72
		21	EQX (EARTHQUAKE LOAD)	-0.22
129	0.9DL-0.72EQZ+0.22EQX	1	SELFWEIGHT	0.90
		2	DL	0.90
		22	EQZ (EARTHQUAKE LOAD)	-0.72
		21	EQX (EARTHQUAKE LOAD)	0.22
130	0.9DL-0.72EQZ-0.22EQX	1	SELFWEIGHT	0.90
		2	DL	0.90
		22	EQZ (EARTHQUAKE LOAD)	-0.72
		21	EQX (EARTHQUAKE LOAD)	-0.22
131	1.0DL+0.75LL+0.54EQX+0.17	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		21	EQX (EARTHQUAKE LOAD)	0.54
		22	EQZ (EARTHQUAKE LOAD)	0.17
132	1.0DL+0.75LL+0.54EQX-0.17	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		21	EQX (EARTHQUAKE LOAD)	0.54
		22	EQZ (EARTHQUAKE LOAD)	-0.17
133	1.0DL+0.75LL-0.54EQX+0.17	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		21	EQX (EARTHQUAKE LOAD)	-0.54
		22	EQZ (EARTHQUAKE LOAD)	0.17
134	1.0DL+0.75LL-0.54EQX-0.17 EQZ	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		21	EQX (EARTHQUAKE LOAD)	-0.54
		22	EQZ (EARTHQUAKE LOAD)	-0.17
135	1.0DL+0.75LL+0.54EQZ+0.17	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		22	EQZ (EARTHQUAKE LOAD)	0.54
		21	EQX (EARTHQUAKE LOAD)	0.17
136	1.0DL+0.75LL+0.54EQZ-0.17	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		22	EQZ (EARTHQUAKE LOAD)	0.54
		21	EQX (EARTHQUAKE LOAD)	-0.17
137	1.0DL+0.75LL-0.54EQZ+0.17	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		22	EQZ (EARTHQUAKE LOAD)	-0.54
		21	EQX (EARTHQUAKE LOAD)	0.17
138	1.0DL+0.75LL-0.54EQZ-0.17 EQX	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		22	EQZ (EARTHQUAKE LOAD)	-0.54
		21	EQX (EARTHQUAKE LOAD)	-0.17
200	1.4DL+1.6LL	1	SELFWEIGHT	1.40
		2	DL	1.40
		3	LL	1.60
201	1.2DL+1.2LL+1.2WL1	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	1.20
		11	WL1 (0 DEG,INTERNAL SUCTION (CPI=-	1.20

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202	1.4DL+1.4WL1	1	SELFWEIGHT	1.40
		2	DL	1.40
		11	WL1 (0 DEG,INTERNAL SUCTION (CPI=-	1.40
203	1DL+1.4WL1	1	SELFWEIGHT	1.00
		2	DL	1.00
		11	WL1 (0 DEG,INTERNAL SUCTION (CPI=-	1.40
204	1.2DL+1.2LL+1.2WL2	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	1.20
		12	WL2 (0 DEG ,INTERNAL PRESSURE	1.20
205	1.4DL+1.4WL2	1	SELFWEIGHT	1.40
		2	DL	1.40
		12	WL2 (0 DEG ,INTERNAL PRESSURE	1.40
206	1DL+1.4WL2	1	SELFWEIGHT	1.00
		2	DL	1.00
		12	WL2 (0 DEG ,INTERNAL PRESSURE	1.40
207	1.2DL+1.2LL+1.2WL3	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	1.20
		13	WL3 (90 DEG ,INTERNAL SUCTION CPI=-	1.20
208	1.4DL+1.4WL3	1	SELFWEIGHT	1.40
		2	DL	1.40
		13	WL3 (90 DEG ,INTERNAL SUCTION CPI=-	1.40
209	1DL+1.4WL3	1	SELFWEIGHT	1.00
		2	DL	1.00
		13	WL3 (90 DEG ,INTERNAL SUCTION CPI=-	1.40
210	1.2DL+1.2LL+1.2WL4	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	1.20
		14	WL4 (90 DEG ,INTERNAL PRESSURE	1.20
211	1.4DL+1.4WL4	1	SELFWEIGHT	1.40
		2	DL	1.40
		14	WL4 (90 DEG ,INTERNAL PRESSURE	1.40
212	1DL+1.4WL4	1	SELFWEIGHT	1.00
		2	DL	1.00
		14	WL4 (90 DEG ,INTERNAL PRESSURE	1.40
213	1.2DL+1.2LL+1.2WL5	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	1.20
		15	WL5 (0 DEG ,INTERNAL SUCTION CPI=-	1.20
214	1.4DL+1.4WL5	1	SELFWEIGHT	1.40
		2	DL	1.40
		15	WL5 (0 DEG ,INTERNAL SUCTION CPI=-	1.40
215	1DL+1.4WL5	1	SELFWEIGHT	1.00
		2	DL	1.00
		15	WL5 (0 DEG ,INTERNAL SUCTION CPI=-	1.40
216	1.2DL+1.2LL+1.2WL6	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	1.20
		16	WL6 (0 DEG ,INTERNAL PRESSURE	1.20
217	1.4DL+1.4WL6	1	SELFWEIGHT	1.40
		2	DL	1.40
		16	WL6 (0 DEG ,INTERNAL PRESSURE	1.40
218	1DL+1.4WL6	1	SELFWEIGHT	1.00
		2	DL	1.00
		16	WL6 (0 DEG ,INTERNAL PRESSURE	1.40
219	1.2DL+0.5LL+1.0EQX+0.3 EQZ	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	0.50
		21	EQX (EARTHQUAKE LOAD)	1.00
		22	EQZ (EARTHQUAKE LOAD)	0.30
220	1.2DL+0.5LL+1.0EQX-0.3 EQZ	1	SELFWEIGHT	1.20
		2	DL	1.20

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		3	LL	0.50
		21	EQX (EARTHQUAKE LOAD)	1.00
		22	EQZ (EARTHQUAKE LOAD)	-0.30
221	1.2DL+0.5LL-1.0EQX+0.3 EQZ	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	0.50
		21	EQX (EARTHQUAKE LOAD)	-1.00
		22	EQZ (EARTHQUAKE LOAD)	0.30
222	1.2DL+0.5LL-1.0EQX-0.3 EQZ	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	0.50
		21	EQX (EARTHQUAKE LOAD)	-1.00
		22	EQZ (EARTHQUAKE LOAD)	-0.30
223	1.2DL+0.5LL+1.0EQZ+0.3 EQX	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	0.50
		22	EQZ (EARTHQUAKE LOAD)	1.00
		21	EQX (EARTHQUAKE LOAD)	0.30
224	1.2DL+0.5LL+1.0EQZ-0.3 EQX	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	0.50
		22	EQZ (EARTHQUAKE LOAD)	1.00
		21	EQX (EARTHQUAKE LOAD)	-0.30
225	1.2DL+0.5LL-1.0EQZ+0.3 EQX	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	0.50
		22	EQZ (EARTHQUAKE LOAD)	-1.00
		21	EQX (EARTHQUAKE LOAD)	0.30
226	1.2DL+0.5LL-1.0EQZ-0.3 EQX	1	SELFWEIGHT	1.00
		2	DL	1.20
		3	LL	0.50
		22	EQZ (EARTHQUAKE LOAD)	-1.00
		21	EQX (EARTHQUAKE LOAD)	-0.30
227	0.9DL+1.0EQX+0.3EQZ	1	SELFWEIGHT	0.90
		2	DL	0.90
		21	EQX (EARTHQUAKE LOAD)	1.00
		22	EQZ (EARTHQUAKE LOAD)	0.30
228	0.9DL+1.0EQX-0.3EQZ	1	SELFWEIGHT	0.90
		2	DL	0.90
		21	EQX (EARTHQUAKE LOAD)	1.00
		22	EQZ (EARTHQUAKE LOAD)	-0.30
229	0.9DL-1.0EQX+0.3EQZ	1	SELFWEIGHT	0.90
		2	DL	0.90
		21	EQX (EARTHQUAKE LOAD)	-1.00
		22	EQZ (EARTHQUAKE LOAD)	0.30
230	0.9DL-1.0EQX+0.3EQZ	1	SELFWEIGHT	0.90
		2	DL	0.90
		21	EQX (EARTHQUAKE LOAD)	-1.00
		22	EQZ (EARTHQUAKE LOAD)	-0.30
231	0.9DL+1.0EQZ+0.3EQX	1	SELFWEIGHT	0.90
		2	DL	0.90
		22	EQZ (EARTHQUAKE LOAD)	1.00
		21	EQX (EARTHQUAKE LOAD)	0.30
232	0.9DL+1.0EQZ-0.3EQX	1	SELFWEIGHT	0.90
		2	DL	0.90
		22	EQZ (EARTHQUAKE LOAD)	1.00
		21	EQX (EARTHQUAKE LOAD)	-0.30
233	0.9DL-1.0EQZ+0.3EQX	1	SELFWEIGHT	0.90
		2	DL	0.90
		22	EQZ (EARTHQUAKE LOAD)	-1.00
		21	EQX (EARTHQUAKE LOAD)	0.30
234	0.9DL-1.0EQZ-0.3EQX	1	SELFWEIGHT	0.90
		2	DL	0.90

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		22	EQZ (EARTHQUAKE LOAD)	-1.00
		21	EQX (EARTHQUAKE LOAD)	-0.30
235	1.0DL+1.0LL+1.0PHX	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	1.00
		4	PH(X-DIR)	1.00
236	1.0DL+1.0LL+1.0PHZ	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	1.00
		5	PH(Z-DIR)	1.00
301	SELF+DL	1	SELFWEIGHT	1.00
		2	DL	1.00
302	LL	3	LL	1.00
303	0.8(LL+WL1)	3	LL	0.80
		11	WL1 (0 DEG,INTERNAL SUCTION (CPI=-	0.80
304	0.8(LL+WL2)	3	LL	0.80
		12	WL2 (0 DEG ,INTERNAL PRESSURE	0.80
305	0.8(LL+WL3)	3	LL	0.80
		13	WL3 (90 DEG ,INTERNAL SUCTION CPI=-	0.80
306	0.8(LL+WL4)	3	LL	0.80
		14	WL4 (90 DEG ,INTERNAL PRESSURE	0.80
307	0.8(LL+WL5)	3	LL	0.80
		15	WL5 (0 DEG ,INTERNAL SUCTION CPI=-	0.80
308	0.8(LL+WL6)	3	LL	0.80
		16	WL6 (0 DEG ,INTERNAL PRESSURE	0.80
309	0.8(LL+EQX/1.4+0.3EQZ/1.4)	3	LL	0.80
		21	EQX (EARTHQUAKE LOAD)	0.57
		22	EQZ (EARTHQUAKE LOAD)	0.17
310	0.8(LL+EQX/1.4-0.3EQZ/1.4)	3	LL	0.80
		21	EQX (EARTHQUAKE LOAD)	0.57
		22	EQZ (EARTHQUAKE LOAD)	-0.17
311	0.8(LL-EQX/1.4+0.3EQZ/1.4)	3	LL	0.80
		21	EQX (EARTHQUAKE LOAD)	-0.57
		22	EQZ (EARTHQUAKE LOAD)	0.17
312	0.8(LL-EQX/1.4-0.3EQZ/1.4)	3	LL	0.80
		21	EQX (EARTHQUAKE LOAD)	-0.57
		22	EQZ (EARTHQUAKE LOAD)	-0.17
313	0.8(LL+EQZ/1.4+0.3EQX/1.4)	3	LL	0.80
		22	EQZ (EARTHQUAKE LOAD)	0.57
		21	EQX (EARTHQUAKE LOAD)	0.17
314	0.8(LL+EQZ/1.4-0.3EQX/1.4)	3	LL	0.80
		22	EQZ (EARTHQUAKE LOAD)	0.57
		21	EQX (EARTHQUAKE LOAD)	-0.17
315	0.8(LL-EQZ/1.4+0.3EQX/1.4)	3	LL	0.80
		22	EQZ (EARTHQUAKE LOAD)	-0.57
		21	EQX (EARTHQUAKE LOAD)	0.17
316	0.8(LL-EQZ/1.4-0.3EQX/1.4)	3	LL	0.80
		22	EQZ (EARTHQUAKE LOAD)	-0.57
		21	EQX (EARTHQUAKE LOAD)	-0.17

DISPLACEMENTS SUMMARY

			Horizontal	Vertical	Horizontal	Resultant	Rotational		
	Node	L/C	X mm	Y mm	Z mm	mm	rX deg	rY deg	rZ deg
Max X	42	310 0.8(LL+EQX/1.4-0.3EQZ/1.4)	1.707	0.007	0.002	1.707	-0.001	0.056	-0.033
Min X	3	312 0.8(LL-EQX/1.4-0.3EQZ/1.4)	-1.707	0.007	0.002	1.707	-0.001	-0.056	0.033
Max Y	59	310 0.8(LL+EQX/1.4-0.3EQZ/1.4)	0.772	2.028	0.024	2.17	-0.004	-0.001	-0.034
Min Y	61	301 SELF+DL	0.015	-4.872	0.008	4.872	-0.002	0	-0.077
Max Z	82	303 0.8(LL+WL1)	-0.001	0.253	2.949	2.96	0.001	0.027	-0.003
Min Z	77	307 0.8(LL+WL5)	0.001	0.253	-2.95	2.96	-0.001	0.027	0.003
Max rX	26	303 0.8(LL+WL1)	0	0	0	0	0.096	0.027	-0.003
Min rX	27	307 0.8(LL+WL5)	0	0	0	0	-0.096	0.027	0.003
Max rY	83	301 SELF+DL	0.003	-0.081	-0.193	0.209	0.001	0.249	-0.03
Min rY	77	301 SELF+DL	0.003	-0.081	0.192	0.209	-0.001	-0.249	-0.03
Max rZ	110	312 0.8(LL-EQX/1.4-0.3EQZ/1.4)	0	0	0	0	-0.003	0	0.416
Min rZ	104	310 0.8(LL+EQX/1.4-0.3EQZ/1.4)	0	0	0	0	-0.003	0	-0.415
Max Rst	61	301 SELF+DL	0.015	-4.872	0.008	4.872	-0.002	0	-0.077

↓ ↓
MAXIMUM DEFLECTION

ALLOWABLE DEFLECTION (IN MM)

Deflection in Y	Span/180	3200/180	17.77
Deflection in X	HEIGHT/300	6700/300	22.33
Deflection in Z	HEIGHT/300	4800/300	16

Column Summary

Ref. node on STAAD	Location on drawing	Size of column				Reinforcement Req.		Reinforcement Prov.	
		Stump to Level - 1		Level - 1 to level - 2		Stump to Level - 1 (cm ²)	Level - 1 to level - 2 (cm ²)	Stump to Level - 1 (cm ²)	Level - 1 to level - 2 (cm ²)
		Width (b) (mm)	Depth (h) (mm)	Width (b) (mm)	Depth (h) (mm)				
1	A-1	550	550	500	500	21	48	50	50
2	A-2	550	550	500	500	18	58	59	59
3	A-3	550	550	500	500	32	60	79	79
4	A-4	550	550	500	500	25	50	50	50
5	B-1	550	550	500	500	47	66	79	79
6	B-2	550	550	500	500	34	73	79	79
7	B-3	550	550	500	500	40	72	79	79
8	B-4	550	550	500	500	43	64	79	79
9	C-1	550	550	500	500	52	68	79	79
10	C-2	550	550	500	500	36	73	79	79
11	C-3	550	550	500	500	26	73	79	79
12	C-4	550	550	500	500	38	65	79	79
13	D-1	550	550	500	500	29	51	59	59
14	D-2	550	550	500	500	20	56	59	59
15	D-3	550	550	500	500	16	55	59	59
16	D-4	550	550	500	500	20	49	50	50

Note : 16T-20 = 50 cm²
 12T-25 = 59 cm²
 16T-25 = 79 cm²

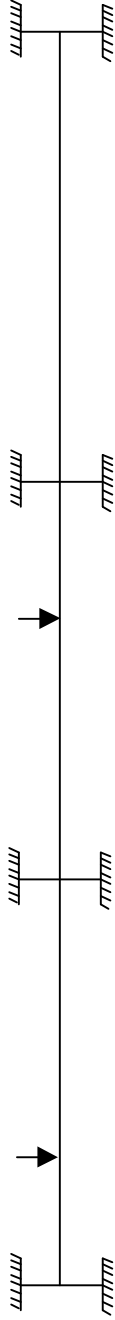
Bar combinations used →

For 400mm x 600mm beams

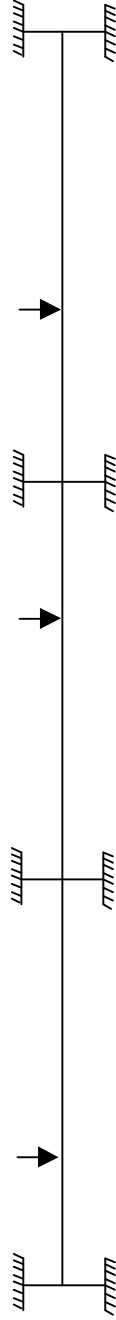
(No + Bar dia)	$A_{st} \text{ (cm}^2\text{)}$
4T20	= 13
4T25	= 20
4T32	= 32
4T25 + 2T20	= 26
4T25 + 4T20	= 32
4T25 + 4T25	= 39
4T25 + 4T20 + 2T20	= 38
4T25 + 4T25 + 4T20	= 52
4T25 + 4T25 + 2T20	= 46
4T25 + 4T25 + 2T25	= 49
4T25 + 4T25 + 4T25	= 59
4T32 + 4T25 + 2T25	= 62
4T32 + 4T25 + 4T25	= 71

For 300mm x 600mm beams

(No + Bar dia)	$A_{st} \text{ (cm}^2\text{)}$
3T20	= 9
3T25	= 15
3T20 + 2T16	= 13
3T25 + 2T16	= 19
3T25 + 2T20	= 21
3T25 + 3T20	= 24

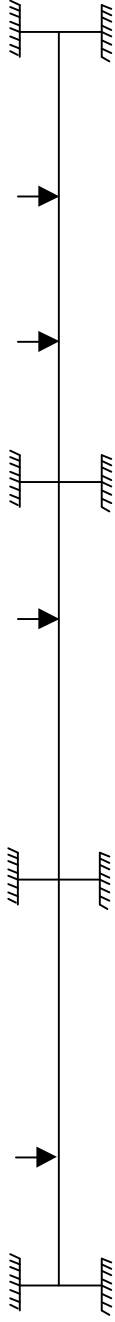


Ref Beam no in STAAD	1	2	3	4	5
Beam marking on Drg.	1BA/1-2		1BA/2-3		1BA/3-4
Reinforcement Req./Prov. (cm ²)	38/39 8/13	0/13 12/13	45/46 5/13	38/46 6/13	30/32 10/13
Stirrups	T10-149 T10-100	T10-195 T10-150	T10-164 T10-100	T10-181 T10-150	T10-270 T10-200

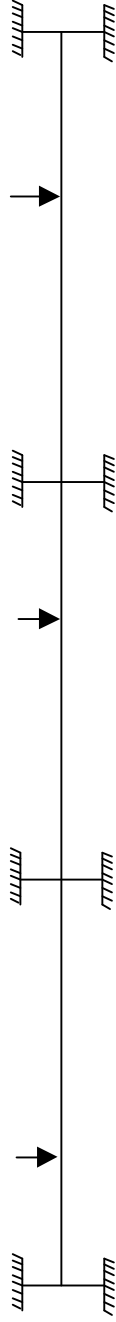


Ref Beam no in STAAD	10	11	12	13	14	15
Beam marking on Drg.	1BB/1-2		1BB/2-3		1BB/3-4	
Reinforcement Req./Prov. (cm ²)	53/62 11/26	3/62 22/26	58/62 16/20	49/62 10/13	54/71 10/13	66/71 24/32
Stirrups	T10-99 2T10-150	T10-179 T10-150	T10-139 T10-100	T10-164 T10-150	T10-102 T10-100	T10-110 T10-100

Note : All beam sections are 400mm x 600mm

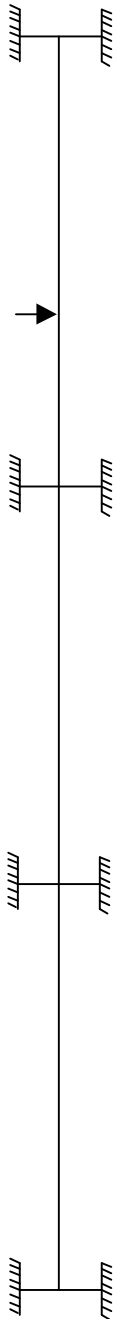


Ref Beam no in STAAD	16	17	18	19	20	21	54
Beam marking on Drg.	1BC/1-2						
Reinforcement Req./Prov. (cm ²)	56/62 14/20	4/62 15/20	61/62 52/62 19/20 10/20	15/20 11/20	56/62 58/62 15/20 16/20	15/52 13/20	49/52 13/20
Stirrups	Req. T10-89 Prov. 2T10-150	T10-208 T10-200	T10-83 T10-120 2T10-150 T10-100	T10-135 T10-100	T10-86 T10-94 2T10-150 2T10-150	T10-187 T10-150	T10-149 T10-100

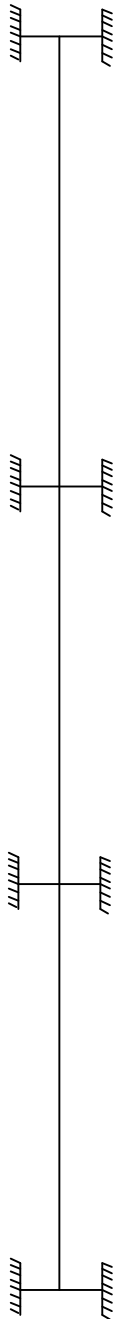


Ref Beam no in STAAD	22	23	24	25	26	55
Beam marking on Drg.	1BD/1-2					
Reinforcement Req./Prov. (cm ²)	46/49 5/13	4/49 11/13	48/49 43/49 6/13 4/13	11/13 9/13	45/49 45/49 4/13 4/13	11/39 12/13
Stirrups	Req. T10-124 Prov. T10-100	T10-312 T10-150	T10-111 T10-128 T10-100 T10-100	T10-189 T10-150	T10-118 T10-173 T10-100 T10-100	T10-249 T10-200

Note : All beam sections are 400mm x 600mm

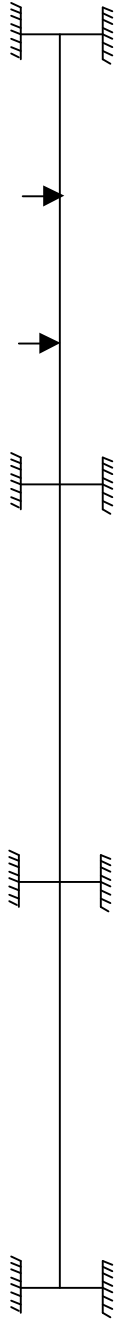


Ref Beam no in STAAD	27	28	29	30
Beam marking on Drg.	1B1/D-C	1B1/C-B	1B1/B-A	
Reinforcement Req./Prov. (cm ²)	25/26 15/17	32/39 10/13 3/13	34/39 28/39 9/13	23/26 5/26 10/17 13/17
Stirrups	Req. T10-279 Prov. T10-200	T10-209 T10-170 T10-150	T10-170 T10-200 T10-150	T10-334 T10-200 T10-200

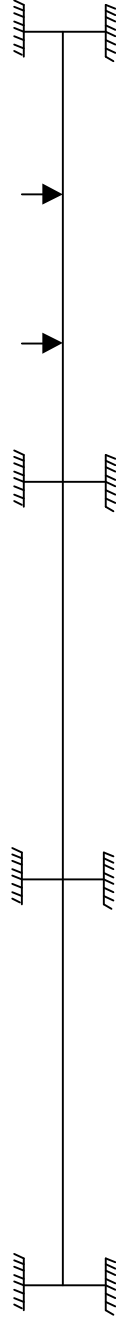


Ref Beam no in STAAD	34	35	36
Beam marking on Drg.	1B2/D-C	1B2/C-B	1B2/B-A
Reinforcement Req./Prov. (cm ²)	31/32 22/24	37/46 19/20 10/20	43/46 18/20 8/24
Stirrups	Req. T10-419 Prov. T10-200	T10-239 T10-201 T10-150	T10-184 T10-201 T10-200

Note : All beam sections are 400mm x 600mm



Ref Beam no in STAAD	40	41	42	43	51
Beam marking on Drg.	1B3/D-C	1B3/C-B		1B3/B-A	
Reinforcement Req./Prov. (cm ²)	30/32 23/24	35/38 20/20	34/38 12/20	43/59 10/20	55/59 14/20
Stirrups	T10-419 T10-200	T10-279 T10-200	T10-237 T10-200	T10-415 T10-200	T10-114 T10-100

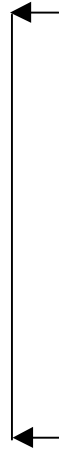


Ref Beam no in STAAD	46	47	48	49	50
Beam marking on Drg.	1B4/D-C	1B4/C-B		1B4/B-A	
Reinforcement Req./Prov. (cm ²)	24/26 13/13	28/32 10/13	28/32 4/13	27/46 6/13	42/46 13/17
Stirrups	T10-279 T10-200	T10-251 T10-200	T10-369 T10-200	T10-126 T10-100	T10-187 T10-100

Note : All beam sections are 400mm x 600mm



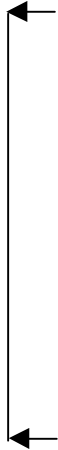
Ref Beam no in STAAD	9			
Beam marking on Drg.	1BB2/1-2a			
Reinforcement Req./Prov. (cm ²)	Top	0/9	0/9	0/9
	Bottom	0/9	3/9	0/9
Stirrups	Req.	T10-179	T10-179	T10-179
	Prov.	T10-150	T10-150	T10-150



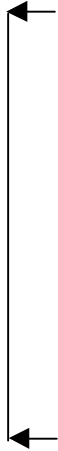
Ref Beam no in STAAD	52			
Beam marking on Drg.	1BB3/3-4			
Reinforcement Req./Prov. (cm ²)	Top	0/9	0/9	0/9
	Bottom	0/15	10/15	0/15
Stirrups	Req.	T10-361	T10-361	T10-361
	Prov.	T10-200	T10-200	T10-200

Ref Beam no in STAAD	8			
Beam marking on Drg.	1BB1/3-4			
Reinforcement Req./Prov. (cm ²)	Top	17/24	17/24	0/9
	Bottom	0/9	0/15	11/15
Stirrups	Req.	T10-117	T10-441	T10-385
	Prov.	T10-100	T10-200	T10-200

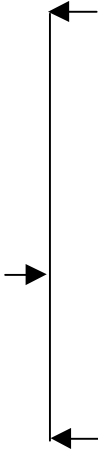
Note : All beam sections are 300mm x 600mm



Ref Beam no in Staad		33	
Beam marking on Drg.		1B2b/D-C	
Reinforcement Req./Prov. (cm ²)	Top	0/9	0/9
	Bottom	0/15	0/15
Stirrups Req.		T10-419 T10-200	T10-559 T10-200

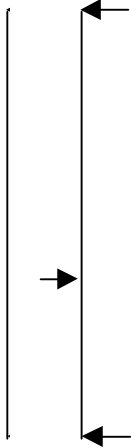


Ref Beam no in Staad		37	
Beam marking on Drg.		1B3a/D-C	
Reinforcement Req./Prov. (cm ²)	Top	0/9	0/9
	Bottom	0/15	0/15
Stirrups Req.		T10-419 T10-200	T10-559 T10-200

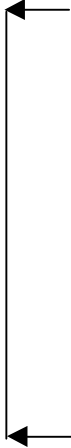


Ref Beam no in Staad		38	
Beam marking on Drg.		1B3b/B-A	
Reinforcement Req./Prov. (cm ²)	Top	0/9	0/9
	Bottom	0/13	0/13
Stirrups Req.		T10-380 T10-200	T10-336 T10-200

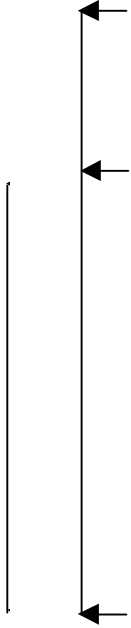
Note : All beam sections are 300mm x 600mm



Ref Beam no in STAAD	31	32
Beam marking on Drg.	1B2a/B-A	
Reinforcement Req./Prov. (cm ²)	Top	0/9 0/9 0/9
	Bottom	0/13 10/13 0/13
Stirrups	Req.	T10-334 T10-275 T10-353
	Prov.	T10-200 T10-200 T10-200



Ref Beam no in Staad		53	
Beam marking on Drg.		1B4a/D-C	
Reinforcement Req./Prov. (cm ²)	Top	0/9	0/9
	Bottom	0/15	9/21
Stirrups Req.	T10-349	T10-349	T10-349
	T10-200	T10-200	T10-200



Ref Beam no in STAAD	44	45
Beam marking on Drg.	1B4b/C-B	
Reinforcement Req./Prov. (cm ²)	Top	0/9 20/24 20/24
	Bottom	0/21 18/21 0/15 0/9
Stirrups	Req.	T10-369 T10-554 T10-221 T10-114
	Prov.	T10-200 T10-200 T10-200 T10-100

Note : All beam sections are 300mm x 600mm

A vertical beam is shown with four fixed supports, indicated by hatched lines, at the top, bottom, and two intermediate points. The beam is divided into three equal segments by these supports.

Ref Beam no in STAAD		101			102		103
Beam marking on Drg.		RBA/1-2			RBA/2-3		RBA/3-4
Reinforcement Req./Prov. (cm ²)	Top	17/21	3/21	14/15	11/15	11/15	15/21
	Bottom	12/13	5/13	7/9	6/9	4/9	7/9
Stirrups	Req.	T10-386	T10-386	T10-361	T10-374	T10-374	T10-361
	Prov.	T10-200	T10-200	T10-200	T10-200	T10-200	T10-200

[illegible]

Note : All beam sections are 300mm x 600mm

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Ref Beam no in STAAD	107	108	109		
Beam marking on Drg.	RBC/1-2	RBC/2-3	RBC/3-4		
Reinforcement Req./Prov.	24/24	3/24	22/24	18/24	3/9
(cm ²)	12/13	8/13	8/9	6/9	4/9
Stirrups	Req.	T10-361	T10-361	T10-363	T10-363
	Prov.	T10-200	T10-200	T10-200	T10-200

mm		mm		mm		mm	
Ref Beam no in STAAD		110		111		112	
Beam marking on Drg.		RBD/1-2		RBD/2-3		RBD/3-4	
Reinforcement Req./Prov. (cm ²)	Top	17/21	3/21	15/15	11/15	11/15	13/15
	Bottom	10/13	7/13	7/9	6/9	3/9	8/9
Stirrups	Req.	T10-386	T10-386	T10-386	T10-363	T10-363	T10-386
	Prov.	T10-200	T10-200	T10-200	T10-200	T10-200	T10-200

Note : All beam sections are 300mm x 600mm

Document No.: DM001-D-GBD-WTS-DR-DCC-545029

Rev A3

Date: 25 Nov 2007

[illegible][illegible]

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Rev A3

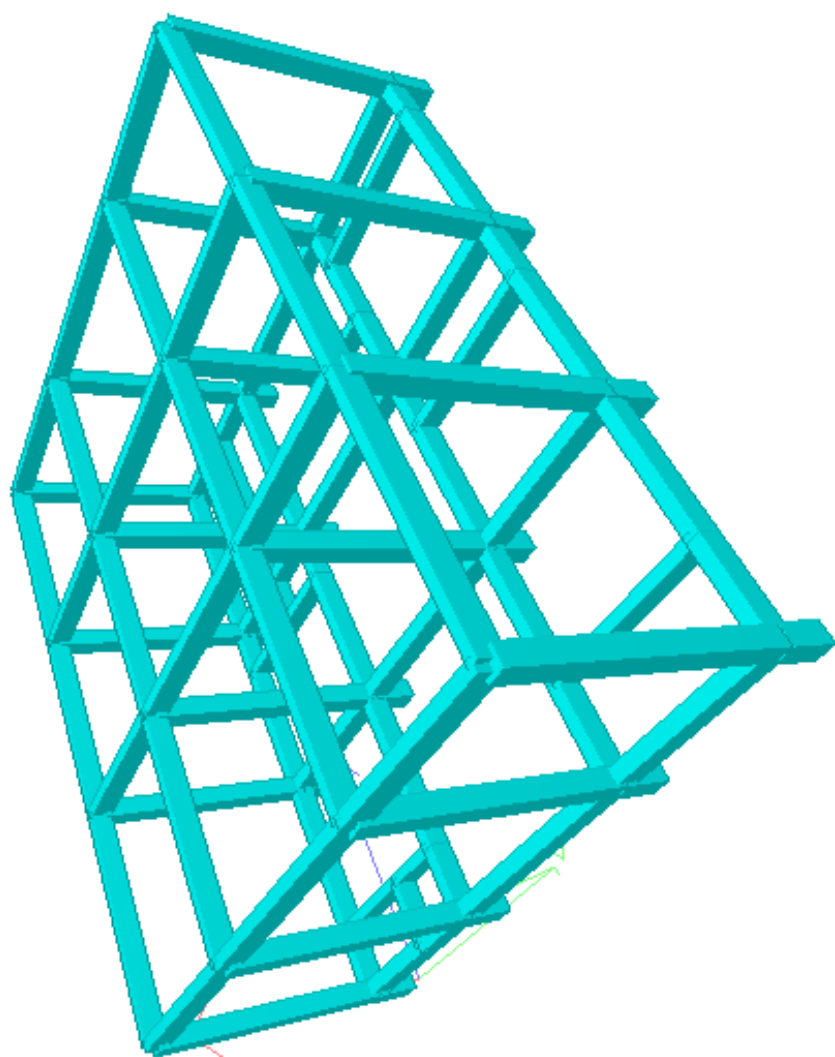
A vertical beam is shown with four fixed supports. The supports are located at the top, bottom, and at two intermediate points. Each support is represented by a horizontal line with diagonal hatching below it, indicating a fixed connection. The beam is a single vertical line connecting these four support points.

[illegible]

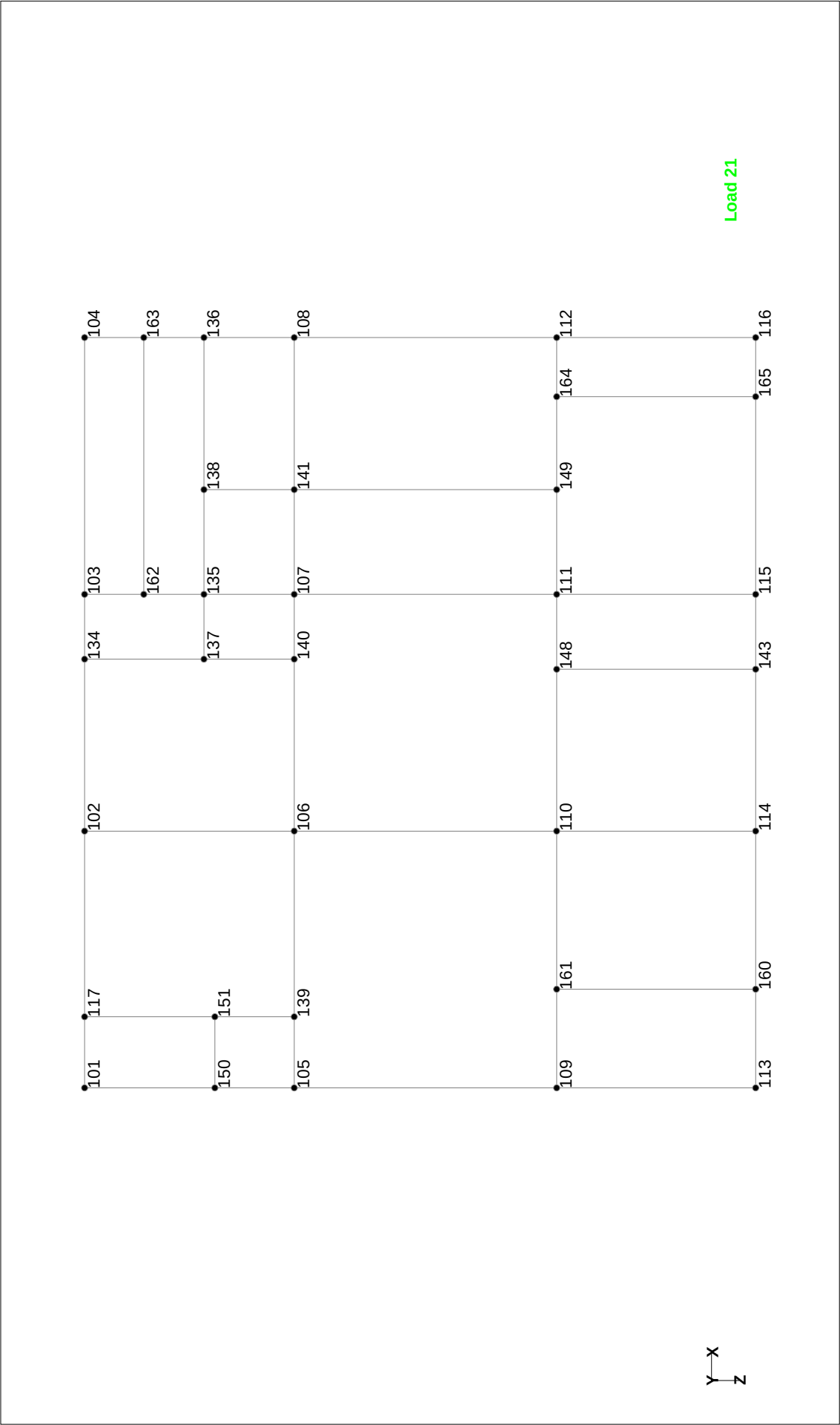
The diagram shows a continuous beam with four supports, labeled 1, 2, 3, and 4 from left to right. The beam is divided into three spans by these supports. Each support is represented by a vertical line with a hatched area below it, indicating a fixed support. The beam is shown as a horizontal line with a slight downward curve in the middle of each span.

Ref Beam no in Staad		122	123	124
Beam marking on Drg.		RB4/D-C	RB4/C-B	RB4/B-A
Reinforcement Req./Prov. (cm ²)	Top	15/15 3/15 15/15	10/15 0/9 10/15	15/15 3/15 15/15
	Bottom	12/13 3/13 8/9 4/9	4/9 4/9 8/9	3/13 3/13 13/13
Stirrups Reqd.		T10-359	T10-387	T10-379
		T10-200	T10-200	T10-200

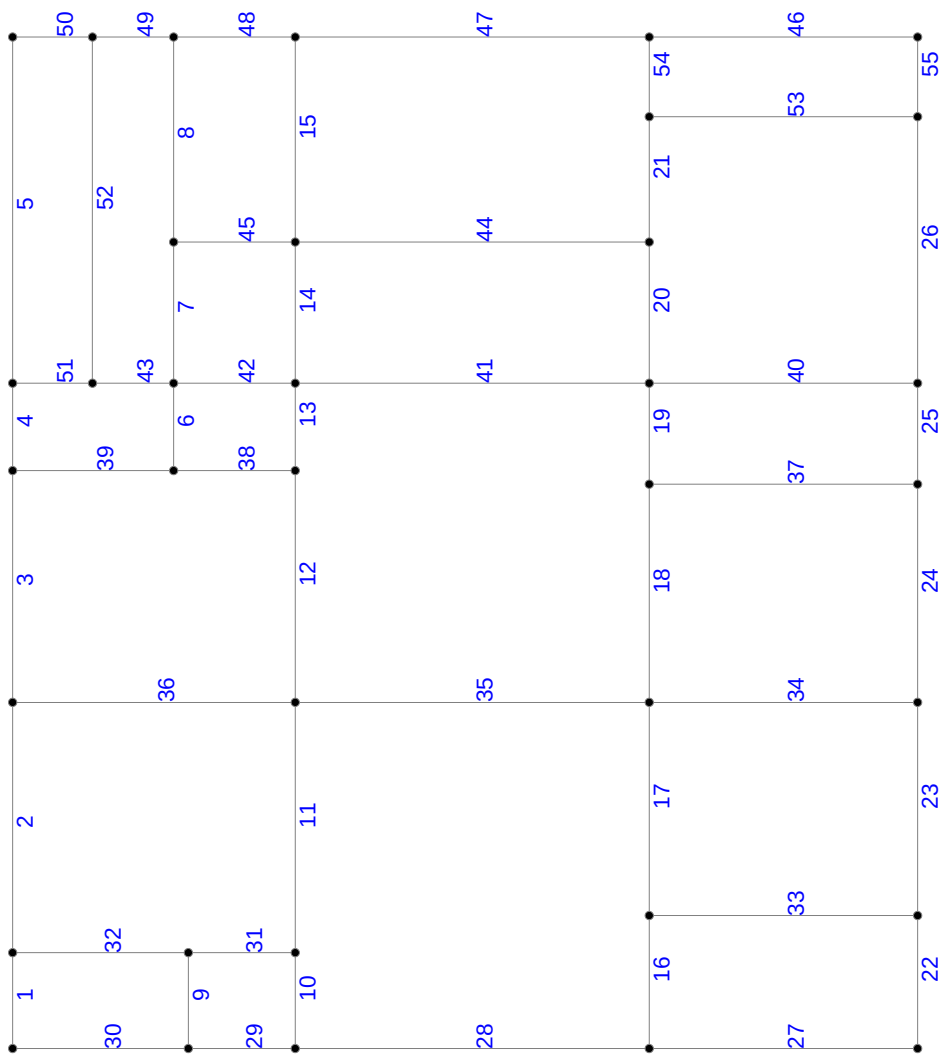
Note : All beam sections are 300mm x 600mm



3D Rendered View



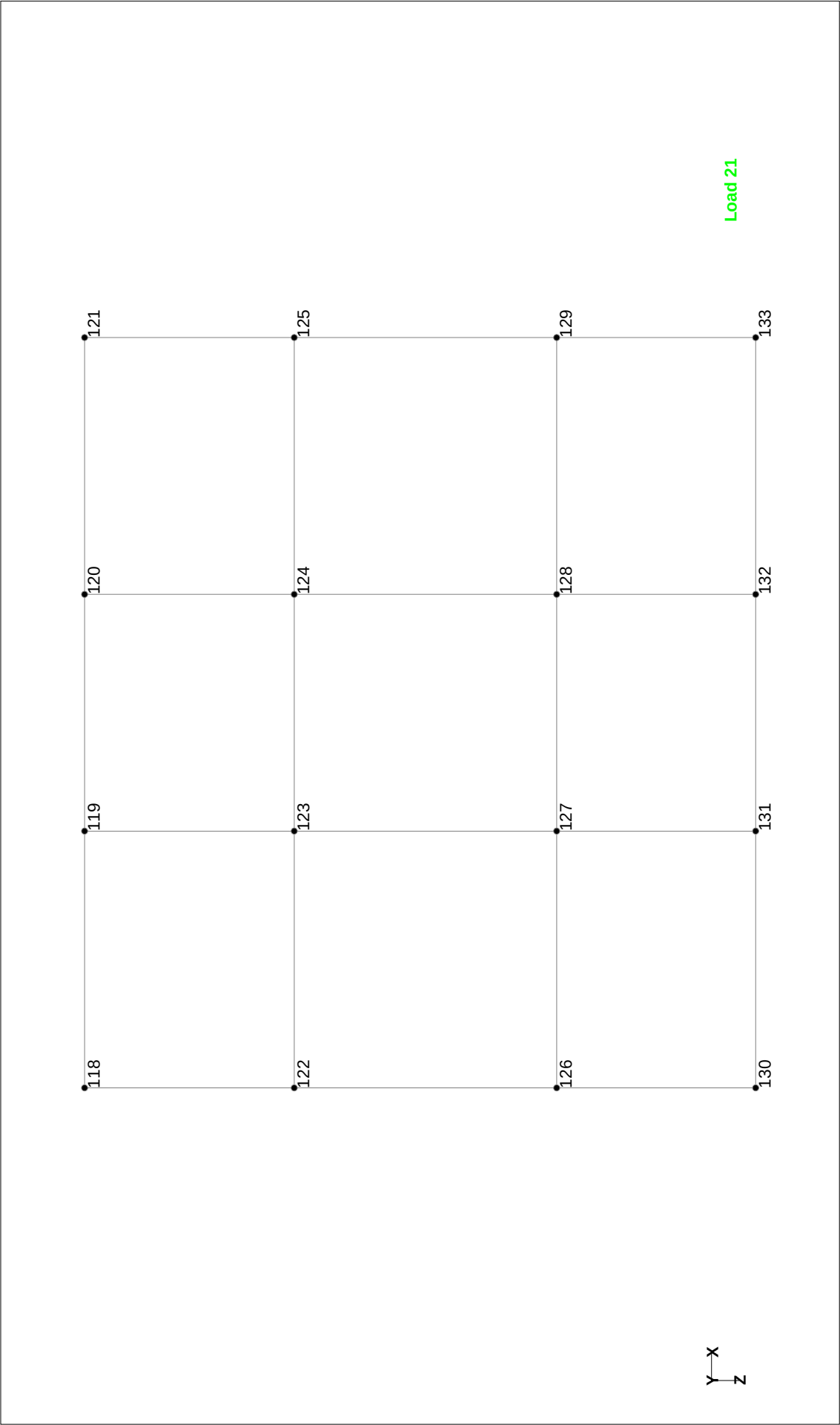
Node No. (Ground Floor)



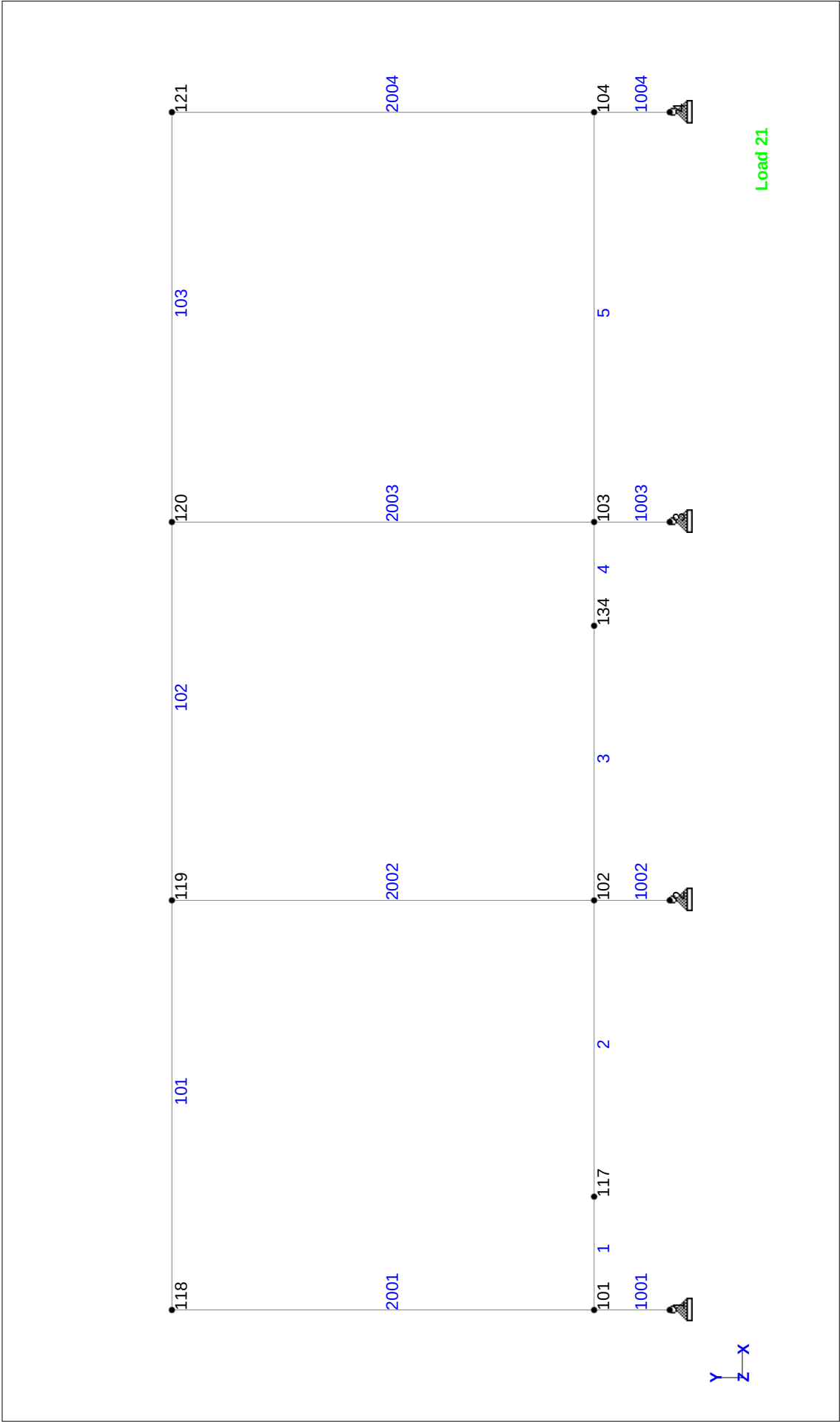
Y-X
Z

Load 21

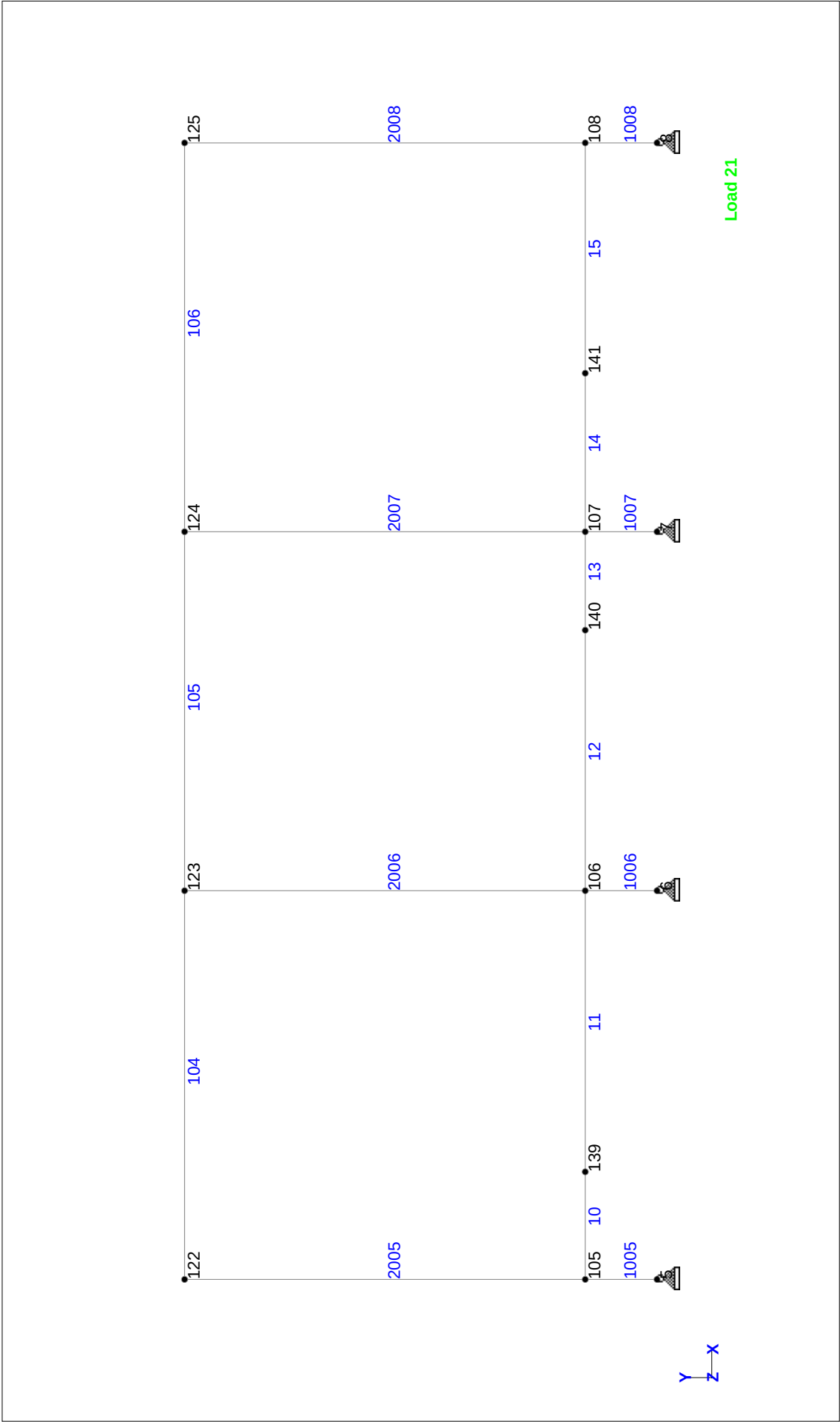
Ground floor (Beam No.)

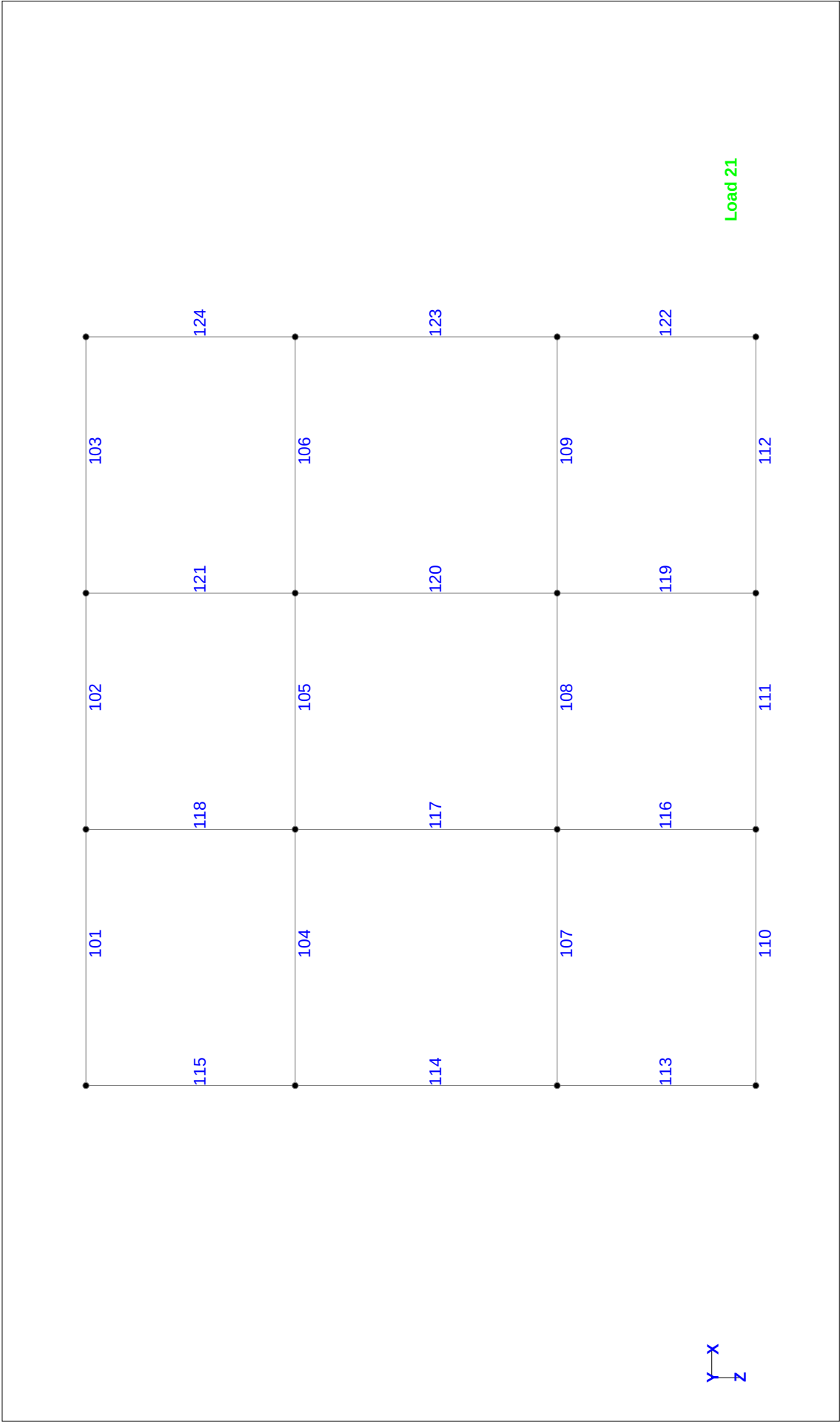


Node no. (Roof Level)

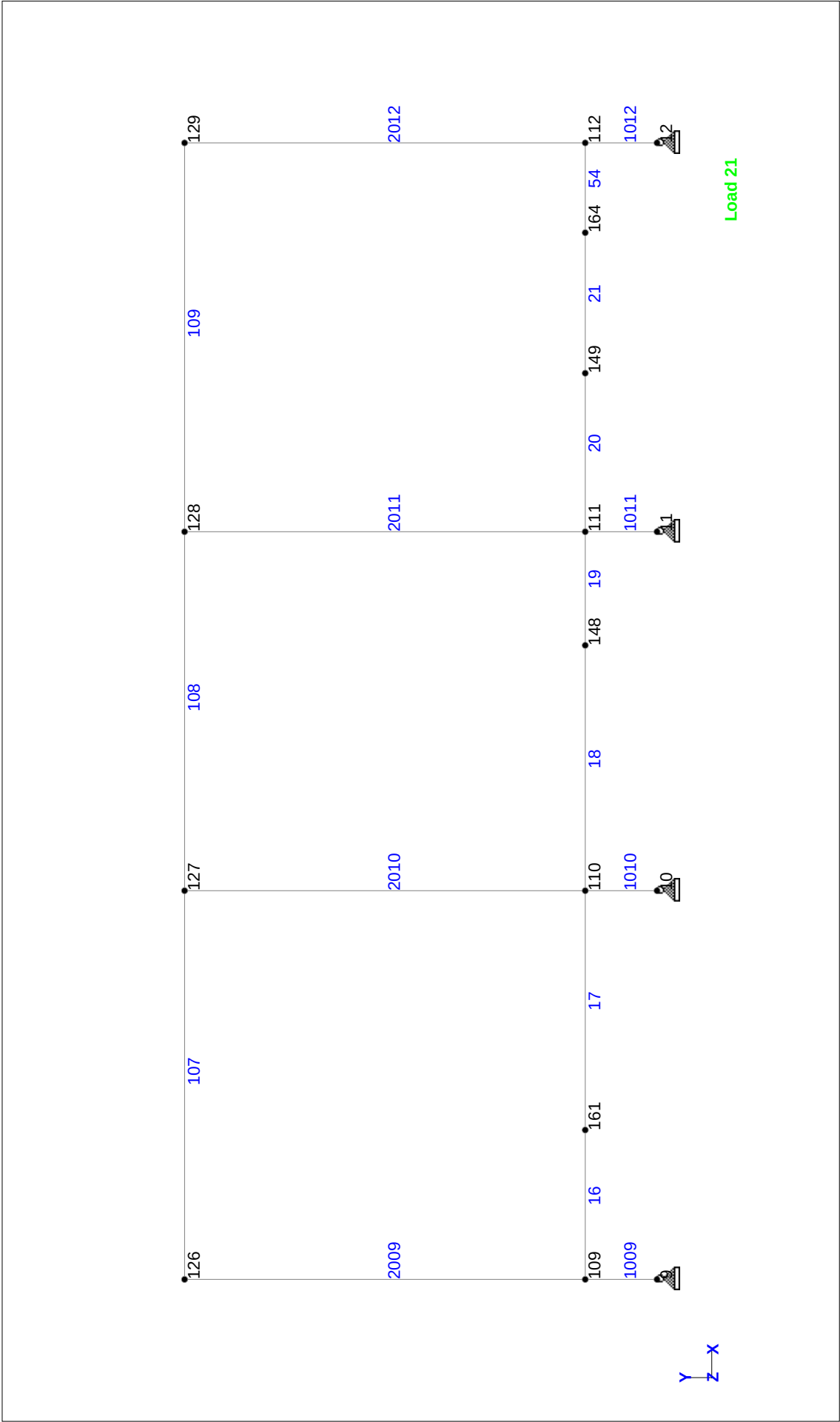


Section about Grid A

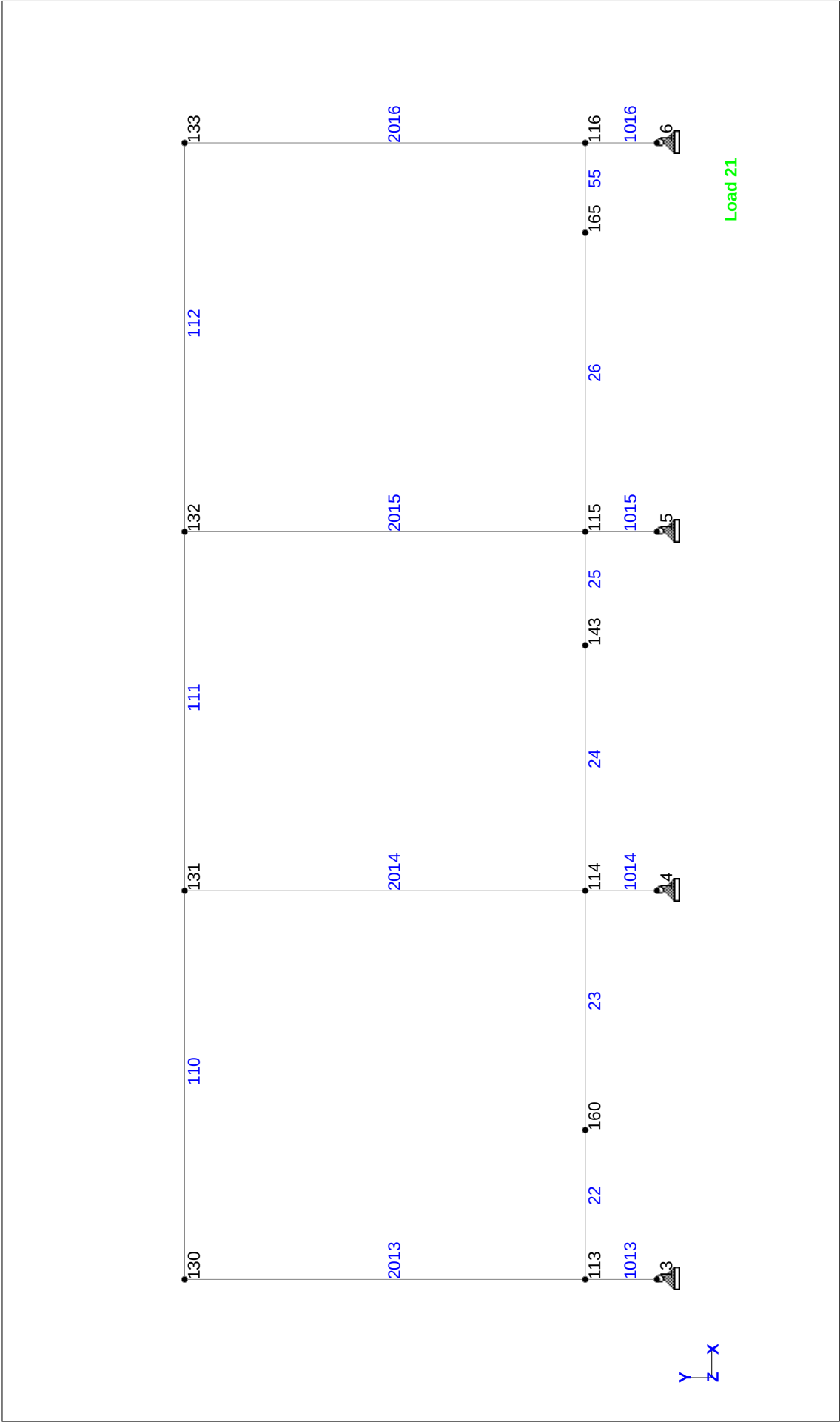




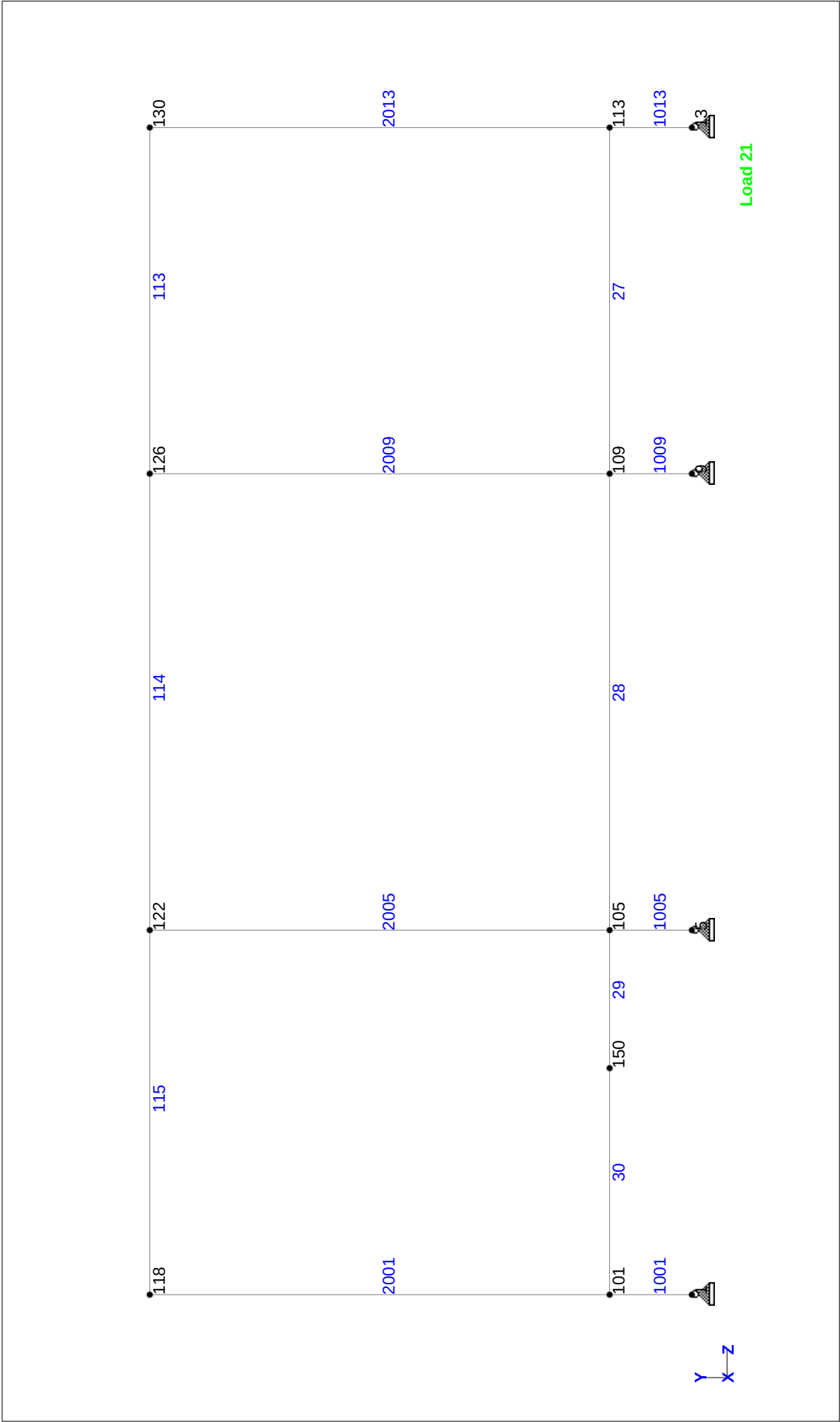
Beam No. (Roof level)



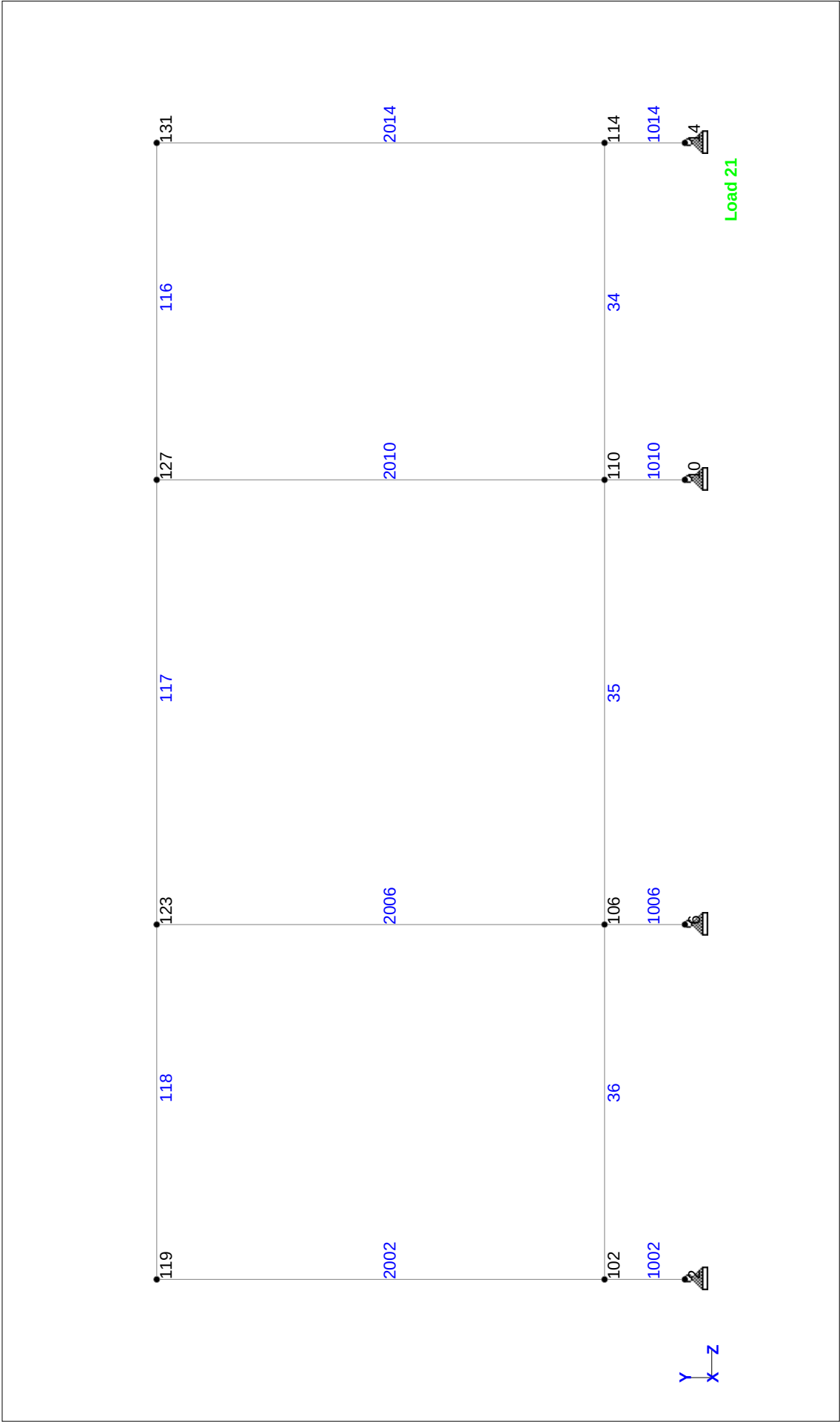
Section About Grid C



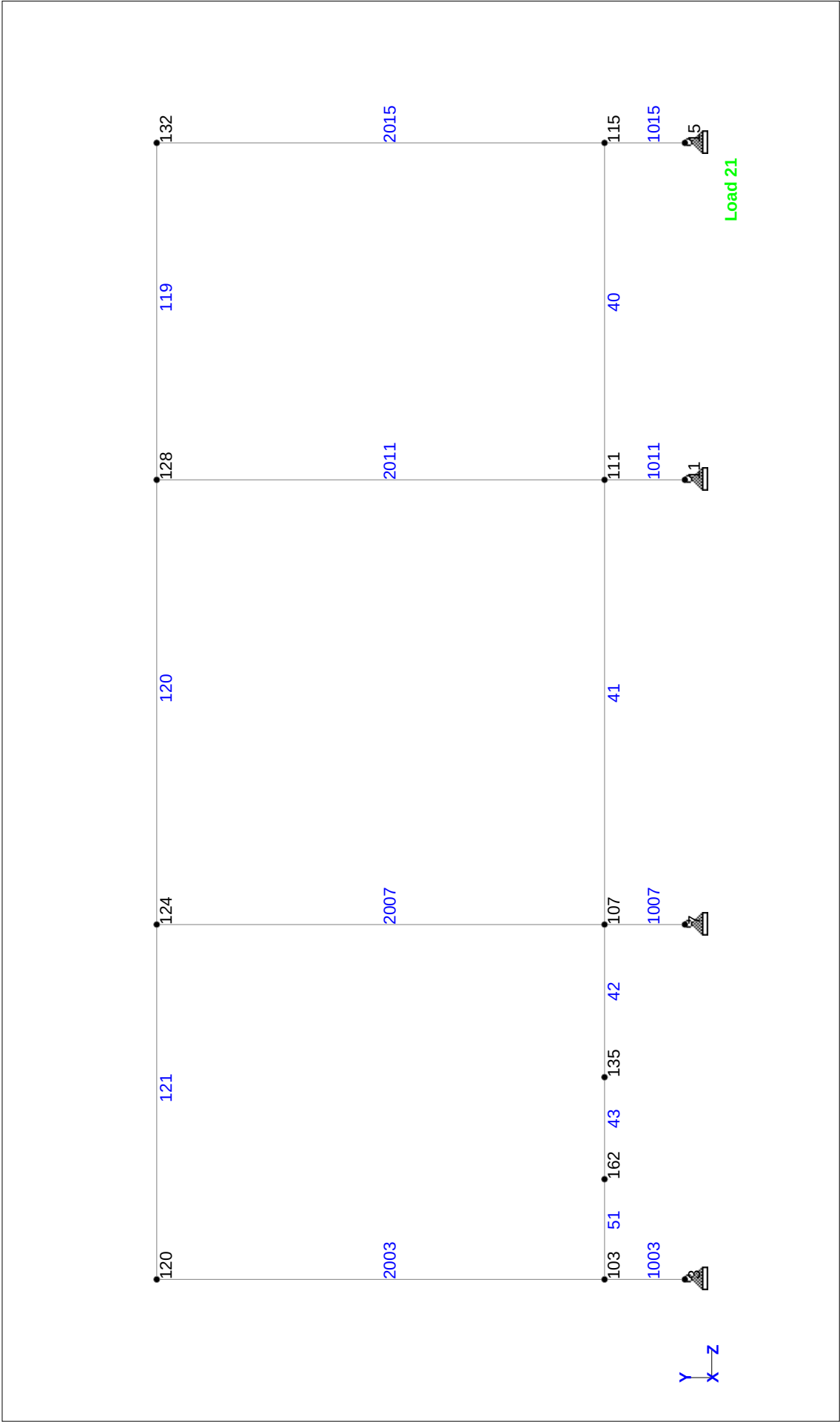
Section about Grid D

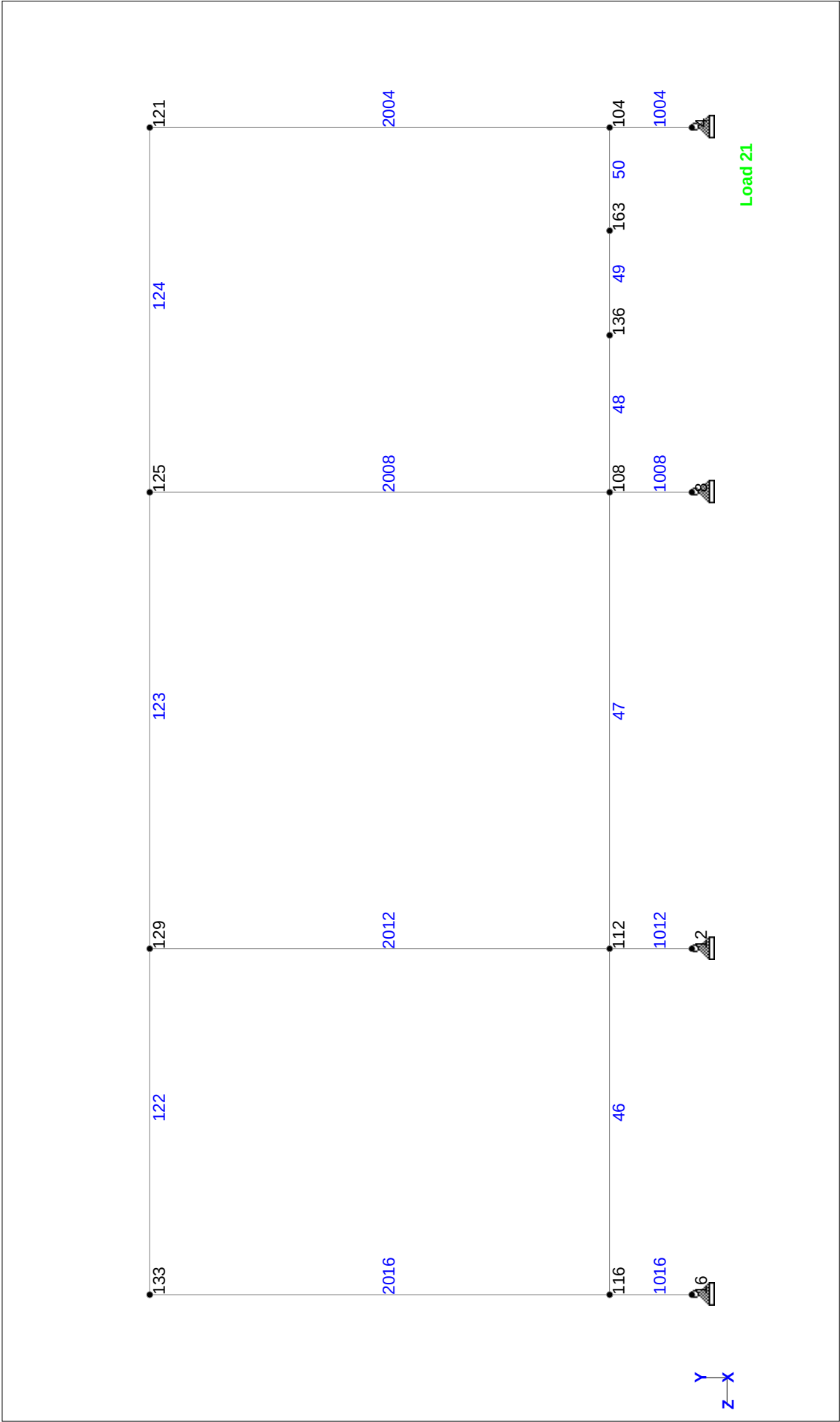


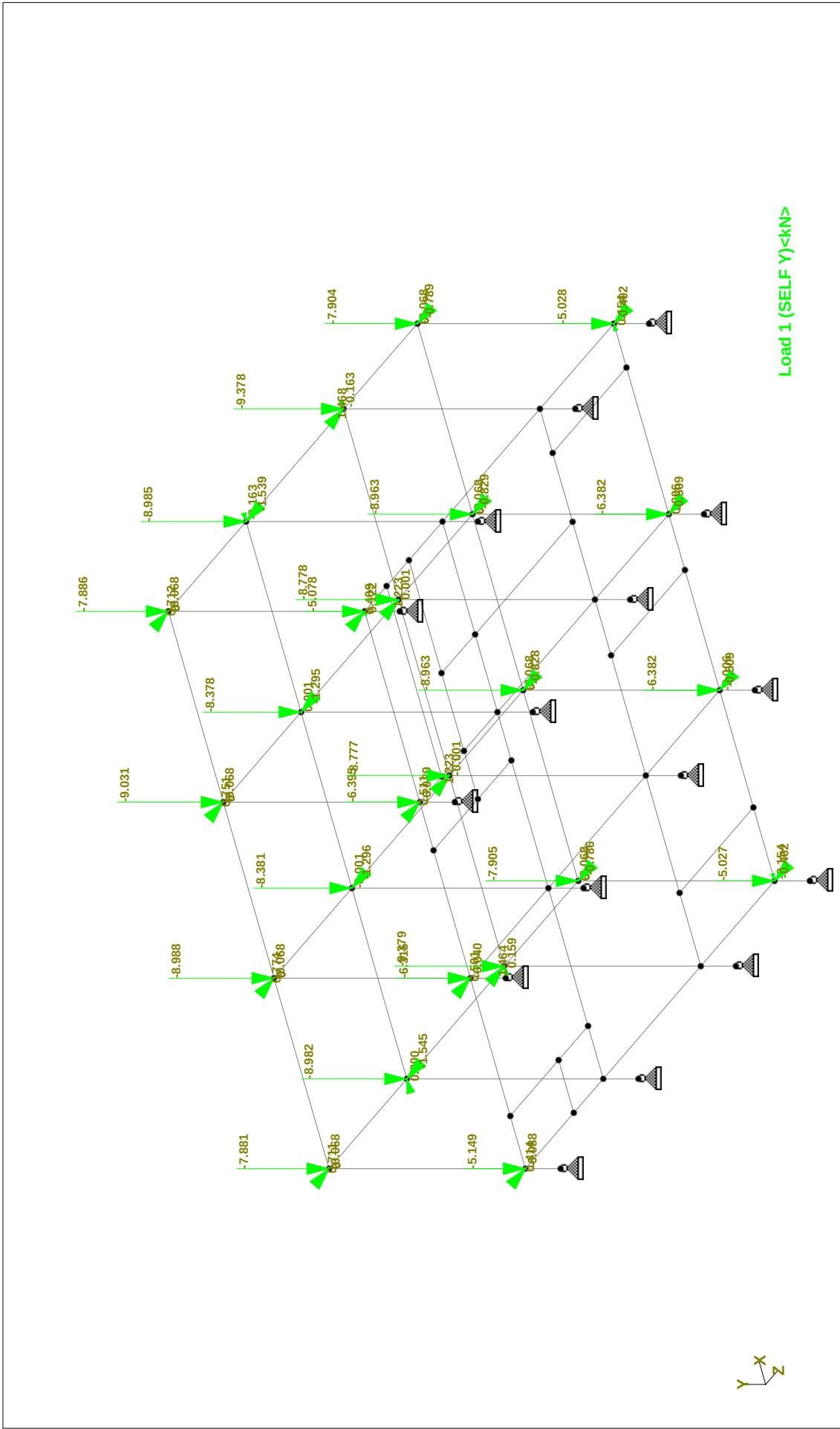
Section about Grid - 1

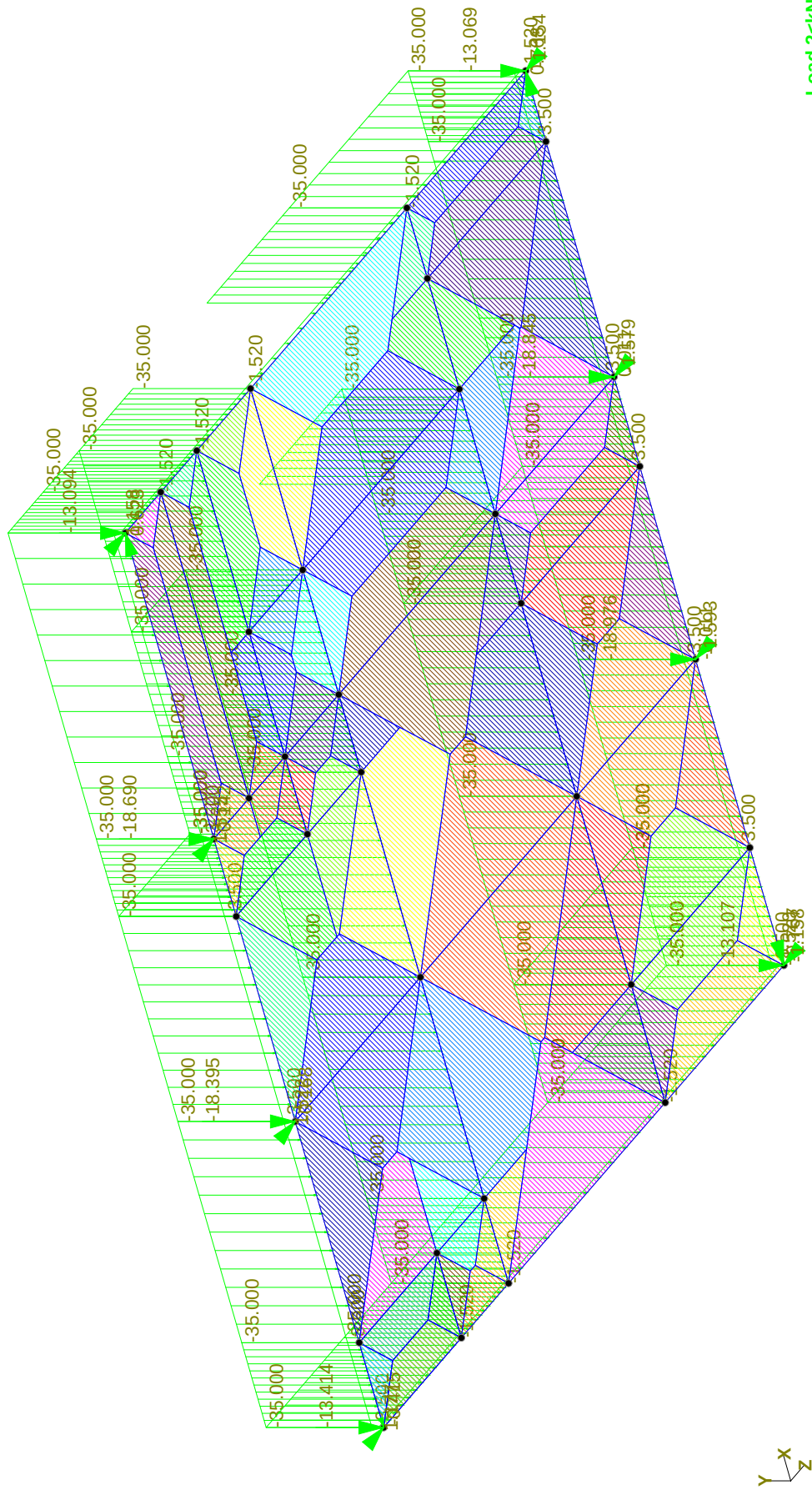


Section about Grid 2



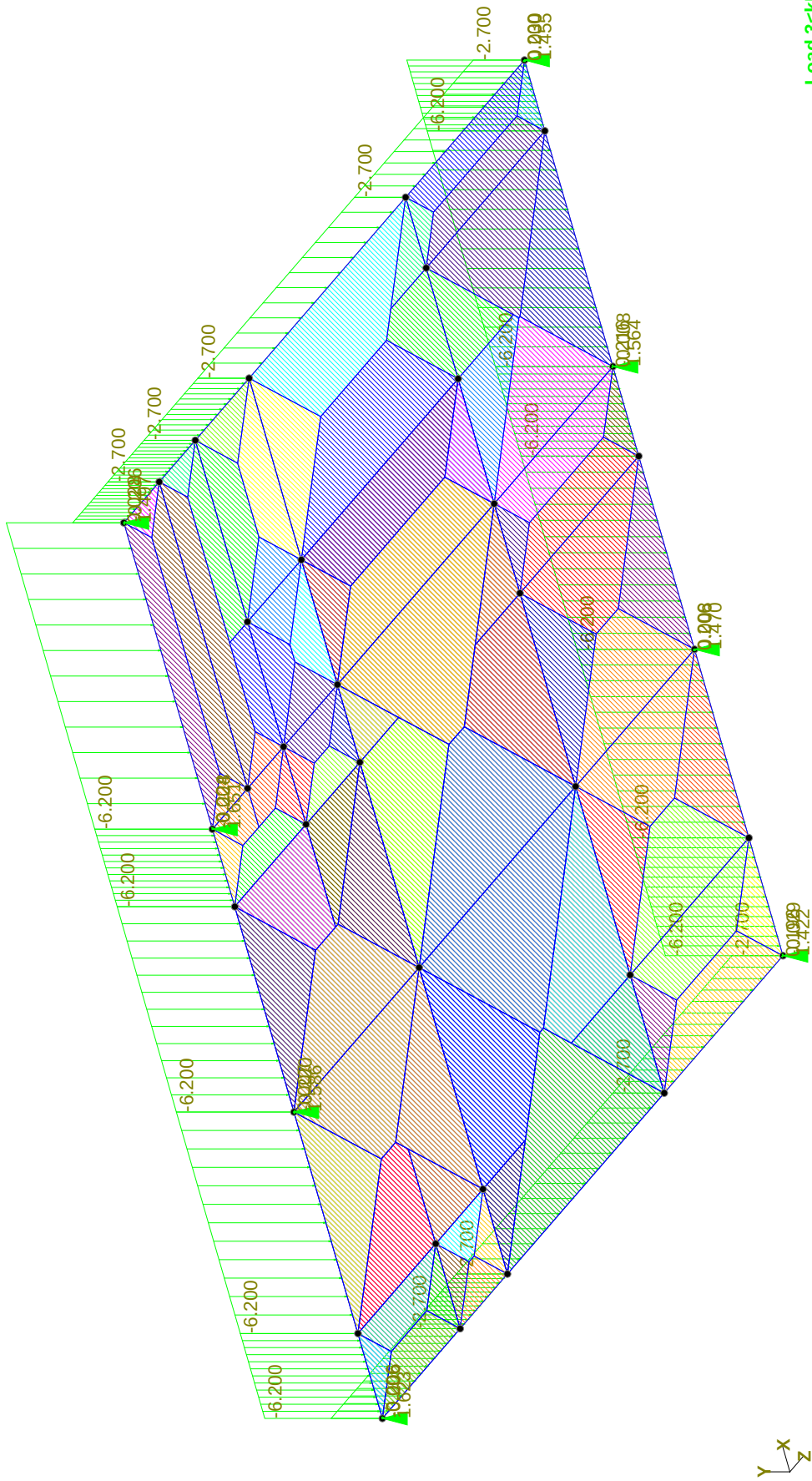




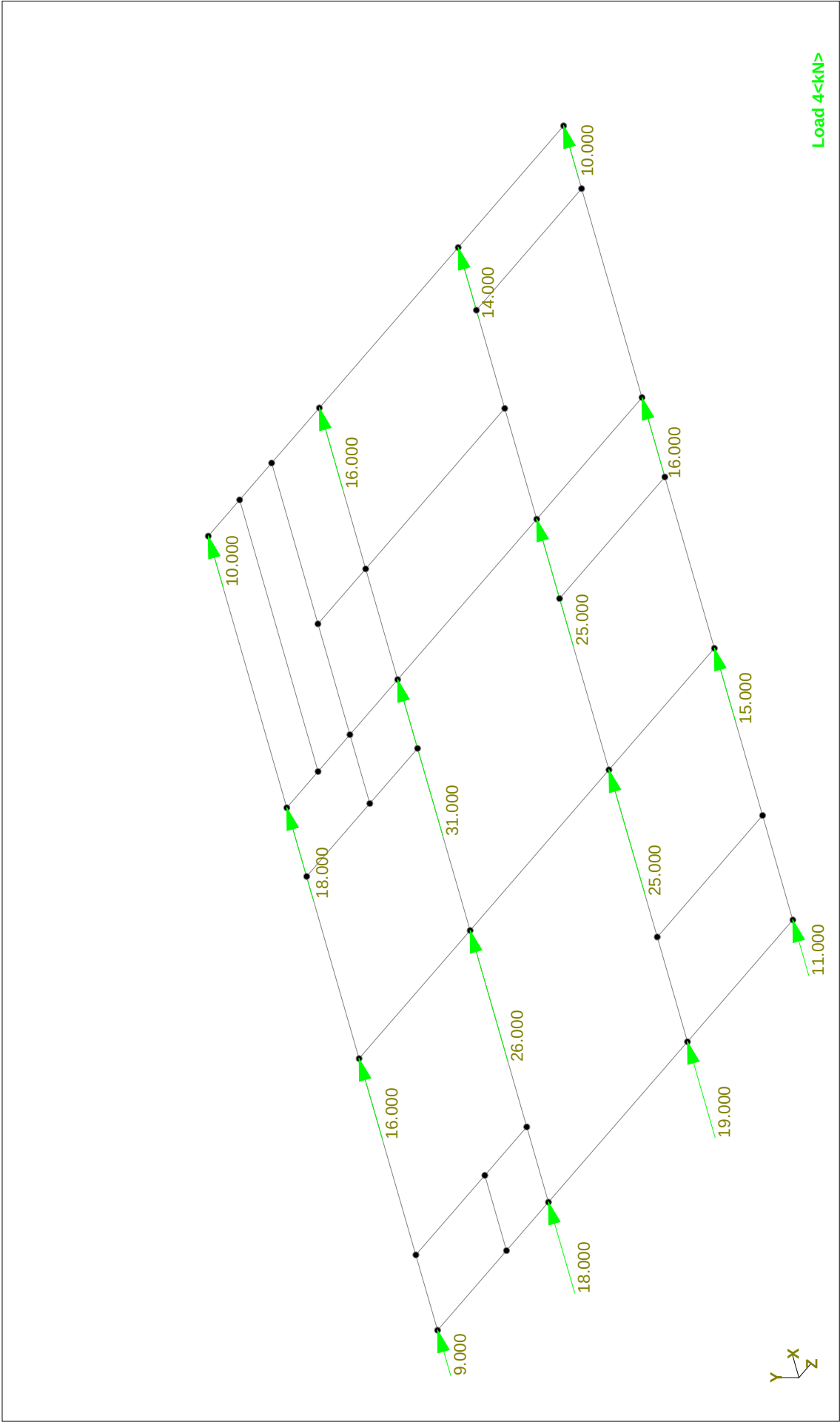


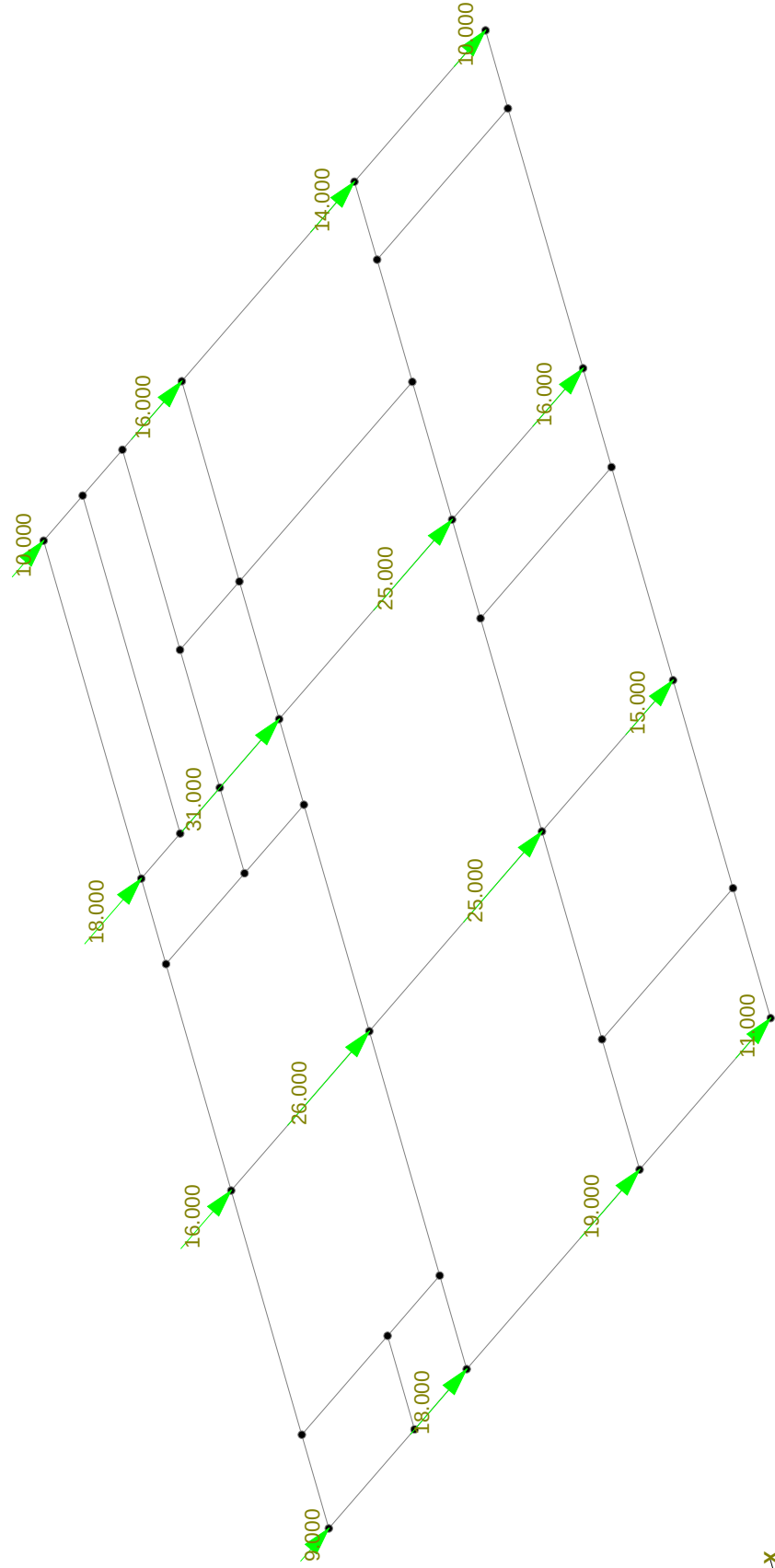
Load 2<kN>

Load - 2 DL (Ground Floor)



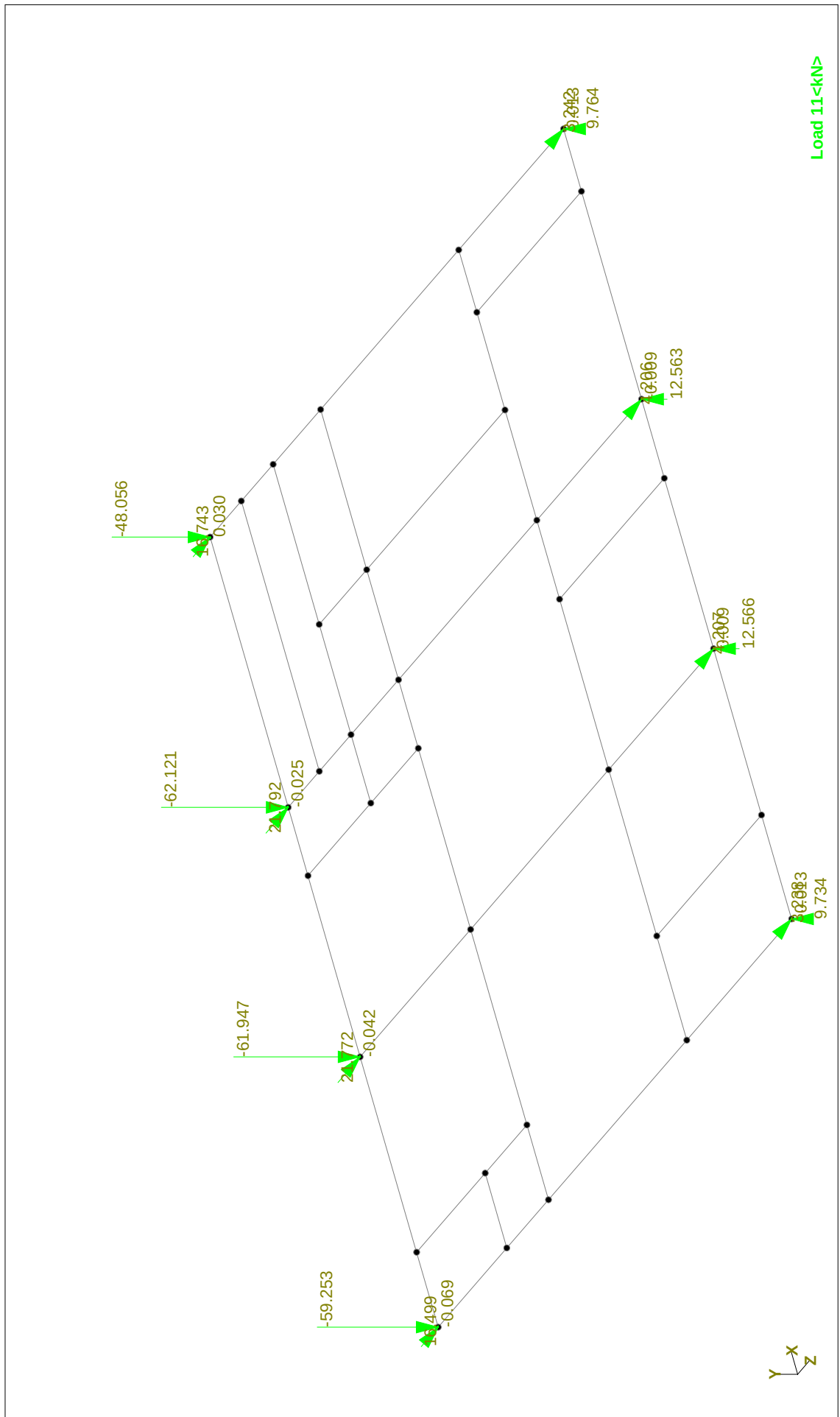
Load 3 Live Load (Ground Floor)





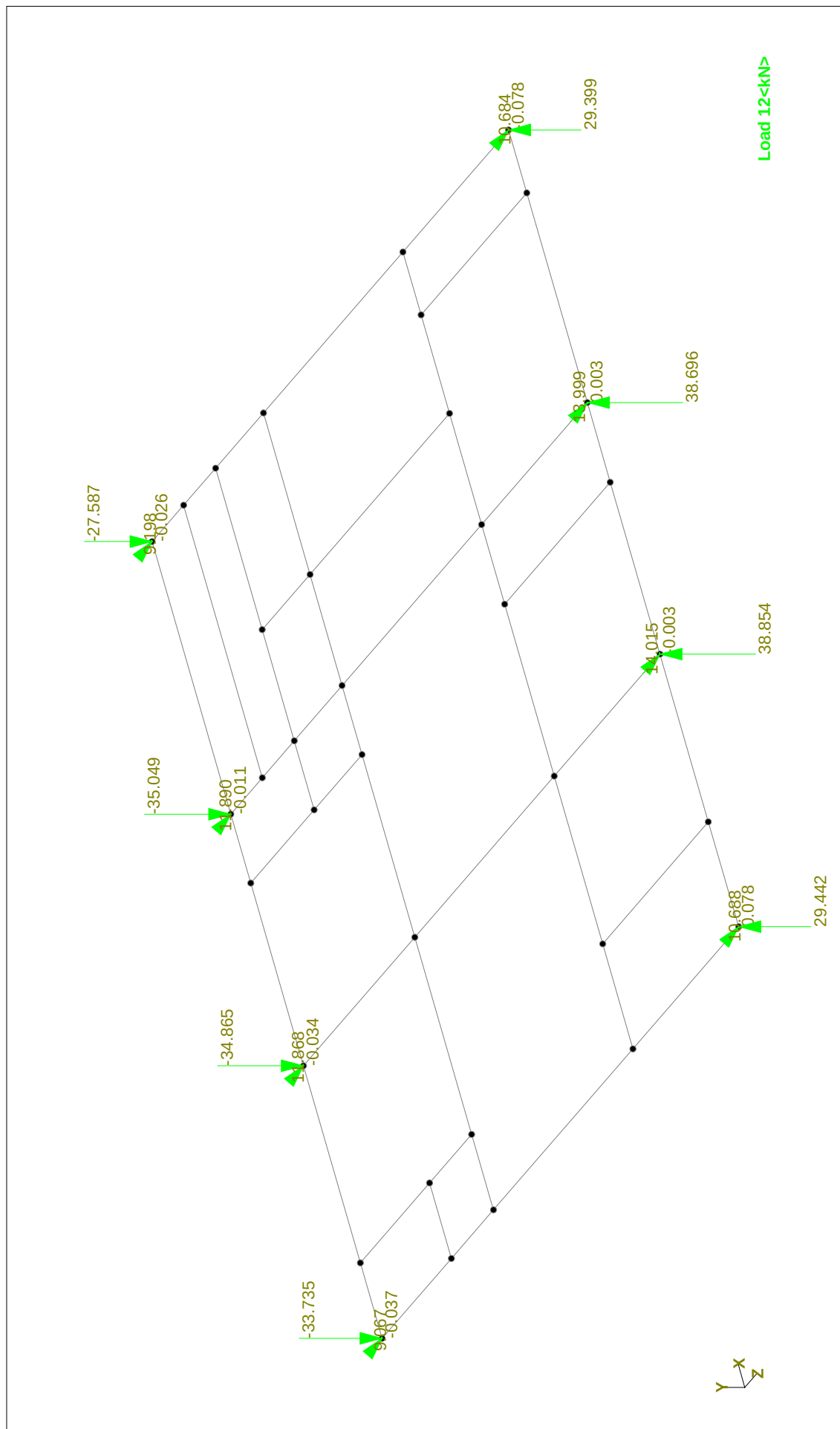
Load 5<kN>

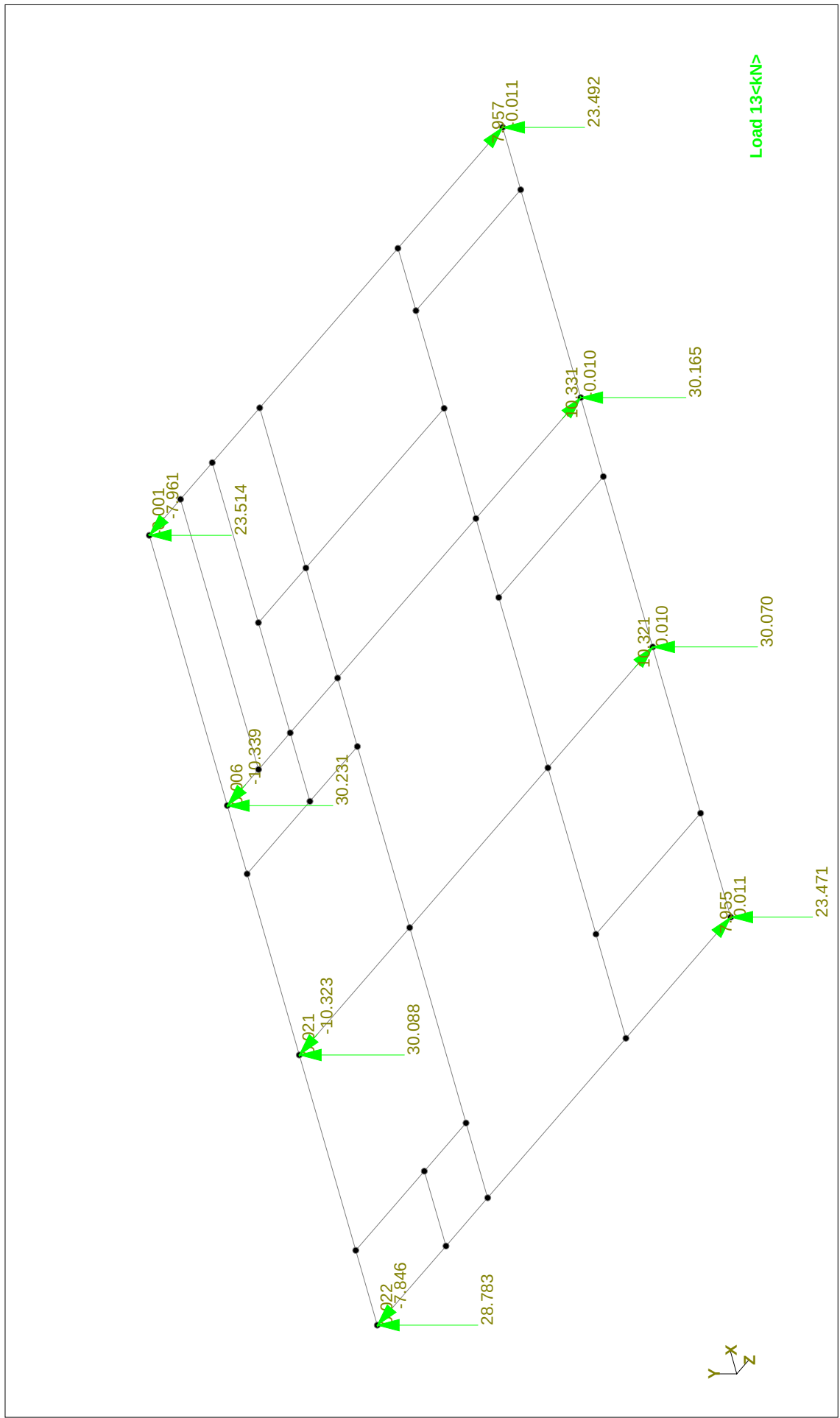
Load 5 PH(Z-DIR) (Ground Floor)



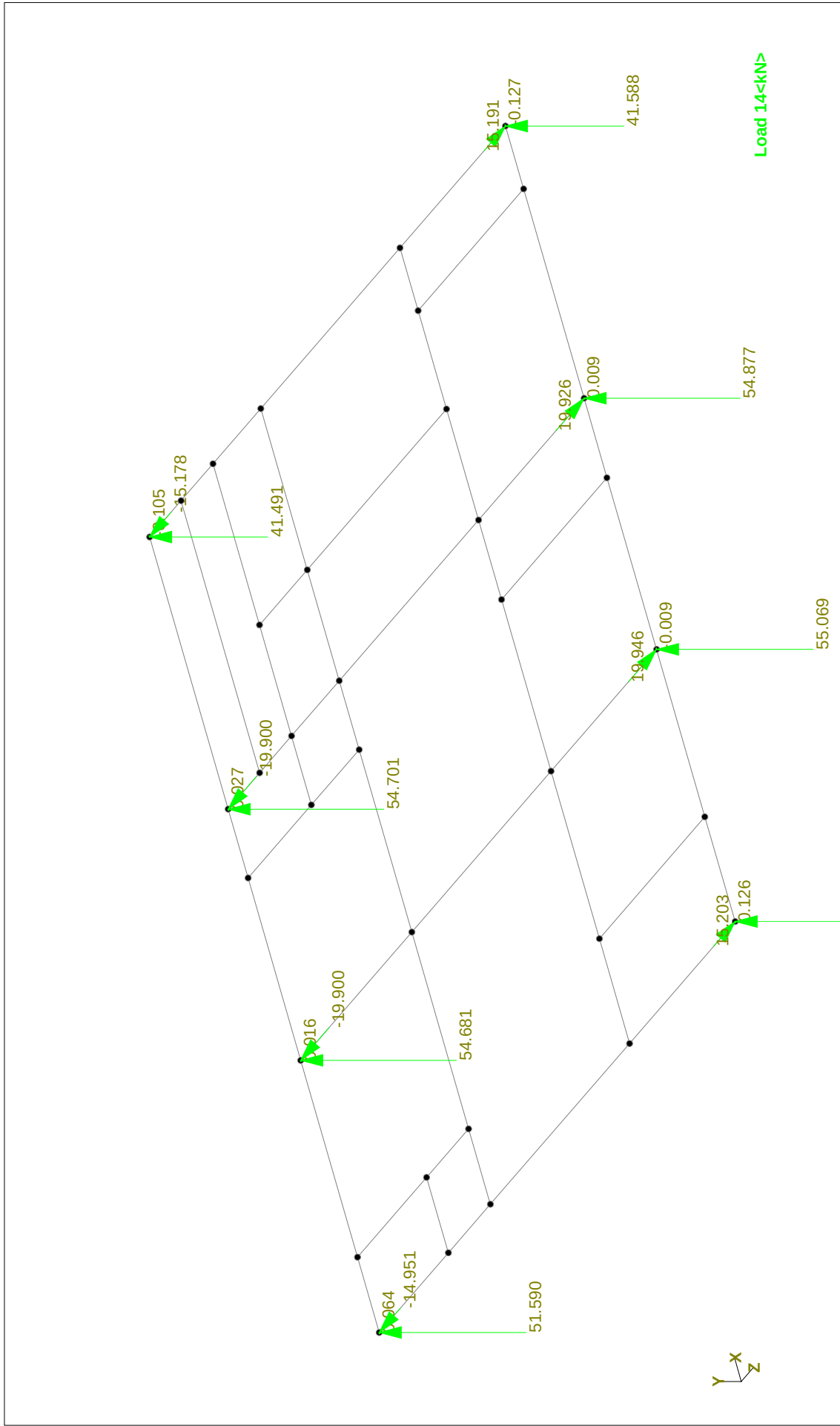
Load - 11 WL1 (0 DEG ,INTERNAL SUCTION -0.3) (Ground Floor)

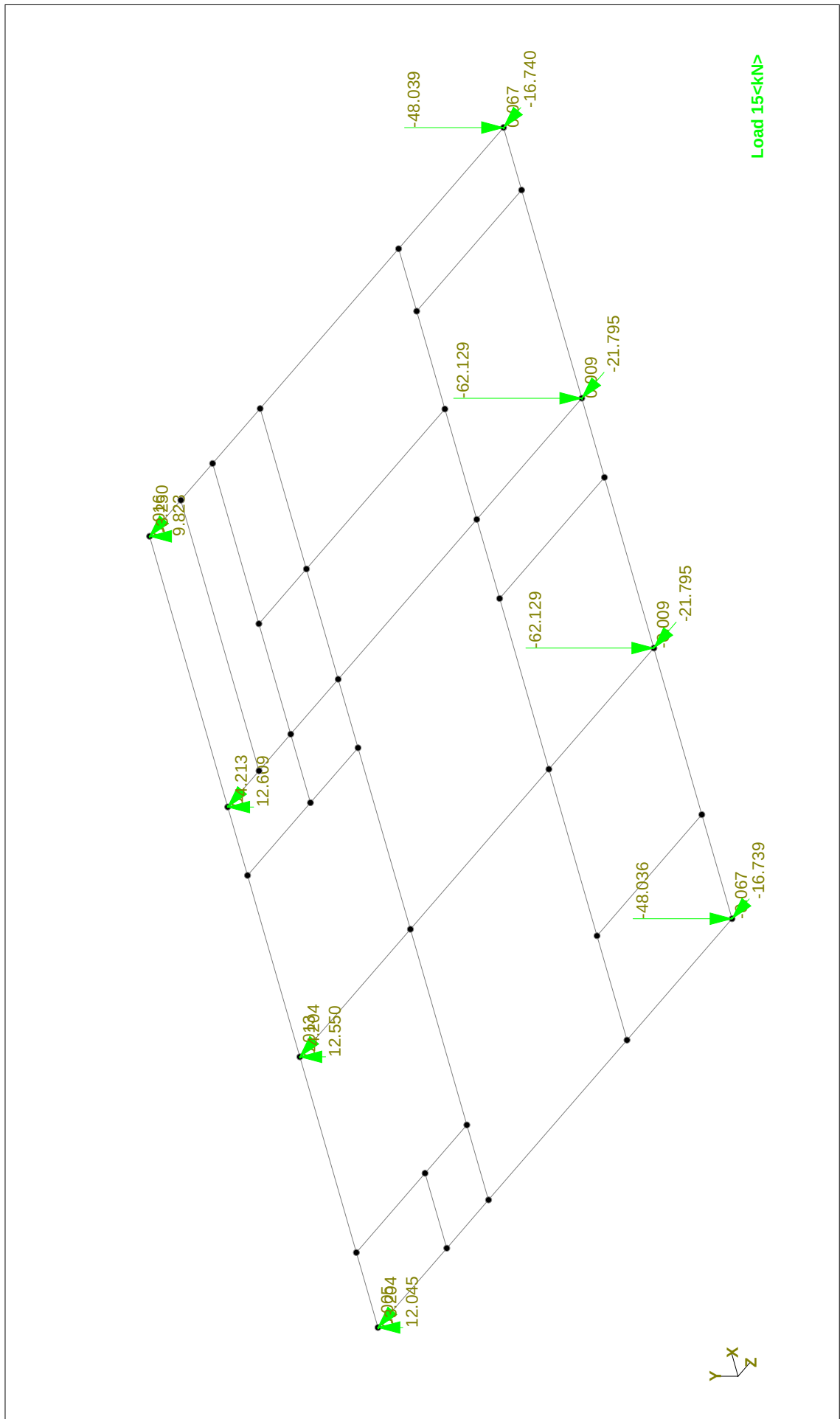
Load 11<kN>

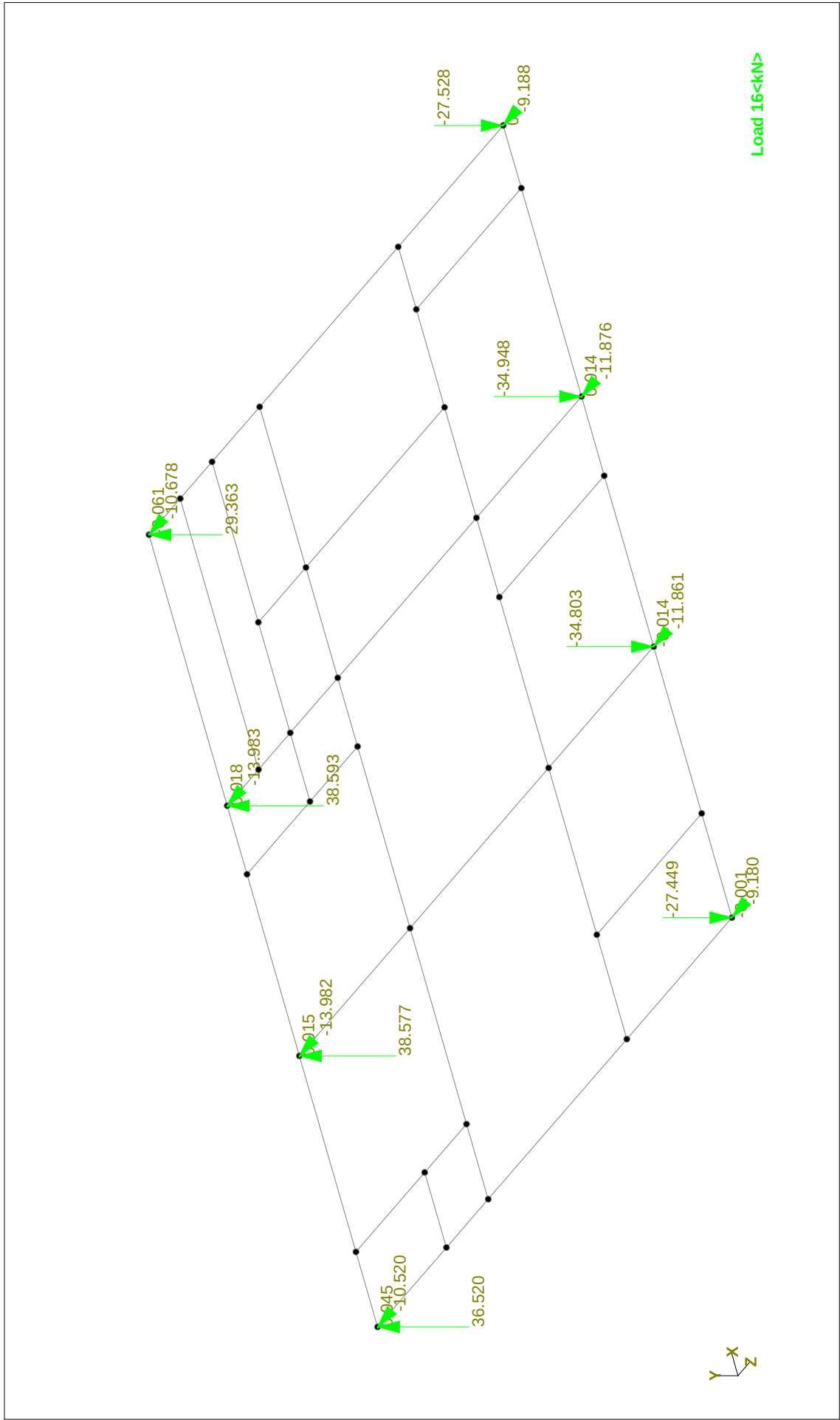




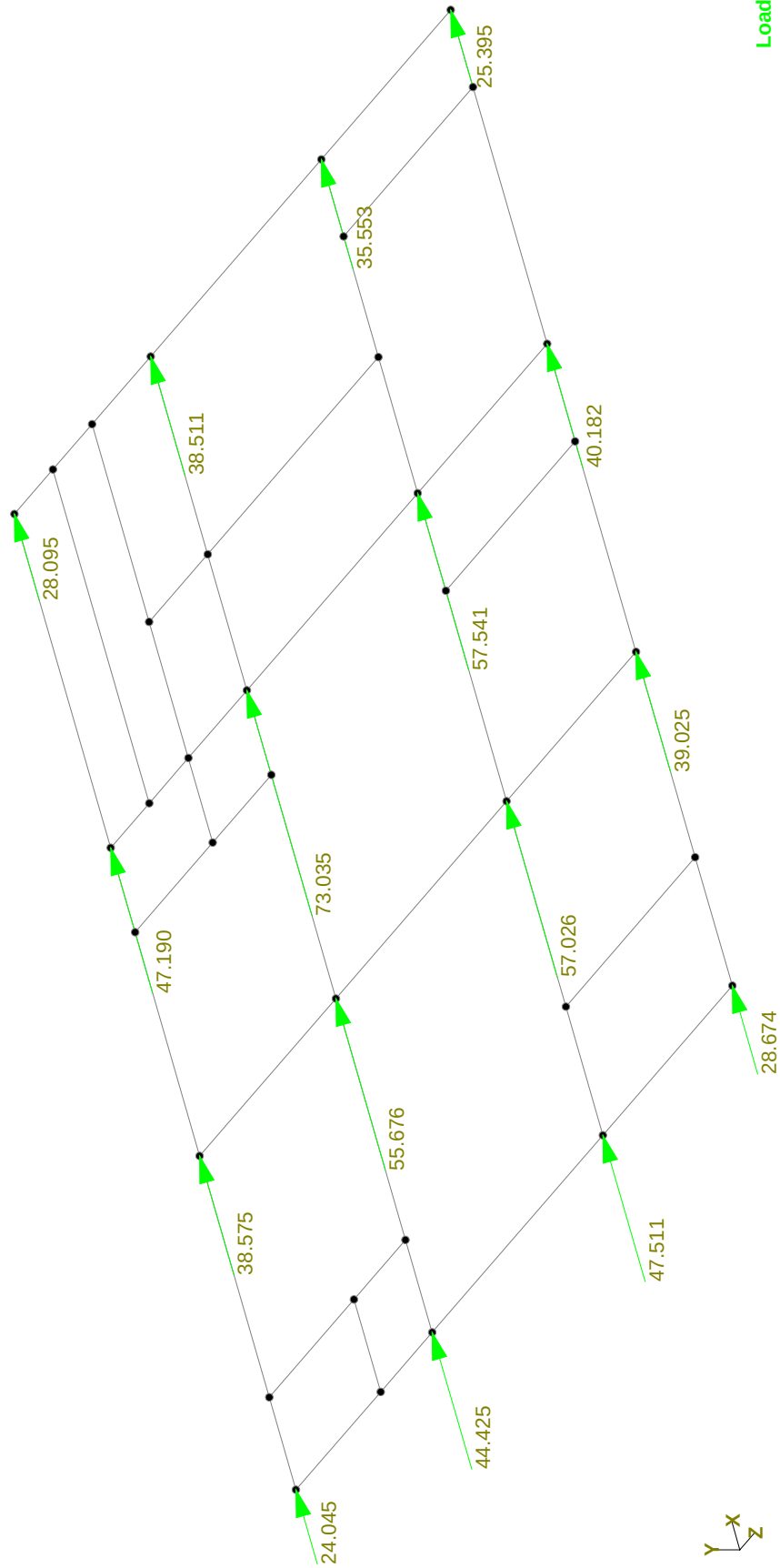
Laod -13 WL3 (90 DEG ,INTERNAL SUCTION -0.3, WW&LW +0.2) (Ground Floor)

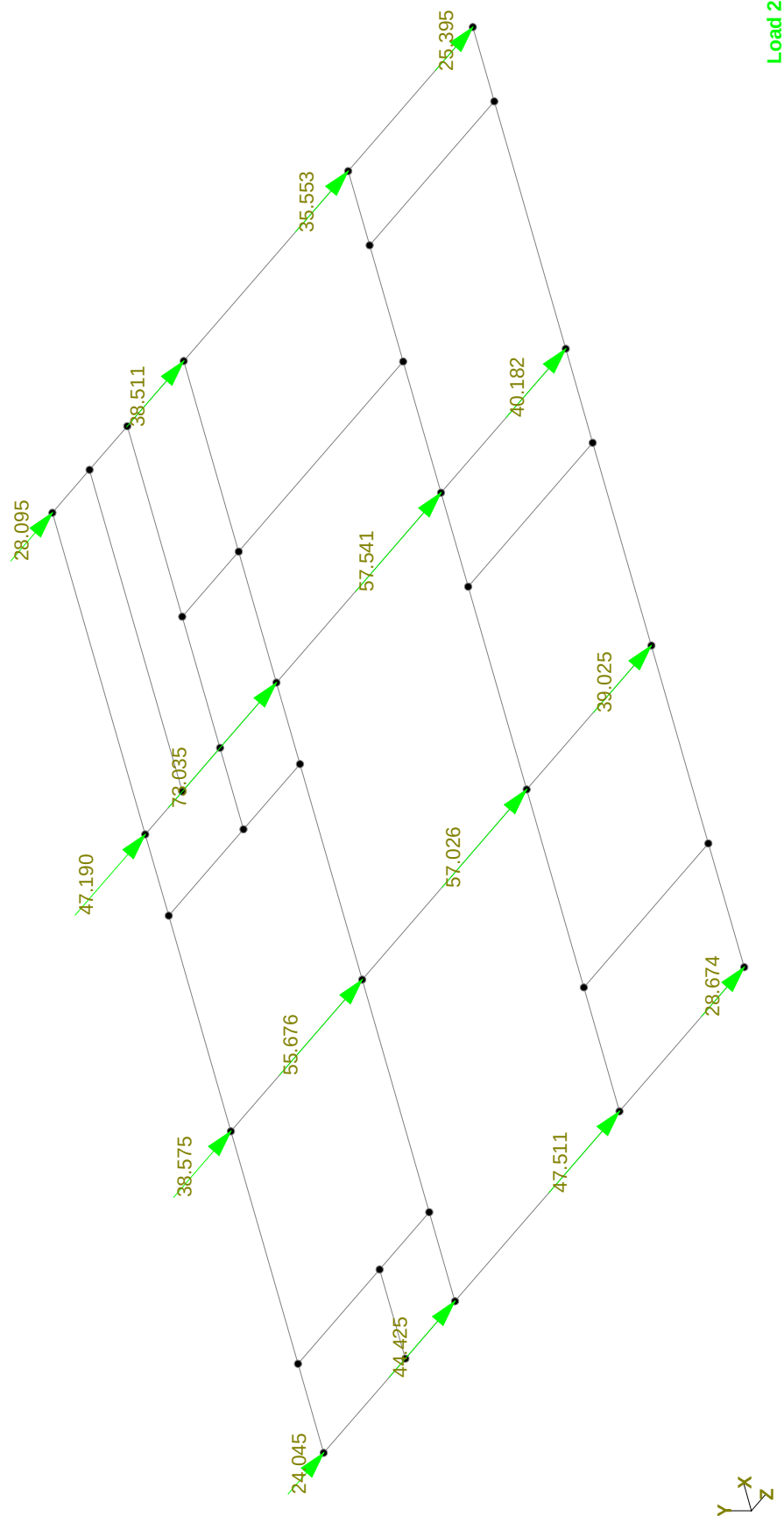




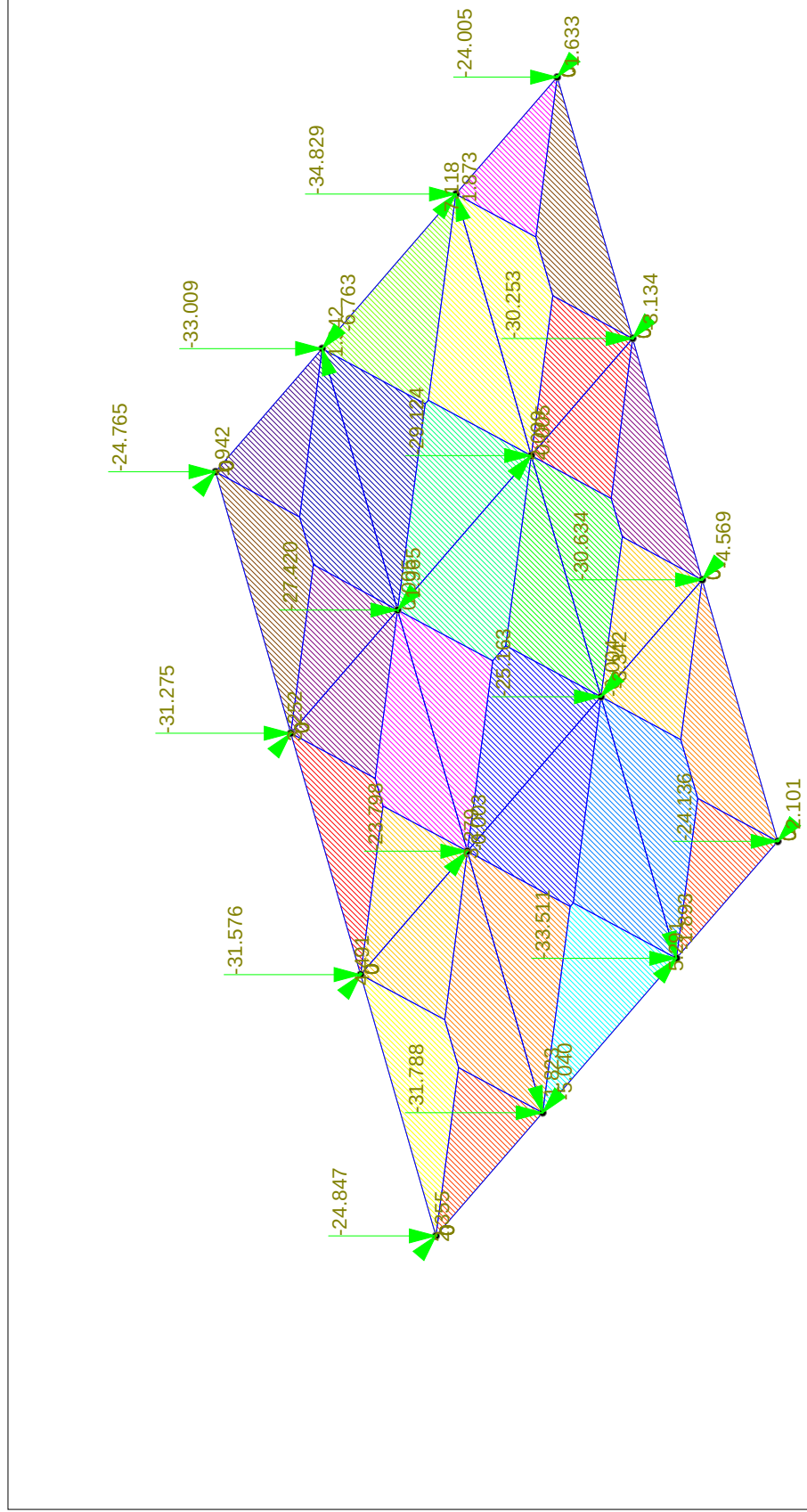


Load - 16 WL6 (0 DEG ,INTERNAL PRESSURE +0.2, LEE WARD -0.2, REVERSE) (Ground Floor)

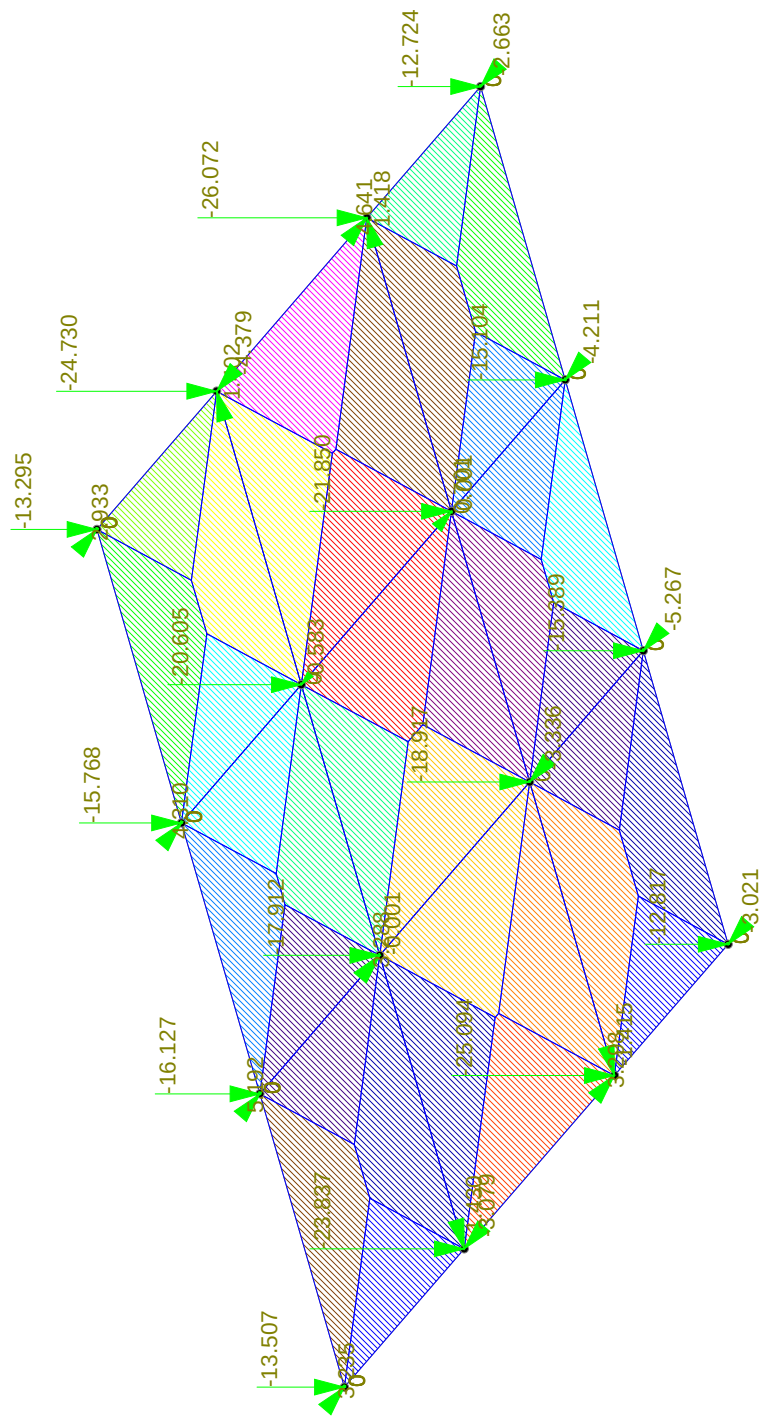




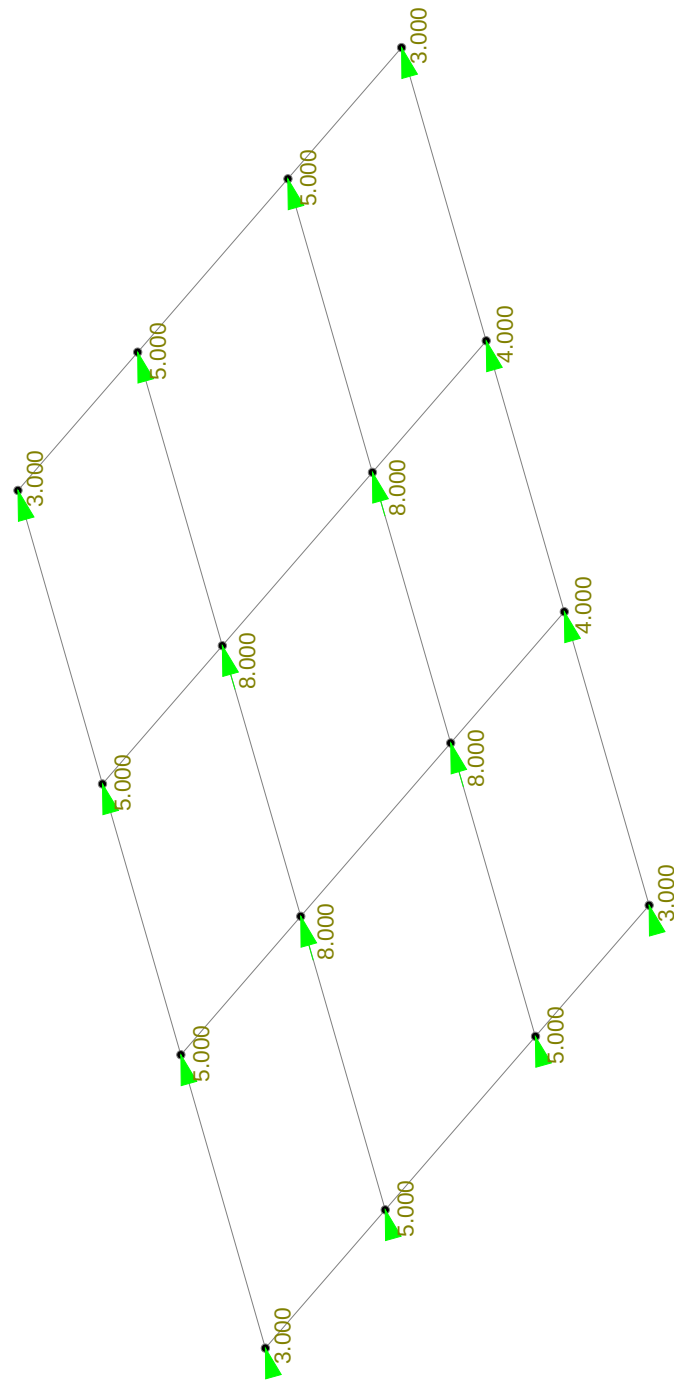
Load 22 EQZ (EARTHQUAKE LOAD) (Ground Floor)



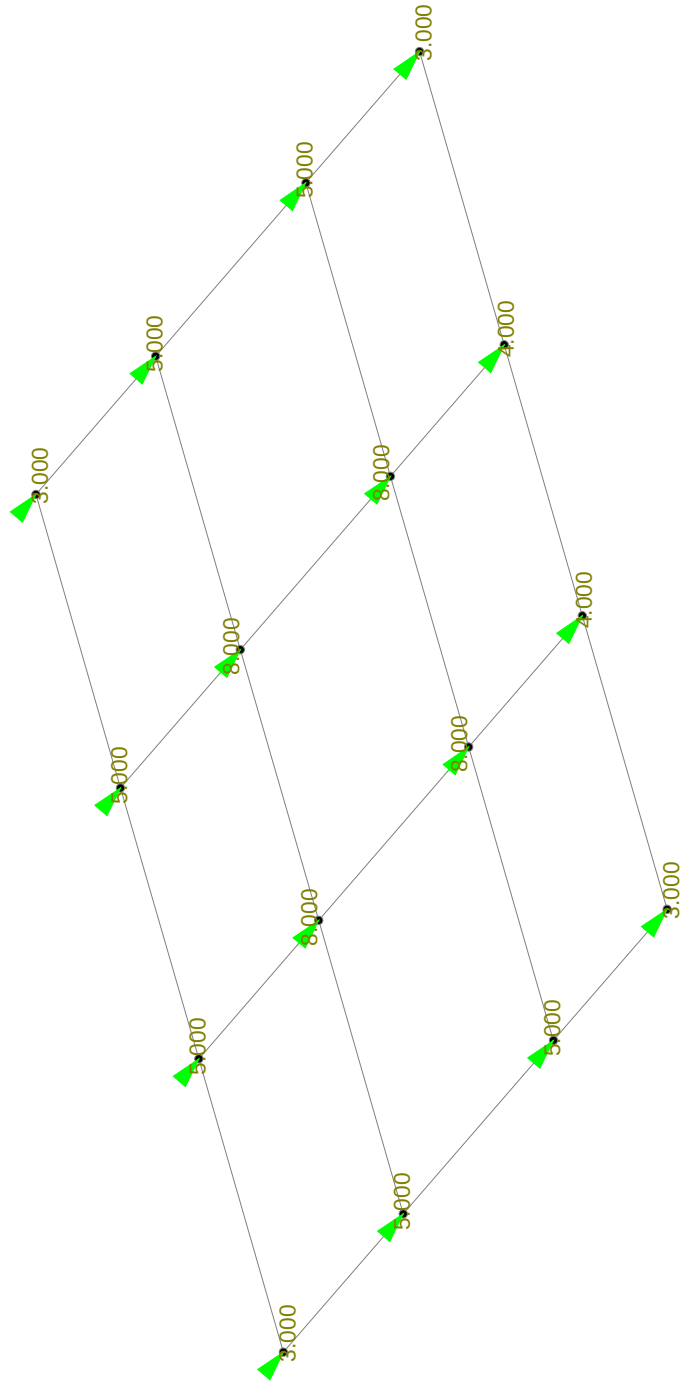
Load 2 Dead Load (Roof Level)



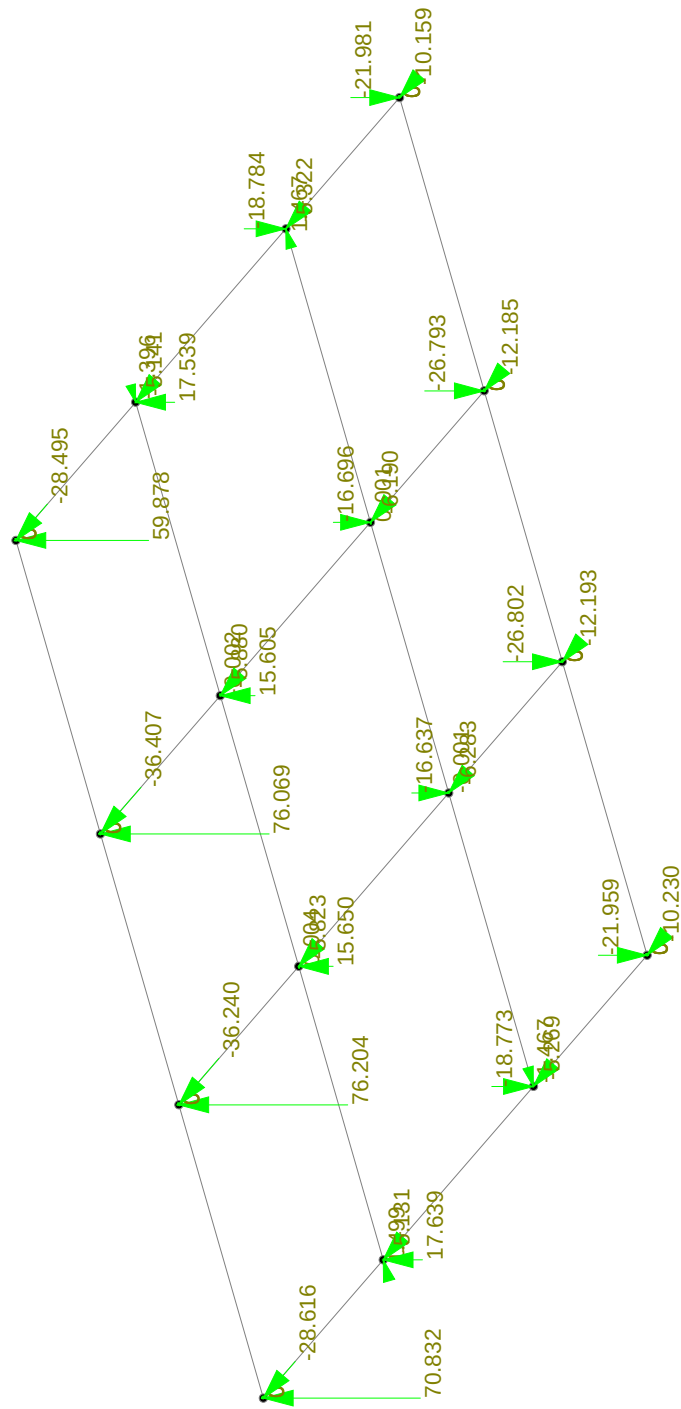
Load 3 Live Load (Roof Level)



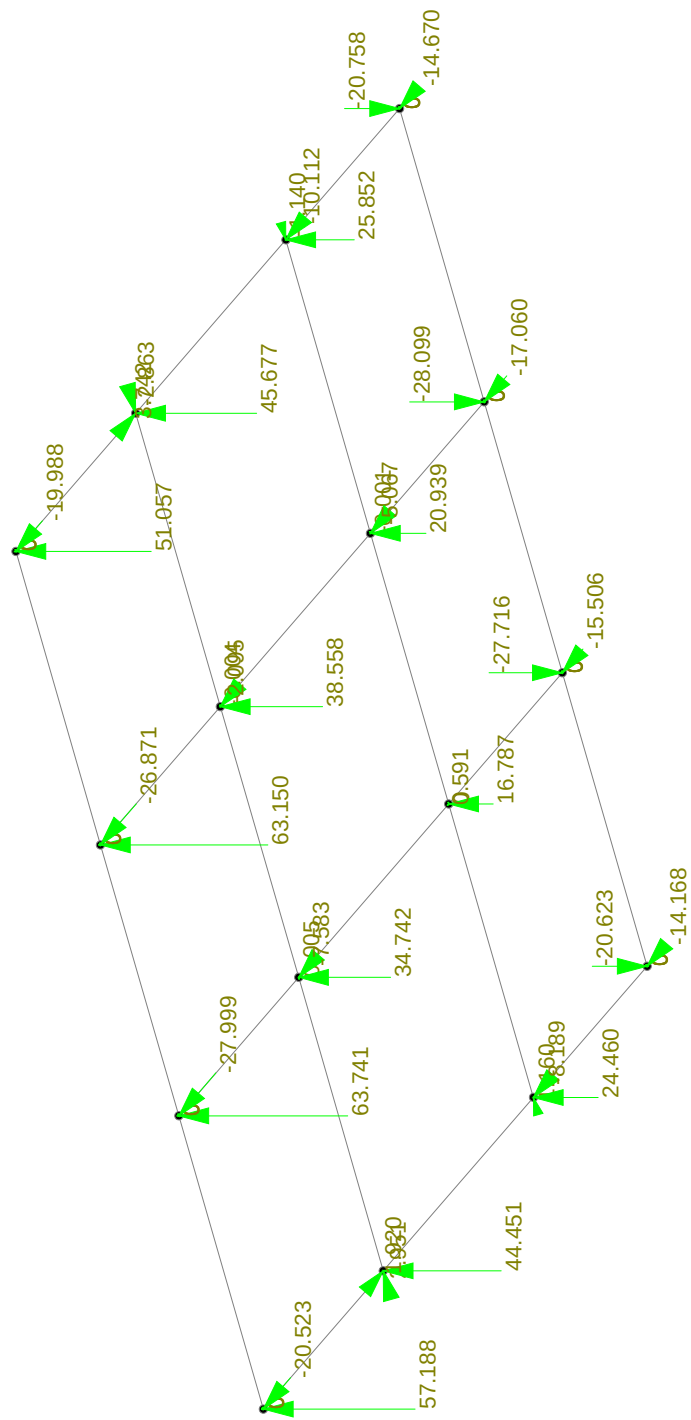
Load 4 PH(X-DIR) (Roof Level)



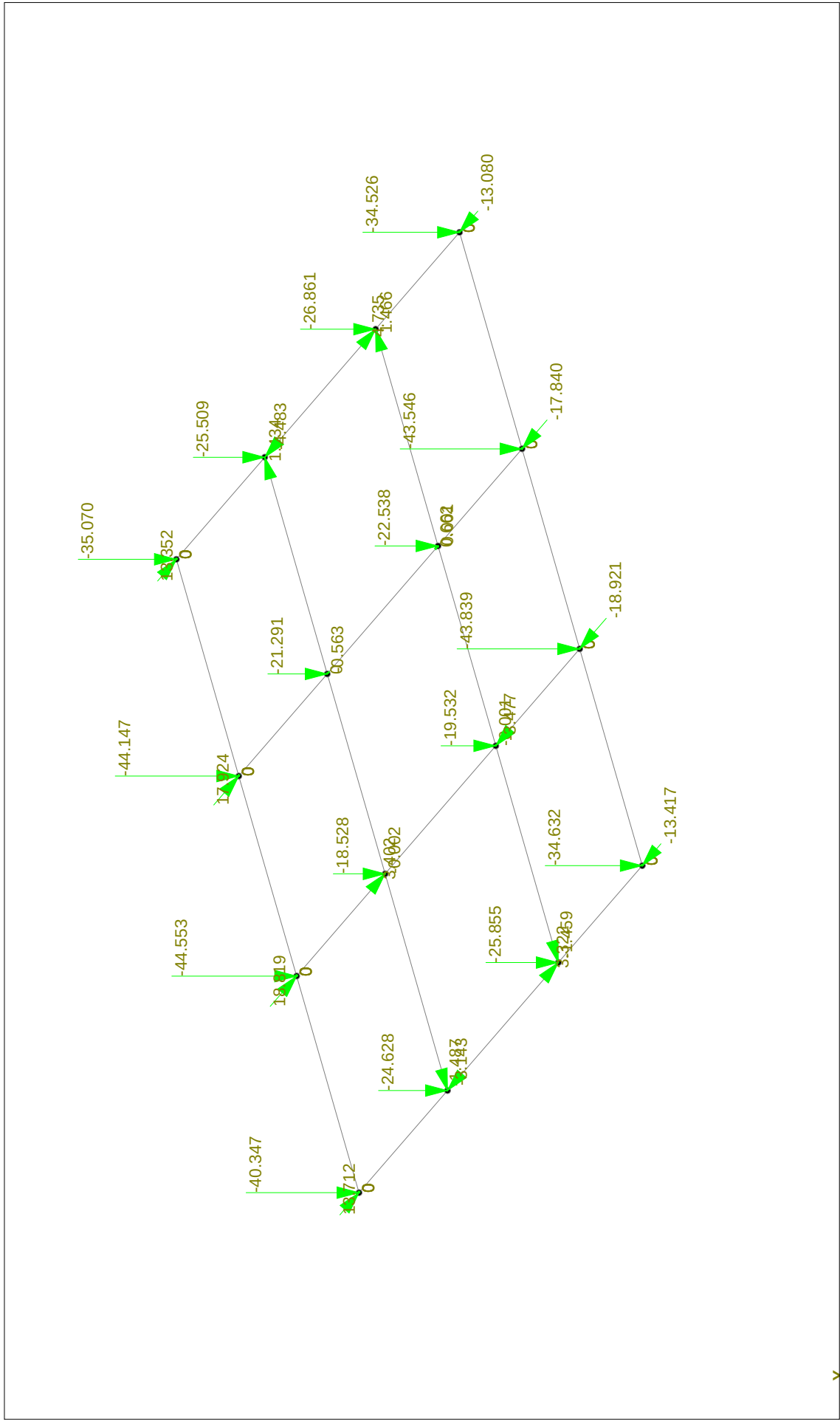
Load 5 PH(Z-DIR) (Roof Level)



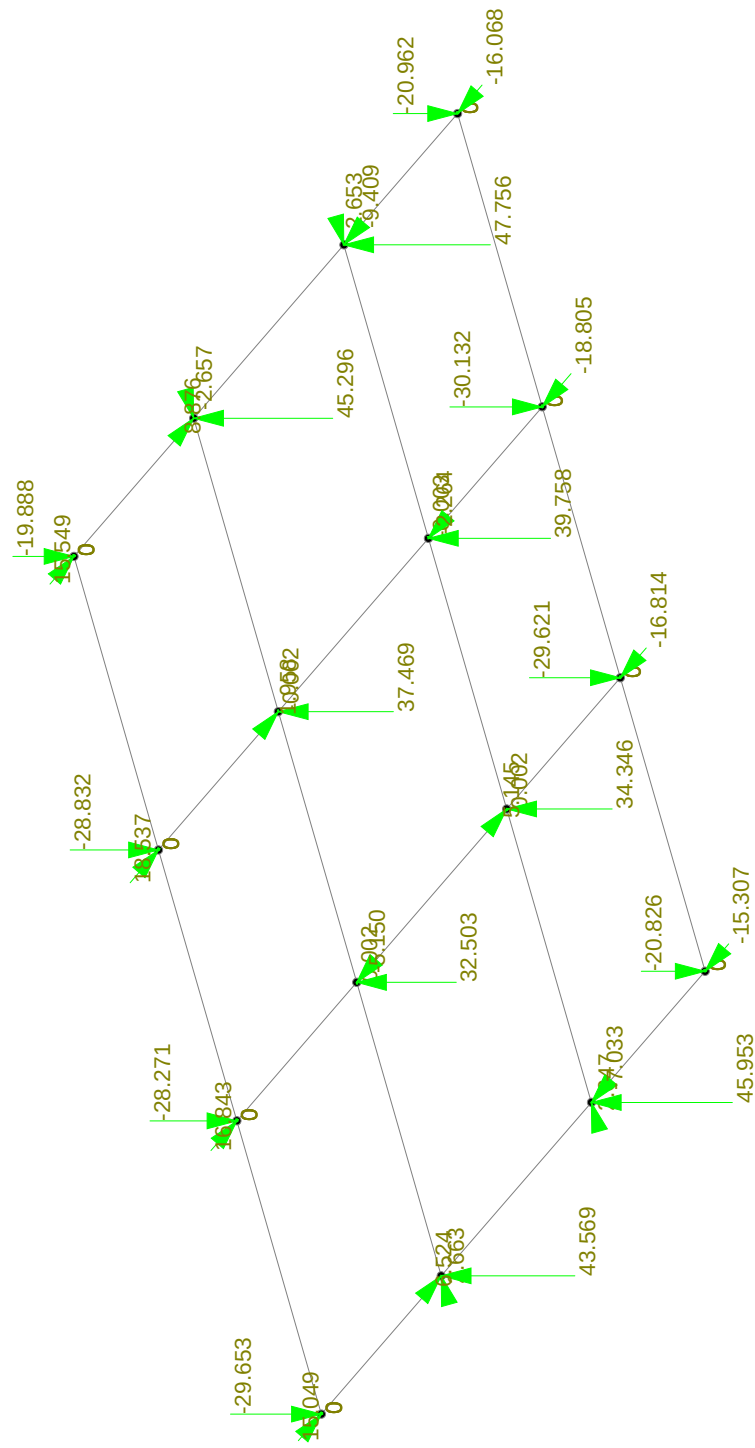
Load 11 WL1 (0 DEG ,INTERNAL SUCTION -0.3) (Roof Level)



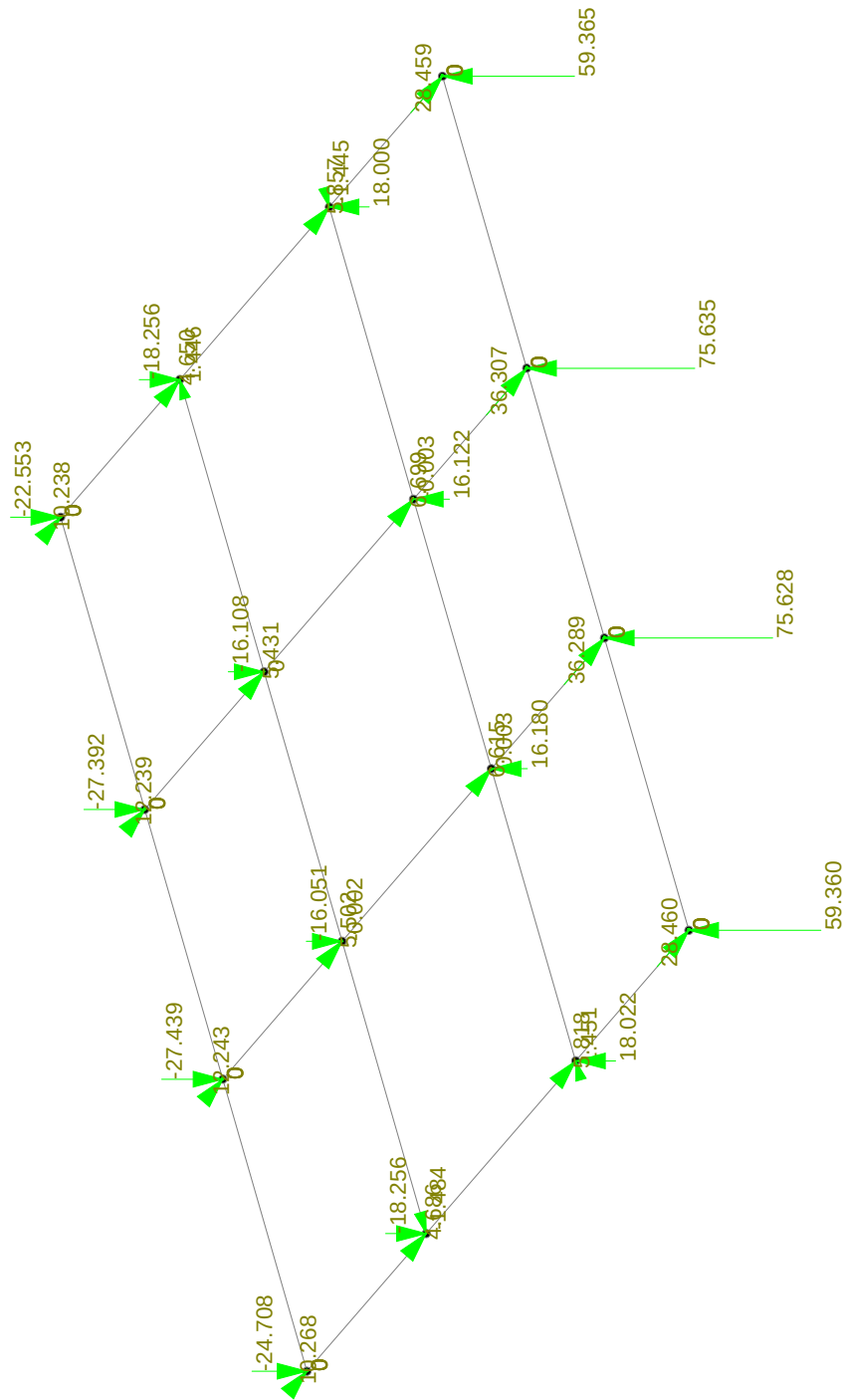
Load 12 WL2 (0 DEG ,INTERNAL PRESSURE +0.2) (Roof Level)



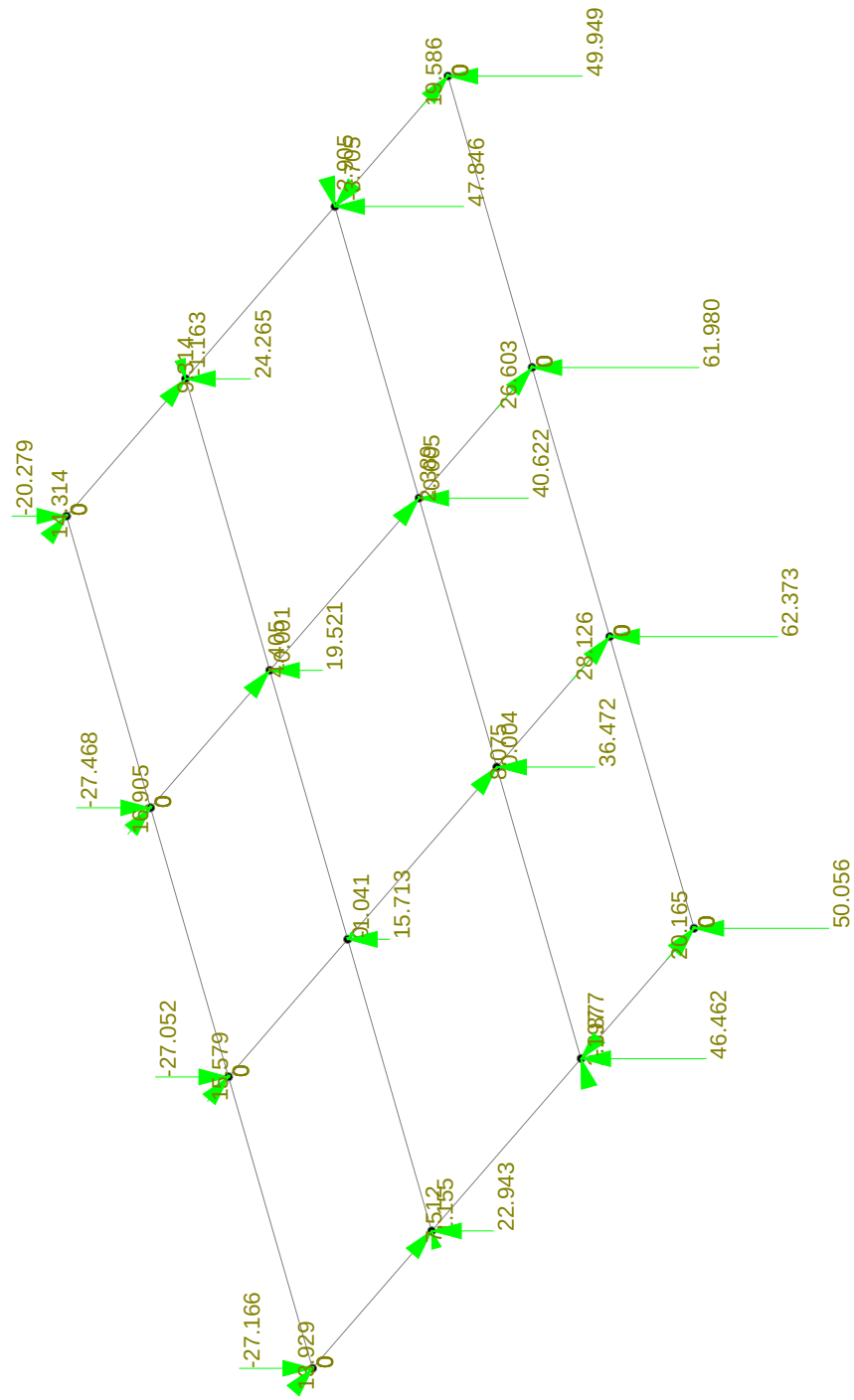
Load 13 WL3 (90 DEG ,INTERNAL SUCTION -0.3, WW&LW +0.2) (Roof Level)



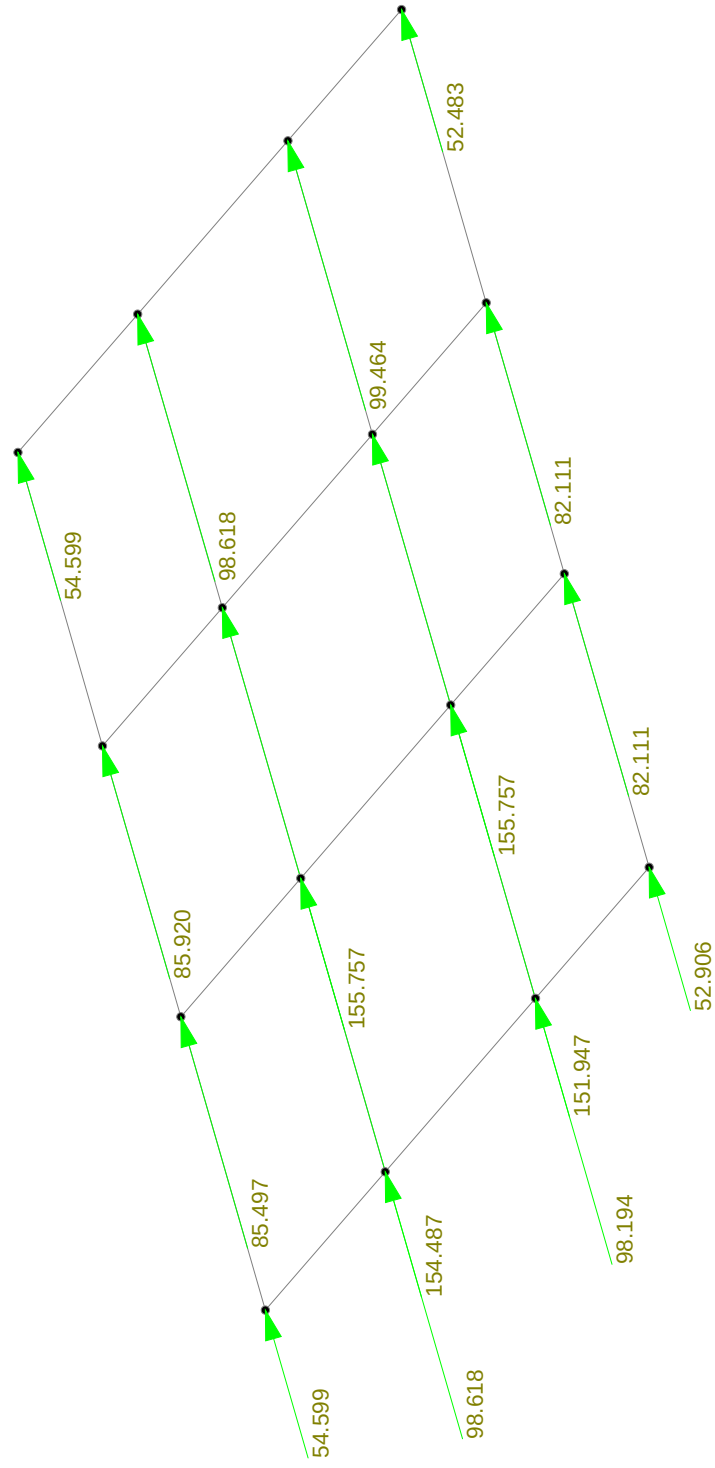
Load 14 WL4 (90 DEG ,INTERNAL PRESSURE +0.2, WW&LW -0.7) (Roof Level)



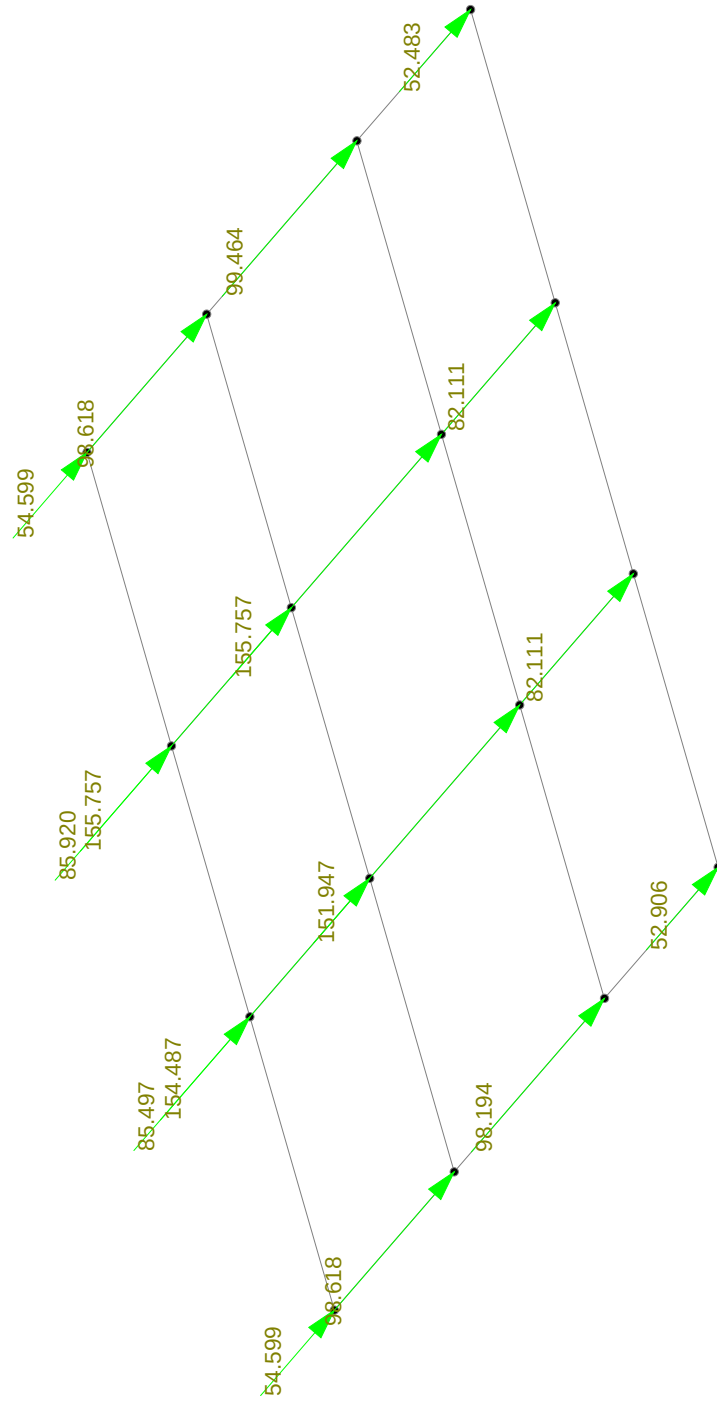
Load 15 WL5 (0 DEG ,INTERNAL SUCTION -0.3, LEE WARD -0.7, REVERSE) (Roof Level)



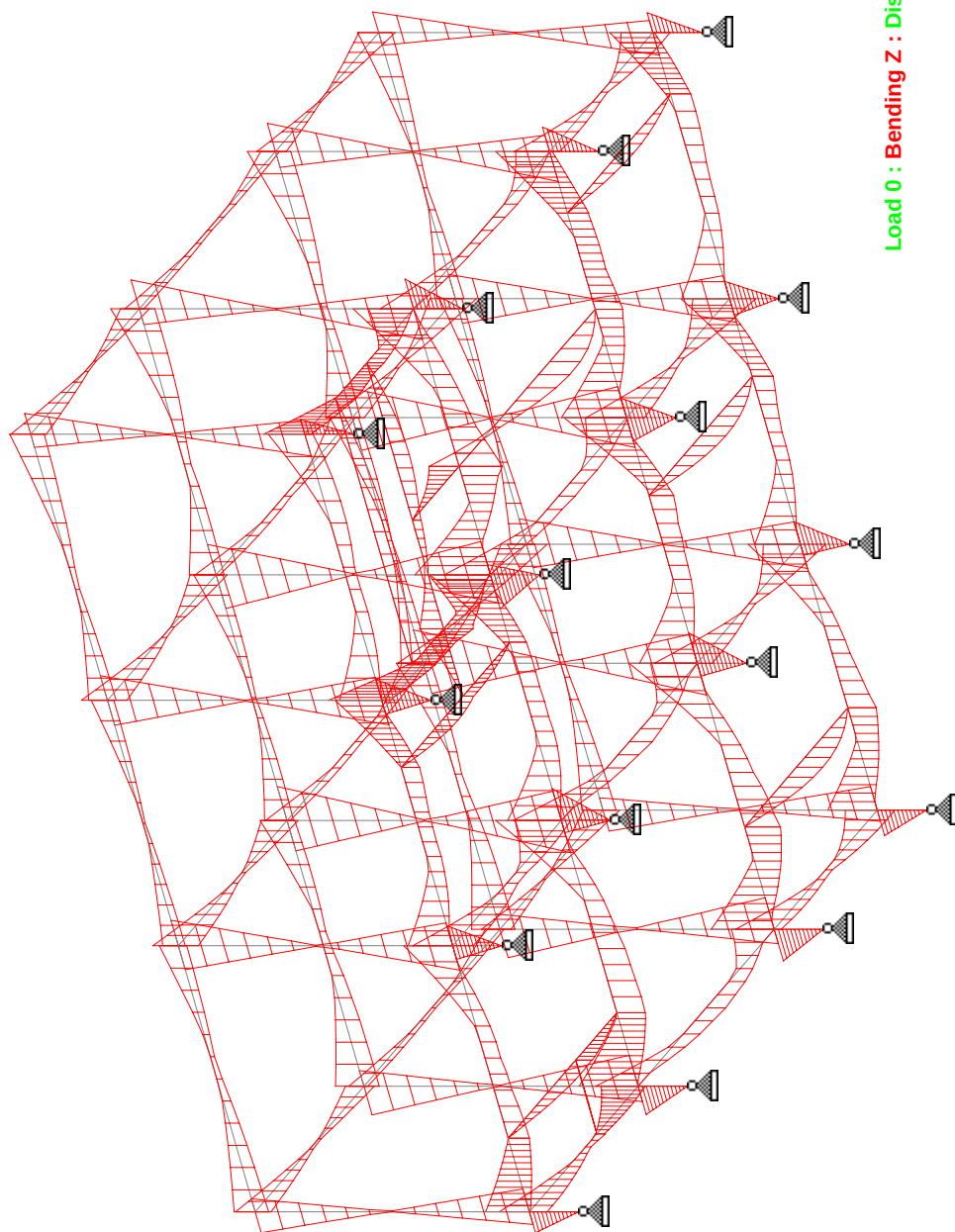
y
Load 16 WL6 (0 DEG ,INTERNAL PRESSURE +0.2, LEE WARD -0.2, REVERSE) (Roof Level)



Load - 21 EQX (EARTHQUAKE LOAD) (Roof Level)



Load 22 EQZ (EARTHQUAKE LOAD) (Roof Level)



Load 0 : Bending Z : Displacement

Bending Moment Envelope

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97. 107 WEIGHT 1136
 98. 108 WEIGHT 599
 99. 109 WEIGHT 739
 100. 110 WEIGHT 887
 101. 111 WEIGHT 895
 102. 112 WEIGHT 553
 103. 113 WEIGHT 446
 104. 114 WEIGHT 607
 105. 115 WEIGHT 625
 106. 116 WEIGHT 395
 107. 118 WEIGHT 129
 108. 119 WEIGHT 202
 109. 120 WEIGHT 203
 110. 121 WEIGHT 129
 111. 122 WEIGHT 233
 112. 123 WEIGHT 365
 113. 124 WEIGHT 368
 114. 125 WEIGHT 233
 115. 126 WEIGHT 232
 116. 127 WEIGHT 359
 117. 128 WEIGHT 368
 118. 129 WEIGHT 235
 119. 130 WEIGHT 125
 120. 131 WEIGHT 194
 121. 132 WEIGHT 194
 122. 133 WEIGHT 124
 123. *****
 124. LOAD 21 EQZ (EARTHQUAKE LOAD)
 125. UBC LOAD X
 126. *****
 127. LOAD 22 EQZ (EARTHQUAKE LOAD)
 128. UBC LOAD Z
 129. *****
 130. LOAD 1 SELFWEIGHT
 131. SELFWEIGHT Y -1
 132. *****SELFWEIGHT TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****
 133. JOINT LOAD
 134. 113 FX -0.154 FY -5.027 FZ -0.402
 135. 101 FX -0.088 FY -5.149 FZ 0.414
 136. 114 FX -0.006 FY -6.382 FZ -0.509
 137. 102 FX -0.04 FY -6.315 FZ 0.501
 138. 115 FX 0.006 FY -6.382 FZ -0.509
 139. 103 FX -0.029 FY -6.395 FZ 0.511
 140. 116 FX 0.154 FY -5.028 FZ -0.402
 141. 104 FX 0.122 FY -5.078 FZ 0.409
 142. 126 FX 0.159 FY -9.379 FZ 1.464
 143. 122 FX 0.2 FY -8.982 FZ -1.545
 144. 127 FX -0.001 FY -8.777 FZ 1.223
 145. 123 FX -0.001 FY -8.381 FZ -1.296
 146. 128 FX 0.001 FY -8.778 FZ 1.223

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147. 124 FX 0.001 FY -8.378 FZ -1.295
 148. 129 FX -0.163 FY -9.378 FZ 1.468
 149. 125 FX -0.163 FY -8.985 FZ -1.539
 150. 130 FX 0 FY -7.905 FZ 0
 151. 118 FX 0 FY -7.881 FZ 0
 152. 131 FX 0 FY -8.963 FZ 0
 153. 119 FX 0 FY -8.988 FZ 0
 154. 132 FX 0 FY -8.963 FZ 0
 155. 120 FX 0 FY -9.031 FZ 0
 156. 133 FX 0 FY -7.904 FZ 0
 157. 121 FX 0 FY -7.886 FZ 0
 158. 130 FX 0 FY -0.068 FZ -0.786
 159. 118 FX 0 FY -0.068 FZ 0.711
 160. 131 FX 0 FY -0.068 FZ -0.828
 161. 119 FX 0 FY -0.068 FZ 0.774
 162. 132 FX 0 FY -0.068 FZ -0.829
 163. 120 FX 0 FY -0.068 FZ 0.751
 164. 133 FX 0 FY -0.068 FZ -0.789
 165. 121 FX 0 FY -0.068 FZ 0.712
 166. LOAD 2 DL
 167. MEMBER LOAD
 168. *****
 169. *****WALL(T=250 MM,DENSITY=23 KN/M^3,H=6.1 M)=0.25X23X6.1=35.0 KN/M
 170. ***PROJECTION OF SLAB (0.62X0.225)=0.62X0.225X25=3.5 KN/M
 171. **PROJECTION OF SLAB (0.27X0.225)=0.27X0.225X25=1.52 KN/M
 172. **FLOOR DL (225 THICK+FIN+M & E)= 7.1 KN/M^2
 173. **ROOF DL (175 THICK+FIN+M & E)= 5.9 KN/M^2
 174. *****
 175. 1 TO 8 10 TO 14 16 TO 20 22 TO 30 33 37 39 45 46 48 TO 50 55 UNI GY -35
 176. 27 TO 30 46 TO 50 UNI GY -1.52
 177. 1 TO 5 22 TO 26 55 UNI GY -3.5
 178. 44 47 UNI GY -35 0 3.5
 179. FLOOR LOAD
 180. YRANGE 1.5 2 FLOAD -7.1 XRange 0 20 ZRange 0 20
 181. YRANGE 1.5 2 FLOAD 7.1 XRange 0 2 ZRange 0 3.3
 182. YRANGE 8 9 FLOAD -5.9 XRange 0 20 ZRange 0 20
 183. *****
 184. JOINT LOAD
 185. *****DEAD LOAD TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****
 186. 113 FX -0.767 FY -13.107 FZ -1.158
 187. 101 FX -0.415 FY -13.414 FZ 1.177
 188. 114 FX -0.011 FY -18.976 FZ -1.593
 189. 102 FX -0.168 FY -18.395 FZ 1.522
 190. 115 FX 0.011 FY -18.845 FZ -1.579
 191. 103 FX -0.142 FY -18.69 FZ 1.555
 192. 116 FX 0.768 FY -13.069 FZ -1.154
 193. 104 FX 0.623 FY -13.094 FZ 1.158
 194. 126 FX -1.893 FY -33.511 FZ 5.291
 195. 122 FX -1.823 FY -31.788 FZ -5.04
 196. 127 FX -0.004 FY -25.163 FZ -3.342
 197. 123 FX -0.003 FY -23.798 FZ 3.279
 198. 128 FX 0.005 FY -29.124 FZ 2.099
 199. 124 FX 0.005 FY -27.42 FZ -1.965

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200. 129 FX 1.873 FY -34.829 FZ 7.118
 201. 125 FX 1.942 FY -33.009 FZ -6.763
 202. 130 FX 0 FY -24.136 FZ 0
 203. 118 FX 0 FY -24.847 FZ 0
 204. 131 FX 0 FY -30.634 FZ 0
 205. 119 FX 0 FY -31.576 FZ 0
 206. 132 FX 0 FY -30.253 FZ 0
 207. 120 FX 0 FY -31.275 FZ 0
 208. 133 FX 0 FY -24.005 FZ 0
 209. 121 FX 0 FY -24.765 FZ 0
 210. 130 FX 0 FY 0 FZ -2.101
 211. 118 FX 0 FY 0 FZ 2.355
 212. 131 FX 0 FY 0 FZ -4.569
 213. 119 FX 0 FY 0 FZ 4.491
 214. 132 FX 0 FY 0 FZ -3.134
 215. 120 FX 0 FY 0 FZ 3.252
 216. 133 FX 0 FY 0 FZ -1.633
 217. 121 FX 0 FY 0 FZ 1.942
 218. *****
 219. LOAD 3 LL
 220. MEMBER LOAD
 221. **PROJECTION OF SLAB =0.62X10=6.2 KN/M
 222. **PROJECTION OF SLAB =0.27X10=2.7 KN/M
 223. **FLOOR LI1 = 15 KN/M^2
 224. **FLOOR LI2 = 10 KN/M^2
 225. **ROOF LI3 = 0.6 KN/M^2
 226. 27 TO 30 46 TO 50 UNI GY -2.7
 227. 1 TO 5 22 TO 26 55 UNI GY -6.2
 228. FLOOR LOAD
 229. YRANGE 1.5 2 FLOAD -10 XRANGE 0 20 ZRANGE 0 20
 230. YRANGE 1.5 2 FLOAD 10 XRANGE 0 2 ZRANGE 0 3.3
 231. YRANGE 1.5 2 FLOAD -5 XRANGE 1.5 11 ZRANGE 0 6
 232. YRANGE 1.5 2 FLOAD -5 XRANGE 0 15.2 ZRANGE 5.25 12
 233. YRANGE 8 9 FLOAD -0.6 XRANGE 0 20 ZRANGE 0 20
 234. JOINT LOAD
 235. ****LIVE LOAD TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****
 236. 113 FX -0.029 FY 1.422 FZ 0.196
 237. 101 FX -0.006 FY 1.623 FZ -0.206
 238. 114 FX 0.008 FY 1.47 FZ 0.206
 239. 102 FX 0.007 FY 1.586 FZ -0.22
 240. 115 FX -0.008 FY 1.564 FZ 0.216
 241. 103 FX -0.004 FY 1.651 FZ -0.228
 242. 116 FX 0.03 FY 1.455 FZ 0.2
 243. 104 FX 0.028 FY 1.497 FZ -0.206
 244. 126 FX -1.415 FY -25.094 FZ 3.298
 245. 122 FX -1.43 FY -23.837 FZ -3.079
 246. 127 FX 0 FY -18.917 FZ -3.336
 247. 123 FX -0.001 FY -17.912 FZ 3.288
 248. 128 FX 0.001 FY -21.85 FZ 0.701
 249. 124 FX 0 FY -20.605 FZ -0.583
 250. 129 FX 1.418 FY -26.072 FZ 4.641
 251. 125 FX 1.402 FY -24.73 FZ -4.379
 252. 130 FX 0 FY -12.817 FZ 0

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253. 118 FX 0 FY -13.507 FZ 0
 254. 131 FX 0 FY -15.389 FZ 0
 255. 119 FX 0 FY -16.127 FZ 0
 256. 132 FX 0 FY -15.104 FZ 0
 257. 120 FX 0 FY -15.768 FZ 0
 258. 133 FX 0 FY -12.724 FZ 0
 259. 121 FX 0 FY -13.295 FZ 0
 260. 130 FX 0 FY 0 FZ -3.021
 261. 118 FX 0 FY 0 FZ 3.235
 262. 131 FX 0 FY 0 FZ -5.267
 263. 119 FX 0 FY 0 FZ 5.192
 264. 132 FX 0 FY 0 FZ -4.211
 265. 120 FX 0 FY 0 FZ 4.31
 266. 133 FX 0 FY 0 FZ -2.663
 267. 121 FX 0 FY 0 FZ 2.933
 268. LOAD 4 PH(X-DIR)
 269. *****
 270. **PH =1.5% (1.4DL+1.6LL)
 271. JOINT LOAD
 272. 101 FX 9
 273. 102 FX 16
 274. 103 FX 18
 275. 104 FX 10
 276. 105 FX 18
 277. 106 FX 26
 278. 107 FX 31
 279. 108 FX 16
 280. 109 FX 19
 281. 110 FX 25
 282. 111 FX 25
 283. 112 FX 14
 284. 113 FX 11
 285. 114 FX 15
 286. 115 FX 16
 287. 116 FX 10
 288. 118 FX 3
 289. 119 FX 5
 290. 120 FX 5
 291. 121 FX 3
 292. 122 FX 5
 293. 123 FX 8
 294. 124 FX 8
 295. 125 FX 5
 296. 126 FX 5
 297. 127 FX 8
 298. 128 FX 8
 299. 129 FX 5
 300. 130 FX 3
 301. 131 FX 4
 302. 132 FX 4
 303. 133 FX 3
 304. LOAD 5 PH(Z-DIR)
 305. *****
 306. **PH =1.5% (1.4DL+1.6LL)
 307. JOINT LOAD
 308. 101 FZ 9

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309. 102 FZ 16
310. 103 FZ 18
311. 104 FZ 10
312. 105 FZ 18
313. 106 FZ 26
314. 107 FZ 31
315. 108 FZ 16
316. 109 FZ 19
317. 110 FZ 25
318. 111 FZ 25
319. 112 FZ 14
320. 113 FZ 11
321. 114 FZ 15
322. 115 FZ 16
323. 116 FZ 10
324. 118 FZ 3
325. 119 FZ 5
326. 120 FZ 5
327. 121 FZ 3
328. 122 FZ 5
329. 123 FZ 8
330. 124 FZ 8
331. 125 FZ 5
332. 126 FZ 5
333. 127 FZ 8
334. 128 FZ 8
335. 129 FZ 5
336. 130 FZ 3
337. 131 FZ 4
338. 132 FZ 4
339. 133 FZ 3
340. *****WIND LOAD TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****
341. LOAD 11 WL1 (0 DEG , INTERNAL SUCTION -0.3)
342. JOINT LOAD
343. 113 FX -0.013 FY 9.734 FZ 3.238
344. 101 FX -0.069 FY -59.253 FZ 16.499
345. 114 FX 0.009 FY 12.566 FZ 4.207
346. 102 FX -0.042 FY -61.947 FZ 21.772
347. 115 FX -0.009 FY 12.563 FZ 4.206
348. 103 FX -0.025 FY -62.121 FZ 21.792
349. 116 FX 0.013 FY 9.764 FZ 3.242
350. 104 FX 0.03 FY -48.056 FZ 16.743
351. 126 FX -1.467 FY -18.773 FZ -5.269
352. 122 FX 1.499 FY 17.639 FZ -5.131
353. 127 FX -0.001 FY -16.637 FZ -6.283
354. 123 FX 0.004 FY 15.65 FZ -5.823
355. 128 FX 0.001 FY -16.696 FZ -6.19
356. 124 FX -0.002 FY 15.605 FZ -5.88
357. 129 FX 1.467 FY -18.784 FZ -5.322
358. 125 FX -1.396 FY 17.539 FZ -5.141
359. 130 FX 0 FY -21.959 FZ 0
360. 118 FX 0 FY 70.832 FZ 0
361. 131 FX 0 FY -26.802 FZ 0
362. 119 FX 0 FY 76.204 FZ 0
363. 132 FX 0 FY -26.793 FZ 0
364. 120 FX 0 FY 76.069 FZ 0

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365. 133 FX 0 FY -21.981 FZ 0
366. 121 FX 0 FY 59.878 FZ 0
367. 130 FX 0 FY 0 FZ -10.23
368. 118 FX 0 FY 0 FZ -28.616
369. 131 FX 0 FY 0 FZ -12.193
370. 119 FX 0 FY 0 FZ -36.24
371. 132 FX 0 FY 0 FZ -12.185
372. 120 FX 0 FY 0 FZ -36.407
373. 133 FX 0 FY 0 FZ -10.159
374. 121 FX 0 FY 0 FZ -28.495
375. *****WIND LOAD TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****
376. LOAD 12 WL2 (0 DEG , INTERNAL PRESSURE +0.2)
377. JOINT LOAD
378. 113 FX 0.078 FY 29.442 FZ 10.688
379. 101 FX -0.037 FY -33.735 FZ 9.067
380. 114 FX -0.003 FY 38.854 FZ 14.015
381. 102 FX -0.034 FY -34.865 FZ 11.868
382. 115 FX 0.003 FY 38.696 FZ 13.999
383. 103 FX -0.011 FY -35.049 FZ 11.89
384. 116 FX -0.078 FY 29.399 FZ 10.684
385. 104 FX -0.026 FY -27.587 FZ 9.198
386. 126 FX 1.16 FY 24.46 FZ -8.189
387. 122 FX 2.951 FY 44.451 FZ 1.92
388. 127 FX 0 FY 16.787 FZ 0.591
389. 123 FX 0.005 FY 34.742 FZ -7.583
390. 128 FX -0.001 FY 20.939 FZ -5.067
391. 124 FX -0.004 FY 38.558 FZ -2.095
392. 129 FX -1.14 FY 25.852 FZ -10.112
393. 125 FX -2.863 FY 45.677 FZ 3.742
394. 130 FX 0 FY -20.623 FZ 0
395. 118 FX 0 FY 57.188 FZ 0
396. 131 FX 0 FY -27.716 FZ 0
397. 119 FX 0 FY 63.741 FZ 0
398. 132 FX 0 FY -28.099 FZ 0
399. 120 FX 0 FY 63.15 FZ 0
400. 133 FX 0 FY -20.758 FZ 0
401. 121 FX 0 FY 51.057 FZ 0
402. 130 FX 0 FY 0 FZ -14.168
403. 118 FX 0 FY 0 FZ -20.523
404. 131 FX 0 FY 0 FZ -15.506
405. 119 FX 0 FY 0 FZ -27.999
406. 132 FX 0 FY 0 FZ -17.06
407. 120 FX 0 FY 0 FZ -26.871
408. 133 FX 0 FY 0 FZ -14.67
409. 121 FX 0 FY 0 FZ -19.988
410. *****WIND LOAD TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****
411. LOAD 13 WL3 (90 DEG , INTERNAL SUCTION -0.3, WW&LW +0.2)
412. JOINT LOAD
413. 113 FX 0.011 FY 23.471 FZ 7.955
414. 101 FX 0.022 FY 28.783 FZ -7.846
415. 114 FX 0.01 FY 30.07 FZ 10.321
416. 102 FX 0.021 FY 30.088 FZ -10.323
417. 115 FX -0.01 FY 30.165 FZ 10.331
418. 103 FX 0.006 FY 30.231 FZ -10.339
419. 116 FX -0.011 FY 23.492 FZ 7.957
420. 104 FX -0.001 FY 23.514 FZ -7.961

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421. 126 FX -1.459 FY -25.855 FZ 3.322
 422. 122 FX -1.487 FY -24.628 FZ -3.143
 423. 127 FX -0.001 FY -19.532 FZ -3.477
 424. 123 FX -0.002 FY -18.528 FZ 3.402
 425. 128 FX 0.001 FY -22.538 FZ 0.662
 426. 124 FX 0 FY -21.291 FZ -0.563
 427. 129 FX 1.466 FY -26.861 FZ 4.735
 428. 125 FX 1.434 FY -25.509 FZ -4.483
 429. 130 FX 0 FY -34.632 FZ 0
 430. 118 FX 0 FY -40.347 FZ 0
 431. 131 FX 0 FY -43.839 FZ 0
 432. 119 FX 0 FY -44.553 FZ 0
 433. 132 FX 0 FY -43.546 FZ 0
 434. 120 FX 0 FY -44.147 FZ 0
 435. 133 FX 0 FY -34.526 FZ 0
 436. 121 FX 0 FY -35.07 FZ 0
 437. 130 FX 0 FY 0 FZ -13.417
 438. 118 FX 0 FY 0 FZ 13.712
 439. 131 FX 0 FY 0 FZ -18.921
 440. 119 FX 0 FY 0 FZ 18.819
 441. 132 FX 0 FY 0 FZ -17.84
 442. 120 FX 0 FY 0 FZ 17.924
 443. 133 FX 0 FY 0 FZ -13.08
 444. 121 FX 0 FY 0 FZ 13.352
 445. *****WIND LOAD TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****
 446. LOAD 14 WL4 (90 DEG , INTERNAL PRESSURE +0.2, WW&LW -0.7)
 447. JOINT LOAD
 448. 113 FX 0.126 FY 41.694 FZ 15.203
 449. 101 FX 0.064 FY 51.59 FZ -14.951
 450. 114 FX -0.009 FY 55.069 FZ 19.946
 451. 102 FX 0.016 FY 54.681 FZ -19.9
 452. 115 FX 0.009 FY 54.877 FZ 19.926
 453. 103 FX 0.027 FY 54.701 FZ -19.9
 454. 116 FX -0.127 FY 41.588 FZ 15.191
 455. 104 FX -0.105 FY 41.491 FZ -15.178
 456. 126 FX 2.647 FY 45.953 FZ -7.033
 457. 122 FX 2.663 FY 43.569 FZ 6.524
 458. 127 FX 0.002 FY 34.346 FZ 5.145
 459. 123 FX 0.002 FY 32.503 FZ -5.15
 460. 128 FX -0.003 FY 39.758 FZ -2.264
 461. 124 FX -0.002 FY 37.469 FZ 1.958
 462. 129 FX -2.653 FY 47.756 FZ -9.409
 463. 125 FX -2.657 FY 45.296 FZ 8.876
 464. 130 FX 0 FY -20.826 FZ 0
 465. 118 FX 0 FY -29.653 FZ 0
 466. 131 FX 0 FY -29.621 FZ 0
 467. 119 FX 0 FY -28.271 FZ 0
 468. 132 FX 0 FY -30.132 FZ 0
 469. 120 FX 0 FY -28.832 FZ 0
 470. 133 FX 0 FY -20.962 FZ 0
 471. 121 FX 0 FY -19.888 FZ 0
 472. 130 FX 0 FY 0 FZ -15.307
 473. 118 FX 0 FY 0 FZ 15.049
 474. 131 FX 0 FY 0 FZ -16.814
 475. 119 FX 0 FY 0 FZ 16.843
 476. 132 FX 0 FY 0 FZ -18.805

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477. 120 FX 0 FY 0 FZ 18.537
 478. 133 FX 0 FY 0 FZ -16.068
 479. 121 FX 0 FY 0 FZ 15.549
 480. *****WIND LOAD TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****
 481. LOAD 15 WL5 (0 DEG , INTERNAL SUCTION -0.3, LEE WARD -0.7, REVERSE)
 482. JOINT LOAD
 483. 113 FX -0.067 FY -48.036 FZ -16.739
 484. 101 FX 0.005 FY 12.045 FZ -3.204
 485. 114 FX -0.009 FY -62.129 FZ -21.795
 486. 102 FX 0.013 FY 12.55 FZ -4.204
 487. 115 FX 0.009 FY -62.129 FZ -21.795
 488. 103 FX 0 FY 12.609 FZ -4.213
 489. 116 FX 0.067 FY -48.039 FZ -16.74
 490. 104 FX 0.016 FY 9.823 FZ -3.25
 491. 126 FX 1.451 FY 18.022 FZ 5.818
 492. 122 FX -1.484 FY -18.256 FZ 4.686
 493. 127 FX 0.003 FY 16.18 FZ 6.615
 494. 123 FX -0.002 FY -16.051 FZ 5.502
 495. 128 FX -0.003 FY 16.122 FZ 6.699
 496. 124 FX 0 FY -16.108 FZ 5.431
 497. 129 FX -1.445 FY 18 FZ 5.857
 498. 125 FX 1.446 FY -18.256 FZ 4.65
 499. 130 FX 0 FY 59.36 FZ 0
 500. 118 FX 0 FY -24.708 FZ 0
 501. 131 FX 0 FY 75.628 FZ 0
 502. 119 FX 0 FY -27.439 FZ 0
 503. 132 FX 0 FY 75.635 FZ 0
 504. 120 FX 0 FY -27.392 FZ 0
 505. 133 FX 0 FY 59.365 FZ 0
 506. 121 FX 0 FY -22.553 FZ 0
 507. 130 FX 0 FY 0 FZ 28.46
 508. 118 FX 0 FY 0 FZ 10.268
 509. 131 FX 0 FY 0 FZ 36.289
 510. 119 FX 0 FY 0 FZ 12.243
 511. 132 FX 0 FY 0 FZ 36.307
 512. 120 FX 0 FY 0 FZ 12.239
 513. 133 FX 0 FY 0 FZ 28.459
 514. 121 FX 0 FY 0 FZ 10.238
 515. *****WIND LOAD TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****
 516. LOAD 16 WL6 (0 DEG , INTERNAL PRESSURE +0.2, LEE WARD -0.2, REVERSE)
 517. JOINT LOAD
 518. 113 FX -0.001 FY -27.449 FZ -9.18
 519. 101 FX 0.045 FY 36.52 FZ -10.52
 520. 114 FX -0.014 FY -34.803 FZ -11.861
 521. 102 FX 0.015 FY 38.577 FZ -13.982
 522. 115 FX 0.014 FY -34.948 FZ -11.876
 523. 103 FX 0.018 FY 38.593 FZ -13.983
 524. 116 FX 0 FY -27.528 FZ -9.188
 525. 104 FX -0.061 FY 29.363 FZ -10.678
 526. 126 FX 2.897 FY 46.462 FZ -1.877
 527. 122 FX 1.155 FY 22.943 FZ 7.512
 528. 127 FX 0.004 FY 36.472 FZ 8.075
 529. 123 FX 0 FY 15.713 FZ -1.041
 530. 128 FX -0.005 FY 40.622 FZ 2.389
 531. 124 FX -0.001 FY 19.521 FZ 4.405
 532. 129 FX -2.905 FY 47.846 FZ -3.705

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533. 125 FX -1.163 FY 24.265 FZ 9.314
534. 130 FX 0 FY 50.056 FZ 0
535. 118 FX 0 FY -27.166 FZ 0
536. 131 FX 0 FY 62.373 FZ 0
537. 119 FX 0 FY -27.052 FZ 0
538. 132 FX 0 FY 61.98 FZ 0
539. 120 FX 0 FY -27.468 FZ 0
540. 133 FX 0 FY 49.949 FZ 0
541. 121 FX 0 FY -20.279 FZ 0
542. 130 FX 0 FY 0 FZ 20.165
543. 118 FX 0 FY 0 FZ 13.929
544. 131 FX 0 FY 0 FZ 28.126
545. 119 FX 0 FY 0 FZ 15.579
546. 132 FX 0 FY 0 FZ 26.603
547. 120 FX 0 FY 0 FZ 16.905
548. 133 FX 0 FY 0 FZ 19.586
549. 121 FX 0 FY 0 FZ 14.314
550. *****
551. ** SERVICEABILITY*
552. *****
553. LOAD COMB 101 1.0DL+1.0LL
554. 1 1.0 2 1.0 3 1.0
555. LOAD COMB 102 1.0DL+0.5LL
556. 1 1.0 2 1.0 3 0.5
557. LOAD COMB 103 0.8DL+0.8LL+0.8WL1
558. 1 0.8 2 0.8 3 0.8 11 0.8
559. LOAD COMB 104 1.0DL+1.0WL1
560. 1 1.0 2 1.0 11 1.0
561. LOAD COMB 105 0.8DL+0.8LL+0.8WL2
562. 1 0.8 2 0.8 3 0.8 12 0.8
563. LOAD COMB 106 1.0DL+1.0WL2
564. 1 1.0 2 1.0 12 1.0
565. LOAD COMB 107 0.8DL+0.8LL+0.8WL3
566. 1 0.8 2 0.8 3 0.8 13 0.8
567. LOAD COMB 108 1.0DL+1.0WL3
568. 1 1.0 2 1.0 13 1.0
569. LOAD COMB 109 0.8DL+0.8LL+0.8WL4
570. 1 0.8 2 0.8 3 0.8 14 0.8
571. LOAD COMB 110 1.0DL+1.0WL4
572. 1 1.0 2 1.0 14 1.0
573. LOAD COMB 111 0.8DL+0.8LL+0.8WL5
574. 1 0.8 2 0.8 3 0.8 15 0.8
575. LOAD COMB 112 1.0DL+1.0WL4
576. 1 1.0 2 1.0 15 1.0
577. LOAD COMB 113 0.8DL+0.8LL+0.8WL6
578. 1 0.8 2 0.8 3 0.8 16 0.8
579. LOAD COMB 114 1.0DL+1.0WL6
580. 1 1.0 2 1.0 16 1.0
581. *****EQX & EQZ *****
582. **DL+EQL/1.4 = 1.0DL+0.72EQX+0.22 EQZ, EQL=EQX+0.3EQZ
583. **0.9 DL+EQL/1.4 =0.9 DL+0.72 EQX+0.22EQZ
584. **DL 0.75(LL+EQL/1.4) =1.0 DL+0.75 EQX+0.17EQZ
585. *****
586. LOAD COMB 115 DL+0.72EQX+0.22EQZ
587. 1 1.0 2 1.0 21 0.72 22 0.22
588. LOAD COMB 116 DL+0.72EQX-0.22EQZ

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589. 1 1.0 2 1.0 21 0.72 22 -0.22
590. LOAD COMB 117 DL-0.72EQX+0.22EQZ
591. 1 1.0 2 1.0 21 -0.72 22 0.22
592. LOAD COMB 118 DL-0.72EQX+0.22EQZ
593. 1 1.0 2 1.0 21 -0.72 22 -0.22
594. LOAD COMB 119 DL+0.72EQX+0.22EQX
595. 1 1.0 2 1.0 22 0.72 21 0.22
596. LOAD COMB 120 DL+0.72EQX-0.22EQX
597. 1 1.0 2 1.0 22 0.72 21 -0.22
598. LOAD COMB 121 DL-0.72EQX+0.22EQX
599. 1 1.0 2 1.0 22 -0.72 21 0.22
600. LOAD COMB 122 DL-0.72EQX-0.22EQX
601. 1 1.0 2 1.0 22 -0.72 21 -0.22
602. LOAD COMB 123 0.9DL+0.72EQX+0.22EQZ
603. 1 0.9 2 0.9 21 0.72 22 0.22
604. LOAD COMB 124 0.9DL+0.72EQX-0.22EQZ
605. 1 0.9 2 0.9 21 0.72 22 -0.22
606. LOAD COMB 125 0.9DL-0.72EQX+0.22EQZ
607. 1 0.9 2 0.9 21 -0.72 22 0.22
608. LOAD COMB 126 0.9DL-0.72EQX+0.22EQZ
609. 1 0.9 2 0.9 21 -0.72 22 -0.22
610. LOAD COMB 127 0.9DL+0.72EQX+0.22EQX
611. 1 0.9 2 0.9 22 0.72 21 0.22
612. LOAD COMB 128 0.9DL+0.72EQX-0.22EQX
613. 1 0.9 2 0.9 22 0.72 21 -0.22
614. LOAD COMB 129 0.9DL-0.72EQX+0.22EQX
615. 1 0.9 2 0.9 22 -0.72 21 0.22
616. LOAD COMB 130 0.9DL-0.72EQX-0.22EQX
617. 1 0.9 2 0.9 22 -0.72 21 -0.22
618. LOAD COMB 131 1.0DL+0.75LL+0.54EQX+0.17 EQZ
619. 1 1.0 2 1.0 3 0.75 21 0.54 22 0.17
620. LOAD COMB 132 1.0DL+0.75LL+0.54EQX-0.17 EQZ
621. 1 1.0 2 1.0 3 0.75 21 0.54 22 -0.17
622. LOAD COMB 133 1.0DL+0.75LL-0.54EQX+0.17 EQZ
623. 1 1.0 2 1.0 3 0.75 21 -0.54 22 0.17
624. LOAD COMB 134 1.0DL+0.75LL-0.54EQX-0.17 EQZ
625. 1 1.0 2 1.0 3 0.75 21 -0.54 22 -0.17
626. LOAD COMB 135 1.0DL+0.75LL+0.54EQZ+0.17 EQX
627. 1 1.0 2 1.0 3 0.75 22 0.54 21 0.17
628. LOAD COMB 136 1.0DL+0.75LL+0.54EQZ-0.17 EQX
629. 1 1.0 2 1.0 3 0.75 22 0.54 21 -0.17
630. LOAD COMB 137 1.0DL+0.75LL-0.54EQZ+0.17 EQX
631. 1 1.0 2 1.0 3 0.75 22 -0.54 21 0.17
632. LOAD COMB 138 1.0DL+0.75LL-0.54EQZ-0.17 EQX
633. 1 1.0 2 1.0 3 0.75 22 -0.54 21 -0.17
634. *****
635. ** CONCRETE DESIGN
636. *****
637. LOAD COMB 200 1.4DL+1.6LL
638. 1 1.4 2 1.4 3 1.6
639. *****WL1*****
640. LOAD COMB 201 1.2DL+1.2LL+1.2WL1
641. 1 1.2 2 1.2 3 1.2 11 1.2
642. LOAD COMB 202 1.4DL+1.4WL1
643. 1 1.4 2 1.4 11 1.4
644. LOAD COMB 203 1DL+1.4WL1

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645. 1 1.0 2 1.0 2 1.0 11 1.4
646. *****WL2*****
647. LOAD COMB 204 1.2DL+1.2LL+1.2WL2
648. 1 1.2 2 1.2 3 1.2 12 1.2
649. LOAD COMB 205 1.4DL+1.4WL2
650. 1 1.4 2 1.4 12 1.4
651. LOAD COMB 206 1DL+1.4WL2
652. 1 1.0 2 1.0 12 1.4
653. *****WL3*****
654. LOAD COMB 207 1.2DL+1.2LL+1.2WL3
655. 1 1.2 2 1.2 3 1.2 13 1.2
656. LOAD COMB 208 1.4DL+1.4WL3
657. 1 1.4 2 1.4 13 1.4
658. LOAD COMB 209 1DL+1.4WL3
659. 1 1.0 2 1.0 13 1.4
660. *****WL4*****
661. LOAD COMB 210 1.2DL+1.2LL+1.2WL4
662. 1 1.2 2 1.2 3 1.2 14 1.2
663. LOAD COMB 211 1.4DL+1.4WL4
664. 1 1.4 2 1.4 14 1.4
665. LOAD COMB 212 1DL+1.4WL4
666. 1 1.0 2 1.0 14 1.4
667. *****WL5*****
668. LOAD COMB 213 1.2DL+1.2LL+1.2WL5
669. 1 1.2 2 1.2 3 1.2 15 1.2
670. LOAD COMB 214 1.4DL+1.4WL5
671. 1 1.4 2 1.4 15 1.4
672. LOAD COMB 215 1DL+1.4WL5
673. 1 1.0 2 1.0 15 1.4
674. *****WL6*****
675. LOAD COMB 216 1.2DL+1.2LL+1.2WL6
676. 1 1.2 2 1.2 3 1.2 16 1.2
677. LOAD COMB 217 1.4DL+1.4WL6
678. 1 1.4 2 1.4 16 1.4
679. LOAD COMB 218 1DL+1.4WL6
680. 1 1.0 2 1.0 16 1.4
681. *****EQX & EQZ *****
682. LOAD COMB 219 1.32DL+1.1LL+1.1EQX+0.33 EQZ
683. 1 1.32 2 1.32 3 1.1 21 1.1 22 0.33
684. LOAD COMB 220 1.32DL+1.1LL+1.1EQX-0.33 EQZ
685. 1 1.32 2 1.32 3 1.1 21 1.1 22 -0.33
686. LOAD COMB 221 1.32DL+1.1LL-1.1EQX+0.33 EQZ
687. 1 1.32 2 1.32 3 1.1 21 -1.1 22 0.33
688. LOAD COMB 222 1.32DL+1.1LL-1.1EQX-0.33 EQZ
689. 1 1.32 2 1.32 3 1.1 21 -1.1 22 -0.33
690. LOAD COMB 223 1.32DL+1.1LL+1.1EQZ+0.33 EQX
691. 1 1.32 2 1.32 3 1.1 22 1.1 21 0.33
692. LOAD COMB 224 1.32DL+1.1LL+1.1EQZ-0.33 EQX
693. 1 1.32 2 1.32 3 1.1 22 1.1 21 -0.33
694. LOAD COMB 225 1.32DL+1.1LL-1.1EQZ+0.33 EQX
695. 1 1.32 2 1.32 3 1.1 22 -1.1 21 0.33
696. LOAD COMB 226 1.32DL+1.1LL-1.1EQZ-0.33 EQX
697. 1 1.32 2 1.32 3 1.1 22 -1.1 21 -0.33
698. LOAD COMB 227 0.99DL+1.1EQX+0.33EQZ
699. 1 0.99 2 0.99 21 1.1 22 0.33
700. LOAD COMB 228 0.99DL+1.1EQX-0.33EQZ

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701. 1 0.99 2 0.99 21 1.1 22 -0.33
702. LOAD COMB 229 0.99DL-1.1EQX+0.33EQZ
703. 1 0.99 2 0.99 21 -1.1 22 0.33
704. LOAD COMB 230 0.99DL-1.1EQX-0.33EQZ
705. 1 0.99 2 0.99 21 -1.1 22 -0.33
706. LOAD COMB 231 0.99DL+1.1EQZ+0.33EQX
707. 1 0.99 2 0.99 22 1.1 21 0.33
708. LOAD COMB 232 0.99DL+1.1EQZ-0.33EQX
709. 1 0.99 2 0.99 22 1.1 21 -0.33
710. LOAD COMB 233 0.99DL-1.1EQZ+0.33EQX
711. 1 0.99 2 0.99 22 -1.1 21 0.33
712. LOAD COMB 234 0.99DL-1.1EQZ-0.33EQX
713. 1 0.99 2 0.99 22 -1.1 21 -0.33
714. LOAD COMB 235 1.0DL+1.0LL+1.0PHX
715. 1 1.0 2 1.0 3 1.0 4 1.0
716. LOAD COMB 236 1.0DL+1.0LL+1.0PHZ
717. 1 1.0 2 1.0 3 1.0 5 1.0
718. *****DEFLECTION*****
719. LOAD COMB 301 SELF+DL
720. 1 1.0 2 1.0
721. LOAD COMB 302 LL
722. 3 1.0
723. LOAD COMB 303 0.8 (LL+WL1)
724. 3 0.8 11 0.8
725. LOAD COMB 304 0.8 (LL+WL2)
726. 3 0.8 12 0.8
727. LOAD COMB 305 0.8 (LL+WL3)
728. 3 0.8 13 0.8
729. LOAD COMB 306 0.8 (LL+WL4)
730. 3 0.8 14 0.8
731. LOAD COMB 307 0.8 (LL+WL5)
732. 3 0.8 15 0.8
733. LOAD COMB 308 0.8 (LL+WL6)
734. 3 0.8 16 0.8
735. LOAD COMB 309 0.8 (LL+EQX/1.4+0.3EQZ/1.4)
736. 3 0.8 21 0.57 22 0.17
737. LOAD COMB 310 0.8 (LL+EQX/1.4-0.3EQZ/1.4)
738. 3 0.8 21 0.57 22 -0.17
739. LOAD COMB 311 0.8 (LL-EQX/1.4+0.3EQZ/1.4)
740. 3 0.8 21 -0.57 22 0.17
741. LOAD COMB 312 0.8 (LL-EQX/1.4-0.3EQZ/1.4)
742. 3 0.8 21 -0.57 22 -0.17
743. LOAD COMB 313 0.8 (LL+EQZ/1.4+0.3EQX/1.4)
744. 3 0.8 22 0.57 21 0.17
745. LOAD COMB 314 0.8 (LL+EQZ/1.4-0.3EQX/1.4)
746. 3 0.8 22 0.57 21 -0.17
747. LOAD COMB 315 0.8 (LL-EQZ/1.4+0.3EQX/1.4)
748. 3 0.8 21 -0.57 22 0.17
749. LOAD COMB 316 0.8 (LL-EQZ/1.4-0.3EQX/1.4)
750. 3 0.8 21 -0.57 22 -0.17
751. PERFORM ANALYSIS

□

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 68/ 111/ 16
ORIGINAL/FINAL BAND-WIDTH= 46/ 13/ 75 DOF
TOTAL PRIMARY LOAD CASES = 13, TOTAL DEGREES OF FREEDOM = 360
SIZE OF STIFFNESS MATRIX = 27 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.6/ 23533.2 MB

*
* X DIRECTION : Ta = 0.344 Tb = 0.610 Tuser = 0.480 *
* T = 0.480, LOAD FACTOR = 1.000 *
* UBC TYPE = 97 *
* UBC FACTOR V = 0.1571 x 14277.00 = 2243.53 KN *
*

*
* Z DIRECTION : Ta = 0.344 Tb = 0.590 Tuser = 0.480 *
* T = 0.480, LOAD FACTOR = 1.000 *
* UBC TYPE = 97 *
* UBC FACTOR V = 0.1571 x 14277.00 = 2243.53 KN *
*

752. LOAD LIST 101 103 TO 118
753. *****
754. LOAD LIST 201 TO 236
755. START CONCRETE DESIGN
[CONCRETE DESIGN]
756. CODE BS8110
PROGRAM CODE REVISION V1.5_8110_97/1
757. UNIT MMS NEWTON

758. FYMAIN 460 ALL

759. FC 40 ALL

760. CLEAR 50 MEMB 1001 TO 1016 2001 TO 2016

761. CLEAR 60 MEMB 1 TO 55 101 TO 124

762. MINSEC 10 ALL

763. TRACK 2 ALL

764. DESIGN COLUMN 1001 TO 1016 2001 TO 2016

C O L U M N N O. 1001 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 2089. SQ. MM.

BAR CONFIGURATION REINF PCT. LOAD LOCATION

28 10 MM 0.727 222 EACH END

(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

|BRACED /SHORT in z E.L.z= 1200 mm (3.8.1.3 & 5)|

|BRACED /SHORT in y E.L.y= 1200 mm (3.8.1.3 & 5)|

|END MOMS. MZ1= 0 MZ2= 316 MY1= 0 MY2= 91|

|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |

|DESIGN LOADS KN METER: MOM.= 399 AXIAL LOAD= 952|

|DESIGNED CAP. KN METER: MOM.= 399 AXIAL CAP.= 955|

C O L U M N N O. 1002 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 1734. SQ. MM.

BAR CONFIGURATION REINF PCT. LOAD LOCATION

16 12 MM 0.598 220 EACH END

(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

|BRACED /SHORT in z E.L.z= 1200 mm (3.8.1.3 & 5)|

|BRACED /SHORT in y E.L.y= 1200 mm (3.8.1.3 & 5)|

|END MOMS. MZ1= 0 MZ2= 302 MY1= 0 MY2= 138|

|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |

|DESIGN LOADS KN METER: MOM.= 423 AXIAL LOAD= 1289|

|DESIGNED CAP. KN METER: MOM.= 423 AXIAL CAP.= 1293|

C O L U M N N O. 1003 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 3103. SQ. MM.

BAR CONFIGURATION REINF PCT. LOAD LOCATION

40 10 MM 1.039 220 EACH END

(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

|BRACED /SHORT in z E.L.z= 1200 mm (3.8.1.3 & 5)|

|BRACED /SHORT in y E.L.y= 1200 mm (3.8.1.3 & 5)|

|END MOMS. MZ1= 0 MZ2= 315 MY1= 0 MY2= 289|

|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |

|DESIGN LOADS KN METER: MOM.= 564 AXIAL LOAD= 1420|

|DESIGNED CAP. KN METER: MOM.= 564 AXIAL CAP.= 1427|

C O L U M N N O. 1004 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 2452. SQ. MM.

BAR CONFIGURATION REINF PCT. LOAD LOCATION

8 20 MM 0.831 225 EACH END

(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

|BRACED /SHORT in z E.L.z= 1200 mm (3.8.1.3 & 5)|

|BRACED /SHORT in y E.L.y= 1200 mm (3.8.1.3 & 5)|

|END MOMS. MZ1= 0 MZ2= 178 MY1= 0 MY2= 292|

|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |

|DESIGN LOADS KN METER: MOM.= 451 AXIAL LOAD= 1072|

|DESIGNED CAP. KN METER: MOM.= 451 AXIAL CAP.= 1075|

C O L U M N N O. 1005 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 4627. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
24 16 MM	1.596	226	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

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|BRACED /SHORT in z E.L.z= 1200 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 1200 mm (3.8.1.3 & 5)|
|END MOMS. M21= 0 M22= 366 MY1= 0 MY2= 366|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 685 AXIAL LOAD= 1315|
|DESIGNED CAP. KN METER: MOM.= 685 AXIAL CAP.= 1330|

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C O L U M N N O. 1006 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 3344. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
44 10 MM	1.142	228	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

```

|BRACED /SHORT in z E.L.z= 1200 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 1200 mm (3.8.1.3 & 5)|
|END MOMS. M21= 0 M22= 397 MY1= 0 MY2= 113|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 501 AXIAL LOAD= 904|
|DESIGNED CAP. KN METER: MOM.= 501 AXIAL CAP.= 908|

```

C O L U M N N O. 1007 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 3938. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
20 16 MM	1.330	229	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

```

|BRACED /SHORT in z E.L.z= 1200 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 1200 mm (3.8.1.3 & 5)|
|END MOMS. M21= 0 M22= 433 MY1= 0 MY2= 208|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 615 AXIAL LOAD= 1267|
|DESIGNED CAP. KN METER: MOM.= 615 AXIAL CAP.= 1274|

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C O L U M N N O. 1008 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 4271. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
40 12 MM	1.495	223	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

```

|BRACED /SHORT in z E.L.z= 1200 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 1200 mm (3.8.1.3 & 5)|
|END MOMS. M21= 0 M22= 389 MY1= 0 MY2= 320|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 665 AXIAL LOAD= 1393|
|DESIGNED CAP. KN METER: MOM.= 665 AXIAL CAP.= 1408|

```

C O L U M N N O. 1009 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 5119. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
28 16 MM	1.862	224	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 1200 mm (3.8.1.3 & 5)|
IBRACED /SHORT in y E.L.y= 1200 mm (3.8.1.3 & 5)|
END MOMS. MZ1= 0 MZ2= 417 MY1= 0 MY2= 364|
SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
DESIGN LOADS KN METER: MOM.= 732 AXIAL LOAD= 1355|
DESIGNED CAP. KN METER: MOM.= 732 AXIAL CAP.= 1357|

C O L U M N N O. 1010 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 3512. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
32 12 MM	1.196	223	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 1200 mm (3.8.1.3 & 5)|
IBRACED /SHORT in y E.L.y= 1200 mm (3.8.1.3 & 5)|
END MOMS. MZ1= 0 MZ2= 231 MY1= 0 MY2= 448|
SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
DESIGN LOADS KN METER: MOM.= 640 AXIAL LOAD= 1732|
DESIGNED CAP. KN METER: MOM.= 640 AXIAL CAP.= 1734|

C O L U M N N O. 1011 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 2567. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
24 12 MM	0.897	231	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 1200 mm (3.8.1.3 & 5)|
IBRACED /SHORT in y E.L.y= 1200 mm (3.8.1.3 & 5)|
END MOMS. MZ1= 0 MZ2= 99 MY1= 0 MY2= 331|
SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
DESIGN LOADS KN METER: MOM.= 422 AXIAL LOAD= 845|
DESIGNED CAP. KN METER: MOM.= 422 AXIAL CAP.= 849|

C O L U M N N O. 1012 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 3744. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
12 20 MM	1.246	223	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 1200 mm (3.8.1.3 & 5)|
IBRACED /SHORT in y E.L.y= 1200 mm (3.8.1.3 & 5)|
END MOMS. MZ1= 0 MZ2= 312 MY1= 0 MY2= 287|
SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
DESIGN LOADS KN METER: MOM.= 569 AXIAL LOAD= 1093|
DESIGNED CAP. KN METER: MOM.= 569 AXIAL CAP.= 1099|

C O L U M N N O. 1013 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 2843. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
20 10 MM	1.039	224	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

```
|BRACED /SHORT in z E.L.z= 1200 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 1200 mm (3.8.1.3 & 5)|
|END MOMS. MZ1= 0 MZ2= 341 MY1= 0 MY2= 166|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 491 AXIAL LOAD= 1099|
|DESIGNED CAP. KN METER: MOM.= 491 AXIAL CAP.= 1103|
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C O L U M N N O. 1014 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 1931. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
4 25 MM	0.638	227	EACH END

```
|BRACED /SHORT in z E.L.z= 1200 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 1200 mm (3.8.1.3 & 5)|
|END MOMS. MZ1= 0 MZ2= 303 MY1= 0 MY2= 60|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 359 AXIAL LOAD= 800|
|DESIGNED CAP. KN METER: MOM.= 359 AXIAL CAP.= 809|
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C O L U M N N O. 1015 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 1551. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
20 10 MM	0.519	227	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

```
|BRACED /SHORT in z E.L.z= 1200 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 1200 mm (3.8.1.3 & 5)|
|END MOMS. MZ1= 0 MZ2= 268 MY1= 0 MY2= 59|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 323 AXIAL LOAD= 792|
|DESIGNED CAP. KN METER: MOM.= 323 AXIAL CAP.= 796|
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C O L U M N N O. 1016 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 1964. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
40 8 MM	0.665	223	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

```
|BRACED /SHORT in z E.L.z= 1200 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 1200 mm (3.8.1.3 & 5)|
|END MOMS. MZ1= 0 MZ2= 255 MY1= 0 MY2= 162|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 402 AXIAL LOAD= 1032|
|DESIGNED CAP. KN METER: MOM.= 402 AXIAL CAP.= 1035|
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C O L U M N N O. 2001 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 4789. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
24	16 MM	1.931	222 EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
END MOMS. M21= 347 M22= 315 MY1= 110 MY2= 106|
SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
DESIGN LOADS KN METER: MOM.= 453 AXIAL LOAD= 307|
DESIGNED CAP. KN METER: MOM.= 453 AXIAL CAP.= 313|

C O L U M N N O. 2002 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 5720. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
12	25 MM	2.356	220 EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
END MOMS. M21= 368 M22= 388 MY1= 151 MY2= 145|
SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
DESIGN LOADS KN METER: MOM.= 533 AXIAL LOAD= 355|
DESIGNED CAP. KN METER: MOM.= 533 AXIAL CAP.= 357|

C O L U M N N O. 2003 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 5958. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
20	20 MM	2.513	220 EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
END MOMS. M21= 370 M22= 373 MY1= 180 MY2= 154|
SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
DESIGN LOADS KN METER: MOM.= 547 AXIAL LOAD= 326|
DESIGNED CAP. KN METER: MOM.= 547 AXIAL CAP.= 327|

C O L U M N N O. 2004 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 4935. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
16	20 MM	2.011	225 EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
END MOMS. M21= 142 M22= 134 MY1= 330 MY2= 299|
SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
DESIGN LOADS KN METER: MOM.= 467 AXIAL LOAD= 321|
DESIGNED CAP. KN METER: MOM.= 467 AXIAL CAP.= 326|

C O L U M N N O. 2005 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 6592. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
24 20 MM	3.016	226	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5) |
IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5) |
END MOMS. M21= 224 M22= 196 MY1= 376 MY2= 380 |
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 596 AXIAL LOAD= 328 |
|DESIGNED CAP. KN METER: MOM.= 596 AXIAL CAP.= 329 |

C O L U M N N O. 2006 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 7254. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
24 20 MM	3.016	228	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5) |
IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5) |
END MOMS. M21= 477 M22= 503 MY1= 146 MY2= 156 |
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 653 AXIAL LOAD= 359 |
|DESIGNED CAP. KN METER: MOM.= 653 AXIAL CAP.= 360 |

C O L U M N N O. 2007 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 7174. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
24 20 MM	3.016	230	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5) |
IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5) |
END MOMS. M21= 489 M22= 505 MY1= 121 MY2= 148 |
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 648 AXIAL LOAD= 359 |
|DESIGNED CAP. KN METER: MOM.= 648 AXIAL CAP.= 364 |

C O L U M N N O. 2008 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 6359. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
32 16 MM	2.574	225	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5) |
IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5) |
END MOMS. M21= 218 M22= 199 MY1= 338 MY2= 368 |
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 578 AXIAL LOAD= 329 |
|DESIGNED CAP. KN METER: MOM.= 578 AXIAL CAP.= 330 |

C O L U M N N O. 2009 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 6793. SQ. MM.

BAR CONFIGURATION REINF PCT. LOAD LOCATION

24 20 MM 3.016 224 EACH END

(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5) |

IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5) |

END MOMS. MZ1= 232 MZ2= 198 MY1= 381 MY2= 387 |

SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |

DESIGN LOADS KN METER: MOM.= 611 AXIAL LOAD= 320 |

DESIGNED CAP. KN METER: MOM.= 611 AXIAL CAP.= 322 |

C O L U M N N O. 2010 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 7289. SQ. MM.

BAR CONFIGURATION REINF PCT. LOAD LOCATION

24 20 MM 3.016 231 EACH END

(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5) |

IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5) |

END MOMS. MZ1= 151 MZ2= 161 MY1= 465 MY2= 488 |

SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |

DESIGN LOADS KN METER: MOM.= 644 AXIAL LOAD= 288 |

DESIGNED CAP. KN METER: MOM.= 644 AXIAL CAP.= 290 |

C O L U M N N O. 2013 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 5060. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
28 16 MM	2.252	224	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

|BRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
|END MOMS. MZ1= 174 MZ2= 140 MY1= 309 MY2= 296|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 476 AXIAL LOAD= 323|
DESIGNED CAP. KN METER: MOM.= 476 AXIAL CAP.= 324

C O L U M N N O. 2014 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 5559. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
28 16 MM	2.252	227	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

|BRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
|END MOMS. MZ1= 362 MZ2= 380 MY1= 124 MY2= 126|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 503 AXIAL LOAD= 252|
DESIGNED CAP. KN METER: MOM.= 503 AXIAL CAP.= 252

C O L U M N N O. 2015 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 5480. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
28 16 MM	2.252	229	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

|BRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
|END MOMS. MZ1= 349 MZ2= 374 MY1= 124 MY2= 126|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 497 AXIAL LOAD= 253|
DESIGNED CAP. KN METER: MOM.= 497 AXIAL CAP.= 254

C O L U M N N O. 2016 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 4896. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
16 20 MM	2.011	223	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

|BRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
|END MOMS. MZ1= 159 MZ2= 137 MY1= 309 MY2= 296|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 463 AXIAL LOAD= 322|
DESIGNED CAP. KN METER: MOM.= 463 AXIAL CAP.= 323

*****END OF COLUMN DESIGN RESULTS*****

=====									
B E A M N O. 1 D E S I G N R E S U L T S - F L E X U R E									
LEN - 1800. mm FY - 460. FC - 40. SIZE - 400. X 600. mm									

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	72.	4- 25 MM	0.	1800.	YES YES
2	513.	3- 40 MM	0.	1800.	YES YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	782.6/ 3709.5	169.76/ 664.57	228/ 221
150.	833.5/ 3240.9	180.81/ 600.77	228/ 221
300.	880.9/ 2819.7	190.87/ 538.43	228/ 221
450.	925.3/ 2436.0	199.95/ 477.56	228/ 221
600.	965.1/ 2083.3	208.04/ 418.16	228/ 221
750.	1000.3/ 1756.8	215.14/ 360.23	228/ 221
900.	1030.6/ 1452.9	221.26/ 303.77	228/ 221
1050.	1056.8/ 1169.4	226.52/ 248.90	227/ 222
1200.	1134.5/ 958.5	241.99/ 206.70	219/ 230
1350.	1246.9/ 800.2	264.11/ 173.59	219/ 230
1500.	1353.4/ 652.2	284.76/ 141.47	219/ 230
1650.	1453.8/ 508.7	303.94/ 110.34	219/ 230
1800.	1547.7/ 369.7	321.65/ 80.20	219/ 230

B E A M N O. 1 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
----- ----- ----- ----- ----- ----- -----						
END 1	899 mm	430.3 kN	221	10 mm	7	149 mm
899	END 2	312.8 kN	221	10 mm	5	224 mm
----- ----- ----- ----- ----- ----- -----						

=====									
101J 1800. X 400. X 600 117J									
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									
3No40 HI513.	10.TO 1800								
7*10c/c149									5*10c/c224
4No25 HI 72.	10.TO 1800								
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									

OOO									
3T40		OOO		OOO		OOO		OOO	
		3T40		3T40		3T40		3T40	
4T25									
oooo		4T25		4T25		4T25		4T25	
		oooo		oooo		oooo		oooo	

B E A M N O. 2 D E S I G N R E S U L T S - F L E X U R E

LEN - 4700. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	72.	4- 25 MM	0.	4700.	YES YES
2	528.	5- 10 MM	0.	1214.	YES NO
**Please Note:					
* Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing. *					
3	517.	3- 32 MM	1951.	4700.	NO YES
4	453.	3- 32 MM	1951.	4700.	NO YES
COMP. 4- 16 MM (REQD. STEEL= 763. SQ. MM)					

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1547.7/ 369.7	321.65/ 80.20	219/ 230
392.	1518.3/ 312.0	316.13/ 21.32	219/ 230
783.	1428.2/ 0.0	299.07/ 0.00	219/ 0
1175.	1270.6/ 0.0	268.72/ 0.00	219/ 0
1567.	1044.3/ 0.0	223.99/ 0.00	219/ 0
1958.	1112.1/ 0.0	237.55/ 0.00	221/ 0
2350.	1108.4/ 0.0	236.80/ 0.00	221/ 0
2742.	1013.1/ 312.0	217.73/ 64.20	221/ 228
3133.	832.9/ 653.6	180.68/ 141.78	222/ 227
3525.	741.0/ 1232.3	160.74/ 261.26	230/ 219
3917.	670.0/ 2022.4	145.35/ 407.57	230/ 219
4308.	564.0/ 3010.0	122.34/ 567.17	230/ 219
4700.	425.0/ 4486.9	92.19/ 738.31	230/ 219

B E A M N O. 2 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2349 mm	199.8 kN	222	10 mm	9	293 mm
2349	2741 mm	267.8 kN	227	10 mm	2	195 mm
2741	END 2	450.3 kN	219	10 mm	14	150 mm

117J 4700.X 400.X 600 102J

5No10 H 528.	0.TO 1214	3No32 H 517.1951.TO 4700							
9*10c/c293						14*10c/c150			
4No25 H 72.	0.TO 4700								

oooo	oooo				ooo	ooo		ooo	ooo
5T10	5T10			3T32	3T32		3T32		3T32
4T25	4T25		4T25	4T25	4T25		4T25		4T25
oooo	oooo	oooo	oooo	oooo	oooo		oooo		oooo

B E A M N O. 3 D E S I G N R E S U L T S - F L E X U R E

LEN - 4355. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
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1	72.	3- 25 MM	0.	4355.	YES	YES
**Please Note:						
* Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing.*						
2	508.	2- 50 MM	0.	3077.	YES	NO
3	528.	4- 10 MM	3170.	4355.	NO	YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
----------------	--------------------------------	-----------------------------	----------------

0.	513.8/ 622.0/	3785.1 2736.4	674.32 525.55	228/ 221
363.				228/ 221

726.	702.1/	1903.4	152.30/	386.59	228/	221
1089.	752.3/	1219.9	163.19/	258.83	228/	221
1452.	806.8/	695.2	175.03/	150.80	220/	229
1815.	941.4/	368.4	203.22/	79.91	220/	229
2177.	1006.9/	312.0	216.48/	16.63	219/	230
2540.	997.8/	0.0	214.64/	0.00	219/	0
2903.	914.7/	0.0	197.77/	0.00	219/	0
3266.	1013.0/	0.0	217.72/	0.00	221/	0
3629.	1168.9/	0.0	248.81/	0.00	221/	0
3992.	1270.4/	312.0	268.69/	7.76	221/	228
4355.	1322.4/	312.0	278.77/	59.77	221/	228

B E A M N O. 3 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1815 mm	422.3 kN	221	10 mm	12	164 mm
1815	2178 mm	256.9 kN	229	10 mm	2	181 mm
2177	END 2	188.1 kN	220	10 mm	8	311 mm

102J 4355.X 400.X 600 134J

2No50 H 508.	0.TO 3077				4No10 H 528.3170.TO 4355	
12*10c/c164					8*10c/c311	
3No25 H 72.	0.TO 4355					

OO	OO	OO	OO	OO	oooo	oooo
2T50	2T50	2T50	2T50	2T50	4T10	4T10
3T25	3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 4 D E S I G N R E S U L T S - F L E X U R E

LEN - 1645. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
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1 72. 3- 25 MM 0. 1645. YES YES

**Please Note:

* Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing.*

2 517. 3- 32 MM 0. 1645. YES YES

3 453. 3- 32 MM 0. 1645. YES YES

COMP. 6- 12 MM (REQD. STEEL= 572. SQ. MM)

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1321.0/ 312.0	278.51/ 59.96	221/ 228
137.	1206.6/ 434.0	256.22/ 94.15	221/ 228
274.	1087.6/ 595.5	232.66/ 129.19	221/ 228
411.	963.7/ 761.0	207.75/ 165.09	221/ 228
548.	867.0/ 967.0	188.02/ 208.40	229/ 220
685.	826.7/ 1252.0	179.34/ 265.11	229/ 220
823.	782.5/ 1556.6	169.74/ 323.31	229/ 220
960.	734.0/ 1883.4	159.22/ 383.02	229/ 220
1097.	681.9/ 2236.5	147.92/ 444.37	230/ 219
1234.	625.9/ 2619.8	135.77/ 507.21	230/ 219
1371.	565.8/ 3038.6	122.75/ 571.41	230/ 219
1508.	501.8/ 3501.4	108.86/ 636.96	230/ 219
1645.	433.9/ 4309.3	94.13/ 703.78	230/ 219

B E A M N O. 4 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	822 mm	370.9 kN	219	10 mm	5	205 mm
822	END 2	492.0 kN	219	10 mm	8	117 mm

134J	1645. X 400. X 600	103J
3No32 H 517.	0. TO 1645	
5*10c/c205		8*10c/c117
3No25 H 72.	0. TO 1645	

000	000	000	000	000	000
3T32	3T32	3T32	3T32	3T32	3T32
3T25	3T25	3T25	3T25	3T25	3T25
000	000	000	000	000	000

B E A M N O. 5 D E S I G N R E S U L T S - F L E X U R E

LEN - 6500. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	70.	4- 20 MM	0.	6500.	YES YES
2	513.	3- 40 MM	0.	2989.	YES NO
3	513.	3- 40 MM	3511.	6500.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	522.6/ 3513.9	113.36/ 638.65	228/ 221
542.	660.3/ 2250.6	143.24/ 446.74	228/ 221
1083.	733.8/ 1313.7	159.19/ 277.10	228/ 221
1625.	762.5/ 625.6	165.41/ 135.71	220/ 229
2167.	827.8/ 312.0	179.58/ 40.29	220/ 229
2708.	784.3/ 0.0	170.13/ 0.00	220/ 0
3250.	762.3/ 0.0	165.37/ 0.00	221/ 0
3792.	1009.1/ 0.0	216.92/ 0.00	221/ 0
4333.	1148.9/ 312.0	244.84/ 47.51	221/ 228
4875.	1170.7/ 610.2	249.14/ 132.37	221/ 228
5417.	1073.3/ 1082.4	229.82/ 231.63	221/ 228
5958.	1045.9/ 1881.6	224.31/ 382.70	229/ 220
6500.	950.1/ 2936.1	204.98/ 556.14	229/ 220

B E A M N O. 5 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1626 mm	372.9 kN	221	10 mm	9	203 mm
1626	4877 mm	247.8 kN	221	10 mm	8	406 mm
4875	5417 mm	257.3 kN	221	10 mm	2	270 mm
5417	END 2	338.8 kN	220	10 mm	5	270 mm

103J		6500.X 400.X 600		104J	
=====		=====		=====	
3No40 H 513. 0.TO 2989		3No40 H 513.3511.TO 6500			
9*10c/c203		5*10c/c270			
4No20 H 70. 0.TO 6500					

OOO	OOO	OOO	OOO	OOO	OOO
3T40	3T40	3T40	3T40	3T40	3T40
4T20	4T20	4T20	4T20	4T20	4T20
oooo	oooo	oooo	oooo	oooo	oooo

B E A M N O. 6 D E S I G N R E S U L T S - F L E X U R E

LEN - 1645. mm FY - 460. mm FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
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1	513.	2- 40 MM	0.	1645.	YES YES
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REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KN-METER)	LOAD(+VE/-VE)
0.	0.0/ 0.0	0.00/ 0.00	0/ 0
137.	0.0/ 234.0	0.00/ 20.50	0/ 221
274.	0.0/ 234.0	0.00/ 42.07	0/ 221
411.	0.0/ 298.9	0.00/ 64.84	0/ 221
548.	0.0/ 409.7	0.00/ 88.88	0/ 221
685.	0.0/ 527.0	0.00/ 114.32	0/ 221
823.	0.0/ 651.6	0.00/ 141.29	0/ 221
960.	0.0/ 792.2	0.00/ 169.79	0/ 221
1097.	0.0/ 944.1	0.00/ 199.82	0/ 221
1234.	0.0/ 1107.7	0.00/ 231.24	0/ 221
1371.	0.0/ 1283.4	0.00/ 263.94	0/ 221
1508.	0.0/ 1472.1	0.00/ 297.84	0/ 221
1645.	0.0/ 1674.6	0.00/ 332.81	0/ 221

B E A M N O. 6 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM - TO mm	SHEAR TORSN kNm	LOAD S T	LINK SIZE	NO. S T	SPACING mm C/T S+T
END 1 822	145.8	22	221225	10mm	1 4 7 571 180 117
822 END 2	258.8	22	221225	10mm	2 4 7 301 180 103

EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 575 mm2 EVENLY DISTRIBUTED
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *

137J	1645.X 300.X 600	135J
2No40 H 513.	0.TO 1645	7*10c/c103
7*10c/c117		

OO	OO	OO	OO	OO	OO
2T40	2T40	2T40	2T40	2T40	2T40

B E A M N O. 7 D E S I G N R E S U L T S - F L E X U R E

LEN - 2650. mm FY - 460. mm FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
-------	--------------	----------	------------	----------	-------------------

1	70.	2- 20 MM	282.	2650.	NO YES
2	513.	2- 40 MM	0.	2368.	YES NO

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KN-METER)	LOAD(+VE/-VE)
0.	0.0/ 1674.6	0.00/ 332.81	0/ 221
221.	0.0/ 1353.6	0.00/ 276.69	0/ 221
442.	0.0/ 1067.2	0.00/ 223.56	0/ 221

662.	0.0/	812.3	0.00/	173.82	0/	221
883.	0.0/	589.8	0.00/	127.95	0/	221
1104.	234.0/	397.4	0.55/	86.22	228/	221
1325.	234.0/	234.0	24.29/	48.88	228/	221
1546.	234.0/	234.0	45.49/	15.93	228/	221
1767.	295.7/	0.0	64.14/	0.00	228/	0
1987.	397.9/	0.0	86.32/	0.00	202/	0
2208.	488.8/	0.0	106.04/	0.00	202/	0
2429.	564.7/	0.0	122.51/	0.00	202/	0
2650.	626.0/	0.0	135.81/	0.00	202/	0

B E A M N O. 7 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	884 mm	260.6 kN	224	10 mm	3	441 mm
884	1326 mm	199.4 kN	224	10 mm	2	220 mm
1324	END 2	53.6 kN	214	10 mm	5	331 mm

135J 2650.X 300.X 600 138J

2No40 H 513.	0.TO 2368				
3*10c/c441					5*10c/c331
2No20 H 70.	282.TO/2650				

OO	OO	OO	OO	OO	OO
2T40	2T40	2T40	2T40	2T40	2T40
	2T20	2T20	2T20	2T20	2T20
	OO	OO	OO	OO	OO

B E A M N O. 8 D E S I G N R E S U L T S - F L E X U R E

LEN - 3850. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
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REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	598.6/	0.0	129.85/ 0.00 202/ 0
321.	740.0/	0.0	159.30/ 0.00 202/ 0
642.	853.4/	0.0	181.99/ 0.00 202/ 0
963.	947.9/	0.0	200.57/ 0.00 220/ 0
1283.	1021.8/	0.0	214.87/ 0.00 220/ 0
1604.	1047.9/	0.0	219.87/ 0.00 220/ 0
1925.	1025.3/	0.0	215.54/ 0.00 220/ 0
2246.	954.6/	0.0	201.86/ 0.00 220/ 0
2567.	837.6/	0.0	178.85/ 0.00 220/ 0
2888.	677.2/	0.0	146.53/ 0.00 220/ 0
3208.	485.3/	0.0	105.28/ 0.00 220/ 0
3529.	258.0/	0.0	55.98/ 0.00 220/ 0
3850.	0.0/	0.0	0.00/ 0.00 0/ 0

B E A M N O. 8 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1283 mm	133.9 kN	221	10 mm	5	320 mm
1925	END 2	183.5 kN	220	10 mm	6	385 mm

138J 3850.X 300.X 600 136J

5*10c/c320						6*10c/c385
3No25 H 72.	0.TO 3850					

3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo

100

BEAM NO. 9 DESIGN RESULTS - FLEXURE

LEN - 1800. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT	BAR INFO	FROM	TO	ANCHOR
	mm		mm	mm	STA END

1	65.	3-10 MM	0.	1800.	YES	YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
-------------------	-----------------------	---------------------------------	----------------

0.	0.0/	0.0	0.00/	0.00	0/
150.	234.0/	0.0	1.93/	0.00	219/
300.	234.0/	0.0	3.67/	0.00	219/
450.	234.0/	0.0	5.14/	0.00	219/
600.	234.0/	0.0	6.28/	0.00	219/
750.	234.0/	0.0	7.01/	0.00	219/
900.	234.0/	0.0	7.26/	0.00	219/
1050.	234.0/	0.0	7.01/	0.00	219/
1200.	234.0/	0.0	6.28/	0.00	219/
1350.	234.0/	0.0	5.14/	0.00	219/
1500.	234.0/	0.0	3.67/	0.00	219/
1650.	234.0/	0.0	1.93/	0.00	219/
800.	0.0/	0.0	0.00/	0.00	0/

BEAM NO. 9 DESIGN RESULTS - SHEAR

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM - TO	SHEAR	TORSN	LOAD	LINK	NO.	SPACING	mm	C/C
mm	kN	kNm	S	T	SIZE	S	T	S+T
END 1	END 2	0 0	19	0236	10mm	0	11	11
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 574 mm2 EVENLY DISTRIBUTED								
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *								

B E A M N O. 10 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	899 mm	580.6 kN	222	10 mm	10	99 mm
END 2	899	425.3 kN	222	10 mm	6	179 mm

105J-----1800.X 400.X 600-----139J-----						

3No40=H=513.===0.T0=1800=====						
2No40cHc433. 0.T0 1800 6*10c/c179						
3No32 H 76. 0.T0 1800						

OOO				OOO		OOO		OOO		OOO
3T40		3T40		3T40		3T40		3T40		3T40
3T32		3T32		3T32		3T32		3T32		3T32
OOO		OOO		OOO		OOO		OOO		OOO

=====

B E A M N O. 11 D E S I G N R E S U L T S - F L E X U R E

LEN - 4700. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	76.	3- 32 MM	0.	4700.	YES YES
2	527.	5- 12 MM	0.	1214.	YES NO
3	513.	3- 40 MM	1951.	4700.	NO YES
4	433.	2- 40 MM	1951.	4700.	NO YES
COMP. 3- 32 MM (REQD. STEEL= 2168. SQ. MM)					

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)

0.	2171.2/	490.4	433.27/	106.39	220/ 229
392.	2206.9/	312.0	439.35/	30.70	220/ 229
783.	2134.6/	0.0	427.00/	0.00	220/ 0
1175.	1939.3/	0.0	392.96/	0.00	220/ 0
1567.	1617.2/	0.0	334.60/	0.00	220/ 0
1958.	1662.0/	0.0	342.89/	0.00	222/ 0
2350.	1596.0/	0.0	330.66/	0.00	222/ 0
2742.	1394.2/	390.2	292.59/	84.64	222/ 227
3133.	1077.0/	844.5	230.55/	183.21	221/ 228
3525.	951.8/	1695.0	205.33/	348.96	229/ 220
3917.	873.2/	2894.1	189.28/	549.79	229/ 220
4308.	762.0/	4625.4	165.30/	765.25	229/ 220
4700.	1596.5/	5791.4	134.30/	992.04	229/ 220

B E A M N O. 11 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1175 mm	293.6 kN	221	10 mm	7	195 mm
1175	2350 mm	146.5 kN	228	10 mm	3	391 mm
2349	END 2	590.8 kN	220	10 mm	25	97 mm

139J-----4700.X 400.X 600-----106J-----

5No12 H 527.	0.T0 1214	3No40 H 513.1951.T0 4700								
7*10c/c195									25*10c/c 97	
3No32 H 176.	0.T0 4700									
=====										

ooooo		ooooo				OOO		OOO		OOO
5T12		5T12				3T40		3T40		3T40
3T32		3T32		3T32		3T32		3T32		3T32
OOO		OOO		OOO		OOO		OOO		OOO

=====									
B E A M N O. 12 D E S I G N R E S U L T S - FLEXURE									
LEN - 4355. mm FY - 460. FC - 40. SIZE - 400. X 600. mm									

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	72.	3- 25 MM	0.	4355.	YES YES
2	513.	3- 40 MM	0.	2968.	YES NO
3	433.	2- 40 MM	0.	2968.	YES NO
	COMP.	4- 20 MM (REQD. STEEL=	1175. SQ. MM)		
4	527.	6- 12 MM	2807.	4355.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)

0.	943.4/	4869.7	203.62/ 812.78
363.	1011.0/	3490.8	217.30/ 635.52
726.	1049.2/	2376.7	224.99/ 467.80
1089.	1054.2/	1497.7	225.99/ 312.25
1452.	1098.0/	857.2	234.74/ 185.96
1815.	1248.6/	487.3	264.43/ 105.71
2177.	1301.1/	312.0	274.66/ 34.40
2540.	1245.5/	0.0	263.84/ 0.00
2903.	1078.6/	0.0	230.87/ 0.00
3266.	1133.4/	0.0	241.78/ 0.00
3629.	1231.7/	312.0	261.13/ 1.97
3992.	1240.0/	322.2	262.77/ 69.90
4355.	1171.4/	669.9	249.30/ 145.33

B E A M N O. 12 D E S I G N R E S U L T S - SHEAR

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
----- ----- ----- ----- ----- ----- -----						
END 1	1815 mm	499.2 kN	222	10 mm	14	139 mm
1815	2178 mm	309.2 kN	230	10 mm	2	181 mm
2177	3266 mm	185.6 kN	230	10 mm	3	362 mm
3266	END 2	316.1 kN	219	10 mm	7	181 mm
----- ----- ----- ----- ----- ----- -----						

106J	4355. X 400. X 600	140J
=====		
3No40H=513. ---0.TO=2968		
2No40chc433. 0.TO 2968 7*10c/c181		
3No25 H 72. 0.TO 4355		
=====		

OOO	OOO	OOO	OOO		
3T40	3T40	3T40	3T40		
3T25	3T25	3T25	3T25		3T25
ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 13 D E S I G N R E S U L T S - FLEXURE

LEN - 1645. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	70.	4- 20 MM	0.	1645.	YES YES
2	513.	3- 40 MM	0.	1645.	YES YES
3	433.	2- 40 MM	0.	1645.	YES YES
	COMP.	4- 25 MM (REQD. STEEL=	1736. SQ. MM)		

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)

0.	1173.1/	669.4	249.63/ 145.20
137.	1067.4/	835.3	228.64/ 181.20
274.	955.0/	1016.5	205.97/ 218.40
411.	951.5/	1331.0	205.27/ 280.44
548.	943.5/	1668.6	203.64/ 344.09
685.	930.6/	2033.8	201.01/ 409.57
823.	913.0/	2432.0	197.43/ 476.90
960.	892.3/	2869.8	193.19/ 546.09
1097.	866.9/	3355.7	187.99/ 616.94
1234.	838.4/	4234.9	181.87/ 689.31
1371.	806.3/	4613.8	174.91/ 762.99
1508.	804.2/	4999.1	167.11/ 837.94
1645.	1195.4/	5390.3	158.51/ 914.03

B E A M N O. 13 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	822 mm	421.5 kN	220	10 mm	6	164 mm
822	END 2	558.8 kN	220	10 mm	9	102 mm

140J_____1645.X 400.X 600_____107J_____						
=====						
3No40=H=513.==0.TO=1645=====						
2No40Chc433. 0.TO 1645 9*10c/c102						
4No20 H 70. 0.TO 1645						

=====						

=====						

B E A M N O. 14 D E S I G N R E S U L T S - S H E A R

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1324 mm	730.1 kN	222	10 mm	21	66 mm
1324	END 2	507.0 kN	222	10 mm	13	110 mm

107J_____2650.X 400.X 600_____141J_____						
=====						
3No40=H=513.==0.TO=2557=====						
3No40Chc433. 10.TO 2557 13*10c/c110						
3No40 H 80. 10.TO 2650						

=====						

=====						

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1324 mm	730.1 kN	222	10 mm	21	66 mm
1324	END 2	507.0 kN	222	10 mm	13	110 mm

107J_____2650.X 400.X 600_____141J_____						
=====						
3No40=H=513.==0.TO=2557=====						
3No40Chc433. 10.TO 2557 13*10c/c110						
3No40 H 80. 10.TO 2650						

=====						

=====						

B E A M N O. 14 D E S I G N R E S U L T S - F L E X U R E

LEN - 2650. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	80.	3- 40 MM	0.	2650.	YES YES
2	513.	3- 40 MM	0.	2557.	YES NO
3	433.	3- 40 MM	0.	2557.	YES NO
COMP. 3- 40 MM (REQD. STEEL= 2993. SQ. MM)					

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	2361.7/ 6556.6	32.34/ 1140.87	227/ 222

B E A M N O. 16 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1250 mm	619.0 kN	222	10 mm	15	89 mm
1250	END 2	371.8 kN	222	10 mm	7	208 mm

109J	2500.X 400.X 600	161J
3No40-H=513.	0.TO=2500	
2No40cHc433.	0.TO 2500	
3No40 H 80.	0.TO 2500	7*10c/c208

LEVEL	HEIGHT	BAR INFO	FROM	TO	ANCHOR
OOO		OOO	OOO	OOO	YES YES
3T40		3T40	3T40	3T40	NO YES
3T40		3T40	3T40	3T40	NO YES
OOO		OOO	OOO	OOO	NO YES

B E A M N O. 17 D E S I G N R E S U L T S - F L E X U R E

LEN - 4000. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT	BAR INFO	FROM	TO	ANCHOR
1	80.	3- 40 MM	0.	4000.	YES YES

1 80. 3- 40 MM 0. 4000. YES YES

**Please Note:

* Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing.*

2 513. 3- 40 MM 1209. 4000. NO YES

3 433. 3- 40 MM 1209. 4000. NO YES

COMP. 2- 40 MM (REQD. STEEL= 2418. SQ. MM)

REQUIRED REINF. STEEL SUMMARY :

SECTION	REINF STEEL (+VE/-VE)	MOMENTS (+VE/-VE)	LOAD (+VE/-VE)
(MM)	(SQ. MM)	(KN-METER)	

0. 2837.5/ 0.0 541.16/ 0.00 219/ 0

333.	2535.7/	0.0	493.75/	0.00	219/	0
667.	2160.0/	0.0	431.35/	0.00	219/	0
1000.	1865.5/	0.0	379.83/	0.00	221/	0
1333.	1803.4/	0.0	368.68/	0.00	222/	0
1667.	1645.3/	0.0	339.80/	0.00	222/	0
2000.	1405.7/	312.0	294.79/	65.05	222/	227
2333.	1096.0/	729.2	234.35/	158.18	222/	227
2667.	872.8/	1371.4	189.20/	288.22	230/	219
3000.	768.1/	2331.5	166.63/	460.30	230/	219
3333.	636.0/	3551.7	137.96/	643.74	230/	219
3667.	797.6/	4992.5	103.74/	836.65	230/	219
4000.	1828.1/	6023.0	64.57/	1037.09	230/	219

B E A M N O. 17 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1999 mm	135.4 kN	227	10 mm	7	333 mm
1999	END 2	611.0 kN	219	10 mm	25	83 mm

161J	4000.X 400.X 600	110J
3No40-H=513.	0.TO=4000	
7*10c/c333	3No40 H 433.1209.TO 4000	25*10c/c 83
3No40 H 80.	0.TO 4000	

3T40		3T40		3T40		3T40
OOO		OOO		OOO		OOO

B E A M N O. 19 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	949 mm	455.9 kN	219	10 mm	8	135 mm
END 2	623.2 kN	219	10 mm	12	86 mm	

148J	1900.X 400.X 600	111J
3No40=H=513.	0.TO=1900	
2No40chc433.	0.TO 1900	12*10c/c 86
4No25 H 72.	0.TO 1900	

FROM	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	949 mm	455.9 kN	219	10 mm	8	135 mm
END 2	623.2 kN	219	10 mm	12	86 mm	

B E A M N O. 20 D E S I G N R E S U L T S - F L E X U R E

LEN - 2650. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
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1	76.	3- 32 MM	0.	2650.	YES	YES
2	513.	3- 40 MM	0.	2650.	YES	YES
3	433.	2- 40 MM	0.	2650.	YES	YES
COMP.	3- 32 MM	(REQD. STEEL= 2142. SQ. MM)				

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
----------------	---------------------------------	------------------------------	----------------

0.	1571.4/	5766.3	99.34/	987.16	228/	221
----	---------	--------	--------	--------	------	-----

221.	897.6/	5092.5	123.77/	856.11	228/	221
442.	673.5/	4434.7	146.10/	728.16	228/	221
662.	765.7/	3263.3	166.11/	603.95	228/	221
883.	847.0/	2473.8	183.74/	483.73	228/	221
1104.	920.0/	1799.3	198.86/	367.94	228/	221
1325.	980.3/	1214.3	211.10/	257.74	228/	221
1546.	1127.4/	796.9	240.58/	172.87	219/	230
1767.	1352.5/	510.0	284.58/	110.64	219/	230
1987.	1554.2/	312.0	322.87/	51.36	219/	230
2208.	1730.9/	0.0	355.52/	0.00	219/	0
2429.	1883.9/	0.0	383.12/	0.00	219/	0
2650.	2008.8/	0.0	405.18/	0.00	219/	0

B E A M N O. 20 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1324 mm	599.6 kN	221	10 mm	15	94 mm
END 2	340.6 kN	221	10 mm	6	264 mm	

111J	2650.X 400.X 600	149J
3No40=H=513.	0.TO=2650	
2No40chc433.	0.TO 2650	6*10c/c264
3No32 H 76.	0.TO 2650	

1	76.	3- 32 MM	0.	2650.	YES	YES
2	513.	3- 40 MM	0.	2650.	YES	YES
3	433.	2- 40 MM	0.	2650.	YES	YES
COMP.	3- 32 MM	(REQD. STEEL= 2142. SQ. MM)				

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
----------------	---------------------------------	------------------------------	----------------

0.	1571.4/	5766.3	99.34/	987.16	228/	221
----	---------	--------	--------	--------	------	-----

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B E A M N O. 21 D E S I G N R E S U L T S - F L E X U R E

LEN - 2350. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	76.	3- 32 MM	0.	2350.	YES YES
2	523.	4- 20 MM	548.	2350.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	2014.5/	0.0	406.20/ 0.00
196.	1816.2/	0.0	370.99/ 0.00
392.	1611.1/	0.0	333.48/ 0.00
588.	1568.4/	0.0	325.51/ 0.00
783.	1609.0/	0.0	333.09/ 0.00
979.	1637.6/	0.0	338.38/ 0.00
1175.	1654.2/	0.0	341.45/ 0.00
1371.	1660.9/	312.0	342.68/ 12.68
1567.	1656.0/	312.0	341.77/ 50.08
1763.	1637.6/	407.7	338.38/ 88.44
1958.	1608.6/	588.3	333.01/ 127.61
2154.	1569.1/	772.5	325.64/ 167.59
2350.	1520.2/	966.4	316.49/ 208.30

B E A M N O. 21 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
----- ----- ----- ----- ----- ----- -----						
END 1	1175 mm	175.4 kN	219	10 mm	5	293 mm
1175	END 2	302.1 kN	219	10 mm	6	235 mm
----- ----- ----- ----- ----- ----- -----						

149J 2350. X 400. X 600 164J

5*10c/c293	4No20 H 523. 548.TO 2350				
3No32 H 76. 0.TO 2350					6*10c/c235
----- ----- ----- ----- ----- -----					

3T32	3T32	3T32	3T32	3T32	3T32
OOO	OOO	OOO	OOO	OOO	OOO
----- ----- ----- ----- ----- -----					

=====

B E A M N O. 22 D E S I G N R E S U L T S - F L E X U R E

LEN - 2500. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
----- ----- ----- ----- ----- -----					

1	76.	3- 32 MM	0.	2500.	YES YES
---	-----	----------	----	-------	---------

**Please Note:

* Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing. *

2	517.	3- 32 MM	0.	2500.	YES YES
3	453.	3- 32 MM	0.	2500.	YES YES
COMP. 5- 16 MM (REQD. STEEL= 825. SQ. MM)					

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
-------------------	------------------------------------	---------------------------------	----------------

0.	442.9/	4544.7	96.07/ 749.56
208.	570.6/	3577.4	123.79/ 647.18
417.	689.3/	2881.1	149.54/ 547.81
625.	798.6/	2279.8	173.25/ 451.66
833.	900.5/	1749.3	194.88/ 358.87
1042.	996.4/	1275.3	214.35/ 269.65
1250.	1139.3/	904.0	242.94/ 195.58
1458.	1345.2/	641.7	283.17/ 139.20
1667.	1536.8/	392.1	319.61/ 85.06
1875.	1714.2/	312.0	352.47/ 33.07
2083.	1877.5/	0.0	381.97/ 0.00
2292.	2026.3/	0.0	408.26/ 0.00
2500.	2161.0/	0.0	431.53/ 0.00

B E A M N O. 22 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM - TO MAX. SHEAR LOAD LINKS NO. SPACING C/C									
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									
END 1 1250 mm 498.3 kN 221 10 mm 11 124 mm									
1250 END 2 303.4 kN 221 10 mm 5 312 mm									
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									
113J 2500.X 400.X 600 160J									
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									
3No32 H 517. 0.TO 2500									
11*10c/c124									
3No32 H 76. 0.TO 2500									
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									

FROM - TO MAX. SHEAR LOAD LINKS NO. SPACING C/C									
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									
END 1 1999 mm 136.2 kN 220 10 mm 7 333 mm									
1999 END 2 478.7 kN 220 10 mm 19 111 mm									
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									

B E A M N O. 23 D E S I G N R E S U L T S - F L E X U R E

LEN - 4000. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
----- ----- ----- ----- ----- -----					

1 76. 3- 32 MM 0. 4000. YES YES

**Please Note:

* Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing.*

2 513. 2- 40 MM 1209. 4000. NO YES

3 433. 2- 40 MM 1209. 4000. NO YES

COMP. 4- 20 MM (REQD. STEEL= 1088. SQ. MM)

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
-------------------	------------------------------------	---------------------------------	----------------

0. 2167.7/ 0.0 432.68/ 0.00 220/ 0

333.	1886.3/	0.0	383.55/	0.00	220/	0
667.	1573.3/	0.0	326.43/	0.00	220/	0
1000.	1344.5/	0.0	283.05/	0.00	222/	0
1333.	1297.2/	0.0	273.90/	0.00	221/	0
1667.	1197.6/	0.0	254.45/	0.00	221/	0
2000.	1043.4/	312.0	223.82/	46.31	221/	228
2333.	838.6/	567.0	181.92/	123.01	221/	228
2667.	680.7/	1048.1	147.67/	224.77	229/	220
3000.	576.3/	1724.7	125.03/	354.38	229/	220
3333.	446.1/	2533.3	96.78/	493.37	229/	220
3667.	312.0/	3532.8	63.12/	641.20	229/	220
4000.	593.8/	4788.8	24.33/	797.03	229/	220

B E A M N O. 23 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM - TO MAX. SHEAR LOAD LINKS NO. SPACING C/C									
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									
END 1 1999 mm 136.2 kN 220 10 mm 7 333 mm									
1999 END 2 478.7 kN 220 10 mm 19 111 mm									
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									

160J	4000.X 400.X 600	114J
----- ----- -----		
=2No40-H=513.1209.TO=4000=		
7*10c/c333	2No40 H 433.1209.TO 4000	19*10c/c111
3No32 H 76.	0.TO 4000	
----- ----- -----		
----- ----- -----		

3T32		3T32		3T32		3T32		3T32	
OOO		OOO		OOO		OOO		OOO	

=====

B E A M N O. 24 D E S I G N R E S U L T S - F L E X U R E

LEN - 4100. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	72.	4- 25 MM	0.	4100.	YES YES
**Please Note:					
* Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing.*					
2	517.	3- 32 MM	0.	2841.	YES NO
3	453.	3- 32 MM	0.	2841.	YES NO
COMP. 5- 12 MM (REQD. STEEL= 505. SQ. MM)					

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KN-METER)	LOAD(+VE/-VE)
0.	369.1/ 4247.2	80.07/ 691.69	228/ 221
342.	495.8/ 2901.9	107.56/ 550.97	228/ 221
683.	597.7/ 2086.2	129.65/ 418.66	228/ 221
1025.	673.2/ 1410.4	146.04/ 295.69	228/ 221
1367.	721.5/ 841.8	156.53/ 182.61	228/ 221
1708.	842.3/ 472.2	182.72/ 102.44	220/ 229
2050.	943.3/ 312.0	203.59/ 35.36	220/ 229
2392.	987.2/ 0.0	212.51/ 0.00	220/ 0
2733.	972.8/ 0.0	209.58/ 0.00	220/ 0
3075.	976.2/ 0.0	210.28/ 0.00	222/ 0
3417.	1205.2/ 0.0	255.95/ 0.00	222/ 0
3758.	1392.6/ 0.0	292.28/ 0.00	222/ 0
4100.	1540.0/ 0.0	320.20/ 0.00	222/ 0

B E A M N O. 24 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C

END 1	2050 mm	423.3 kN	221	10 mm	17	128 mm
2050	END 2	133.0 kN	220	10 mm	7	341 mm

=====

B E A M N O. 25 D E S I G N R E S U L T S - F L E X U R E

LEN - 1900. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	72.	4- 25 MM	0.	1900.	YES YES
**Please Note:					
* Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing.*					
2	517.	3- 32 MM	0.	1900.	YES YES
3	453.	3- 32 MM	0.	1900.	YES YES
COMP. 4- 16 MM (REQD. STEEL= 752. SQ. MM)					

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KN-METER)	LOAD(+VE/-VE)
0.	1531.1/ 0.0	318.53/ 0.00	222/ 0
158.	1390.9/ 312.0	291.96/ 30.27	222/ 227
317.	1244.7/ 333.0	263.67/ 72.23	222/ 227
475.	1092.2/ 531.8	233.58/ 115.36	222/ 227
633.	933.6/ 736.0	201.63/ 159.65	222/ 227
792.	831.7/ 1013.9	180.43/ 217.88	230/ 219
950.	764.5/ 1360.0	165.85/ 286.03	230/ 219
1108.	691.5/ 1734.9	150.02/ 356.25	230/ 219
1267.	612.8/ 2143.4	132.95/ 428.52	230/ 219
1425.	528.6/ 2591.9	114.66/ 502.76	230/ 219
1583.	438.8/ 3089.2	95.20/ 578.86	230/ 219
1742.	343.8/ 3649.8	74.58/ 656.77	230/ 219
1900.	312.0/ 4477.0	52.84/ 736.39	230/ 219

B E A M N O. 25 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -		TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1		949 mm	365.8 kN	220	10 mm	6	189 mm
949		END 2	508.1 kN	220	10 mm	9	118 mm

143J 1900.X 400.X 600 115J

3No32 H 517.	0.TO 1900								
6*10c/c189									9*10c/c118
4No25 H 72.	0.TO 1900								
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									

OOO		OOO		OOO		OOO		OOO	
3T32		3T32		3T32		3T32		3T32	
4T25		4T25		4T25		4T25		4T25	
oooo		oooo		oooo		oooo		oooo	

=====

B E A M N O. 26 D E S I G N R E S U L T S - F L E X U R E

LEN - 5000. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
----- ----- ----- ----- ----- -----					

1	72.	4- 25 MM	0.	5000.	YES YES
**Please Note:					
* Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing.*					
2	517.	3- 32 MM	0.	2874.	YES NO
3	453.	3- 32 MM	0.	2874.	YES NO
COMP. 4- 16 MM (REQD. STEEL= 707. SQ. MM)					
4	527.	6- 12 MM	3344.	5000.	NO YES

REQUIRED REINF. STEEL SUMMARY :

----- ----- ----- -----			
SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)

0.	352.5/	4435.0	76.47/	728.21	227/	222
417.	508.0/	2918.0	110.19/	553.41	227/	222
833.	626.0/	1930.4	135.81/	391.38	227/	222
1250.	704.1/	1143.6	152.74/	243.79	227/	222
1667.	808.2/	582.5	175.32/	126.35	219/	230
2083.	960.8/	312.0	207.16/	45.20	219/	230
2500.	1024.5/	0.0	220.02/	0.00	219/	0
2917.	993.0/	0.0	213.67/	0.00	220/	0
3333.	1089.6/	0.0	233.07/	0.00	222/	0
3750.	1294.3/	0.0	273.33/	0.00	222/	0
4167.	1423.1/	312.0	298.11/	1.79	222/	227
4583.	1477.6/	312.0	308.46/	61.80	222/	227
5000.	1464.8/	598.9	306.03/	129.92	222/	227

B E A M N O. 26 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM - TO		MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
----- ----- ----- ----- ----- ----- -----						
END 1	2084 mm	433.6 kN	222	10 mm	13	173 mm
2084	2501 mm	249.0 kN	222	10 mm	2	208 mm
2500	END 2	219.0 kN	219	10 mm	9	312 mm
----- ----- ----- ----- ----- ----- -----						

115J 5000.X 400.X 600 165J

3No32 H 517.	0.TO 2874								
13*10c/c173									9*10c/c312
4No25 H 72.	0.TO 5000								
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									

OOO		OOO		OOO		OOO		OOO	
3T32		3T32		3T32		3T32		3T32	
4T25		4T25		4T25		4T25		4T25	
oooo		oooo		oooo		oooo		oooo	

=====

B E A M N O. 27 D E S I G N R E S U L T S - F L E X U R E

LEN - 5040. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1 72. 3- 25 MM 0. 5040. YES YES

**Please Note:

* Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing.*

2 517. 4- 32 MM 0. 2923. YES NO

3 513. 3- 40 MM 2117. 5040. NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1438.0/ 2472.8	300.94/ 483.58	233/ 224
420.	1346.1/ 1741.4	283.35/ 357.43	233/ 224
840.	1244.1/ 1169.2	263.56/ 248.86	225/ 232
1260.	1148.7/ 752.3	244.80/ 163.20	225/ 232
1680.	982.0/ 398.2	211.44/ 86.38	225/ 232
2100.	752.4/ 312.0	163.22/ 18.49	225/ 232
2520.	461.6/ 0.0	100.15/ 0.00	225/ 0
2940.	523.2/ 312.0	113.51/ 0.82	224/ 233
3360.	646.8/ 365.7	140.31/ 79.33	224/ 233
3780.	754.7/ 821.7	163.73/ 178.26	232/ 225
4200.	862.3/ 1435.8	187.04/ 300.52	232/ 225
4620.	935.0/ 2188.4	201.91/ 436.21	232/ 225
5040.	969.3/ 3122.9	208.88/ 583.78	232/ 225

B E A M N O. 27 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING	C/C

END 1	840 mm	313.3 kN	224	10 mm	4	279	mm
840	4200 mm	257.7 kN	233	10 mm	8	419	mm
4199	END 2	364.3 kN	225	10 mm	5	209	mm

113J 5040.X 400.X 600 109J

B E A M N O. 28 D E S I G N R E S U L T S - F L E X U R E

LEN - 6650. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1 72. 3- 25 MM 0. 6650. YES YES

2 513. 3- 40 MM 0. 3039. YES NO

3 513. 3- 40 MM 3611. 6650. NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	312.0/ 3560.1	48.36/ 644.86	233/ 224
554.	455.9/ 2123.9	98.91/ 425.17	233/ 224
1108.	624.1/ 1064.2	135.39/ 228.00	233/ 224
1662.	877.2/ 421.5	190.10/ 91.43	225/ 232
2217.	1159.7/ 312.0	246.99/ 3.40	225/ 232
2771.	1259.5/ 0.0	266.57/ 0.00	225/ 0
3325.	1163.4/ 0.0	247.71/ 0.00	223/ 0
3879.	1267.9/ 0.0	268.21/ 0.00	224/ 0
4433.	1171.1/ 312.0	249.22/ 4.45	224/ 233
4988.	891.0/ 426.3	192.94/ 92.48	224/ 233
5542.	628.7/ 1057.3	136.39/ 226.61	232/ 225
6096.	460.5/ 2112.3	99.91/ 423.18	232/ 225
6650.	312.0/ 3540.8	49.35/ 642.27	232/ 225

B E A M N O. 28 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2217 mm	413.8 kN	224	10 mm	14	170 mm
2217	3880 mm	230.2 kN	224	10 mm	4	415 mm
2771	4434 mm	229.1 kN	224	10 mm	4	415 mm
4434	END 2	412.7 kN	225	10 mm	14	170 mm

109J	6650.X 400.X 600	105J
3No40 H 513.	10.TO 3039	3No40 H 513.3611.TO 6650
14*10c/c170		14*10c/c170
3No25 H 172.	10.TO 6650	

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
000		000		000		000
3T40		3T40		3T40		3T40
3T25		3T25		3T25		3T25
000		000		000		000

B E A M N O. 29 D E S I G N R E S U L T S - F L E X U R E

LEN - 2010. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
1	68.	5- 16 MM	0.	2010.		YES YES
2	513.	3- 40 MM	0.	2010.		YES YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	903.8/ 2948.0	195.55/ 557.92	233/ 224
167.	901.0/ 2588.1	194.98/ 502.16	233/ 224
335.	892.3/ 2259.0	193.20/ 448.16	233/ 224

502.	877.6/ 1956.4	190.19/ 395.99	233/ 224
670.	856.9/ 1677.7	185.90/ 345.77	233/ 224
837.	831.2/ 1420.2	180.31/ 297.56	233/ 224
1005.	799.2/ 1182.6	173.36/ 251.50	233/ 224
1172.	761.0/ 962.9	165.08/ 207.57	233/ 224
1340.	716.6/ 764.3	155.45/ 165.81	233/ 224
1507.	666.2/ 581.0	144.53/ 126.04	233/ 224
1675.	610.0/ 406.8	132.32/ 88.24	233/ 224
1842.	562.5/ 312.0	122.02/ 55.40	225/ 232
2010.	514.3/ 312.0	111.58/ 25.38	225/ 232

B E A M N O. 29 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1004 mm	338.0 kN	223	10 mm	6	200 mm
1004	END 2	199.5 kN	223	10 mm	4	334 mm

105J	2010.X 400.X 600	150J
3No40 H 513.	0.TO 2010	
6*10c/c200		4*10c/c334
5No16 H 168.	0.TO 2010	

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
000		000		000		000
3T40		3T40		3T40		3T40
5T16		5T16		5T16		5T16
00000		00000		00000		00000

B E A M N O. 30 D E S I G N R E S U L T S - F L E X U R E

LEN - 3300. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
1	72.	3- 25 MM	0.	3300.		YES YES

2	528.	4- 10 MM	0.	823.	YES	NO
3	517.	3- 32 MM	277.	3300.	NO	YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	602.5/ 312.0	130.71/ 38.58	226/ 231
275.	489.8/ 0.0	106.25/ 0.00	226/ 0
550.	356.5/ 0.0	77.34/ 0.00	226/ 0
825.	500.8/ 0.0	108.63/ 0.00	224/ 0
1100.	664.8/ 312.0	144.21/ 15.88	224/ 233
1375.	808.3/ 312.0	175.35/ 57.00	224/ 233
1650.	935.7/ 466.9	202.05/ 101.28	224/ 233
1925.	1045.7/ 685.6	224.29/ 148.72	224/ 233
2200.	1135.0/ 922.3	242.10/ 199.33	224/ 233
2475.	1202.7/ 1190.8	255.45/ 253.11	224/ 233
2750.	1268.2/ 1506.7	268.26/ 313.95	231/ 226
3025.	1336.7/ 1882.5	281.53/ 382.86	231/ 226
3300.	1389.3/ 2307.0	291.64/ 456.22	231/ 226

B E A M N O. 30 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING	C/C
150J			3300.X 400.X 600			101J	
4No103Ho528.	510.To77823	3300					
2*10c/c549			2*10c/c549				
3No25 H 72.	0.TO 3300						
ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo
4T10	4T10	3T32	3T32	3T32	3T32	3T32	3T32
3T25	3T25	3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 31 D E S I G N R E S U L T S - F L E X U R E

LEN - 2010. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	72.	2- 25 MM	0.	2010.	YES YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	0.0/ 0.0	0.00/ 0.00	0/ 0
167.	234.0/ 0.0	22.29/ 0.00	226/ 0
335.	234.0/ 0.0	44.18/ 0.00	226/ 0
502.	301.8/ 0.0	65.47/ 0.00	226/ 0
670.	396.5/ 0.0	86.00/ 0.00	226/ 0
837.	486.3/ 0.0	105.48/ 0.00	226/ 0
1005.	569.8/ 0.0	123.60/ 0.00	226/ 0
1172.	648.0/ 0.0	140.55/ 0.00	226/ 0
1340.	723.0/ 0.0	155.85/ 0.00	226/ 0
1507.	791.0/ 0.0	169.56/ 0.00	226/ 0
1675.	853.3/ 0.0	181.97/ 0.00	226/ 0
1842.	907.6/ 0.0	192.68/ 0.00	226/ 0
2010.	953.3/ 0.0	201.62/ 0.00	226/ 0

B E A M N O. 31 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING	C/C
END 1	1004 mm	133.6 kN	226 10 mm	4	334	mm	
1004	END 2	49.6 kN	226 10 mm	4	334	mm	
Nominal links should be provided through out the length.							

139J		2010.X 300.X 600		151J	
4*10c/c334					
2No25 H 72. 0.TO 2010					
4*10c/c334					
2T25	2T25	2T25	2T25	2T25	2T25
oo	oo	oo	oo	oo	oo

B E A M N O. 32 D E S I G N R E S U L T S - F L E X U R E

LEN - 3300. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
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1	72.	2- 25 MM	0.	3300.	YES YES
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REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	939.8/	0.0	223/ 0
275.	972.0/	0.0	223/ 0
550.	979.2/	0.0	223/ 0
825.	960.5/	0.0	223/ 0
1100.	919.1/	0.0	223/ 0
1375.	850.6/	0.0	223/ 0
1650.	761.9/	0.0	223/ 0
1925.	656.9/	0.0	223/ 0
2200.	543.7/	0.0	223/ 0
2475.	419.3/	0.0	223/ 0
2750.	285.6/	0.0	223/ 0
3025.	234.0/	0.0	223/ 0
3300.	0.0/	0.0	0/ 0

B E A M N O. 32 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	550 mm	36.2 kN	226	10 mm	3	275 mm
825	END 2	115.5 kN	223	10 mm	8	353 mm
151J 3300.X 300.X 600 117J						
3*10c/c275 8*10c/c353						
2No25 H 72. 0.TO 3300						
2T25	2T25	2T25	2T25	2T25	2T25	2T25
oo	oo	oo	oo	oo	oo	oo

B E A M N O. 33 D E S I G N R E S U L T S - F L E X U R E

LEN - 5040. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
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1	80.	2- 40 MM	0.	5040.	YES YES
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REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	0.0/	0.0	0.00/ 0.00
420.	443.3/	0.0	96.17/ 0.00
840.	844.3/	0.0	180.18/ 0.00
1260.	1201.6/	0.0	248.85/ 0.00

1680.	1483.0/	0.0	299.76/	0.00	219/	0
2100.	1662.8/	0.0	330.82/	0.00	219/	0
2520.	1724.7/	0.0	341.23/	0.00	219/	0
2940.	1662.8/	0.0	330.82/	0.00	219/	0
3360.	1483.0/	0.0	299.76/	0.00	219/	0
3780.	1201.6/	0.0	248.85/	0.00	219/	0
4200.	844.3/	0.0	180.18/	0.00	219/	0
4620.	443.3/	0.0	96.17/	0.00	219/	0
5040.	0.0/	0.0	0.00/	0.00	0/	0

B E A M N O. 33 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	840 mm	241.0 kN	226	10 mm	3	419 mm
840	2520 mm	185.4 kN	219	10 mm	3	559 mm
2520	4200 mm	185.4 kN	219	10 mm	3	559 mm
4200	END 2	241.0 kN	226	10 mm	3	419 mm

160J	5040.X 300.X 600	161J
------	------------------	------

3*10c/c419
2No40 H 80. 0.TO 5040

2T40	2T40	2T40	2T40	2T40	2T40
OO	OO	OO	OO	OO	OO

B E A M N O. 34 D E S I G N R E S U L T S - F L E X U R E

LEN - 5040. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
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1	76.	3- 32 MM	0.	5040.	YES	YES
2	513.	3- 40 MM	0.	2923.	YES	NO

3	513.	3- 40 MM	2117.	5040.	NO	YES
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REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
----------------	---------------------------------	------------------------------	----------------

0.	2166.7/	2991.8	432.51/	564.47	234/	223
420.	1851.3/	2207.4	377.30/	439.44	234/	223
840.	1555.9/	1549.4	323.19/	321.97	226/	231
1260.	1353.1/	1078.1	284.70/	230.77	226/	231
1680.	1105.5/	661.8	236.24/	143.57	226/	231
2100.	800.8/	312.0	173.73/	61.75	226/	231
2520.	442.2/	0.0	95.92/	0.00	226/	0
2940.	545.1/	312.0	118.24/	31.79	223/	234
3360.	741.5/	549.7	160.85/	119.24	223/	234
3780.	960.7/	1073.3	207.14/	229.80	231/	226
4200.	1235.7/	1771.1	261.91/	362.82	231/	226
4620.	1504.6/	2595.6	313.55/	503.35	231/	226
5040.	1772.8/	3585.7	363.13/	648.28	231/	226

B E A M N O. 34 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1260 mm	300.3 kN	223	10 mm	4	419 mm
1260	2520 mm	258.1 kN	231	10 mm	3	419 mm
2520	3360 mm	271.0 kN	231	10 mm	2	419 mm
3360	END 2	347.7 kN	226	10 mm	8	239 mm

114J	5040.X 400.X 600	110J
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3No40 H|513. 10.TO 2923 3No40 H 513.2117.TO 5040
4*10c/c419 8*10c/c239
3No32 H| 76. 10.TO 5040

OOO	OOO	OOO	OOO	OOO	OOO
3T40	3T40	3T40	3T40	3T40	3T40
3T32	3T32	3T32	3T32	3T32	3T32
OOO	OOO	OOO	OOO	OOO	OOO

B E A M N O. 36 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2213 mm	388.6 kN	224	10 mm	12	201 mm
2213	2656 mm	237.1 kN	232	10 mm	2	221 mm
2654	3540 mm	251.4 kN	232	10 mm	3	295 mm
3540	END 2	339.6 kN	225	10 mm	8	252 mm

106J	5310.X 400.X 600	102J
3No32 H 517.	0.TO 3003	8*10c/c252
12*10c/c201		
3No32 H 76.	0.TO 5310	

000	000	000				
3T32	3T32	3T32				
3T32	3T32	3T32	3T32	3T32	3T32	3T32
000	000	000	000	000	000	000

B E A M N O. 37 D E S I G N R E S U L T S - F L E X U R E

LEN - 5040. mm FY - 460. mm FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
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1	80.	2- 40 MM	0.	5040.	YES YES
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REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KN-METER)	LOAD(+VE/-VE)
0.	0.0/	0.00/	0/ 0
420.	427.6/	92.75/	0.00 219/ 0
840.	809.5/	173.26/	0.00 219/ 0
1260.	1147.0/	238.66/	0.00 219/ 0

1680.	1412.3/	0.0	287.22/	0.00	219/ 0
2100.	1580.4/	0.0	316.73/	0.00	219/ 0
2520.	1638.6/	0.0	326.69/	0.00	219/ 0
2940.	1580.4/	0.0	316.73/	0.00	219/ 0
3360.	1412.3/	0.0	287.22/	0.00	219/ 0
3780.	1147.0/	0.0	238.66/	0.00	219/ 0
4200.	809.5/	0.0	173.26/	0.00	219/ 0
4620.	427.6/	0.0	92.75/	0.00	219/ 0
5040.	0.0/	0.0	0.00/	0.00	0/ 0

B E A M N O. 37 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	840 mm	232.8 kN	226	10 mm	3	419 mm
840	2520 mm	176.2 kN	219	10 mm	3	559 mm
2520	4200 mm	176.2 kN	219	10 mm	3	559 mm
4200	END 2	232.8 kN	226	10 mm	3	419 mm

143J	5040.X 300.X 600	148J
3*10c/c419		3*10c/c419
2No40 H 80.	0.TO 5040	

2T40	2T40	2T40	2T40	2T40	2T40
OO	OO	OO	OO	OO	OO

B E A M N O. 38 D E S I G N R E S U L T S - F L E X U R E

LEN - 2285. mm FY - 460. mm FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
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1	70.	3- 20 MM	0.	2285.	YES YES
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REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	0.0/	0.00/	0/ 0
190.	234.0/	20.07/	0.00 220/ 0
381.	234.0/	39.62/	0.00 220/ 0
571.	268.6/	58.27/	0.00 220/ 0
762.	349.4/	75.80/	0.00 220/ 0
952.	424.1/	92.00/	0.00 220/ 0
1142.	489.4/	106.17/	0.00 220/ 0
1333.	546.6/	118.58/	0.00 220/ 0
1523.	595.6/	129.21/	0.00 220/ 0
1714.	632.9/	137.29/	0.00 220/ 0
1904.	660.4/	143.09/	0.00 220/ 0
2095.	679.3/	146.96/	0.00 220/ 0
2285.	685.1/	148.14/	0.00 220/ 0

B E A M N O. 38 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1904 mm	106.1 kN	220	10 mm	6	380 mm
1904	END 2	16.4 kN	221	10 mm	2	380 mm
Nominal links should be provided through out the length.						

140J	2285.X 300.X 600	137J
6*10c/c380		2*10c/c380
3No20 H 70.	0.TO 2285	

3T20	3T20	3T20	3T20	3T20	3T20
ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 39 D E S I G N R E S U L T S - F L E X U R E

LEN - 3025. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
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1	72.	2- 25 MM	0.	3025.	YES YES
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REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
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0.	667.6/	0.0	144.56/	0.00	219/ 0
252.	799.3/	0.0	171.23/	0.00	219/ 0
504.	897.5/	0.0	190.69/	0.00	219/ 0
756.	958.6/	0.0	202.64/	0.00	219/ 0
1008.	981.3/	0.0	207.06/	0.00	219/ 0
1260.	963.3/	0.0	203.56/	0.00	219/ 0
1512.	907.5/	0.0	192.67/	0.00	219/ 0
1765.	817.6/	0.0	174.87/	0.00	219/ 0
2017.	697.6/	0.0	150.68/	0.00	219/ 0
2269.	555.6/	0.0	120.52/	0.00	219/ 0
2521.	391.2/	0.0	84.86/	0.00	219/ 0
2773.	234.0/	0.0	44.45/	0.00	219/ 0
3025.	0.0/	0.0	0.00/	0.00	0/ 0

B E A M N O. 39 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1008 mm	131.3 kN	222	10 mm	4	336 mm
1260	2521 mm	150.6 kN	222	10 mm	3	420 mm
2521	END 2	183.3 kN	219	10 mm	2	504 mm

0.	1085.2/	3395.2	232.20/	622.42	234/	223
554.	1019.6/	2178.0	219.04/	434.44	234/	223
1108.	933.6/	1215.5	201.64/	257.96	234/	223
1662.	1015.2/	649.3	218.15/	140.85	226/	231
2217.	1132.2/	312.0	241.54/	66.09	226/	231
2771.	1103.2/	312.0	235.77/	0.49	225/	232
3325.	923.3/	0.0	199.54/	0.00	213/	0
3879.	1051.4/	312.0	225.42/	10.53	223/	234
4433.	1019.5/	388.7	219.02/	84.31	223/	234
4988.	845.2/	770.6	183.34/	167.17	223/	234
5542.	789.5/	1481.0	171.26/	309.10	231/	226
6096.	832.4/	2561.2	180.57/	497.86	231/	226
6650.	855.7/	4280.2	185.63/	698.11	231/	226

B E A M N O. 41 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1662 mm	343.6 kN	223	10 mm	8	237 mm
1662	4988 mm	289.7 kN	234	10 mm	8	415 mm
4988	END 2	365.8 kN	226	10 mm	8	237 mm
111J		6650.X 400.X 600				107J
3No40 H 513.	0.TO 3593	3No32 H 517.3089	TO 6650			
8*10c/c237			8*10c/c237			
4No20 H 70.	0.TO 6650					

OOO		OOO		OOO		OOO
3T40		3T40		3T32		3T32
4T20		4T20		4T20		4T20
OOOO		OOOO		OOOO		OOOO

B E A M N O. 42 D E S I G N R E S U L T S - F L E X U R E

LEN - 2285. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	76.	3- 32 MM	0.	2285.	YES YES
2	513.	3- 40 MM	0.	2285.	YES YES
3	433.	2- 40 MM	0.	2285.	YES YES
COMP.		4- 25 MM	(REQD. STEEL=	1840. SQ. MM)	

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1291.4/	5486.3	170.98/ 932.70
190.	838.3/	4898.2	181.84/ 818.31
381.	888.4/	4312.9	192.40/ 704.48
571.	938.2/	3176.2	202.56/ 591.49
762.	985.8/	2448.5	212.22/ 479.60
952.	1030.8/	1805.9	221.29/ 369.13
1142.	1076.7/	1231.3	230.48/ 261.06
1333.	1155.4/	745.7	246.13/ 161.76
1523.	1354.6/	405.3	284.98/ 87.92
1714.	1551.8/	312.0	322.41/ 14.67
1904.	1748.6/	0.0	358.74/ 0.00
2095.	1946.4/	0.0	394.21/ 0.00
2285.	2147.0/	0.0	429.14/ 0.00

B E A M N O. 42 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1142 mm	601.7 kN	224	10 mm	13	95 mm
1142	END 2	533.2 kN	224	10 mm	11	114 mm

1662.	1675.8/	0.0	333.01/	0.00	224/	0
2217.	1951.7/	0.0	378.32/	0.00	224/	0
2771.	1995.0/	0.0	385.18/	0.00	224/	0
3325.	1798.8/	0.0	353.54/	0.00	224/	0
3879.	1409.4/	0.0	286.71/	0.00	224/	0
4433.	928.7/	0.0	196.81/	0.00	213/	0
4988.	384.2/	0.0	83.35/	0.00	213/	0
5542.	0.0/	388.2	0.00/	84.21	0/	202
6096.	0.0/	1049.3	0.00/	220.13	0/	225
6650.	0.0/	2001.4	0.00/	386.19	0/	225

B E A M N O. 44 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1109 mm	264.6 kN	224	10 mm	4	369 mm
1109	2772 mm	175.0 kN	224	10 mm	3	554 mm
2771	4434 mm	188.4 kN	213	10 mm	3	554 mm
4434	END 2	303.5 kN	225	10 mm	11	221 mm

149J 6650.X 300.X 600 141J

4*10c/c369	2No40 H 513.4719.TO 6650	11*10c/c221				
2No40 H 80.	0.TO 5810					
2T40	2T40	2T40	2T40	2T40	2T40	2T40
OO	OO	OO	OO	OO	OO	OO

B E A M N O. 45 D E S I G N R E S U L T S - F L E X U R E

LEN - 2285. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
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1	513.	2- 40 MM	0.	2285.	YES	YES
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REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	0.0/ 2001.4	0.00/ 386.19	0/ 225
190.	0.0/ 1709.2	0.00/ 338.63	0/ 225
381.	0.0/ 1446.1	0.00/ 293.23	0/ 225
571.	0.0/ 1209.4	0.00/ 250.31	0/ 225
762.	0.0/ 996.9	0.00/ 210.06	0/ 225
952.	0.0/ 807.4	0.00/ 172.84	0/ 225
1142.	0.0/ 640.6	0.00/ 138.96	0/ 225
1333.	0.0/ 500.0	0.00/ 108.48	0/ 225
1523.	0.0/ 374.9	0.00/ 81.33	0/ 225
1714.	0.0/ 263.8	0.00/ 57.22	0/ 225
1904.	0.0/ 234.0	0.00/ 35.77	0/ 225
2095.	0.0/ 234.0	0.00/ 16.81	0/ 225
2285.	0.0/ 0.0	0.00/ 0.00	0/ 0

B E A M N O. 45 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM -	TO	SHEAR TORSN mm	LOAD kNm	LINK S T	NO. S T	SPACING mm C/C
END 1	953	254.9	21	225227/ 10mm	3 6 10	317 180 114
953	1144	185.1	21	225227/ 10mm	2 2 3	95 180 62
1142	END 2	83.1	21	225227/ 10mm	1 6 8	571 180 127

EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 64 mm2 EVENLY DISTRIBUTED
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *

141J 2285.X 300.X 600 138J

2No40 H 513. 0.TO 2285
10*10c/c114 8*10c/c127

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B E A M N O. 46 D E S I G N R E S U L T S - F L E X U R E

LEN - 5040. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	72.	3- 25 MM	0.	5040.	YES YES
**Please Note:					
* Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing.*					
2	513.	2- 40 MM	0.	2923.	YES NO
3	513.	3- 40 MM	2117.	5040.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KN-METER)	LOAD(+VE/-VE)
0.	1439.4/ 2453.6	301.20/ 480.43	234/ 223
420.	1342.9/ 1741.5	282.74/ 357.45	234/ 223
840.	1217.1/ 1166.3	258.28/ 248.28	226/ 231
1260.	1111.0/ 755.1	237.32/ 163.80	226/ 231
1680.	941.8/ 404.0	203.29/ 87.64	226/ 231
2100.	720.0/ 312.0	156.20/ 19.78	226/ 231
2520.	442.7/ 0.0	96.05/ 0.00	226/ 0
2940.	524.6/ 312.0	113.81/ 0.00	223/ 234
3360.	676.0/ 351.3	146.65/ 76.22	223/ 234
3780.	776.7/ 750.5	168.49/ 162.80	231/ 226
4200.	900.0/ 1304.0	194.77/ 275.22	231/ 226
4620.	988.6/ 1982.4	212.79/ 400.56	231/ 226
5040.	1038.8/ 2815.5	222.89/ 537.80	231/ 226

B E A M N O. 46 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C

END 1	840 mm	305.7 kN	223	10 mm	4	279 mm
840	2520 mm	249.3 kN	234	10 mm	4	419 mm
2520	3780 mm	252.1 kN	234	10 mm	3	419 mm
3780	END 2	339.7 kN	226	10 mm	6	251 mm

116J 5040.X 400.X 600 112J

2No40 H 513.	0.TO 2923	3No40 H 513.2117.TO 5040				
4*10c/c279						6*10c/c251
3No25 H 72.	0.TO 5040					
=====						

OO		OO		OO		OO
2T40		2T40		2T40		3T40
3T25		3T25		3T25		3T25
ooo		ooo		ooo		ooo
4*10c/c279						
=====						

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B E A M N O. 47 D E S I G N R E S U L T S - F L E X U R E

LEN - 6650. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	70.	3- 20 MM	0.	6650.	YES YES
2	513.	3- 40 MM	0.	3039.	YES NO
3	513.	3- 40 MM	3611.	6650.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KN-METER)	LOAD(+VE/-VE)
0.	467.7/ 2665.8	101.45/ 514.49	234/ 223
554.	622.0/ 1653.9	134.94/ 341.38	234/ 223
1108.	711.7/ 875.5	154.39/ 189.75	234/ 223
1662.	846.6/ 407.5	183.66/ 88.39	226/ 231
2217.	922.1/ 312.0	199.29/ 18.06	226/ 231
2771.	852.3/ 0.0	184.90/ 0.00	226/ 0
3325.	651.0/ 0.0	141.23/ 0.00	224/ 0
3879.	703.1/ 0.0	152.53/ 0.00	224/ 0
4433.	678.7/ 312.0	147.22/ 57.61	224/ 233
4988.	581.2/ 606.7	126.08/ 131.61	224/ 233
5542.	552.3/ 1126.0	119.80/ 240.30	232/ 225
6096.	543.8/ 1805.7	117.96/ 369.08	232/ 225
6650.	519.7/ 2606.8	112.73/ 505.14	232/ 225

B E A M N O. 47 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1109 mm	329.6 kN	224	10 mm	5	277 mm
1109	2218 mm	249.8 kN	224	10 mm	3	369 mm
2216	END 2	250.0 kN	225	10 mm	13	369 mm

112J	6650.X 400.X 600	108J
3No40 H 513.	0.TO 3039	3No40 H 513.3611.TO 6650
5*10c/c277		13*10c/c369
3No20 H 70.	0.TO 6650	

FROM	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1142 mm	463.9 kN	224	10 mm	10	126 mm
1142	END 2	303.0 kN	224	10 mm	4	380 mm

B E A M N O. 48 D E S I G N R E S U L T S - F L E X U R E

LEN - 2285. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
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1 70. 4- 20 MM 0. 2285. YES YES

**Please Note: * Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing. *

2 517. 3- 32 MM 0. 2285. YES YES

3 453. 3- 32 MM 0. 2285. YES YES

COMP. 4- 12 MM (REQD. STEEL= 451. SQ. MM)

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
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0. 554.1/ 4197.8 120.19/ 682.10 233/ 224

190.	620.4/ 3199.5	134.58/ 594.85	233/ 224
381.	679.4/ 2636.6	147.39/ 509.88	233/ 224
571.	731.1/ 2136.5	158.59/ 427.33	233/ 224
762.	775.1/ 1686.1	168.13/ 347.32	233/ 224
952.	811.1/ 1277.1	175.96/ 270.00	233/ 224
1142.	839.1/ 903.8	182.02/ 195.55	233/ 224
1333.	890.0/ 601.2	192.73/ 130.41	225/ 232
1523.	984.9/ 361.9	212.05/ 78.51	225/ 232
1714.	1067.5/ 312.0	228.66/ 28.33	225/ 232
1904.	1138.3/ 0.0	242.75/ 0.00	225/ 0
2095.	1197.4/ 0.0	254.40/ 0.00	225/ 0
2285.	1245.3/ 0.0	263.79/ 0.00	225/ 0

B E A M N O. 48 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1142 mm	463.9 kN	224	10 mm	10	126 mm
1142	END 2	303.0 kN	224	10 mm	4	380 mm

108J	2285.X 400.X 600	136J
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3No32 H 517.	0.TO 2285				
10*10c/c126					4*10c/c380
4No20 H 70.	0.TO 2285				

FROM	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1142 mm	463.9 kN	224	10 mm	10	126 mm
1142	END 2	303.0 kN	224	10 mm	4	380 mm

100

BEAM NO. 49 DESIGN RESULTS - FLEXURE

LEN - 1525. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT	BAR INFO	FROM	TO	ANCHOR
	mm		mm	mm	STA END

1	72.	3- 25 MM	0.	1525.	YES	YES
2	527.	4- 12 MM	321.	1525.	NO	YES

REQUIRED REINF. STEEL SUMMARY :

.....

SECTION	REINF STEEL (+VE/-VE)	MOMENTS (+VE/-VE)	LOAD (+VE/-VE)
(MM)	(SQ. MM)	(KN-METER)	

0.	1243.9/	0.0	263.53/	0.00	225/	0
127.	1151.1/	0.0	245.29/	0.00	225/	0
254.	1054.5/	0.0	226.06/	0.00	225/	0
381.	1110.1/	0.0	237.14/	0.00	223/	0
508.	1169.0/	0.0	248.81/	0.00	224/	0
635.	1222.7/	0.0	259.37/	0.00	224/	0
762.	1270.8/	0.0	268.76/	0.00	224/	0
890.	1313.1/	0.0	276.98/	0.00	224/	0
1017.	1349.6/	0.0	284.03/	0.00	224/	0
1144.	1380.5/	312.0	289.97/	11.75	224/	233
1271.	1405.9/	312.0	294.83/	36.89	224/	233
1398.	1426.0/	312.0	298.66/	62.73	224/	233
1525.	1440.9/	411.5	301.50/	89.26	224/	233

BEAM NO. 49 DESIGN RESULTS - SHEAR

PROVIDE SHEAR LINKS AS FOLLOWS

FROM	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING	C/C
END 1	762 mm	143.5 kN	233	10 mm	3	381	mm
762	1144 mm	217.3 kN	225	10 mm	2	190	mm
1144	END 2	241.2 kN	225	10 mm	2	381	mm

162J 6500.X 300.X 600 163J

B E A M N O. 52 D E S I G N R E S U L T S - F L E X U R E

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
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1 70. 3- 20 MM 0. 6500. YES YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KN-METER)	LOAD(+VE/-VE)
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0.	0.0/	0.0	0.00/	0.00	0/	0
542.	258.4/	0.0	56.06/	0.00	219/	0
1083.	480.3/	0.0	104.20/	0.00	219/	0
1625.	653.6/	0.0	141.70/	0.00	219/	0
2167.	785.6/	0.0	168.48/	0.00	219/	0
2708.	866.3/	0.0	184.55/	0.00	219/	0
3250.	893.5/	0.0	189.91/	0.00	219/	0
3792.	866.3/	0.0	184.55/	0.00	219/	0
4333.	785.6/	0.0	168.48/	0.00	219/	0
4875.	653.6/	0.0	141.70/	0.00	219/	0
5417.	480.3/	0.0	104.20/	0.00	219/	0
5958.	258.4/	0.0	56.06/	0.00	219/	0
6500.	0.0/	0.0	0.00/	0.00	0/	0

B E A M N O. 52 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2166 mm	107.0 kN	226	10 mm	7	361 mm
4333	END 2	107.0 kN	226	10 mm	7	361 mm

B E A M N O. 53 D E S I G N R E S U L T S - F L E X U R E

LEN - 5040. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
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1 70. 3- 20 MM 0. 5040. YES YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KN-METER)	LOAD(+VE/-VE)
-------------------	-----------------------------------	--------------------------------	---------------

0.	0.0/	0.0	0.00/	0.00	0/	0
420.	234.0/	0.0	46.01/	0.00	219/	0
840.	405.6/	0.0	87.99/	0.00	219/	0
1260.	568.1/	0.0	123.24/	0.00	219/	0
1680.	697.0/	0.0	150.58/	0.00	219/	0
2100.	782.7/	0.0	167.89/	0.00	219/	0
2520.	812.4/	0.0	173.85/	0.00	219/	0
2940.	782.7/	0.0	167.89/	0.00	219/	0
3360.	697.0/	0.0	150.58/	0.00	219/	0
3780.	568.1/	0.0	123.24/	0.00	219/	0
4200.	405.6/	0.0	87.99/	0.00	219/	0
4620.	234.0/	0.0	46.01/	0.00	219/	0
5040.	0.0/	0.0	0.00/	0.00	0/	0

B E A M N O. 54 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2099 mm	111.8 kN	226	10 mm	7	349 mm
END 2	2939 mm	111.8 kN	226	10 mm	7	349 mm
165J 5040.X 300.X 600 164J						
7*10c/c349 7*10c/c349						
3No20 H 70. 0.TO 5040						
3T20	3T20	3T20	3T20	3T20	3T20	3T20
ooo	ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 54 D E S I G N R E S U L T S - S H E A R

125.	1435.7/	1119.9	300.50/	239.10	221/	228
250.	1325.1/	1283.4	279.30/	271.22	221/	228
375.	1257.1/	1497.7	266.09/	312.25	229/	220
500.	1251.0/	1788.8	264.91/	366.03	229/	220
625.	1243.6/	2096.5	263.46/	420.45	229/	220
750.	1234.8/	2423.4	261.74/	475.50	229/	220
875.	1224.9/	2772.0	259.80/	531.08	230/	219
1000.	1213.8/	3146.5	257.63/	587.20	230/	219
1125.	1201.7/	3552.0	255.25/	643.78	230/	219
1250.	1188.8/	4293.4	252.71/	700.68	230/	219
1375.	1175.1/	4587.5	250.02/	757.88	230/	219
1500.	1160.9/	4882.6	247.21/	815.28	230/	219

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	749 mm	413.9 kN	219	10 mm	5	187 mm
END 2	749 mm	459.7 kN	219	10 mm	6	149 mm
164J 1500.X 400.X 600 112J						
3No40=H=513. 0.TO=1500						
2No40c/c433. 0.TO 1500						
4No25 H 72. 0.TO 1500						
6*10c/c149						

B E A M N O. 54 D E S I G N R E S U L T S - F L E X U R E

LEN - 1500. mm FY - 460. FC - 40. SIZE - 400. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
-------	-----------	----------	---------	-------	----------------

1	72.	4- 25 MM	0.	1500.	YES YES
2	513.	3- 40 MM	0.	1500.	YES YES
3	433.	2- 40 MM	0.	1500.	YES YES
COMP. 4- 20 MM (REQD. STEEL= 1189. SQ. MM)					

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
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0.	1544.5/	961.3	321.05/	207.26	221/	228
----	---------	-------	---------	--------	------	-----

=====									
B E A M N O. 55 D E S I G N R E S U L T S - F L E X U R E									
LEN - 1500. mm FY - 460. FC - 40. SIZE - 400. X 600. mm									

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	72.	3- 25 MM	0.	1500.	YES YES
2	513.	3- 40 MM	0.	1500.	YES YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1440.7/ 603.7	301.46/ 130.96	222/ 227
125.	1352.0/ 730.4	284.48/ 158.45	222/ 227
250.	1258.8/ 860.4	266.44/ 186.65	222/ 227
375.	1161.3/ 1002.3	247.29/ 215.55	222/ 227
500.	1077.3/ 1168.8	230.62/ 248.78	230/ 219
625.	1049.9/ 1407.5	225.13/ 295.12	230/ 219
750.	1018.8/ 1660.9	218.89/ 342.69	230/ 219
875.	984.1/ 1931.0	211.88/ 391.48	230/ 219
1000.	945.9/ 2219.6	204.13/ 441.51	230/ 219
1125.	904.2/ 2529.2	195.63/ 492.71	230/ 219
1250.	859.3/ 2862.9	186.40/ 545.05	230/ 219
1375.	813.5/ 3225.0	176.47/ 598.49	230/ 219
1500.	764.5/ 3621.2	165.83/ 652.99	230/ 219

B E A M N O. 55 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
----- ----- ----- ----- ----- ----- -----						
END 1	749 mm	330.8 kN	219	10 mm	4	249 mm
749	END 2	440.1 kN	219	10 mm	7	124 mm
----- ----- ----- ----- ----- ----- -----						

165J	1500.X 400.X 600	116J
=====		
3No40 H 513.	0.TO 1500	7*10c/c124
4*10c/c249		
3No25 H 72.	0.TO 1500	
=====		

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

000		000		000	000
3T40		3T40		3T40	3T40
3T25		3T25		3T25	3T25
000		000		000	000

B E A M N O. 101 D E S I G N R E S U L T S - F L E X U R E

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	72.	3- 25 MM	0.	6500.	YES YES
2	517.	2- 32 MM	0.	3531.	YES NO
3	521.	3- 25 MM	3511.	6500.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1071.5/ 1590.0	224.37/ 318.37	227/ 222
542.	978.7/ 1201.4	206.54/ 248.82	227/ 222
1083.	876.1/ 855.1	186.49/ 182.34	227/ 222
1625.	783.7/ 576.3	168.10/ 125.01	219/ 230
2167.	677.7/ 340.7	146.63/ 73.91	219/ 230
2708.	542.3/ 234.0	117.65/ 28.10	219/ 230
3250.	372.6/ 0.0	80.83/ 0.00	220/ 0
3792.	276.3/ 0.0	59.93/ 0.00	222/ 0
4333.	392.4/ 234.0	85.13/ 26.74	222/ 227
4875.	473.9/ 374.4	102.81/ 81.22	222/ 227
5417.	548.8/ 662.7	119.05/ 143.57	230/ 219
5958.	625.7/ 1016.5	135.72/ 213.84	230/ 219
6500.	695.1/ 1412.0	150.18/ 287.18	230/ 219

B E A M N O. 101 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	5416 mm	130.3 kN	222	10 mm	15	386 mm
1083	END 2	137.3 kN	219	10 mm	15	386 mm

118J 6500.X 300.X 600 119J

2No32	H	517.	0.TO	3531	1	1	3No25	H	521.3511	TO	6500	1	1
15*10c	/c386	1	1	1	1	1	15*10c	/c386	1	1	1	1	1
3No25	H	72.	0.TO	6500	1	1	1	1	1	1	1	1	1

OO	1	1	OO	1	OO	1	OO	1	OO	1	OO	1	OO
2T32	1	2T32	1	2T32	1	3T25	1	3T25	1	3T25	1	3T25	1
3T25	1	3T25	1	3T25	1	3T25	1	3T25	1	3T25	1	3T25	1
ooo	1	ooo	1	ooo	1	ooo	1	ooo	1	ooo	1	ooo	1

B E A M N O. 102 D E S I G N R E S U L T S - F L E X U R E

LEN - 6000. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
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1	68.	3- 16 MM	0.	6000.	YES	YES
2	521.	3- 25 MM	0.	3323.	YES	NO
3	521.	3- 25 MM	2677.	6000.	NO	YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
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0.	562.8/ 1142.3	122.08/ 237.78	228/ 221
500.	517.0/ 860.6	112.15/ 183.42	228/ 221

1000.	462.6/ 100.35/ 131.64	228/ 221
1500.	404.4/ 390.7 87.74/ 84.74	220/ 229
2000.	342.3/ 234.0 74.26/ 45.42	220/ 229
2500.	256.2/ 234.0 55.58/ 9.79	220/ 229
3000.	234.0/ 0.0 29.89/ 0.00	226/ 0
3500.	251.3/ 234.0 54.51/ 10.33	222/ 227
4000.	331.8/ 234.0 71.98/ 46.37	222/ 227
4500.	392.7/ 401.2 85.18/ 87.03	230/ 219
5000.	456.2/ 630.3 98.97/ 136.73	230/ 219
5500.	508.9/ 892.6 110.39/ 189.74	230/ 219
6000.	552.9/ 1182.7 119.95/ 245.33	230/ 219

B E A M N O. 102 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	3999 mm	110.4 kN	222	10 mm	12	363 mm
1999	END 2	112.9 kN	219	10 mm	12	363 mm

119J 6000.X 300.X 600 120J

3No25	H	521.	0.TO	3323	1	1	3No25	H	521.2677	TO	6000	1	1
12*10c	/c363	1	1	1	1	1	12*10c	/c363	1	1	1	1	1
3No16	H	68.	0.TO	6000	1	1	1	1	1	1	1	1	1

ooo	1	ooo	1	ooo	1	ooo	1	ooo	1	ooo	1	ooo	1
3T25	1	3T25	1	3T25	1	3T25	1	3T25	1	3T25	1	3T25	1
3T16	1	3T16	1	3T16	1	3T16	1	3T16	1	3T16	1	3T16	1
ooo	1	ooo	1	ooo	1	ooo	1	ooo	1	ooo	1	ooo	1

B E A M N O. 103 D E S I G N R E S U L T S - F L E X U R E

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
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1	72.	3- 25 MM	0.	6500.	YES	YES
2	521.	3- 25 MM	0.	2989.	YES	NO
3	517.	2- 32 MM	2969.	6500.	NO	YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	705.0/ 1392.7	152.19/ 283.73	228/ 221
542.	634.1/ 999.6	137.56/ 210.59	228/ 221
1083.	556.5/ 647.9	120.71/ 140.52	228/ 221
1625.	488.2/ 368.7	105.90/ 79.98	220/ 229
2167.	405.7/ 234.0	88.02/ 25.68	220/ 229
2708.	288.7/ 0.0	62.63/ 0.00	220/ 0
3250.	383.0/ 0.0	83.08/ 0.00	222/ 0
3792.	551.8/ 234.0	119.69/ 27.32	221/ 228
4333.	686.7/ 337.9	148.47/ 73.30	221/ 228
4875.	791.9/ 574.3	169.75/ 124.58	221/ 228
5417.	876.2/ 846.6	186.50/ 180.64	229/ 220
5958.	977.8/ 1193.4	206.37/ 247.33	229/ 220
6500.	1069.7/ 1582.5	224.03/ 317.08	229/ 220

B E A M N O. 103 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	5416 mm	136.9 kN	221	10 mm	15	386 mm
1083	END 2	130.6 kN	220	10 mm	15	386 mm
120J	6500.X 300.X 600					121J
3No25/H 521.	0.TO 2989		2No32 H/517.2969	TO 6500		
15*10c/c386					15*10c/c386	
3No25/H 72.	0.TO 6500					

ooo	ooo	ooo	ooo	oo	oo	oo
3T25	3T25	3T25	2T32	2T32	2T32	2T32
3T25	3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 104 D E S I G N R E S U L T S - F L E X U R E

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	72.	3- 25 MM	0.	6500.	YES YES
2	513.	2- 40 MM	0.	3531.	YES NO
3	513.	2- 40 MM	3511.	6500.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
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0.	1310.4/ 2283.1	268.86/ 429.17	228/ 221
542.	1236.5/ 1641.2	255.33/ 327.13	228/ 221
1083.	1147.1/ 1098.3	238.67/ 229.47	228/ 221
1625.	1078.1/ 683.1	225.63/ 147.73	220/ 229
2167.	984.0/ 353.4	207.57/ 76.67	220/ 229
2708.	825.9/ 234.0	176.54/ 14.68	220/ 229
3250.	602.4/ 0.0	130.67/ 0.00	219/ 0
3792.	469.4/ 0.0	101.82/ 0.00	221/ 0
4333.	576.0/ 234.0	124.96/ 23.44	221/ 228
4875.	622.8/ 466.9	135.11/ 101.29	221/ 228
5417.	668.2/ 915.7	144.69/ 194.27	229/ 220
5958.	716.6/ 1483.4	154.56/ 299.83	229/ 220
6500.	749.9/ 2153.6	161.30/ 409.77	229/ 220

B E A M N O. 104 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1083 mm	190.6 kN	221	10 mm	4	361 mm
1083	END 2	205.2 kN	220	10 mm	15	386 mm

122J									
6500.X 300.X 600									
123J									
2No40/H 513. 0.TO 3531 2No40/H 513.3511.TO 6500									
4*10c/c361 15*10c/c386									
3No25/H 72. 0.TO 6500									

OO	OO	OO	OO	OO	OO	OO	OO	OO	OO
2T40	2T40	2T40	2T40	2T40	2T40	2T40	2T40	2T40	2T40
3T25	3T25	3T25	3T25	3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo

=====

B E A M N O. 105 D E S I G N R E S U L T S - F L E X U R E

LEN - 6000. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	70.	3- 20 MM	0.	6000.	YES YES
2	513.	2- 40 MM	0.	3323.	YES NO
3	513.	2- 40 MM	2677.	6000.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)

0.	615.9/ 1719.9	133.60/ 340.44	228/ 221
500.	594.7/ 1252.4	129.01/ 258.25	227/ 222
1000.	561.6/ 841.7	121.84/ 179.68	227/ 222
1500.	521.6/ 502.7	113.15/ 109.05	219/ 230
2000.	484.1/ 241.8	105.02/ 52.44	219/ 230
2500.	402.7/ 234.0	87.36/ 2.50	219/ 230
3000.	263.2/ 0.0	57.09/ 2.03	223/ 0
3500.	404.9/ 234.0	87.83/ 2.03	221/ 228
4000.	488.4/ 237.4	105.95/ 51.51	221/ 228
4500.	528.1/ 496.2	114.56/ 107.65	221/ 228
5000.	570.5/ 832.5	123.75/ 177.84	229/ 220
5500.	605.7/ 1239.9	131.41/ 255.95	229/ 220
6000.	629.1/ 1703.5	136.48/ 337.68	229/ 220

B E A M N O. 105 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C

END 1	3999 mm	166.4 kN	221	10 mm	12	363 mm
1999	END 2	165.4 kN	220	10 mm	12	363 mm

123J	6000.X 300.X 600	124J

2No40/H 513. 0.TO 3323 2No40 H 513.2677.TO 6000		
12*10c/c363 12*10c/c363		
3No20/H 70. 0.TO 6000		

OO	OO	OO	OO	OO	OO	OO
2T40	2T40	2T40	2T40	2T40	2T40	2T40
3T20	3T20	3T20	3T20	3T20	3T20	3T20
ooo	ooo	ooo	ooo	ooo	ooo	ooo

=====

B E A M N O. 106 D E S I G N R E S U L T S - F L E X U R E

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	72.	3- 25 MM	0.	6500.	YES YES
2	513.	2- 40 MM	0.	2989.	YES NO
3	513.	2- 40 MM	2969.	6500.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)

0.	756.2/ 2142.9	162.56/ 408.13	227/ 222
542.	721.3/ 1475.8	155.51/ 298.48	227/ 222

1083.	671.3/	910.3	145.33/	193.21	227/	222
1625.	626.4/	465.5	135.89/	100.98	219/	230
2167.	577.9/	234.0	125.36/	23.36	219/	230
2708.	469.5/	0.0	101.86/	0.00	219/	0
3250.	601.8/	0.0	130.54/	0.00	221/	0
3792.	823.8/	234.0	176.12/	15.60	222/	227
4333.	980.3/	359.1	206.86/	77.90	222/	227
4875.	1072.8/	690.6	224.62/	149.27	222/	227
5417.	1140.0/	1108.0	237.34/	231.30	230/	219
5958.	1228.1/	1654.1	253.77/	329.33	230/	219
6500.	1300.6/	2300.5	267.07/	431.74	230/	219

B E A M N O. 106 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	5416 mm	204.7 kN	222	10 mm	15	386 mm
5416	END 2	191.3 kN	219	10 mm	4	361 mm

124J	6500.X 300.X 600	125J
2No40 H 513.	0.TO 2989	2No40 H 513.2969.TO 6500
15*10c/c361		4*10c/c361
3No25 H 72.	0.TO 6500	

OO	OO	OO	OO	OO	OO
2T40	2T40	2T40	2T40	2T40	2T40
3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 107 D E S I G N R E S U L T S - F L E X U R E

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT	BAR INFO	FROM	TO	ANCHOR
	mm		mm	mm	STA END

1	72.	3- 25 MM	0.	6500.	YES	YES
2	513.	2- 40 MM	0.	3531.	YES	NO
3	513.	2- 40 MM	3511.	6500.	NO	YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1273.0/ 2277.6	262.03/ 428.36	227/ 222
542.	1203.8/ 1639.0	249.26/ 326.76	227/ 222
1083.	1119.2/ 1098.3	233.42/ 229.46	227/ 222
1625.	1053.7/ 684.5	220.98/ 148.03	219/ 230
2167.	962.7/ 357.4	203.44/ 77.53	219/ 230
2708.	810.1/ 234.0	173.39/ 15.78	219/ 230
3250.	592.3/ 0.0	128.50/ 0.00	220/ 0
3792.	463.3/ 0.0	100.50/ 0.00	222/ 0
4333.	570.7/ 234.0	123.79/ 23.09	222/ 227
4875.	620.3/ 458.4	134.57/ 99.44	222/ 227
5417.	664.4/ 895.9	143.91/ 190.38	230/ 219
5958.	713.4/ 1452.9	153.90/ 294.44	230/ 219
6500.	747.6/ 2108.1	160.83/ 402.80	230/ 219

B E A M N O. 107 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1083 mm	189.8 kN	222	10 mm	4	361 mm
1083	END 2	202.3 kN	219	10 mm	15	386 mm

126J	6500.X 300.X 600	127J
2No40 H 513.	0.TO 3531	2No40 H 513.3511.TO 6500
4*10c/c361		15*10c/c386
3No25 H 72.	0.TO 6500	

OO	OO	OO	OO	OO	OO
2T40	2T40	2T40	2T40	2T40	2T40
3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo

=====

B E A M N O. 108 D E S I G N R E S U L T S - F L E X U R E

LEN - 6000. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	70.	2- 20 MM	0.	6000.	YES YES
2	513.	2- 40 MM	0.	3323.	YES NO
3	513.	2- 40 MM	2677.	6000.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)

0.	616.8/ 1682.2	133.81/ 334.10	228/ 221
500.	594.4/ 1226.0	128.94/ 253.38	228/ 221
1000.	560.1/ 824.5	121.50/ 176.25	228/ 221
1500.	517.1/ 493.1	112.18/ 106.97	220/ 229
2000.	478.3/ 237.0	103.75/ 51.41	220/ 229
2500.	392.8/ 234.0	85.21/ 2.92	220/ 229
3000.	252.4/ 0.0	54.75/ 0.00	225/ 0
3500.	387.9/ 234.0	84.15/ 3.60	222/ 227
4000.	468.5/ 243.3	101.64/ 52.77	222/ 227
4500.	502.6/ 502.6	109.02/ 109.03	222/ 227
5000.	547.5/ 845.7	118.77/ 180.46	230/ 219
5500.	578.6/ 1254.6	125.52/ 258.65	230/ 219
6000.	597.9/ 1719.8	129.71/ 340.42	230/ 219

B E A M N O. 108 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING	C/C

END 1	3999 mm	163.4 kN	222	10 mm	12	363	mm
1999	END 2	165.5 kN	219	10 mm	12	363	mm

127J 6000.X 300.X 600 128J

2No40 H 513.	0.TO 3323		2No40 H 513.2677.TO 6000		
12*10c/c363					12*10c/c363
2No20 H 70.	0.TO 6000				
=====					
OO		OO		OO	
2T40		2T40		2T40	
2T20		2T20		2T20	
oo		oo		oo	
=====					

B E A M N O. 109 D E S I G N R E S U L T S - F L E X U R E

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	72.	3- 25 MM	0.	6500.	YES YES
2	513.	2- 40 MM	0.	2989.	YES NO
3	513.	2- 40 MM	2969.	6500.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)

0.	754.9/ 2095.5	162.30/ 400.86	228/ 221
542.	720.9/ 1441.1	155.42/ 292.35	228/ 221
1083.	672.0/ 884.4	145.47/ 188.13	228/ 221
1625.	631.7/ 451.3	137.04/ 97.89	220/ 229
2167.	582.4/ 234.0	126.33/ 21.41	220/ 229
2708.	475.3/ 0.0	103.11/ 0.00	220/ 0
3250.	605.6/ 0.0	131.37/ 0.00	222/ 0
3792.	825.3/ 234.0	176.41/ 13.97	221/ 228
4333.	979.1/ 348.8	206.63/ 75.68	221/ 228
4875.	1071.2/ 675.3	224.32/ 146.13	221/ 228
5417.	1132.3/ 1082.6	235.88/ 226.48	229/ 220
5958.	1217.8/ 1621.1	251.85/ 323.71	229/ 220
6500.	1287.9/ 2256.5	264.75/ 425.23	229/ 220

B E A M N O. 109 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS									
FROM	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING	C/C		
END 1	5416 mm	202.6 kN	221	10 mm	15	386	mm		
5416	END 2	189.7 kN	220	10 mm	4	361	mm		
128J 6500.X 300.X 600 129J									
2No40 H	513.	0.TO 2989		2No40 H	513.2969	TO 6500			
15*10c/c386								4*10c/c361	
3No25 H	72.	0.TO 6500							
OO		OO		OO		OO		OO	
2T40		2T40		2T40		2T40		2T40	
3T25		3T25		3T25		3T25		3T25	
ooo		ooo		ooo		ooo		ooo	

B E A M N O. 110 D E S I G N R E S U L T S - F L E X U R E

LEN - 6500. mm					FY - 460. FC - 40. SIZE - 300. X 600. mm				
LEVEL	HEIGHT	BAR INFO	FROM	TO	ANCHOR				
	mm		mm	mm	STA	END			
1	72.	3- 25 MM	0.	6500.	YES	YES			
2	517.	2- 32 MM	0.	3531.	YES	NO			
3	521.	3- 25 MM	3511.	6500.	NO	YES			

REQUIRED REINF. STEEL SUMMARY :

SECTION	REINF STEEL (+VE/-VE)	MOMENTS (+VE/-VE)	LOAD (+VE/-VE)
(MM)	(SQ. MM)	(KN-METER)	
0.	1035.5/	1597.0	217.49/ 319.57
542.	946.5/	1209.7	200.29/ 250.36
			228/ 221
			228/ 221

1083.	848.0/	864.1	180.93/	184.12	228/	221
1625.	753.5/	580.9	162.02/	126.02	220/	229
2167.	650.2/	347.8	141.00/	75.44	220/	229
2708.	520.6/	234.0	112.94/	29.83	220/	229
3250.	355.1/	0.0	77.04/	0.00	219/	0
3792.	264.4/	0.0	57.35/	0.00	221/	0
4333.	381.7/	234.0	82.81/	27.51	221/	228
4875.	466.7/	371.6	101.23/	80.62	221/	228
5417.	543.1/	652.8	117.82/	141.54	229/	220
5958.	620.7/	998.5	134.65/	210.37	229/	220
6500.	690.9/	1384.1	149.33/	282.19	229/	220

B E A M N O. 110 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS									
FROM	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING	C/C		
END 1	5416 mm	129.7 kN	221	10 mm	15	386	mm		
1083	END 2	134.5 kN	220	10 mm	15	386	mm		

130J 6500.X 300.X 600 131J									
2No32 H	517.	0.TO 3531		3No25 H	521.3511	TO 6500			
15*10c/c386								15*10c/c386	
3No25 H	72.	0.TO 6500							

OO		OO		OO		ooo		ooo	
2T32		2T32		2T32		3T25		3T25	
3T25		3T25		3T25		3T25		3T25	
ooo		ooo		ooo		ooo		ooo	

B E A M N O. 111 D E S I G N R E S U L T S - F L E X U R E

LEN - 6000. mm					FY - 460. FC - 40. SIZE - 300. X 600. mm				
LEVEL	HEIGHT	BAR INFO	FROM	TO	ANCHOR				
	mm		mm	mm	STA	END			

1	68.	3- 16 MM	0.	6000.	YES	YES
2	521.	3- 25 MM	0.	3323.	YES	NO
3	521.	3- 25 MM	2677.	6000.	NO	YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	564.2/ 1123.7	122.40/ 234.26	227/ 222
500.	518.3/ 845.4	112.44/ 180.40	227/ 222
1000.	463.9/ 595.1	100.64/ 129.10	227/ 222
1500.	404.6/ 381.3	87.78/ 82.71	219/ 230
2000.	344.4/ 234.0	74.71/ 43.91	219/ 230
2500.	257.4/ 234.0	55.84/ 9.21	219/ 230
3000.	234.0/ 0.0	30.59/ 0.00	224/ 0
3500.	252.5/ 234.0	54.78/ 9.99	221/ 228
4000.	334.6/ 234.0	72.58/ 45.46	221/ 228
4500.	389.9/ 392.0	84.58/ 85.03	221/ 228
5000.	449.8/ 614.9	97.58/ 133.40	229/ 220
5500.	500.7/ 872.5	108.62/ 185.78	229/ 220
6000.	543.1/ 1158.0	117.81/ 240.71	229/ 220

B E A M N O. 111 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	3999 mm	109.4 kN	221	10 mm	12	363 mm
1999	END 2	111.6 kN	220	10 mm	12	363 mm
131J		6000.X 300.X 600				132J
3No25/H 521.	0.TO 3323		3No25 H 521.2677.TO 6000			
12*10c/c363						12*10c/c363
3No16/H 68.	0.TO 6000					
ooo	ooo	ooo	ooo	ooo	ooo	ooo
3T25	3T25	3T25	3T25	3T25	3T25	3T25
3T16	3T16	3T16	3T16	3T16	3T16	3T16
ooo	ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 112 D E S I G N R E S U L T S - F L E X U R E

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	72.	3- 25 MM	0.	6500.	YES YES
2	521.	3- 25 MM	0.	2989.	YES NO
3	517.	2- 32 MM	2969.	6500.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
-------------------	------------------------------------	---------------------------------	----------------

0.	698.2/ 1373.7	150.81/ 280.32	227/ 222
542.	627.7/ 988.4	136.16/ 208.42	227/ 222
1083.	550.2/ 643.1	119.36/ 139.50	227/ 222
1625.	476.6/ 364.6	103.39/ 79.08	219/ 230
2167.	392.0/ 234.0	85.03/ 25.93	219/ 230
2708.	274.9/ 0.0	59.63/ 0.00	219/ 0
3250.	366.1/ 0.0	79.43/ 0.00	221/ 0
3792.	532.0/ 234.0	115.41/ 28.16	222/ 227
4333.	662.7/ 339.9	143.56/ 73.74	222/ 227
4875.	766.6/ 573.0	164.67/ 124.30	222/ 227
5417.	857.5/ 851.0	182.81/ 181.53	230/ 219
5958.	956.4/ 1195.4	202.22/ 247.70	230/ 219
6500.	1045.8/ 1581.2	219.48/ 316.85	230/ 219

B E A M N O. 112 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	5416 mm	134.7 kN	222	10 mm	15	386 mm
1083	END 2	129.6 kN	219	10 mm	15	386 mm

132J	6500.X 300.X 600	133J
=====		
3No25 H 521.	0.TO 2989 2No32 H 517.2969.TO 6500	
15*10c/c386		15*10c/c386
3No25 H 72.	0.TO 6500	
=====		

130J		5040.X 300.X 600		126J	
PROVIDE SHEAR LINKS AS FOLLOWS					
FROM	TO	MAX. SHEAR	LOAD	LINKS	NO. SPACING C/C
END 1	2519 mm	133.9 kN	224	10 mm	8 359 mm
2519	END 2	145.2 kN	225	10 mm	8 359 mm

B E A M N O. 113 D E S I G N R E S U L T S - F L E X U R E

LEN - 5040. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR
1	72.	3- 25 MM	0.	5040.	YES YES
2	517.	2- 32 MM	0.	5040.	YES YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1249.6/ 1480.9	257.74/ 299.38	233/ 224
420.	1081.4/ 1174.1	226.26/ 243.74	233/ 224
840.	912.2/ 892.5	193.60/ 189.71	233/ 224
1260.	754.3/ 649.9	162.18/ 140.93	225/ 232
1680.	592.3/ 436.1	128.48/ 94.60	225/ 232
2100.	420.5/ 234.1	91.22/ 50.78	225/ 232
2520.	234.0/ 234.0	49.70/ 9.98	225/ 232
2940.	234.0/ 234.0	34.15/ 2.56	224/ 233
3360.	307.2/ 234.0	66.65/ 50.46	224/ 233
3780.	440.6/ 465.0	95.59/ 100.88	224/ 233
4200.	576.0/ 726.0	124.96/ 156.46	232/ 225
4620.	711.6/ 1023.8	153.53/ 215.25	232/ 225
5040.	848.0/ 1347.9	180.93/ 275.66	232/ 225

B E A M N O. 113 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM	TO	MAX. SHEAR	LOAD	LINKS	NO. SPACING C/C
END 1	2519 mm	133.9 kN	224	10 mm	8 359 mm
2519	END 2	145.2 kN	225	10 mm	8 359 mm

130J	5040.X 300.X 600	126J
=====		
2No32 H 517.	0.TO 5040	
8*10c/c359		
3No25 H 72.	0.TO 5040	
=====		
8*10c/c359		

OO	OO	OO	OO	OO	OO
2T32	2T32	2T32	2T32	2T32	2T32
3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 114 D E S I G N R E S U L T S - F L E X U R E

LEN - 6650. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR
1	66.	4- 12 MM	0.	6650.	YES YES
2	521.	3- 25 MM	0.	3039.	YES NO
3	521.	3- 25 MM	3611.	6650.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	397.6/ 1054.0	86.26/ 221.04	233/ 224
554.	402.7/ 754.9	87.35/ 162.30	233/ 224

1108.	397.0/	492.2	86.12/	106.77	233/	224
1662.	392.5/	274.4	85.14/	59.52	225/	232
2217.	381.8/	234.0	82.83/	20.74	225/	232
2771.	334.3/	0.0	72.52/	0.00	226/	0
3325.	253.6/	0.0	55.01/	0.00	214/	0
3879.	337.2/	0.0	73.15/	0.00	224/	0
4433.	382.4/	234.0	82.96/	22.59	224/	233
4988.	390.6/	284.5	84.73/	61.71	224/	233
5542.	396.3/	507.4	85.98/	110.08	232/	225
6096.	400.5/	774.0	86.88/	166.14	232/	225
6650.	393.9/	1077.0	85.45/	225.42	232/	225

B E A M N O. 114 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	3879 mm	107.9 kN	224	10 mm	11	387 mm
2770	END 2	108.9 kN	225	10 mm	11	387 mm

126J	6650. X 300. X 600	122J
3No25 H 521.	0.TO 3039	3No25 H 521.3611.TO 6650
11*10c/c387		11*10c/c387
4No12 H 66.	0.TO 6650	

ooo	ooo	ooo	ooo	ooo	ooo	ooo
3T25	3T25	3T25	3T25	3T25	3T25	3T25
4T12	4T12	4T12	4T12	4T12	4T12	4T12
oooo	oooo	oooo	oooo	oooo	oooo	oooo

B E A M N O. 115 D E S I G N R E S U L T S - F L E X U R E

LEN - 5310. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT	BAR INFO	FROM	TO	ANCHOR
	mm		mm	mm	STA END

1	72.	3- 25 MM	0.	5310.	YES	YES
2	521.	3- 25 MM	0.	2593.	YES	NO
3	521.	3- 25 MM	1832.	5310.	NO	YES

REQUIRED REINF. STEEL SUMMARY :

SECTION	REINF STEEL (+VE/-VE)	MOMENTS (+VE/-VE)	LOAD (+VE/-VE)
(MM)	(SQ. MM)	(KN-METER)	
0.	805.3/	1360.8	172.43/ 278.00
442.	683.5/	1024.8	147.81/ 215.43
885.	561.8/	717.3	121.86/ 154.71
1328.	439.4/	451.6	95.33/ 97.96
1770.	318.8/	234.0	69.15/ 47.01
2212.	234.0/	0.0	38.90/ 0.00
2655.	253.1/	234.0	54.91/ 5.62
3098.	441.4/	234.0	95.75/ 45.65
3540.	606.9/	410.9	131.66/ 89.15
3982.	760.7/	624.8	163.47/ 135.55
4425.	904.6/	864.6	192.10/ 184.21
4868.	1058.0/	1146.2	221.80/ 238.50
5310.	1213.0/	1458.5	250.97/ 295.43

B E A M N O. 115 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2654 mm	142.9 kN	224	10 mm	8	379 mm
2654	END 2	130.2 kN	225	10 mm	8	379 mm

122J	5310. X 300. X 600	118J
3No25 H 521.	0.TO 2593 3No25 H 521.1832.TO 5310	
8*10c/c379		8*10c/c379
3No25 H 72.	0.TO 5310	

ooo	ooo	ooo	ooo	ooo	ooo	ooo
3T25	3T25	3T25	3T25	3T25	3T25	3T25
3T25	3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo	ooo

=====

B E A M N O. 116 D E S I G N R E S U L T S - F L E X U R E

LEN - 5040. mm FY - 460. mm FC - 40. mm SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	80.	2- 40 MM	0.	5040.	YES YES
2	513.	2- 40 MM	0.	3343.	YES NO
3	513.	2- 40 MM	2537.	5040.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)

0.	1651.2/	1995.2	328.84/	385.21	233/ 224
420.	1434.4/	1541.0	291.16/	309.89	233/ 224
840.	1219.3/	1138.6	252.14/	237.08	225/ 232
1260.	1026.9/	809.3	215.85/	173.22	225/ 232
1680.	818.3/	518.8	175.02/	112.54	225/ 232
2100.	590.4/	258.8	128.09/	56.15	225/ 232
2520.	339.4/	234.0	73.63/	5.01	225/ 232
2940.	236.0/	0.0	51.20/	0.00	223/ 0
3360.	402.3/	296.0	87.26/	64.20	223/ 234
3780.	541.5/	617.5	117.46/	133.97	232/ 225
4200.	696.6/	1018.7	150.49/	214.27	232/ 225
4620.	848.0/	1474.3	180.93/	298.22	232/ 225
5040.	995.4/	1990.1	209.77/	384.41	232/ 225

B E A M N O. 116 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C

END 1	2519 mm	181.0 kN	224	10 mm	8	359 mm
2520	4200 mm	194.5 kN	224	10 mm	3	559 mm
4200	END 2	206.8 kN	225	10 mm	2	839 mm

131J 5040.X 300.X 600 127J

2No40 H 513.	0.TO 3343	12No40/H 513.2537.TO 5040	2*10c/c839
8*10c/c359			
2No40 H 80.	0.TO 5040		

OO	OO	OO	OO	OO	OO
2T40	2T40	2T40	2T40	2T40	2T40
2T40	2T40	2T40	2T40	2T40	2T40
OO	OO	OO	OO	OO	OO

=====

B E A M N O. 117 D E S I G N R E S U L T S - F L E X U R E

LEN - 6650. mm FY - 460. mm FC - 40. mm SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	68.	3- 16 MM	0.	6650.	YES YES
2	517.	2- 32 MM	0.	3039.	YES NO
3	513.	2- 40 MM	3611.	6650.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)

0.	401.6/	1604.2	87.12/	320.81	233/ 224
554.	456.0/	1099.7	98.92/	229.72	233/ 224
1108.	494.7/	661.9	107.32/	143.40	233/ 224
1662.	540.6/	327.9	117.26/	71.13	225/ 232
2217.	575.5/	234.0	124.84/	14.14	225/ 232
2771.	548.7/	0.0	119.02/	0.00	225/ 0
3325.	453.0/	0.0	98.27/	0.00	223/ 0
3879.	553.6/	0.0	120.10/	0.00	224/ 0
4433.	578.8/	234.0	125.56/	16.57	224/ 233
4988.	542.2/	341.4	117.62/	74.05	224/ 233
5542.	492.2/	675.9	106.78/	146.27	232/ 225
6096.	451.2/	1116.7	97.88/	232.95	232/ 225
6650.	394.5/	1625.2	85.59/	324.40	232/ 225

B E A M N O. 117 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	3879 mm	166.6 kN	224	10 mm	11	387 mm
2770	END 2	167.3 kN	225	10 mm	11	387 mm

127J	6650.X 300.X 600	123J
2No32 H 517.	0.TO 3039	2No40 H 513.3611.TO 6650
11*10c/c387		11*10c/c387
3No16 H 68.	0.TO 6650	

OO	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
2T32	2T32	2T32	2T40	2T40	2T40	2T40
3T16	3T16	3T16	3T16	3T16	3T16	3T16
ooo	ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 118 D E S I G N R E S U L T S - F L E X U R E

LEN - 5310. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
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1	76.	2- 32 MM	0.	5310.	YES	YES
2	513.	2- 40 MM	0.	2593.	YES	NO
3	513.	2- 40 MM	2275.	5310.	NO	YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
----------------	---------------------------------	------------------------------	----------------

0.	933.6/ 2022.4	197.78/ 389.49	233/ 224
442.	809.2/ 1482.8	173.19/ 299.72	233/ 224

885.	678.4/ 1009.5	146.78/ 212.51	233/ 224
1328.	541.1/ 597.0	117.39/ 129.50	233/ 224
1770.	420.9/ 270.7	91.31/ 58.72	225/ 232
2212.	268.6/ 0.0	58.27/ 0.00	225/ 0
2655.	373.0/ 0.0	80.91/ 0.00	224/ 0
3098.	617.3/ 234.0	133.92/ 48.26	224/ 233
3540.	834.3/ 482.9	178.20/ 104.75	224/ 233
3982.	1024.6/ 774.3	215.39/ 166.21	224/ 233
4425.	1196.9/ 1111.5	247.97/ 231.97	232/ 225
4868.	1391.9/ 1529.8	283.59/ 307.95	232/ 225
5310.	1584.1/ 2003.2	317.36/ 386.48	232/ 225

B E A M N O. 118 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2654 mm	204.6 kN	224	10 mm	8	379 mm
2654	END 2	179.2 kN	225	10 mm	8	379 mm

123J	5310.X 300.X 600	119J
2No40 H 513.	0.TO 2593	2No40 H 513.2275.TO 5310
8*10c/c379		8*10c/c379
2No32 H 76.	0.TO 5310	

OO	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
2T40	2T40	2T40	2T40	2T40	2T40	2T40
2T32	2T32	2T32	2T32	2T32	2T32	2T32
OO	OO	OO	OO	OO	OO	OO

B E A M N O. 119 D E S I G N R E S U L T S - F L E X U R E

LEN - 5040. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
-------	-----------	----------	---------	-------	------------	-----

1	80.	2- 40 MM	0.	5040.	YES	YES
2	513.	2- 40 MM	0.	3343.	YES	NO
3	513.	2- 40 MM	2537.	5040.	NO	YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1662.6/ 2000.4	330.77/ 386.04	234/ 223
420.	1443.0/ 1546.0	292.70/ 310.77	234/ 223
840.	1226.4/ 1144.4	253.44/ 238.17	226/ 231
1260.	1030.8/ 814.3	216.60/ 174.21	226/ 231
1680.	819.4/ 522.9	175.23/ 113.44	226/ 231
2100.	588.9/ 262.5	127.75/ 56.94	226/ 231
2520.	335.4/ 234.0	72.75/ 5.72	225/ 232
2940.	234.0/ 0.0	50.00/ 0.00	223/ 0
3360.	396.5/ 301.7	86.01/ 65.44	223/ 234
3780.	539.6/ 629.2	117.05/ 136.50	231/ 226
4200.	695.1/ 1034.7	150.18/ 217.35	231/ 226
4620.	847.0/ 1494.9	180.72/ 301.85	231/ 226
5040.	994.8/ 2016.6	209.66/ 388.59	231/ 226

B E A M N O. 119 D E S I G N R E S U L T S - SHEAR

PROVIDE SHEAR LINKS AS FOLLOWS					
FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO. SPACING C/C
END 1	2519 mm	180.8 kN	223	10 mm	8 359 mm
2520	4200 mm	195.8 kN	223	10 mm	3 559 mm
4200	END 2	208.1 kN	226	10 mm	2 839 mm
132J		5040.X 300.X 600			128J
2No40 H 513.	0.TO 3343			12No40/H 513.2537.TO 5040	
8*10c/c359					2*10c/c839
2No40 H 80.	0.TO 5040				

OO		OO		OO		OO
2T40		2T40		2T40		2T40
2T40		2T40		2T40		2T40
OO		OO		OO		OO

B E A M N O. 120 D E S I G N R E S U L T S - FLEXURE

LEN - 6650. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	68.	3- 16 MM	0.	6650.	YES YES
2	513.	2- 40 MM	0.	3039.	YES NO
3	517.	2- 32 MM	3611.	6650.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
-------------------	------------------------------------	---------------------------------	----------------

0.	397.1/ 1622.2	86.14/ 323.89	233/ 224
554.	453.6/ 1110.8	98.40/ 231.83	234/ 223
1108.	494.5/ 667.5	107.26/ 144.54	234/ 223
1662.	543.8/ 330.1	117.96/ 71.60	226/ 231
2217.	581.9/ 234.0	126.24/ 13.89	226/ 231
2771.	558.4/ 0.0	121.13/ 0.00	226/ 0
3325.	465.8/ 0.0	101.04/ 0.00	224/ 0
3879.	570.9/ 0.0	123.85/ 0.00	224/ 0
4433.	600.6/ 234.0	130.28/ 13.92	224/ 233
4988.	568.5/ 327.1	123.32/ 70.96	224/ 233
5542.	513.8/ 648.9	111.46/ 140.73	232/ 225
6096.	476.2/ 1083.8	103.30/ 226.72	232/ 225
6650.	422.9/ 1584.8	91.74/ 317.48	232/ 225

B E A M N O. 120 D E S I G N R E S U L T S - SHEAR

PROVIDE SHEAR LINKS AS FOLLOWS					
FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO. SPACING C/C
END 1	3879 mm	168.4 kN	224	10 mm	11 387 mm
2770	END 2	166.0 kN	225	10 mm	11 387 mm

128J	6650.X 300.X 600		124J
=====			
2No40 H 513.	0.TO 3039	2No32 H 517.3611	TO 6650
11*10c/c387			11*10c/c387
3No16 H 68.	0.TO 6650		

OO	OO	OO	OO	OO	OO
2T40	2T40	2T32	2T32	2T32	2T32
3T16	3T16	3T16	3T16	3T16	3T16
ooo	ooo	ooo	ooo	ooo	ooo

BEAM NO. 121 DESIGN RESULTS - FLEXURE

LEN - 5310. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	76.	2- 32 MM	0.	5310.	YES YES
2	513.	2- 40 MM	0.	2593.	YES NO
3	513.	2- 40 MM	1832.	5310.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	928.6/ 2054.6	196.80/ 394.52	234/ 223
442.	801.7/ 1511.7	171.70/ 304.79	233/ 224
885.	668.6/ 1036.3	144.78/ 217.64	233/ 224
1328.	529.6/ 620.9	114.89/ 134.70	233/ 224
1770.	399.4/ 287.2	86.64/ 62.30	225/ 232
2212.	245.0/ 0.0	53.15/ 0.00	225/ 0
2655.	348.7/ 234.0	75.64/ 1.97	223/ 234
3098.	592.9/ 243.8	128.61/ 52.88	223/ 234
3540.	807.4/ 506.6	172.85/ 109.89	223/ 234
3982.	996.6/ 802.5	210.00/ 171.87	223/ 234
4425.	1175.1/ 1151.6	243.92/ 239.53	231/ 226
4868.	1368.8/ 1576.1	279.44/ 315.97	231/ 226
5310.	1559.6/ 2057.5	313.13/ 394.97	231/ 226

BEAM NO. 121 DESIGN RESULTS - SHEAR

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2654 mm	204.5 kN	223	10 mm	8	379 mm
2654	END 2	180.3 kN	226	10 mm	8	379 mm

124J	5310.X 300.X 600	120J
2No40 H 513. 0.TO 2593 2No40 H 513.1832.TO 5310		
8*10c/c379		
2No32 H 76. 0.TO 5310		

OO	OO	OO	OO	OO	OO
2T40	2T40	2T40	2T40	2T40	2T40
2T32	2T32	2T32	2T32	2T32	2T32
OO	OO	OO	OO	OO	OO

BEAM NO. 122 DESIGN RESULTS - FLEXURE

LEN - 5040. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	72.	3- 25 MM	0.	5040.	YES YES
2	517.	2- 32 MM	0.	5040.	YES YES

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1262.4/ 1481.8	260.09/ 299.55	234/ 223
420.	1091.2/ 1176.4	228.12/ 244.16	234/ 223
840.	919.3/ 896.0	194.97/ 190.40	234/ 223

REQUIRED REINF. STEEL SUMMARY :

1260.	757.5/	653.4	162.84/	141.65	226/	231
1680.	592.1/	439.7	128.44/	95.38	226/	231
2100.	417.1/	238.0	90.49/	51.62	226/	231
2520.	234.0/	234.0	48.28/	10.88	226/	231
2940.	234.0/	234.0	32.15/	3.64	223/	234
3360.	296.8/	239.8	64.39/	52.02	223/	234
3780.	431.6/	477.1	93.63/	103.50	231/	226
4200.	570.7/	746.7	123.81/	160.65	231/	226
4620.	705.6/	1049.2	152.33/	220.12	231/	226
5040.	841.6/	1378.7	179.66/	281.22	231/	226

B E A M N O. 122 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS						
FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2519 mm	133.3 kN	223	10 mm	8	359 mm
2519	END 2	146.9 kN	226	10 mm	8	359 mm

133J	5040.X 300.X 600	129J
2No32 H 517.	0.TO 5040	
8*10c/c359		8*10c/c359
3No25 H 72.	0.TO 5040	

OO	OO	OO	OO	OO	OO
2T32	2T32	2T32	2T32	2T32	2T32
3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 123 D E S I G N R E S U L T S - F L E X U R E

LEN - 6650. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
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1	66.	4- 12 MM	0.	6650.	YES	YES
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SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	397.2/ 1065.1	86.17/ 223.15	234/ 223
554.	403.6/ 760.9	87.55/ 163.51	234/ 223
1108.	399.3/ 493.6	86.61/ 107.09	234/ 223
1662.	398.3/ 274.0	86.40/ 59.44	226/ 231
2217.	390.6/ 234.0	84.74/ 20.11	226/ 231
2771.	346.1/ 0.0	75.07/ 0.00	225/ 0
3325.	266.6/ 0.0	57.83/ 0.00	223/ 0
3879.	356.3/ 0.0	77.30/ 0.00	223/ 0
4433.	405.6/ 234.0	88.00/ 20.37	223/ 234
4988.	417.9/ 272.9	90.66/ 59.20	223/ 234
5542.	414.1/ 480.9	89.84/ 104.31	231/ 226
6096.	420.7/ 742.2	91.27/ 159.73	231/ 226
6650.	416.6/ 1040.0	90.38/ 218.37	231/ 226

B E A M N O. 123 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS						
FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	3879 mm	109.6 kN	223	10 mm	11	387 mm
2770	END 2	107.7 kN	226	10 mm	11	387 mm

129J	6650.X 300.X 600	125J
3No25 H 521.	0.TO 3039	3No25 H 521.3611.TO 6650
11*10c/c387		11*10c/c387
4No12 H 66.	0.TO 6650	

ooo	ooo	ooo	ooo	ooo	ooo
3T25	3T25	3T25	3T25	3T25	3T25
4T12	4T12	4T12	4T12	4T12	4T12
oooo	oooo	oooo	oooo	oooo	oooo

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B E A M N O. 124 D E S I G N R E S U L T S - F L E X U R E

LEN - 5310. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	72.	3- 25 MM	0.	5310.	YES YES
2	517.	2- 32 MM	0.	5310.	YES YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	795.2/ 1401.7	170.40/ 285.34	234/ 223
442.	672.5/ 1062.0	145.57/ 222.57	234/ 223
885.	550.4/ 751.6	119.40/ 161.64	234/ 223
1328.	421.0/ 476.5	91.33/ 103.36	234/ 223
1770.	291.9/ 234.0	63.31/ 50.59	225/ 232
2212.	234.0/ 234.0	32.94/ 2.45	225/ 232
2655.	234.0/ 234.0	48.80/ 8.95	223/ 234
3098.	414.2/ 234.0	89.84/ 49.19	223/ 234
3540.	580.6/ 428.3	125.96/ 92.91	223/ 234
3982.	733.5/ 643.2	157.98/ 139.53	223/ 234
4425.	885.7/ 893.8	188.38/ 189.98	231/ 226
4868.	1043.3/ 1181.9	219.00/ 245.19	231/ 226
5310.	1198.5/ 1497.2	248.27/ 302.25	231/ 226

B E A M N O. 124 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C

END 1	2654 mm	143.3 kN	223	10 mm	8	379 mm
2654	END 2	130.4 kN	226	10 mm	8	379 mm

125J	5310. X 300. X 600	121J

2No32 H 517.	0.TO 5310	
8*10c/c379		8*10c/c379
3No25 H 72.	0.TO 5310	

OO	OO	OO	OO	OO	OO
2T32	2T32	2T32	2T32	2T32	2T32
3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo

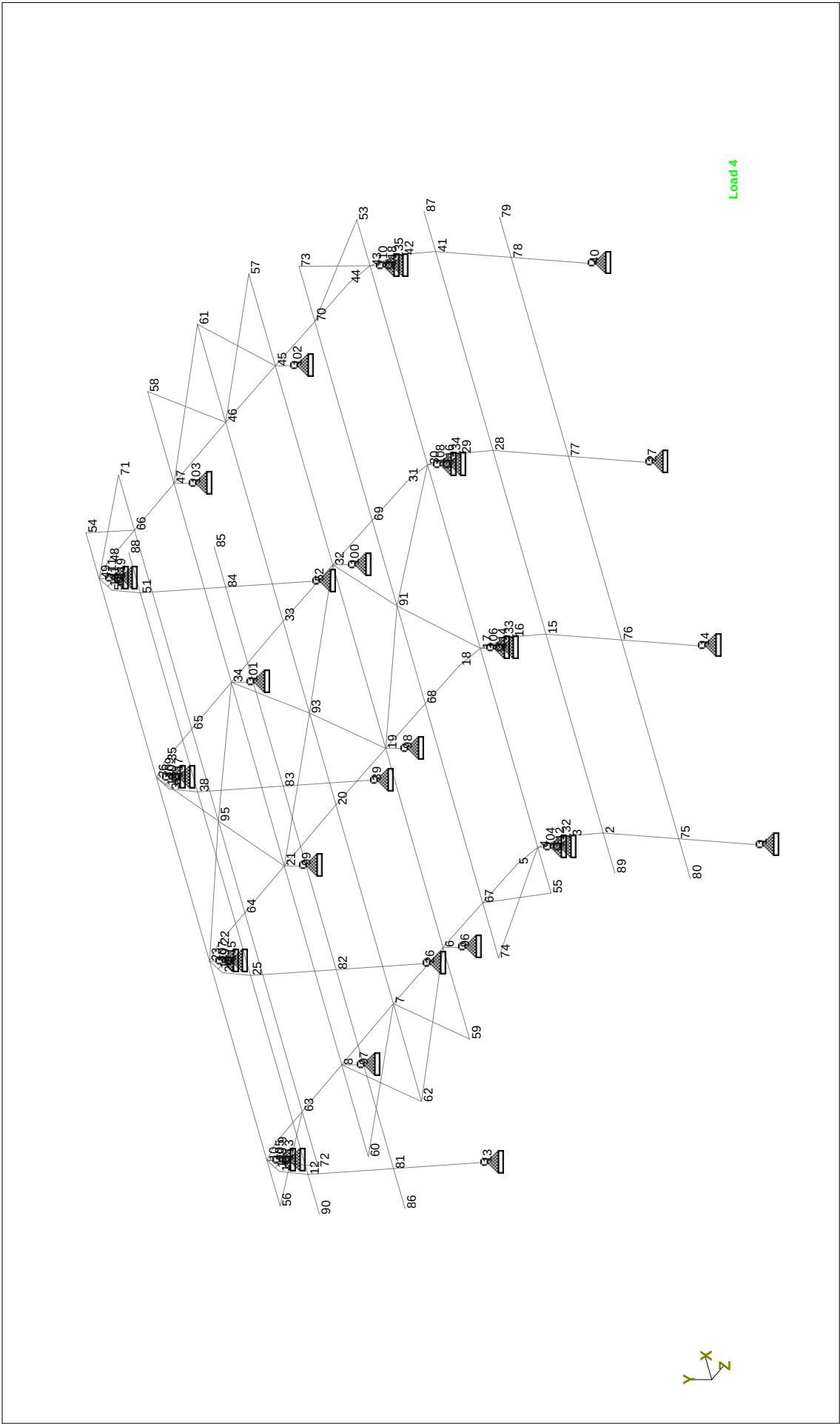
*****END OF BEAM DESIGN*****

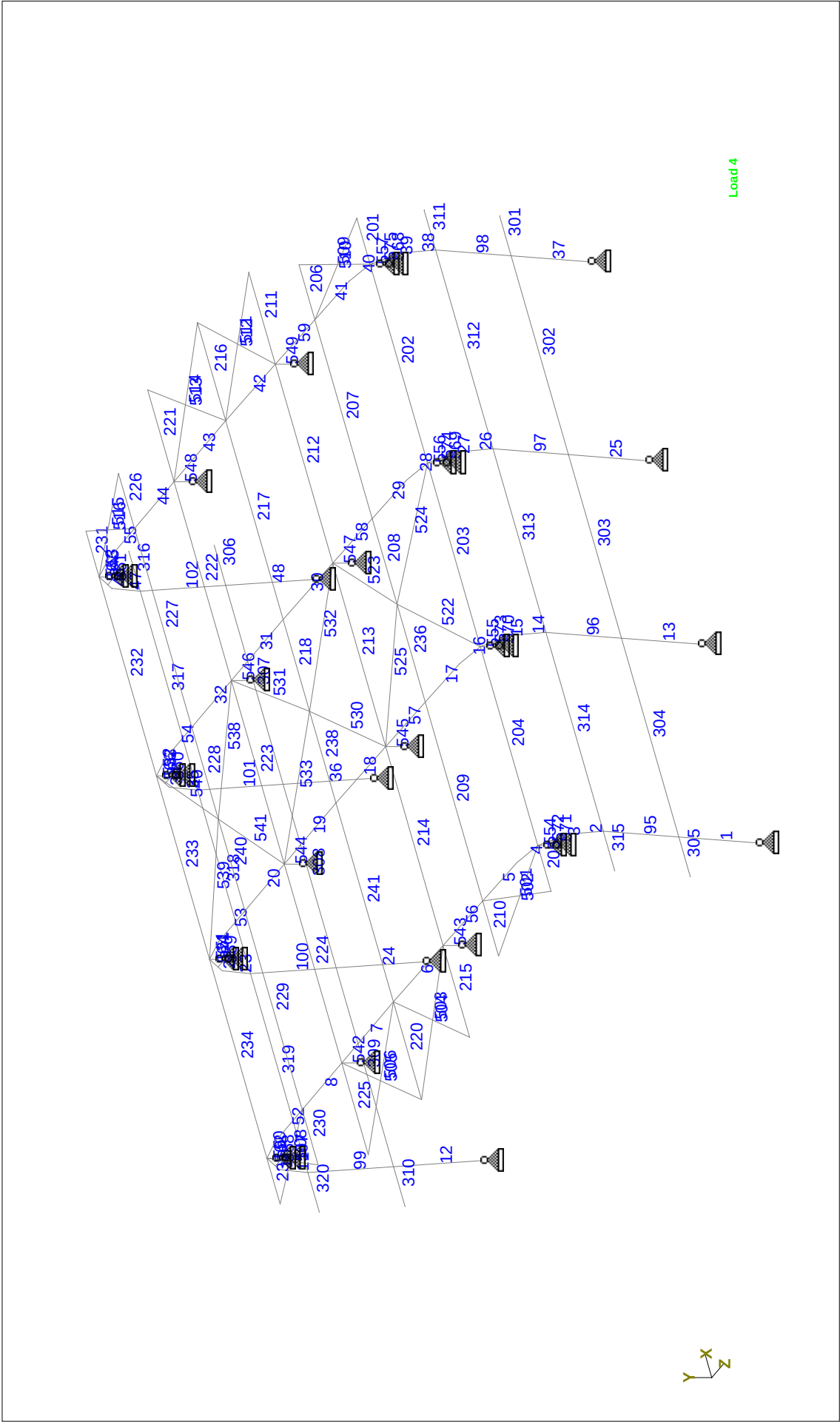
766. END CONCRETE DESIGN

767. FINISH

***** END OF THE STAAD.Pro RUN *****

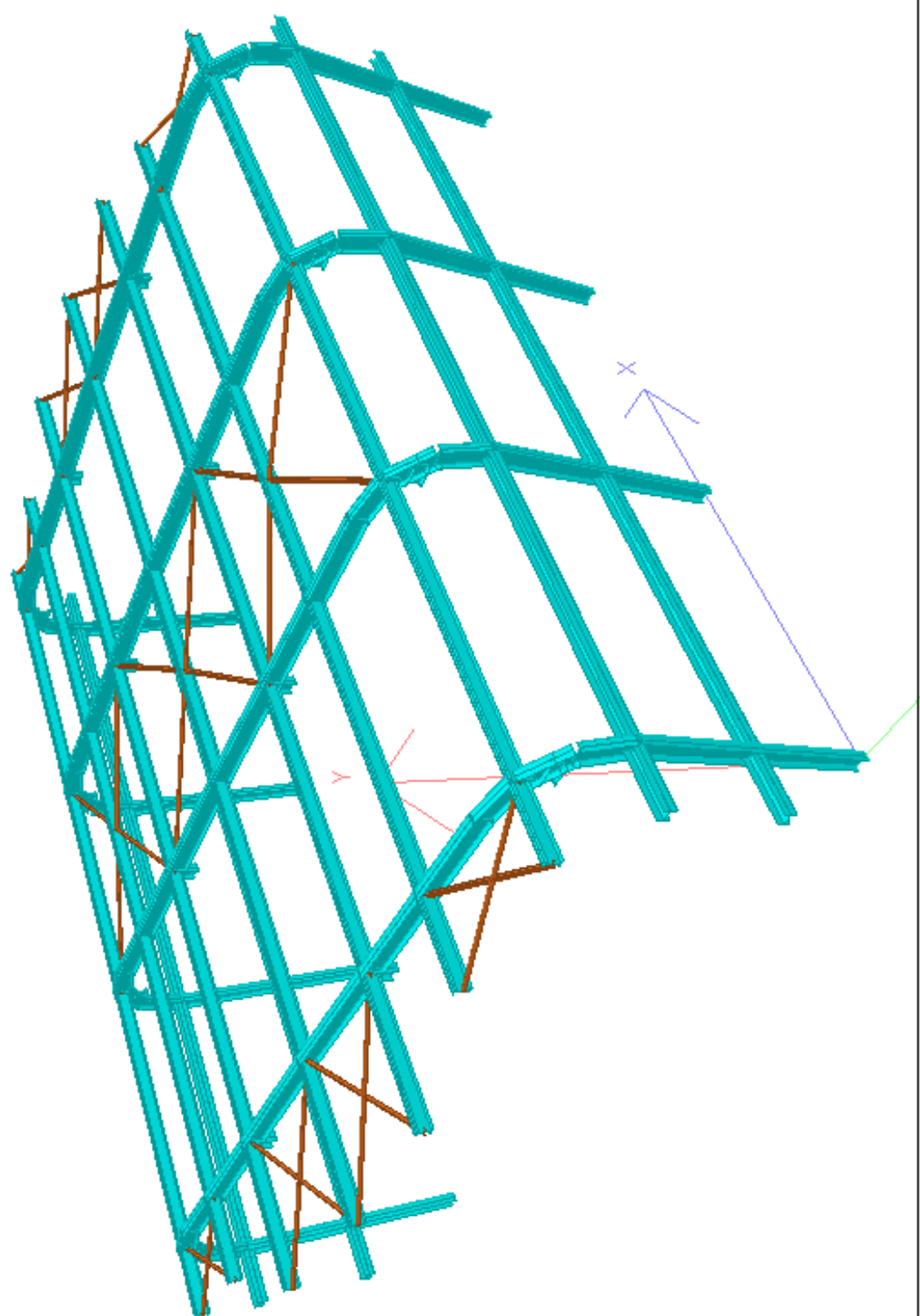
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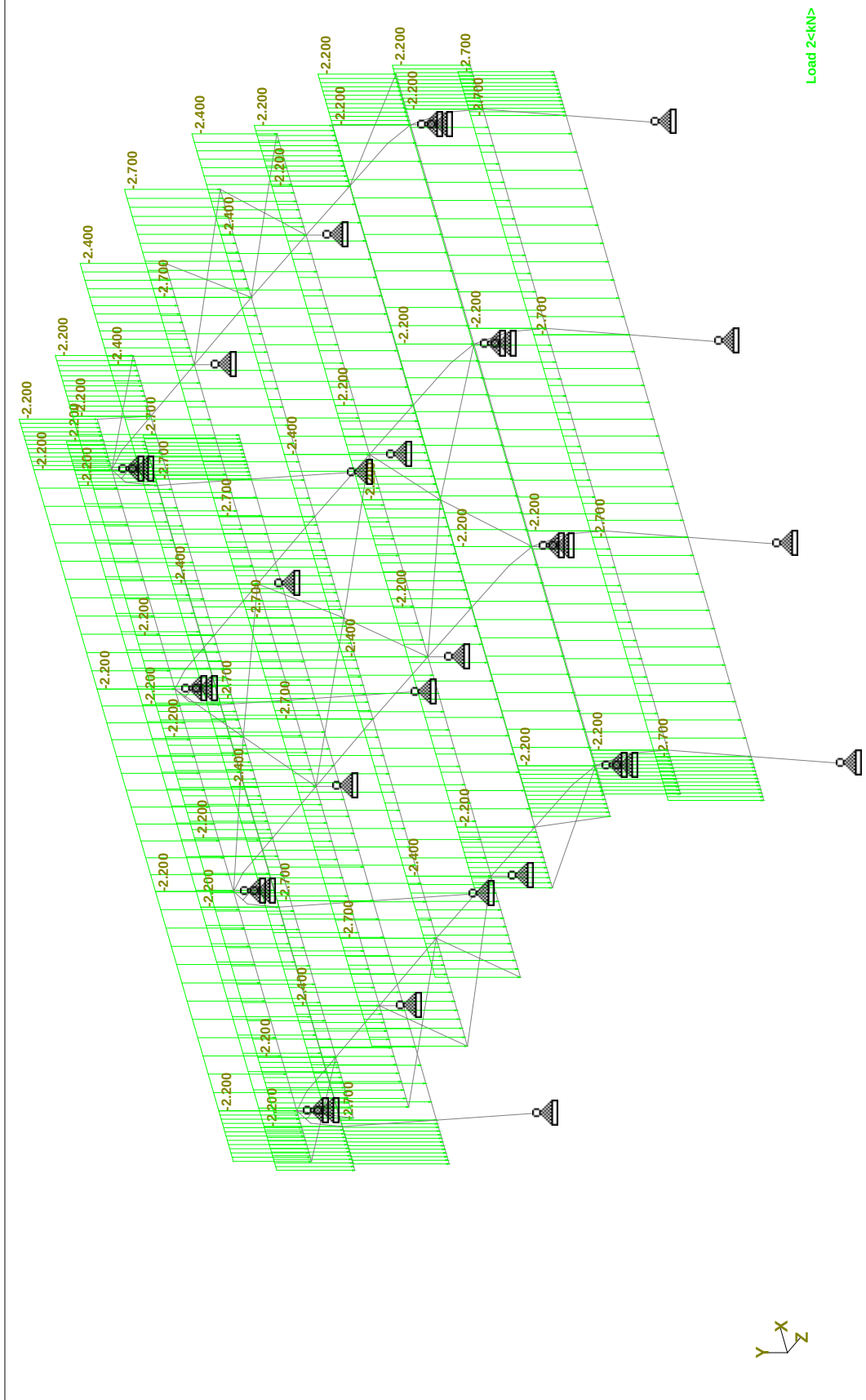


Load 4

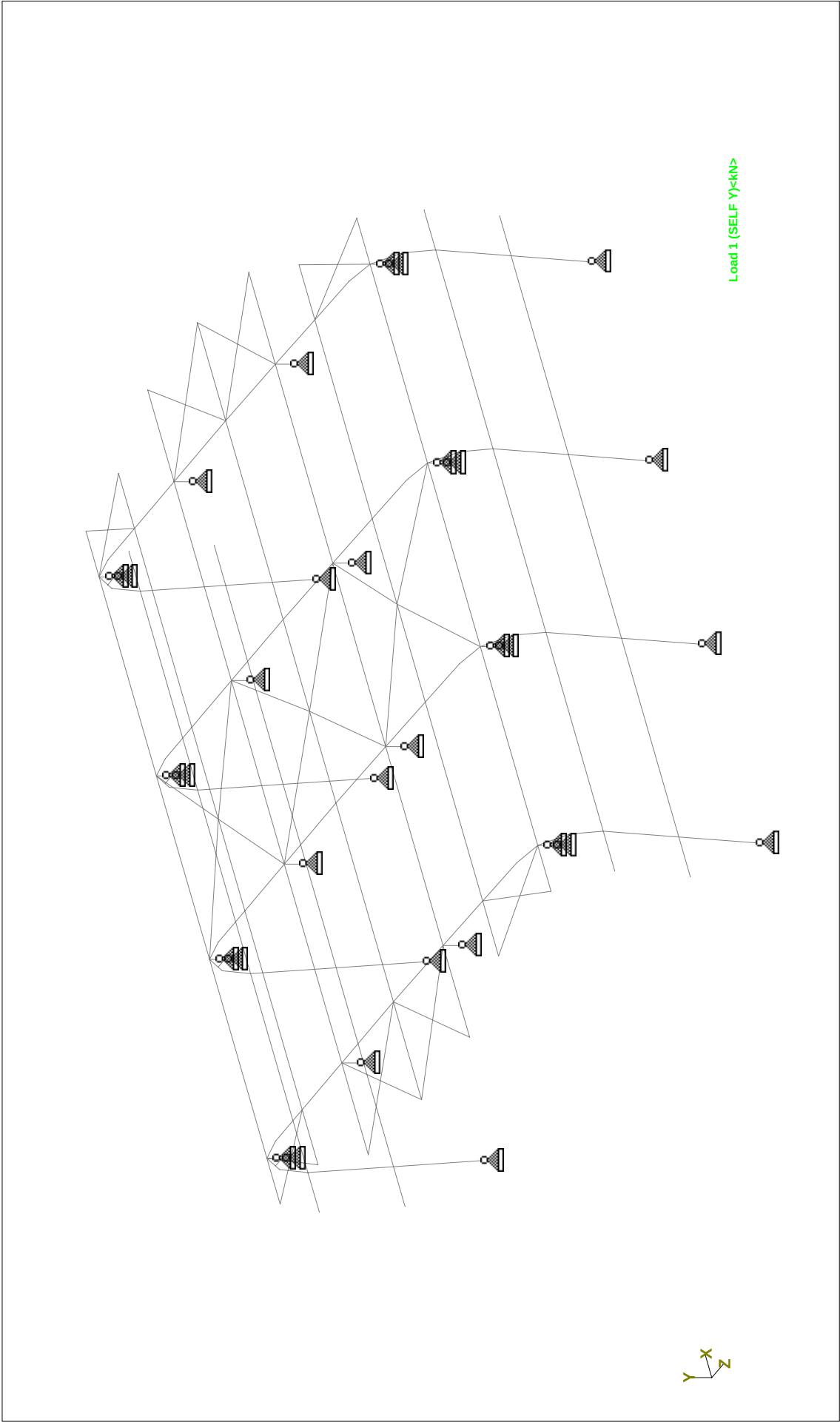
Beam No.



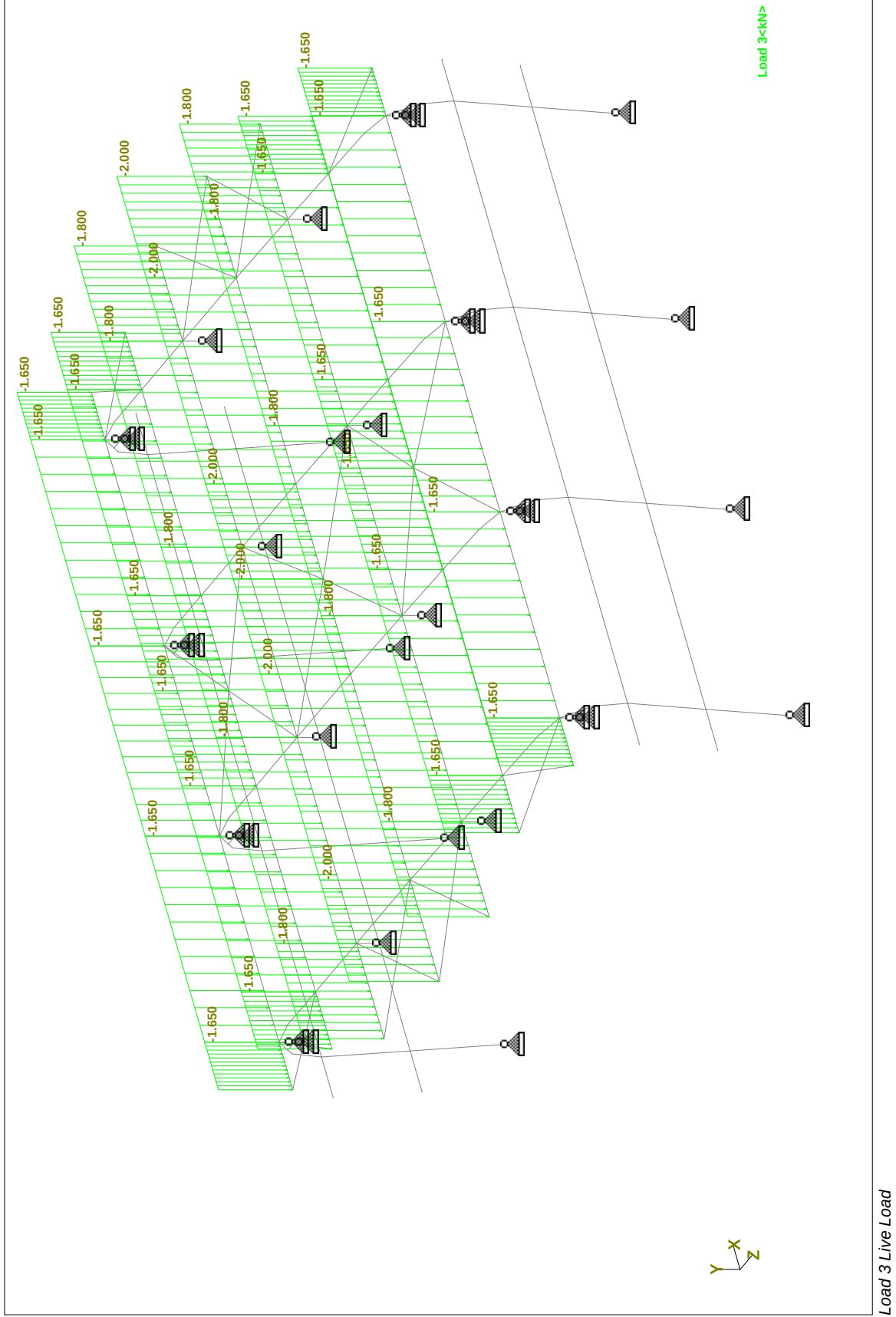
3D Rendered View

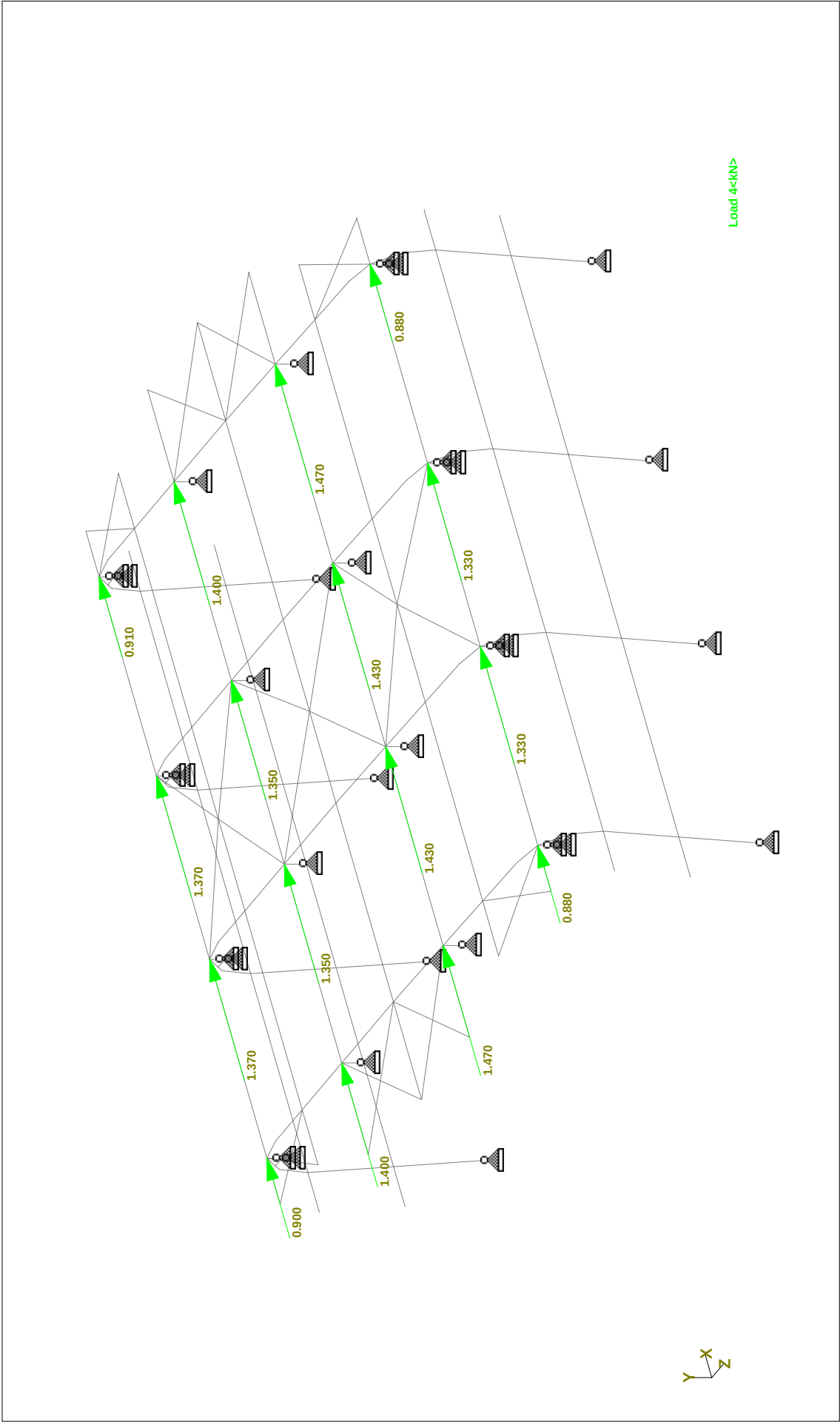


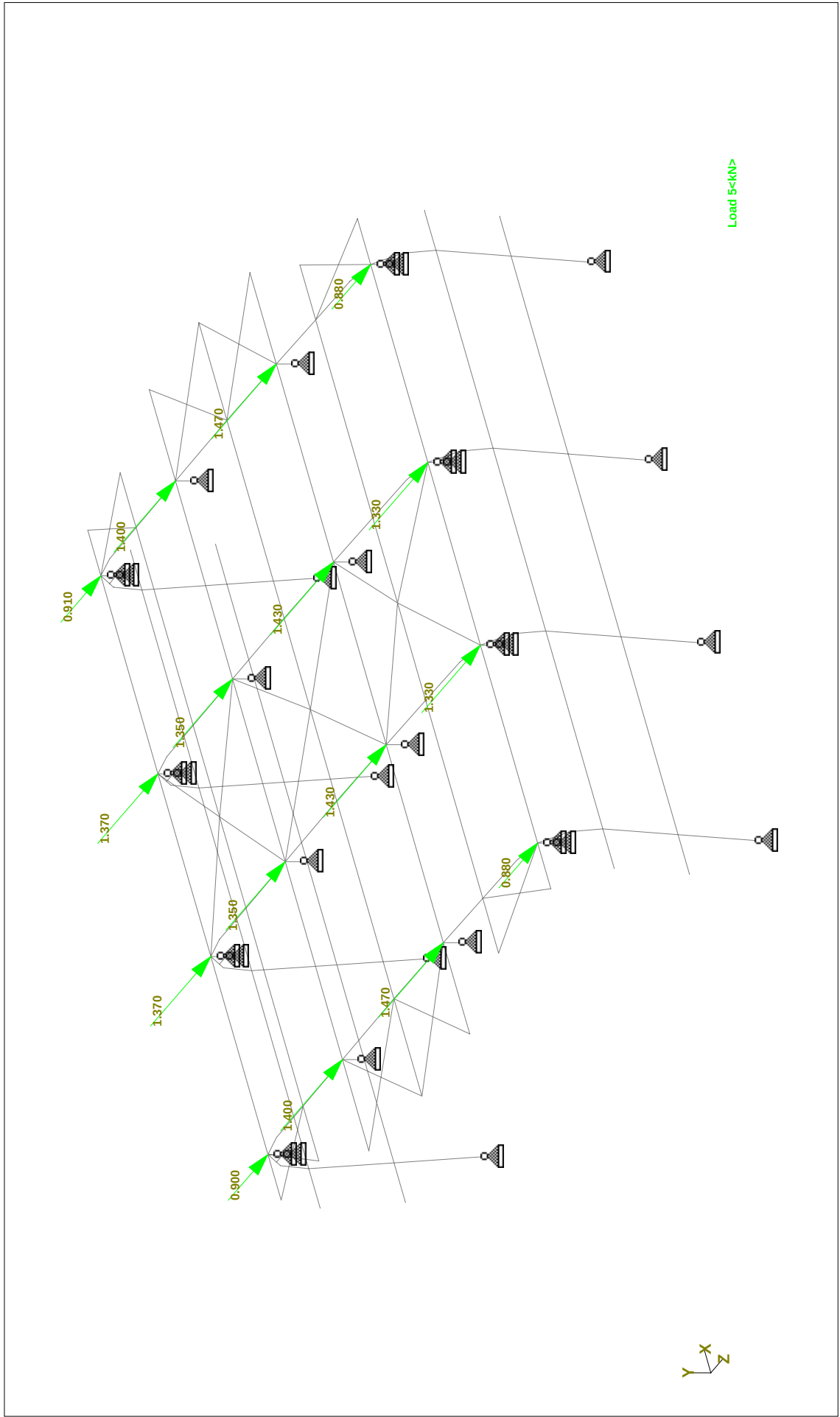
Load 2 Dead Load



Load 1 SelfWeight

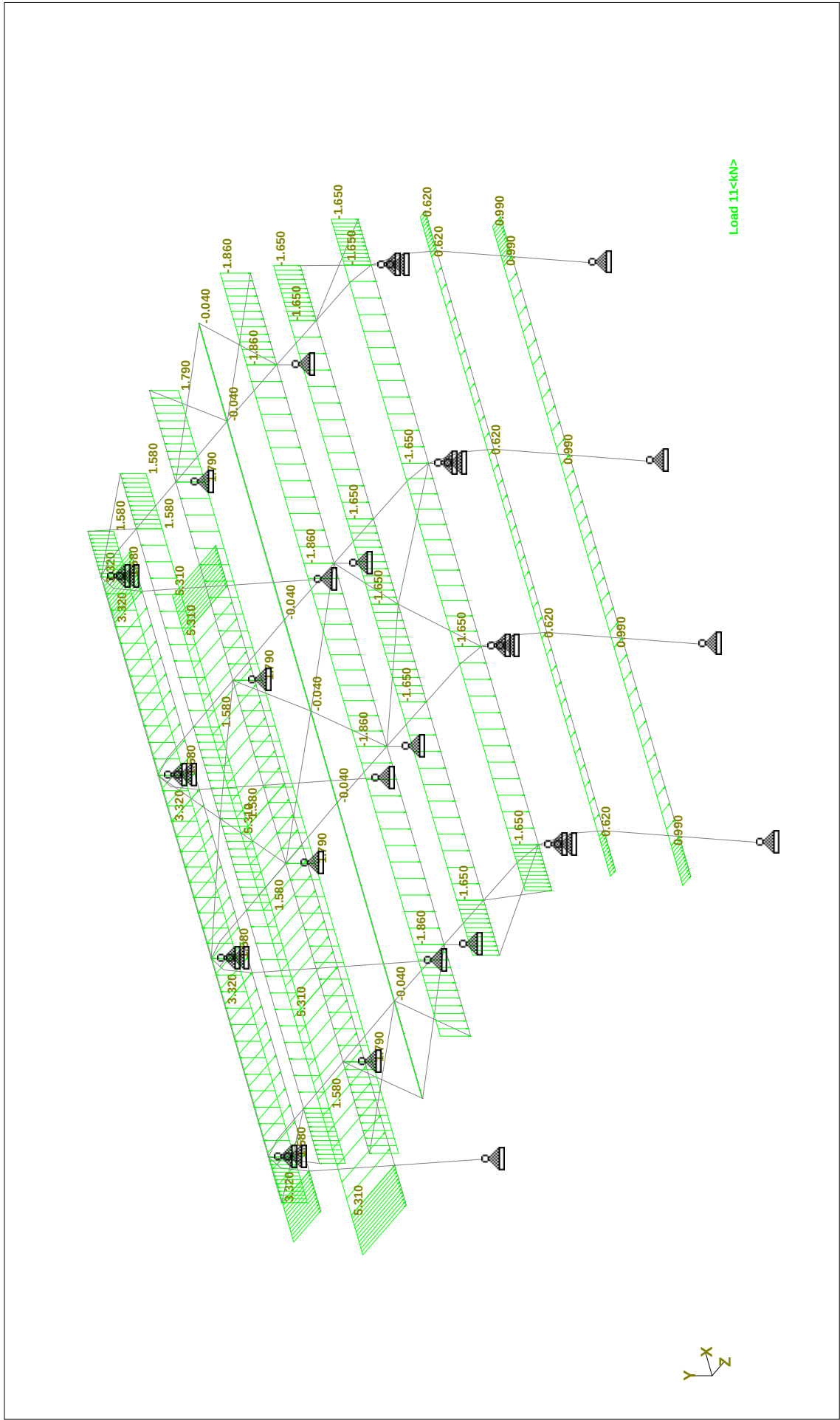




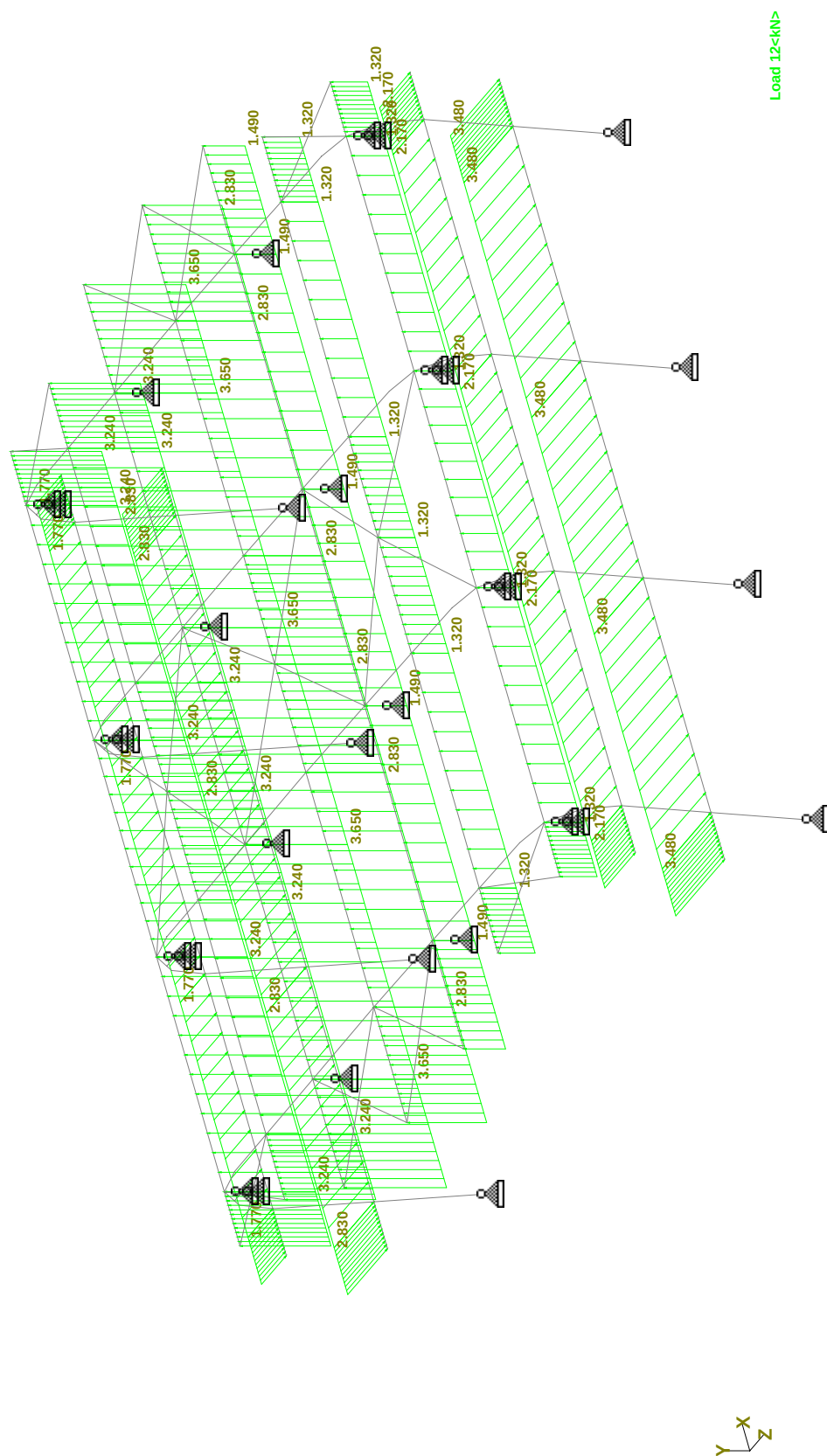


Load 5<KN>

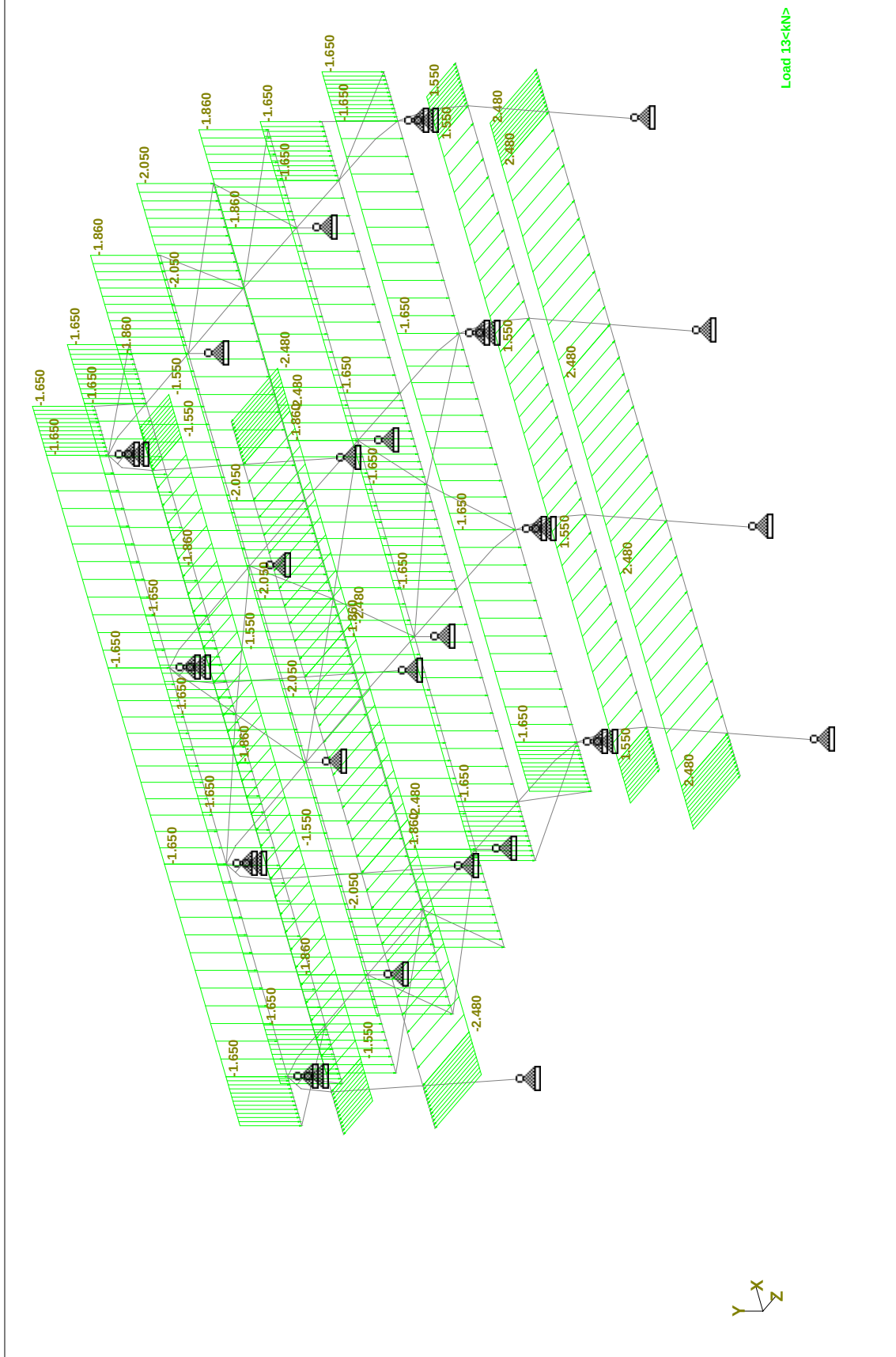
Load 5 PH (Z - dir)



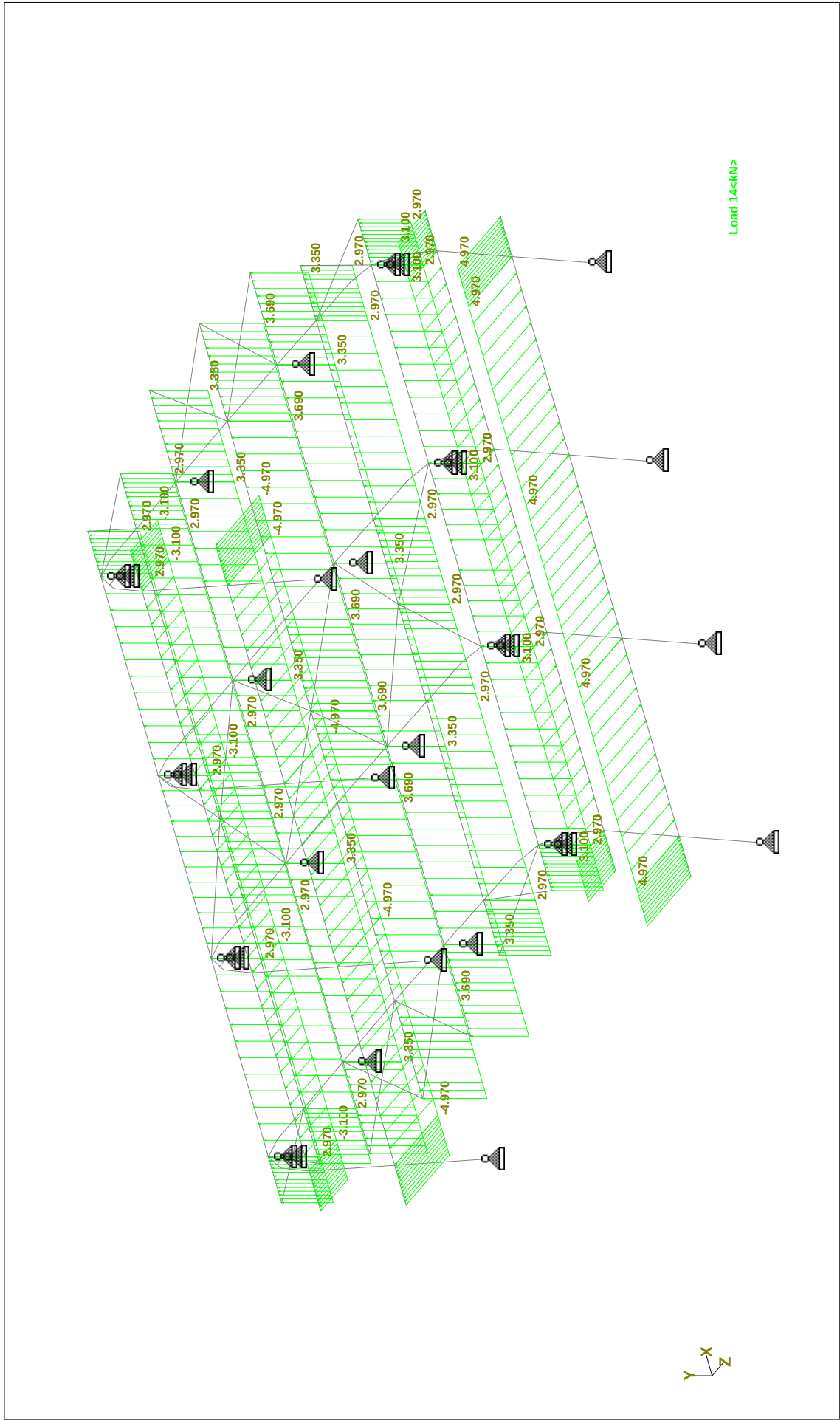
Load 11 WL1 (0 DEG, INTERNAL SUCTION (CPI=-0.3)



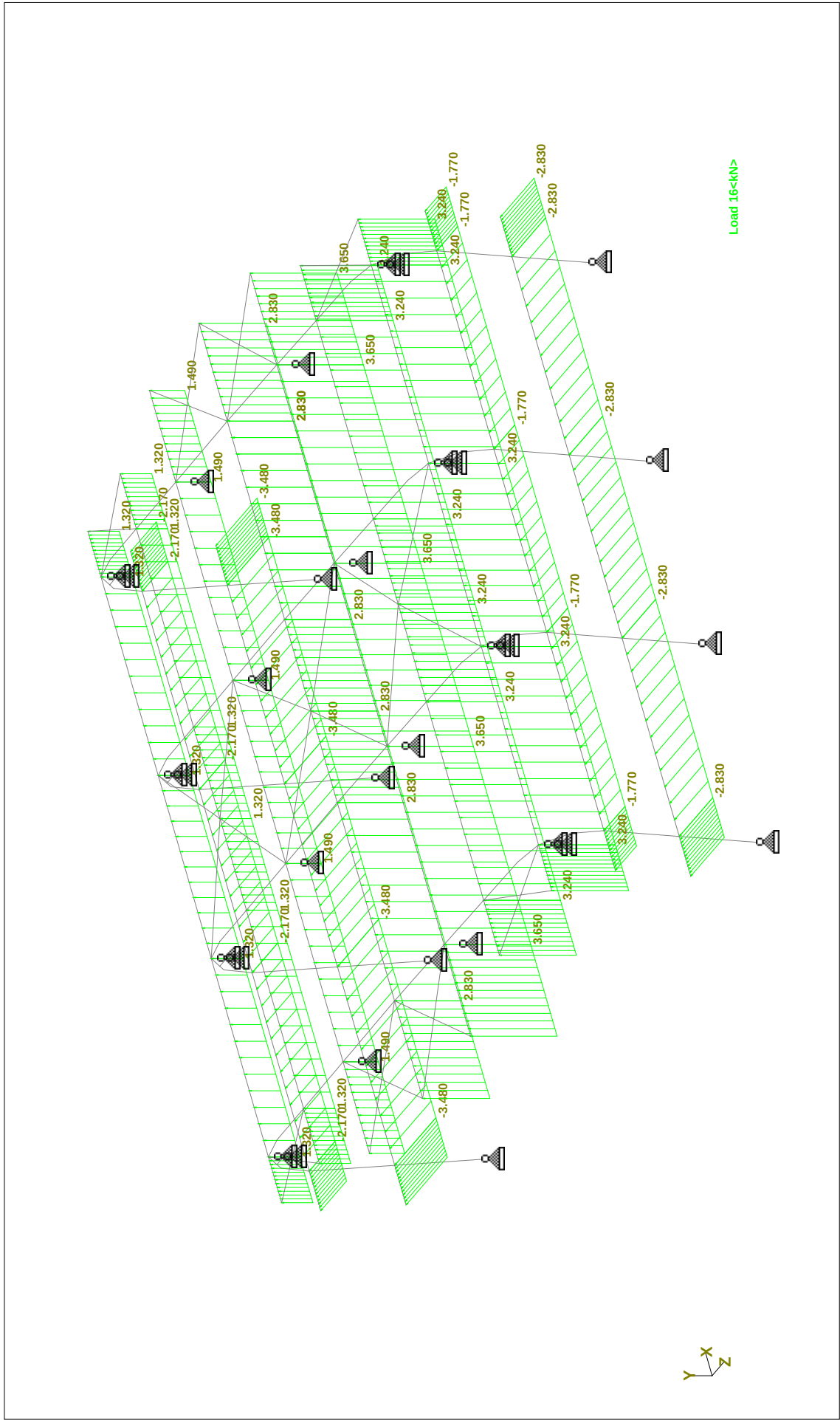
Load 12 WL2 (0 DEG ,INTERNAL PRESSURE $CPI=+0.2$)



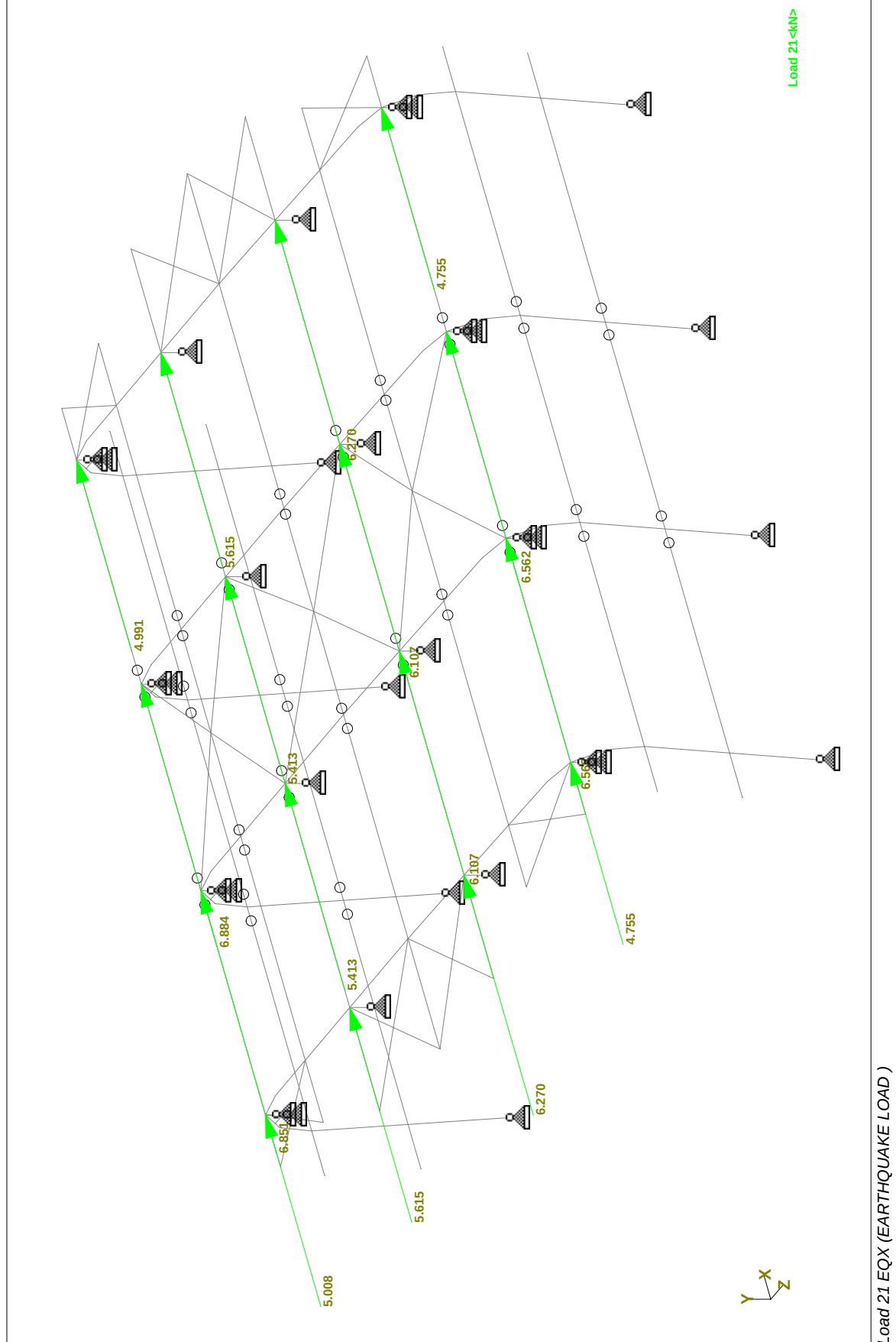
Load 13 WL.3 (90 DEG ,INTERNAL SUCTION CPI=-0.3, WW&LW CPE=+0.2)

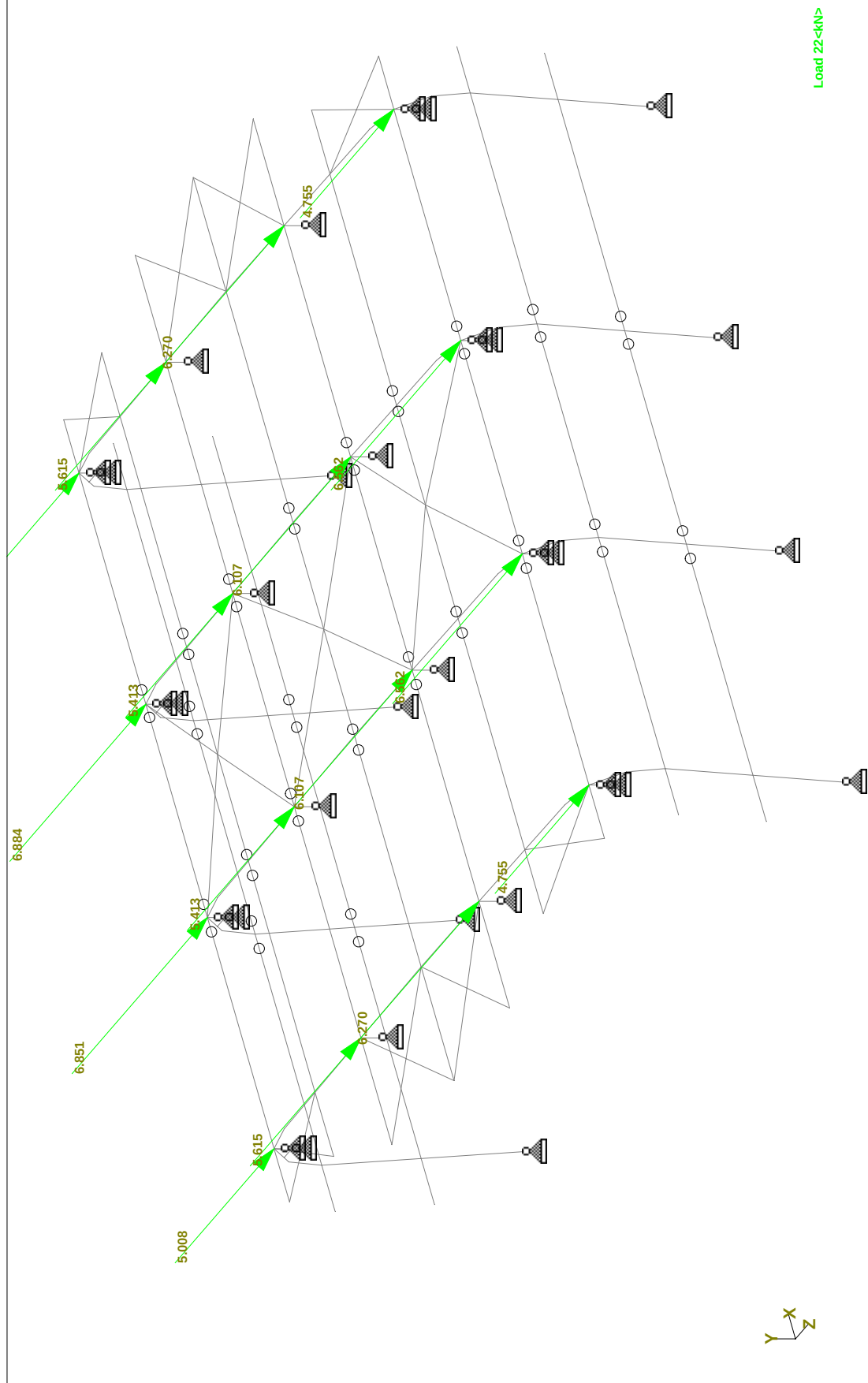


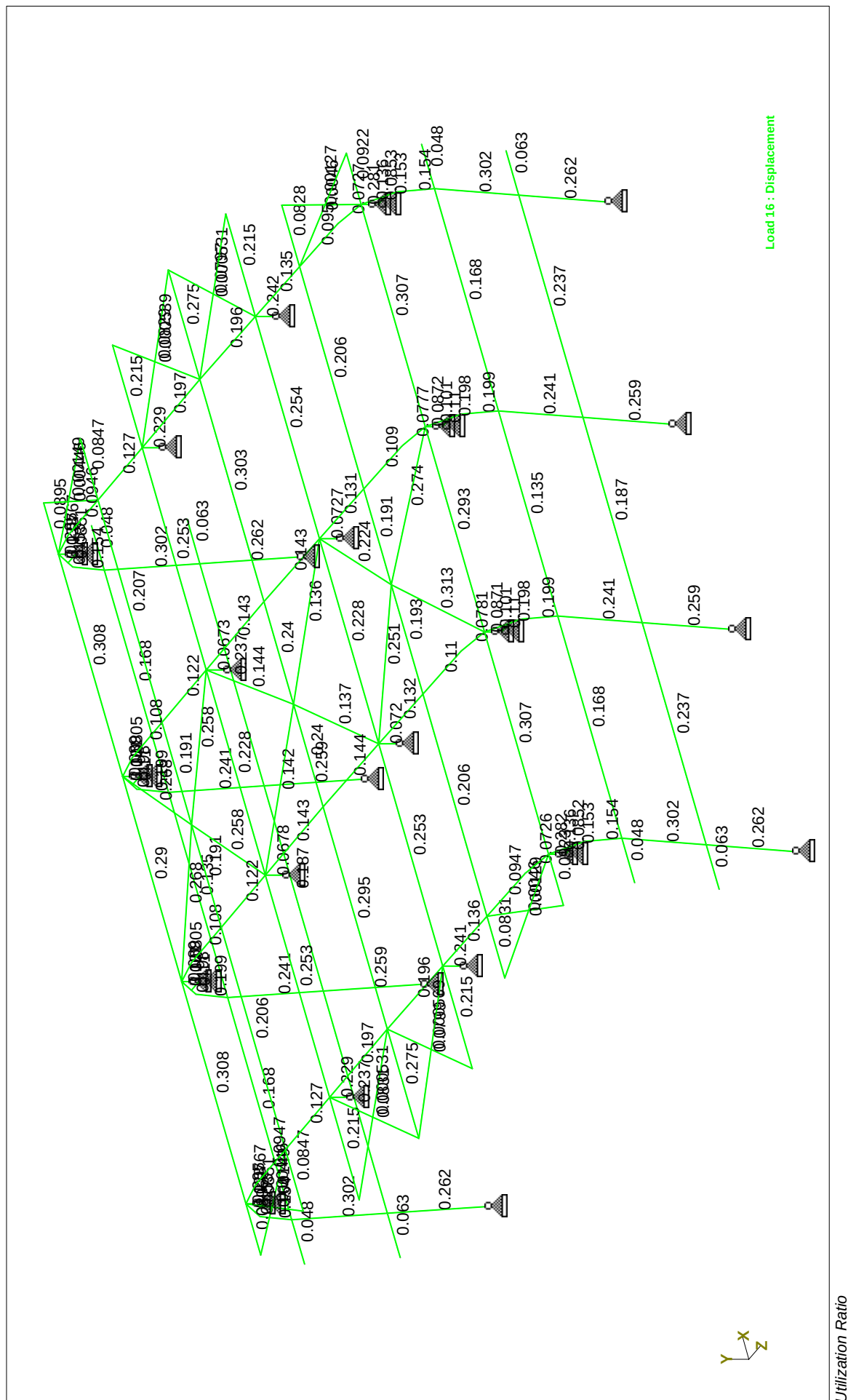
Load 14 WL 4 (90 DEG ,INTERNAL PRESSURE CPI=+0.2, WW&LW CPE=-0.7)



Load 16 WL6 (0 DEG ,INTERNAL PRESSURE CPI=+0.2, LEE WARD -0.2, REVERSE)







```
*****
*
*          STAAD.Pro
*          Version 2006      Bld 1002.US
*          Proprietary Program of
*          Research Engineers, Intl.
*          Date= NOV 16, 2007
*          Time= 16:52:21
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*          USER ID: roark
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1. STAAD SPACE BLDG -26(STEEL)
INPUT FILE: BLDG-26 STEEL Structure.STD
2. START JOB INFORMATION
3. ENGINEER DATE 29-SEPT-07
4. END JOB INFORMATION
5. INPUT WIDTH 79
6. *****
7. UNIT METER KN
8. JOINT COORDINATES
9. 1 0 0 0; 2 0 5.312 0.662; 3 0 6.317 0.503; 4 0 7.056 -0.14; 5 0 7.253 -1.111
10. 6 0 7.317 -5.809; 7 0 7.347 -8.999; 8 0 7.306 -12.459; 9 0 7.253 -16.887
11. 10 0 7.056 -17.856; 11 0 6.317 -18.5; 12 0 5.312 -18.659; 13 0 0 -17.997
12. 14 6 5 0 0; 15 6 5 5.312 0.662; 16 6 5 6.317 0.503; 17 6 5 7.056 -0.14
13. 18 6 5 7.253 -1.111; 19 6 5 7.317 -5.809; 20 6 5 7.347 -8.999
14. 21 6 5 7.306 -12.459; 22 6 5 7.253 -16.887; 23 6 5 7.056 -17.856
15. 24 6 5 6.317 -18.5; 25 6 5 5.312 -18.659; 26 6 5 0 -17.997; 27 12.5 0 0
16. 28 12.5 5.312 0.662; 29 12.5 6.317 0.503; 30 12.5 7.056 -0.14
17. 31 12.5 7.253 -1.111; 32 12.5 7.317 -5.809; 33 12.5 7.347 -8.999
18. 34 12.5 7.306 -12.459; 35 12.5 7.253 -16.887; 36 12.5 7.056 -17.856
19. 37 12.5 6.317 -18.5; 38 12.5 5.312 -18.659; 39 12.5 0 -17.997; 40 19 0 0
20. 41 19 5.312 0.662; 42 19 6.317 0.503; 43 19 7.056 -0.14; 44 19 7.253 -1.111
21. 45 19 7.317 -5.809; 46 19 7.347 -8.999; 47 19 7.306 -12.459
22. 48 19 7.253 -16.887; 49 19 7.056 -17.856; 50 19 6.317 -18.5
23. 51 19 5.312 -18.659; 52 19 0 -17.997; 53 20.5 7.056 -0.14
24. 54 20.5 7.056 -17.856; 55 -1.5 7.056 -0.14; 56 -1.5 7.056 -17.856
25. 57 22 7.317 -5.809; 58 22 7.306 -12.459; 59 -3 7.317 -5.809
26. 60 -3 7.306 -12.459; 61 22.2 7.347 -8.999; 62 -3.2 7.347 -8.999
27. 63 0 7.27428 -15.1088; 64 6.5 7.27428 -15.1088; 65 12.5 7.27428 -15.1088
28. 66 19 7.27428 -15.1088; 67 0 7.28269 -3.2908; 68 6.5 7.28269 -3.2908
29. 69 12.5 7.28269 -3.2908; 70 19 7.28269 -3.2908; 71 20.8 7.27428 -15.1088
30. 72 -1.8 7.27428 -15.1088; 73 20.8 7.28269 -3.2908; 74 -1.8 7.28269 -3.2908
31. 75 0 2.67927 0.333901; 76 6.5 2.67927 0.333901; 77 12.5 2.67927 0.333901
32. 78 19 2.67927 0.333901; 79 20.3 2.67927 0.333901; 80 -1.3 2.67927 0.333901
33. 81 0 2.67927 -18.3309; 82 6.5 2.67927 -18.3309; 83 12.5 2.67927 -18.3309
34. 84 19 2.67927 -18.3309; 85 20.3 2.67927 -18.3309; 86 -1.3 2.67927 -18.3309
35. 87 20.3 5.312 0.662; 88 20.3 5.312 -18.659; 89 -1.3 5.312 0.662
36. 90 -1.3 5.312 -18.659; 91 9.7 7.28269 -3.2908; 93 9.5 7.347 -8.999
37. 95 9.5 7.27428 -15.1088; 96 0 6.817 -5.809; 97 0 6.806 -12.459
38. 98 6.5 6.817 -5.809; 99 6.5 6.806 -12.459; 100 12.5 6.817 -5.809
39. 101 12.5 6.806 -12.459; 102 19 6.817 -5.809; 103 19 6.806 -12.459
40. 104 0 6.856 -0.14; 105 0 6.856 -17.856; 106 6.5 6.856 -0.14
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41. 107 6.5 6.856 -17.856; 108 12.5 6.856 -0.14; 109 12.5 6.856 -17.856
42. 110 19 6.856 -0.14; 111 19 6.856 -17.856; 112 0 6.556 -0.14
43. 113 0 6.556 -17.856; 114 6.5 6.556 -0.14; 115 6.5 6.556 -17.856
44. 116 12.5 6.556 -0.14; 117 12.5 6.556 -17.856; 118 19 6.556 -0.14
45. 119 19 6.556 -17.856; 120 0 6.556 -18.2917; 129 6.5 6.556 -18.2917
46. 130 12.5 6.556 -18.2917; 131 19 6.556 -18.2917; 132 0 6.556 0.295047
47. 133 6.5 6.556 0.295047; 134 12.5 6.556 0.295047; 135 19 6.556 0.295047
48. *****
49. MEMBER INCIDENCES
50. 1 1 75; 2 2 3; 3 3 132; 4 4 5; 5 5 67; 6 6 7; 7 7 8; 8 8 63; 9 10 9; 10 11 128
51. 11 12 11; 12 13 81; 13 14 76; 14 15 16; 15 16 133; 16 17 18; 17 18 68
52. 18 19 20; 19 20 21; 20 21 64; 21 23 22; 22 24 129; 23 25 24; 24 26 82
53. 25 27 77; 26 28 29; 27 29 134; 28 30 31; 29 31 69; 30 32 33; 31 33 34
54. 32 34 65; 33 36 35; 34 37 130; 35 38 37; 36 39 83; 37 40 78; 38 41 42
55. 39 42 135; 40 43 44; 41 44 70; 42 45 46; 43 46 47; 44 47 66; 45 49 48
56. 46 50 131; 47 51 50; 48 52 84; 52 63 9; 53 64 22; 54 65 35; 55 66 48; 56 67 6
57. 57 68 19; 58 69 32; 59 70 45; 95 75 2; 96 76 15; 97 77 28; 98 78 41; 99 81 12
58. 100 82 25; 101 83 38; 102 84 51; 201 53 43; 202 43 30; 203 30 17; 204 17 4
59. 205 4 55; 206 73 70; 207 70 69; 208 69 91; 209 68 67; 210 67 74; 211 57 45
60. 212 45 32; 213 32 19; 214 19 6; 215 6 59; 216 61 46; 217 46 33; 218 33 93
61. 220 7 62; 221 58 47; 222 47 34; 223 34 21; 224 21 8; 225 8 60; 226 71 66
62. 227 66 65; 228 65 95; 229 64 63; 230 63 72; 231 54 49; 232 49 36; 233 36 23
63. 234 23 10; 235 10 56; 236 91 68; 238 93 20; 240 95 64; 241 20 7; 301 79 78
64. 302 78 77; 303 77 76; 304 76 75; 305 75 80; 306 85 84; 307 84 83; 308 83 82
65. 309 82 81; 310 81 86; 311 87 41; 312 41 28; 313 28 15; 314 15 2; 315 2 89
66. 316 88 51; 317 51 38; 318 38 25; 319 25 12; 320 12 90; 501 55 67; 502 4 74
67. 503 59 7; 504 6 62; 505 62 8; 506 7 60; 507 72 10; 508 63 56; 509 43 73
68. 510 53 70; 511 45 61; 512 57 46; 513 46 58; 514 61 47; 515 71 49; 516 54 66
69. 522 17 91; 523 91 32; 524 30 91; 525 91 19; 530 19 93; 531 93 34; 532 32 93
70. 533 93 21; 538 34 95; 539 95 23; 540 36 95; 541 95 21; 542 97 8; 543 96 6
71. 544 99 21; 545 98 19; 546 101 34; 547 100 32; 548 103 47; 549 102 45
72. 550 105 10; 551 107 23; 552 109 36; 553 111 49; 554 104 4; 555 106 17
73. 556 108 30; 557 110 43; 558 113 128; 559 115 129; 560 117 130; 561 119 131
74. 562 128 10; 564 129 23; 566 130 36; 567 131 49; 568 118 135; 569 116 134
75. 570 114 133; 571 113 132; 572 132 4; 573 133 17; 574 134 30; 575 135 43
76. START GROUP DEFINITION
77. MEMBER
78. _COLUMN 1 TO 3 10 TO 15 22 TO 27 34 TO 39 46 TO 48 95 TO 102 562 564 566 567 -
79. 572 TO 575
80. _RAFTER 4 TO 9 16 TO 21 28 TO 33 40 TO 45 52 TO 59
81. _SEC_BEAM ROOF 201 TO 218 220 TO 236 238 240 241
82. _SIDE_BEAM 301 TO 320
83. _BRACING 501 TO 516 522 TO 525 530 TO 533 538 TO 541
84. _SUBCOLUMN 542 TO 561 568 TO 571
85. END GROUP DEFINITION
86. *****
87. DEFINE MATERIAL START
88. ISOTROPIC STEEL
89. E 2.05E+008
90. POISSON 0.3
91. DENSITY 78
92. ALPHA 1.2E-005
93. DAMP 0.03
94. END DEFINE MATERIAL
95. MEMBER PROPERTY BRITISH
96. *****MAIN COLUMN*****
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97. 1 12 13 24 25 36 37 48 TAPERED 0.3 0.006 0.35 0.2 0.01 0.2 0.01 0.2 0.01
98. 95 TO 102 TAPERED 0.35 0.006 0.4 0.2 0.01 0.2 0.01
99. 2 3 10 11 14 15 22 23 26 27 34 35 38 39 46 47 562 564 566 567 572 TO 574 -
100. 575 TAPERED 0.4 0.006 0.4 0.2 0.01 0.2 0.01
101. *****MAIN RAFTER *****
102. 4 TO 9 16 TO 21 28 TO 33 40 TO 45 52 TO 58 -
103. 59 TAPERED 0.4 0.006 0.4 0.2 0.01 0.2 0.01
104. *****SEC BEAM *****
105. 201 TO 218 220 TO 236 238 240 241 TAPERED 0.3 0.006 0.3 0.15 0.01 0.01 0.15 0.01
106. *****SIDE BEAM *****
107. 301 TO 320 TAPERED 0.25 0.006 0.25 0.25 0.012 0.25 0.012
108. *****STUB COLUMN *****
109. 542 TO 549 TAPERED 0.3 0.006 0.3 0.2 0.008 0.2 0.008
110. 550 TO 561 568 TO 571 TAPERED 0.3 0.006 0.3 0.2 0.008 0.2 0.008
111. *550 TO 561 568 TO 571 PRIS YD 0.4 ZD 0.01
112. *****BRACING *****
113. 501 TO 516 522 TO 525 530 TO 533 538 TO 541 TABLE ST UA70X70X6
114. *****
115. CONSTANTS
116. BETA 90 MEMB 301 TO 320 542 TO 557
117. MATERIAL STEEL ALL
118. *****
119. SUPPORTS
120. 1 13 14 26 27 39 40 52 PINNED
121. 96 TO 119 PINNED
122. *4 6 10 17 19 21 23 30 32 34 36 43 45 47 49 PINNED
123. *4 6 10 17 19 21 23 30 32 34 36 43 45 47 49 128 TO 134 PINNED
124. *****
125. MEMBER RELEASE
126. *558 TO 561 568 TO 571 END MX MY MZ
127. *550 TO 557 END MX MY MZ
128. 203 204 208 209 213 214 218 223 224 228 229 233 234 241 303 304 308 309 313 -
129. 314 318 319 START MZ
130. 202 203 207 212 213 217 222 223 227 232 233 236 238 240 302 303 307 308 312 -
131. 313 317 318 END MZ
132. MEMBER TRUSS
133. 501 TO 516 522 TO 525 530 TO 533 538 TO 541
134. DEFINE UBC LOAD
135. ZONE 0.15 I 1 RMX 4.5 STYP 4 CT 0.0853 PX 0.381 PZ 0.381 NA 1 NV 1
136. JOINT WEIGHT
137. 4 WEIGHT 39.57
138. 6 WEIGHT 50.32
139. 8 WEIGHT 45.126
140. 10 WEIGHT 41.672
141. 17 WEIGHT 54.606
142. 19 WEIGHT 49.012
143. 21 WEIGHT 43.508
144. 23 WEIGHT 57.012
145. 30 WEIGHT 54.606
146. 32 WEIGHT 49.012
147. 34 WEIGHT 43.508
148. 36 WEIGHT 57.29
149. 43 WEIGHT 39.57
150. 45 WEIGHT 50.32
151. 47 WEIGHT 45.126
152. 49 WEIGHT 41.537

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153. LOAD 21 EQX (EARTHQUAKE LOAD)
154. UBC LOAD X
155. LOAD 22 EQZ (EARTHQUAKE LOAD)
156. UBC LOAD Z
157. LOAD 1 SELFWEIGHT
158. SELFWEIGHT Y -1
159. LOAD 2 DL
160. **DEAD LOAD = 0.8 KN/M^2
161. **BEAM SPACING = 3.3 M , 3.0 M & 2.7 M
162. **DL1 = 3.3X0.8 =2.66=2.7 KN/M
163. **DL2 = 3.0X0.8 =2.4 KN/M
164. **DL3 = 2.7X0.8 =2.16=2.2 KN/M
165. MEMBER LOAD
166. 216 TO 218 220 238 241 301 TO 310 UNI GY -2.7
167. 211 TO 215 221 TO 225 UNI GY -2.4
168. 206 TO 210 226 TO 230 236 240 UNI GY -2.2
169. 201 TO 205 231 TO 235 311 TO 320 UNI GY -2.2
170. *****
171. LOAD 3 LL
172. MEMBER LOAD
173. **LIVE LOAD = 0.6 KN/M^2
174. **BEAM SPACING = 3.3 M , 3.0 M & 2.7 M
175. **LL1 = 3.3X0.6 =1.98=2.0 KN/M
176. **LL2 = 3.0X0.6 =1.8 KN/M
177. **LL3 = 2.7X0.6 =1.62=1.65 KN/M
178. 216 TO 218 220 238 241 UNI GY -2
179. 211 TO 215 221 TO 225 UNI GY -1.8
180. 201 TO 210 226 TO 236 240 UNI GY -1.65
181. *****A/C EQUIPMENT S/W = 5 KN (POINT LOAD)
182. *JOINT LOAD
183. *20 33 FY -5
184. LOAD 4 PH(X-DIR)
185. *****
186. **PH =1.5% (1.4DL+1.6LL)
187. JOINT LOAD
188. 4 FX 0.88
189. 6 FX 1.47
190. 8 FX 1.4
191. 10 FX 0.9
192. 17 FX 1.33
193. 19 FX 1.43
194. 21 FX 1.35
195. 23 FX 1.37
196. 30 FX 1.33
197. 32 FX 1.43
198. 34 FX 1.35
199. 36 FX 1.37
200. 43 FX 0.88
201. 45 FX 1.47
202. 47 FX 1.4

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203. 49 FX 0.91
 204. LOAD 5 PH(Z-DIR)
 205. *****
 206. **PH =1.5% (1.4DL+1.6LL)
 207. JOINT LOAD
 208. 4 FZ 0.88
 209. 6 FZ 1.47
 210. 8 FZ 1.4
 211. 10 FZ 0.9
 212. 17 FZ 1.33
 213. 19 FZ 1.43
 214. 21 FZ 1.35
 215. 23 FZ 1.37
 216. 30 FZ 1.33
 217. 32 FZ 1.43
 218. 34 FZ 1.35
 219. 36 FZ 1.37
 220. 43 FZ 0.88
 221. 45 FZ 1.47
 222. 47 FZ 1.4
 223. 49 FZ 0.91
 224. *****
 225. LOAD 11 WL1 (0 DEG, INTERNAL SUCTION (CPI=-0.3))
 226. **WIND LOAD(Q) = 1.24 KN/M^2
 227. **CPE(WINDWARD ROOF)=-0.78, CPE(LEEWARD ROOF=0.2)
 228. **CPE(WINDWARD WALL)=0.77, CPE(LEEWARD WALL=-0.5)
 229. **BEAM SPACING (ROOF) = 3.3 M , 3.0 M & 2.7 M
 230. **BEAM SPACING (WALL) = 4.0M , 2.5 M
 231. **WIND LOAD = QX(CPE-CPI)XBEAM SPACING
 232. **WL1(WINDWARD WALL)= 1.24(0.77+0.3)X4.0 =5.31 KN/M
 233. **WL2(WINDWARD WALL)= 1.24(0.77+0.3)X2.5 =3.32 KN/M
 234. **WL3(LEEWARD WALL)= 1.24(-0.5+0.3)X4.0 =0.99 KN/M
 235. **WL4(LEEWARD WALL)= 1.24(-0.5+0.3)X2.5 =0.62 KN/M
 236. **WL5(WINDWARD ROOF)= 1.24(-0.78+0.3)X3.0 =1.79(UP) KN/M
 237. **WL6(WINDWARD ROOF)= 1.24(-0.78+0.3)X2.66 =1.58(UP) KN/M
 238. **WL7(LEEWARD ROOF)= 1.24(0.2+0.3)X3.0 =1.86(DN) KN/M
 239. **WL8(LEEWARD ROOF)= 1.24(0.2+0.3)X2.66 =1.65(DN)KN/M
 240. MEMBER LOAD
 241. 306 TO 310 UNI GZ 5.31
 242. 221 TO 225 UNI Y 1.79
 243. 211 TO 215 UNI Y -1.86
 244. 301 TO 305 UNI GZ 0.99
 245. 316 TO 320 UNI GZ 3.32
 246. 311 TO 315 UNI GZ 0.62
 247. 226 TO 235 240 UNI Y 1.58
 248. 201 TO 210 236 UNI Y -1.65
 249. 216 TO 218 220 238 241 UNI Y -0.04
 250. *****
 251. LOAD 12 WL2 (0 DEG, INTERNAL PRESSURE CPI=+0.2)
 252. **WIND LOAD(Q) = 1.24 KN/M^2
 253. **CPE(WINDWARD ROOF)=-0.78, CPE(LEEWARD ROOF)=-0.2
 254. **CPE(WINDWARD WALL)=+0.77, CPE(LEEWARD WALL)=-0.5
 255. **BEAM SPACING (ROOF) = 3.3 M , 3.0 M & 2.7 M
 256. **BEAM SPACING (WALL) = 4.0M , 2.5 M
 257. **WIND LOAD = QX(CPE-CPI)XBEAM SPACING
 258. **WL1(WINDWARD WALL)= 1.24(0.77-0.2)X4.0 =2.83 KN/M

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259. **WL2(WINDWARD WALL)= 1.24(0.77-0.2)X2.5 =1.77 KN/M
 260. **WL3(LEEWARD WALL)= 1.24(-0.5-0.2)X4.0 =-3.48 KN/M
 261. **WL4(LEEWARD WALL)= 1.24(-0.5-0.2)X2.5 =-2.17 KN/M
 262. **WL5(WINDWARD ROOF)= 1.24(-0.78-0.2)X3.0 =3.65(UP) KN/M
 263. **WL6(WINDWARD ROOF)= 1.24(-0.78-0.2)X2.66 =3.24(UP) KN/M
 264. **WL7(LEEWARD ROOF)= 1.24(-0.2-0.2)X3.0 =-1.65(UP) KN/M
 265. **WL8(LEEWARD ROOF)= 1.24(-0.2-0.2)X2.66 =-1.49(UP)KN/M
 266. MEMBER LOAD
 267. 306 TO 310 UNI GZ 2.83
 268. 301 TO 305 UNI GZ 3.48
 269. 316 TO 320 UNI GZ 1.77
 270. 311 TO 315 UNI GZ 2.17
 271. 221 TO 225 UNI Y 3.65
 272. 211 TO 215 UNI Y 1.49
 273. 226 TO 235 240 UNI Y 3.24
 274. 201 TO 210 236 UNI Y 1.32
 275. 216 TO 218 220 238 241 UNI Y 2.83
 276. *****
 277. LOAD 13 WL3 (90 DEG, INTERNAL SUCTION CPI=-0.3, WM&LW CPE=+0.2)
 278. **WIND LOAD(Q) = 1.24 KN/M^2
 279. **CPE(WINDWARD ROOF)=0.2, CPE(LEEWARD ROOF=0.2)
 280. **CPE(WINDWARD WALL)=-0.8, CPE(LEEWARD WALL=-0.8)
 281. **BEAM SPACING (ROOF) = 3.3 M , 3.0 M & 2.7 M
 282. **BEAM SPACING (WALL) = 4.0M , 2.5 M
 283. **WIND LOAD = QX(CPE-CPI)XBEAM SPACING
 284. **WL1(WINDWARD WALL)= 1.24(-0.8+0.3)X4.0 =-2.48 KN/M
 285. **WL2(WINDWARD WALL)= 1.24(-0.8+0.3)X2.5 =-1.55 KN/M
 286. **WL3(LEEWARD WALL)= 1.24(-0.8+0.3)X4.0 =-2.48 KN/M
 287. **WL4(LEEWARD WALL)= 1.24(-0.8+0.3)X2.5 =-1.55 KN/M
 288. **WL5(WINDWARD ROOF)= 1.24(0.2+0.3)X3.0 =1.86(DN) KN/M
 289. **WL6(WINDWARD ROOF)= 1.24(0.2+0.3)X2.66 =1.65(DN) KN/M
 290. **WL7(LEEWARD ROOF)= 1.24(0.2+0.3)X3.0 =1.86(DN) KN/M
 291. **WL8(LEEWARD ROOF)= 1.24(0.2+0.3)X2.66 =1.65(DN)KN/M
 292. MEMBER LOAD
 293. 306 TO 310 UNI GZ -2.48
 294. 301 TO 305 UNI GZ 2.48
 295. 316 TO 320 UNI GZ -1.55
 296. 311 TO 315 UNI GZ 1.55
 297. 221 TO 225 UNI Y -1.86
 298. 211 TO 215 UNI Y -1.86
 299. 226 TO 235 240 UNI Y -1.65
 300. 201 TO 210 236 UNI Y -1.65
 301. 216 TO 218 220 238 241 UNI Y -2.05
 302. *****
 303. LOAD 14 WL4 (90 DEG, INTERNAL PRESSURE CPI=+0.2, WM&LW CPE=-0.7)
 304. **WIND LOAD(Q) = 1.24 KN/M^2
 305. **CPE(WINDWARD ROOF)=-0.7, CPE(LEEWARD ROOF=-0.7)
 306. **CPE(WINDWARD WALL)=-0.8, CPE(LEEWARD WALL=-0.8)
 307. **BEAM SPACING (ROOF) = 3.3 M , 3.0 M & 2.7 M
 308. **BEAM SPACING (WALL) = 4.0M , 2.5 M
 309. **WIND LOAD = QX(CPE-CPI)XBEAM SPACING
 310. **WL1(WINDWARD WALL)= 1.24(-0.8-0.2)X4.0 =-4.97 KN/M
 311. **WL2(WINDWARD WALL)= 1.24(-0.8-0.2)X2.5 =-3.1 KN/M
 312. **WL3(LEEWARD WALL)= 1.24(-0.8-0.2)X4.0 =-4.97 KN/M
 313. **WL4(LEEWARD WALL)= 1.24(-0.8-0.2)X2.5 =-3.1 KN/M
 314. **WL5(WINDWARD ROOF)= 1.24(-0.7-0.2)X3.0 =-3.35(UP) KN/M

315. **WL6(WINDWARD ROOF)= 1.24(-0.7-0.2)X2.66 =2.97(UP) KN/M
 316. **WL7(LIEWARD ROOF)= 1.24(-0.7-0.2)X3.0 =3.35(UP) KN/M
 317. **WL8(LIEWARD ROOF)= 1.24(-0.7-0.2)X2.66=2.97(UP) KN/M
 318. MEMBER LOAD
 319. 306 TO 310 UNI GZ -4.97
 320. 301 TO 305 UNI GZ 4.97
 321. 316 TO 320 UNI GZ -3.1
 322. 311 TO 315 UNI GZ 3.1
 323. 221 TO 225 UNI Y 3.35
 324. 211 TO 215 UNI Y 3.35
 325. 226 TO 235 240 UNI Y 2.97
 326. 201 TO 210 236 UNI Y 2.97
 327. 216 TO 218 220 238 241 UNI Y 3.69
 328. *****
 329. LOAD 15 WL5 (0 DEG , INTERNAL SUCTION CPI=-0.3, LEE WARD -0.7, REVERSE)
 330. MEMBER LOAD
 331. **WIND LOAD(Q) = 1.24 KN/M^2
 332. **CPE(WINDWARD ROOF)=-0.78, CPE(LIEWARD ROOF=0.2)
 333. **CPE(WINDWARD WALL)=0.77, CPE(LIEWARD WALL=-0.5)
 334. **BEAM SPACING (ROOF) = 3.3 M , 3.0 M & 2.7 M
 335. **BEAM SPACING (WALL) = 4.0M , 2.5 M
 336. **WIND LOAD = QX(CPE-CPI)XBEAM SPACING
 337. **WL1(WINDWARD WALL)= 1.24(0.77+0.3)X4.0 =5.31 KN/M
 338. **WL2(WINDWARD WALL)= 1.24(0.77+0.3)X2.5 =3.32 KN/M
 339. **WL3(LIEWARD WALL)= 1.24(-0.5+0.3)X4.0 =0.99 KN/M
 340. **WL4(LIEWARD WALL)= 1.24(-0.5+0.3)X2.5 =0.62 KN/M
 341. **WL5(WINDWARD ROOF)= 1.24(-0.78+0.3)X3.0 =1.79(UP) KN/M
 342. **WL6(WINDWARD ROOF)= 1.24(-0.78+0.3)X2.66 =1.58(UP) KN/M
 343. **WL7(LIEWARD ROOF)= 1.24(0.2+0.3)X3.0 =1.86(DN) KN/M
 344. **WL8(LIEWARD ROOF)= 1.24(0.2+0.3)X2.66 =1.65(DN)KN/M
 345. 306 TO 310 UNI GZ -0.99
 346. 301 TO 305 UNI GZ -5.31
 347. 316 TO 320 UNI GZ -0.62
 348. 311 TO 315 UNI GZ -3.32
 349. 221 TO 225 UNI Y -1.86
 350. 211 TO 215 UNI Y 1.79
 351. 226 TO 235 240 UNI Y -1.65
 352. 201 TO 210 236 UNI Y 1.58
 353. 216 218 220 238 UNI Y -0.04
 354. 241 UNI Y -0.04
 355. 217 UNI Y -0.04
 356. 217 UNI Y -0.04
 357. *****
 358. LOAD 16 WL6 (0 DEG , INTERNAL PRESSURE CPI=+0.2, LEE WARD -0.2, REVERSE)
 359. MEMBER LOAD
 360. **WIND LOAD(Q) = 1.24 KN/M^2
 361. **CPE(WINDWARD ROOF)=-0.78, CPE(LIEWARD ROOF)=-0.2
 362. **CPE(WINDWARD WALL)=+0.77, CPE(LIEWARD WALL)=-0.5
 363. **BEAM SPACING (ROOF) = 3.3 M , 3.0 M & 2.7 M
 364. **BEAM SPACING (WALL) = 4.0M , 2.5 M
 365. **WIND LOAD = QX(CPE-CPI)XBEAM SPACING
 366. **WL1(WINDWARD WALL)= 1.24(0.77-0.2)X4.0 =2.83 KN/M
 367. **WL2(WINDWARD WALL)= 1.24(0.77-0.2)X2.5 =1.77 KN/M
 368. **WL3(LIEWARD WALL)= 1.24(-0.5-0.2)X4.0 =3.48 KN/M
 369. **WL4(LIEWARD WALL)= 1.24(-0.5-0.2)X2.5 =2.17 KN/M
 370. **WL5(WINDWARD ROOF)= 1.24(-0.78-0.2)X3.0 =-3.65(UP) KN/M

371. **WL6(WINDWARD ROOF)= 1.24(-0.78-0.2)X2.66 =3.24(UP) KN/M
 372. **WL7(LIEWARD ROOF)= 1.24(-0.2-0.2)X3.0 =1.65(UP) KN/M
 373. **WL8(LIEWARD ROOF)= 1.24(-0.2-0.2)X2.66 =1.49(UP)KN/M
 374. 306 TO 310 UNI GZ -3.48
 375. 301 TO 305 UNI GZ -2.83
 376. 316 TO 320 UNI GZ -2.17
 377. 311 TO 315 UNI GZ -1.77
 378. 221 TO 225 UNI Y 1.49
 379. 211 TO 215 UNI Y 3.65
 380. 226 TO 235 240 UNI Y 1.32
 381. 201 TO 210 236 UNI Y 3.24
 382. 216 218 220 238 UNI Y 2.83
 383. 241 UNI Y 2.83
 384. 217 UNI Y 2.83
 385. 217 UNI Y 2.83
 386. *****
 387. ** SERVICEABILITY*
 388. *****
 389. LOAD COMB 101 1.0DL+1.0LL
 390. 1 1.0 2 1.0 3 1.0
 391. LOAD COMB 102 1.0DL+0.5LL
 392. 1 1.0 2 1.0 3 0.5
 393. LOAD COMB 103 0.8DL+0.8LL+0.8WL1
 394. 1 0.8 2 0.8 3 0.8 11 0.8
 395. LOAD COMB 104 1.0DL+1.0WL1
 396. 1 1.0 2 1.0 11 1.0
 397. LOAD COMB 105 0.8DL+0.8LL+0.8WL2
 398. 1 0.8 2 0.8 3 0.8 12 0.8
 399. LOAD COMB 106 1.0DL+1.0WL2
 400. 1 1.0 2 1.0 12 1.0
 401. LOAD COMB 107 0.8DL+0.8LL+0.8WL3
 402. 1 0.8 2 0.8 3 0.8 13 0.8
 403. LOAD COMB 108 1.0DL+1.0WL3
 404. 1 1.0 2 1.0 13 1.0
 405. LOAD COMB 109 0.8DL+0.8LL+0.8WL4
 406. 1 0.8 2 0.8 3 0.8 14 0.8
 407. LOAD COMB 110 1.0DL+1.0WL4
 408. 1 1.0 2 1.0 14 1.0
 409. LOAD COMB 111 0.8DL+0.8LL+0.8WL5
 410. 1 0.8 2 0.8 3 0.8 15 0.8
 411. LOAD COMB 112 1.0DL+1.0WL4
 412. 1 1.0 2 1.0 15 1.0
 413. LOAD COMB 113 0.8DL+0.8LL+0.8WL6
 414. 1 0.8 2 0.8 3 0.8 16 0.8
 415. LOAD COMB 114 1.0DL+1.0WL6
 416. 1 1.0 2 1.0 16 1.0
 417. *****EQX & EQZ *****
 418. **DL+EQL/1.4 = 1.0DL+0.72EQX+0.22 EQZ, EQL=EQX+0.3EQZ
 419. **0.9 DL+EQL/1.4 =0.9 DL+0.72 EQX+0.22EQZ
 420. **DL 0.75(LL+EQL/1.4) =1.0 DL+0.75 LL+0.54 EQX+0.17EQZ
 421. *****
 422. LOAD COMB 115 DL+0.72EQX+0.22EQZ
 423. 1 1.0 2 1.0 21 0.72 22 0.22
 424. LOAD COMB 116 DL+0.72EQX-0.22EQZ
 425. 1 1.0 2 1.0 21 0.72 22 -0.22
 426. LOAD COMB 117 DL-0.72EQX+0.22EQZ

BLDG -26(STEEL)

427.	1 1 0 2 1 0 21 -0.72 22 0.22	LOAD COMB 118 DL-0.72EQX+0.22EQZ
428.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 119 DL+0.72EQX+0.22EQX
429.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 120 DL+0.72EQZ-0.22EQX
430.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 121 DL-0.72EQZ+0.22EQX
431.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 122 DL-0.72EQZ-0.22EQX
432.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 123 0.9DL+0.72EQX+0.22EQZ
433.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 124 0.9DL+0.72EQX-0.22EQZ
434.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 125 0.9DL-0.72EQX+0.22EQZ
435.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 126 0.9DL-0.72EQX-0.22EQZ
436.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 127 0.9DL+0.72EQZ+0.22EQX
437.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 128 0.9DL+0.72EQZ-0.22EQX
438.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 129 0.9DL-0.72EQZ+0.22EQX
439.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 130 0.9DL-0.72EQZ-0.22EQX
440.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 131 1.0DL+0.75LL-0.54EQX+0.17 EQZ
441.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 132 1.0DL+0.75LL+0.54EQX-0.17 EQZ
442.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 133 1.0DL+0.75LL-0.54EQX+0.17 EQZ
443.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 134 1.0DL+0.75LL-0.54EQX-0.17 EQZ
444.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 135 1.0DL+0.75LL+0.54EQZ+0.17 EQX
445.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 136 1.0DL+0.75LL+0.54EQZ-0.17 EQX
446.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 137 1.0DL+0.75LL-0.54EQZ+0.17 EQX
447.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 138 1.0DL+0.75LL-0.54EQZ-0.17 EQX
448.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 200 1.4DL+1.6LL
449.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 201 1.2DL+1.2LL+1.2WL1
450.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 202 1.4DL+1.4WL1
451.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 203 1DL+1.4WL1
452.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 204 1DL+1.4WL1
453.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 205 1DL+1.4WL1
454.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 206 1DL+1.4WL1
455.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 207 1DL+1.4WL1
456.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 208 1DL+1.4WL1
457.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 209 1DL+1.4WL1
458.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 210 1DL+1.4WL1
459.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 211 1DL+1.4WL1
460.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 212 1DL+1.4WL1
461.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 213 1DL+1.4WL1
462.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 214 1DL+1.4WL1
463.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 215 1DL+1.4WL1
464.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 216 1DL+1.4WL1
465.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 217 1DL+1.4WL1
466.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 218 1DL+1.4WL1
467.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 219 1DL+1.4WL1
468.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 220 1DL+1.4WL1
469.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 221 1DL+1.4WL1
470.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 222 1DL+1.4WL1
471.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 223 1DL+1.4WL1
472.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 224 1DL+1.4WL1
473.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 225 1DL+1.4WL1
474.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 226 1DL+1.4WL1
475.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 227 1DL+1.4WL1
476.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 228 1DL+1.4WL1
477.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 229 1DL+1.4WL1
478.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 230 1DL+1.4WL1
479.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 231 1DL+1.4WL1
480.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 232 1DL+1.4WL1
481.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 233 1DL+1.4WL1
482.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 234 1DL+1.4WL1
483.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 235 1DL+1.4WL1
484.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 236 1DL+1.4WL1
485.	1 1 0 2 1 0 21 -0.72 22 -0.22	LOAD COMB 237 1DL+1.4WL1
486.		

539. 1 0.9 2 0.9 21 -1.0 22 0.3
540. LOAD COMB 230 0.9DL-1.0EQX+0.3EQZ
541. 1 0.9 2 0.9 21 -1.0 22 -0.3
542. LOAD COMB 231 0.9DL+1.0EQZ+0.3EQX
543. 1 0.9 2 0.9 22 1.0 21 0.3
544. LOAD COMB 232 0.9DL+1.0EQZ-0.3EQX
545. 1 0.9 2 0.9 22 1.0 21 -0.3
546. LOAD COMB 233 0.9DL-1.0EQZ+0.3EQX
547. 1 0.9 2 0.9 22 -1.0 21 0.3
548. LOAD COMB 234 0.9DL-1.0EQZ-0.3EQX
549. 1 0.9 2 0.9 22 -1.0 21 -0.3
550. LOAD COMB 235 1.0DL+1.0LL+1.0PHX
551. 1 1.0 2 1.0 3 1.0 4 1.0
552. LOAD COMB 236 1.0DL+1.0LL+1.0PHZ
553. 1 1.0 2 1.0 3 1.0 5 1.0
554. *****DEFLECTION*****
555. LOAD COMB 301 SELF+DL
556. 1 1.0 2 1.0
557. LOAD COMB 302 LL
558. 3 1.0
559. LOAD COMB 303 0.8(LL+WL1)
560. 3 0.8 11 0.8
561. LOAD COMB 304 0.8(LL+WL2)
562. 3 0.8 12 0.8
563. LOAD COMB 305 0.8(LL+WL3)
564. 3 0.8 13 0.8
565. LOAD COMB 306 0.8(LL+WL4)
566. 3 0.8 14 0.8
567. LOAD COMB 307 0.8(LL+WL5)
568. 3 0.8 15 0.8
569. LOAD COMB 308 0.8(LL+WL6)
570. 3 0.8 16 0.8
571. LOAD COMB 309 0.8(LL+EQX/1.4+0.3EQZ/1.4)
572. 3 0.8 21 0.57 22 0.17
573. LOAD COMB 310 0.8(LL+EQX/1.4-0.3EQZ/1.4)
574. 3 0.8 21 0.57 22 -0.17
575. LOAD COMB 311 0.8(LL-EQX/1.4+0.3EQZ/1.4)
576. 3 0.8 21 -0.57 22 0.17
577. LOAD COMB 312 0.8(LL-EQX/1.4-0.3EQZ/1.4)
578. 3 0.8 21 -0.57 22 -0.17
579. LOAD COMB 313 0.8(LL+EQZ/1.4+0.3EQX/1.4)
580. 3 0.8 22 0.57 21 0.17
581. LOAD COMB 314 0.8(LL+EQZ/1.4-0.3EQX/1.4)
582. 3 0.8 22 0.57 21 -0.17
583. LOAD COMB 315 0.8(LL-EQZ/1.4+0.3EQX/1.4)
584. 3 0.8 22 -0.57 21 0.17
585. LOAD COMB 316 0.8(LL-EQZ/1.4-0.3EQX/1.4)
586. 3 0.8 22 -0.57 21 -0.17
587. *****
588. PERFORM ANALYSIS

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 125/ 182/ 32
ORIGINAL/FINAL BAND-WIDTH= 119/ 15/ 81 DOF
TOTAL PRIMARY LOAD CASES = 13, TOTAL DEGREES OF FREEDOM = 654
SIZE OF STIFFNESS MATRIX = 53 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 13.0/ 23735.8 MB

**WARNING: IF THIS UBC/IBC ANALYSIS HAS TENSION/COMPRESSION
OR REPEAT LOAD OR RE-ANALYSIS OR SELECT OPTIMIZE, THEN EACH
UBC/IBC CASE SHOULD BE FOLLOWED BY PERFORM ANALYSIS & CHANGE.

* X DIRECTION : Ta = 0.381 Tb = 0.115 Tuser = 0.381 *
* T = 0.381, LOAD FACTOR = 1.000 *
* UBC TYPE = 97 *
* UBC FACTOR V = 0.1222 x 761.80 = 93.11 KN *
* *****

* Z DIRECTION : Ta = 0.381 Tb = 0.028 Tuser = 0.381 *
* T = 0.381, LOAD FACTOR = 1.000 *
* UBC TYPE = 97 *
* UBC FACTOR V = 0.1222 x 761.80 = 93.11 KN *
* *****

589. LOAD LIST 101 TO 138
590. PRINT SUPPORT REACTION
SUPPORT REACTION

SUPPORT REACTIONS -UNIT KN										STRUCTURE TYPE = SPACE					
-----										METE					
JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z	-----							
								JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
1	101	0.92	15.43	1.24	0.00	0.00	0.00								
	102	0.91	16.23	1.34	0.00	0.00	0.00		111	0.74	3.63	1.69	0.00	0.00	0.00
	103	0.74	3.75	-1.67	0.00	0.00	0.00		112	0.91	6.20	1.89	0.00	0.00	0.00
	104	0.90	6.29	-1.88	0.00	0.00	0.00		113	0.68	-11.75	7.55	0.00	0.00	0.00
	105	0.68	-11.74	-7.55	0.00	0.00	0.00		114	0.84	-13.03	9.21	0.00	0.00	0.00
	106	0.84	-13.07	-9.22	0.00	0.00	0.00		115	0.94	16.62	-1.40	0.00	0.00	0.00
	107	0.72	-6.74	-5.35	0.00	0.00	0.00		116	0.94	17.40	-1.50	0.00	0.00	0.00
	108	0.88	-6.83	-6.47	0.00	0.00	0.00		117	0.87	16.56	-1.39	0.00	0.00	0.00
	109	0.65	-21.01	-11.06	0.00	0.00	0.00		118	0.87	17.34	-1.49	0.00	0.00	0.00
	110	0.80	-24.66	-13.62	0.00	0.00	0.00		119	0.92	15.72	-1.28	0.00	0.00	0.00
	111	0.79	51.90	14.39	0.00	0.00	0.00		120	0.90	15.70	-1.28	0.00	0.00	0.00
	112	0.98	66.47	18.19	0.00	0.00	0.00		121	0.92	18.26	-1.60	0.00	0.00	0.00
	113	0.75	35.33	8.38	0.00	0.00	0.00		122	0.90	18.24	-1.60	0.00	0.00	0.00
	114	0.92	45.77	10.69	0.00	0.00	0.00		123	0.85	14.92	-1.25	0.00	0.00	0.00
	115	0.94	17.45	1.50	0.00	0.00	0.00		124	0.85	15.70	-1.35	0.00	0.00	0.00
	116	0.94	16.68	1.40	0.00	0.00	0.00		125	0.78	14.86	-1.24	0.00	0.00	0.00
	117	0.87	17.39	1.49	0.00	0.00	0.00		126	0.78	15.64	-1.34	0.00	0.00	0.00
	118	0.87	16.61	1.40	0.00	0.00	0.00		127	0.83	14.02	-1.14	0.00	0.00	0.00
	119	0.92	18.31	1.61	0.00	0.00	0.00		128	0.81	14.00	-1.14	0.00	0.00	0.00
	120	0.90	18.29	1.61	0.00	0.00	0.00		129	0.83	16.56	-1.46	0.00	0.00	0.00
	121	0.92	15.77	1.29	0.00	0.00	0.00		130	0.81	16.55	-1.46	0.00	0.00	0.00
	122	0.90	15.75	1.29	0.00	0.00	0.00		131	0.94	15.46	-1.24	0.00	0.00	0.00
	123	0.85	15.75	1.36	0.00	0.00	0.00		132	0.94	16.06	-1.32	0.00	0.00	0.00
	124	0.85	14.97	1.26	0.00	0.00	0.00		133	0.89	15.41	-1.24	0.00	0.00	0.00
	125	0.78	15.68	1.35	0.00	0.00	0.00		134	0.89	16.01	-1.31	0.00	0.00	0.00
	126	0.78	14.91	1.25	0.00	0.00	0.00		135	0.93	14.79	-1.16	0.00	0.00	0.00
	127	0.83	16.61	1.47	0.00	0.00	0.00		136	0.91	14.77	-1.16	0.00	0.00	0.00
	128	0.81	16.59	1.46	0.00	0.00	0.00		137	0.93	16.70	-1.40	0.00	0.00	0.00
	129	0.83	14.07	1.14	0.00	0.00	0.00		138	0.91	16.68	-1.40	0.00	0.00	0.00
13	130	0.81	14.05	1.14	0.00	0.00	0.00	14	101	0.02	22.29	1.74	0.00	0.00	0.00
	131	0.94	16.15	1.33	0.00	0.00	0.00		102	0.02	23.14	1.86	0.00	0.00	0.00
	132	0.94	15.55	1.26	0.00	0.00	0.00		103	0.01	6.92	-2.04	0.00	0.00	0.00
	133	0.89	16.11	1.32	0.00	0.00	0.00		104	0.02	10.35	-2.32	0.00	0.00	0.00
	134	0.89	15.51	1.25	0.00	0.00	0.00		105	0.02	-13.72	-9.77	0.00	0.00	0.00
	135	0.92	16.79	1.41	0.00	0.00	0.00		106	0.03	-15.45	-11.99	0.00	0.00	0.00
	136	0.91	16.78	1.41	0.00	0.00	0.00		107	0.01	-6.64	-6.83	0.00	0.00	0.00
	137	0.92	14.89	1.17	0.00	0.00	0.00		108	0.02	-6.60	-8.31	0.00	0.00	0.00
	138	0.91	14.87	1.17	0.00	0.00	0.00		109	0.02	-26.02	-14.39	0.00	0.00	0.00
	139	0.92	15.32	-1.22	0.00	0.00	0.00		110	0.03	-30.83	-17.77	0.00	0.00	0.00
	140	0.91	16.15	-1.33	0.00	0.00	0.00		111	0.02	68.96	18.83	0.00	0.00	0.00
	141	0.79	51.80	-14.37	0.00	0.00	0.00		112	0.04	87.90	23.77	0.00	0.00	0.00
	142	0.98	66.41	-18.19	0.00	0.00	0.00		113	0.03	47.10	10.96	0.00	0.00	0.00
	143	0.75	35.30	-8.38	0.00	0.00	0.00		114	0.04	60.58	13.92	0.00	0.00	0.00
	144	0.92	45.78	-10.69	0.00	0.00	0.00		115	0.05	23.94	1.96	0.00	0.00	0.00
	145	0.72	-6.86	5.36	0.00	0.00	0.00		116	0.05	23.03	1.85	0.00	0.00	0.00
	146	0.88	-6.91	6.48	0.00	0.00	0.00		117	0.00	24.95	2.09	0.00	0.00	0.00
	147	0.65	-20.98	11.06	0.00	0.00	0.00		118	0.00	24.04	1.98	0.00	0.00	0.00
	148	0.80	-24.56	13.60	0.00	0.00	0.00		119	0.03	25.33	2.14	0.00	0.00	0.00
	149								120	0.02	25.63	2.18	0.00	0.00	0.00

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE									

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z		
26	121	0.03	22.35	1.76	0.00	0.00	0.00		
	122	0.02	22.65	1.80	0.00	0.00	0.00		
	123	0.05	21.55	1.77	0.00	0.00	0.00		
	124	0.05	20.64	1.65	0.00	0.00	0.00		
	125	0.00	22.55	1.89	0.00	0.00	0.00		
	126	0.00	21.64	1.78	0.00	0.00	0.00		
	127	0.03	22.93	1.94	0.00	0.00	0.00		
	128	0.02	23.23	1.98	0.00	0.00	0.00		
	129	0.03	19.95	1.57	0.00	0.00	0.00		
	130	0.02	20.25	1.60	0.00	0.00	0.00		
	131	0.04	22.69	1.80	0.00	0.00	0.00		
	132	0.04	21.99	1.71	0.00	0.00	0.00		
	133	0.01	23.44	1.89	0.00	0.00	0.00		
	134	0.01	22.74	1.80	0.00	0.00	0.00		
	135	0.03	23.71	1.93	0.00	0.00	0.00		
	136	0.02	23.95	1.96	0.00	0.00	0.00		
	137	0.03	21.48	1.65	0.00	0.00	0.00		
	138	0.02	21.72	1.67	0.00	0.00	0.00		
	101	0.02	22.07	-1.72	0.00	0.00	0.00		
	102	0.02	22.96	-1.83	0.00	0.00	0.00		
	103	0.02	68.75	-18.81	0.00	0.00	0.00		
	104	0.04	87.73	-23.75	0.00	0.00	0.00		
	105	0.03	47.02	-10.95	0.00	0.00	0.00		
	106	0.04	60.57	-13.92	0.00	0.00	0.00		
	107	0.01	-6.87	6.86	0.00	0.00	0.00		
	108	0.02	-6.79	8.33	0.00	0.00	0.00		
	109	0.02	-26.01	14.39	0.00	0.00	0.00		
	110	0.03	-30.72	17.75	0.00	0.00	0.00		
	111	0.01	6.72	2.06	0.00	0.00	0.00		
	112	0.02	10.19	2.34	0.00	0.00	0.00		
	113	0.02	-13.78	9.78	0.00	0.00	0.00		
	114	0.03	-15.43	11.99	0.00	0.00	0.00		
	115	0.05	22.92	-1.83	0.00	0.00	0.00		
	116	0.05	23.93	-1.95	0.00	0.00	0.00		
	117	0.00	23.89	-1.96	0.00	0.00	0.00		
	118	0.00	24.80	-2.07	0.00	0.00	0.00		
	119	0.03	22.22	-1.75	0.00	0.00	0.00		
	120	0.02	22.52	-1.78	0.00	0.00	0.00		
	121	0.03	25.21	-2.12	0.00	0.00	0.00		
	122	0.02	25.50	-2.16	0.00	0.00	0.00		
	123	0.05	20.53	-1.64	0.00	0.00	0.00		
	124	0.05	21.45	-1.75	0.00	0.00	0.00		
	125	0.00	21.51	-1.76	0.00	0.00	0.00		
	126	0.00	22.42	-1.88	0.00	0.00	0.00		
	127	0.03	19.84	-1.55	0.00	0.00	0.00		
	128	0.02	20.13	-1.59	0.00	0.00	0.00		
	129	0.03	22.82	-1.93	0.00	0.00	0.00		
	130	0.02	23.12	-1.96	0.00	0.00	0.00		
27	131	0.04	21.80	-1.68	0.00	0.00	0.00		
	132	0.04	22.50	-1.77	0.00	0.00	0.00		
	133	0.01	22.53	-1.78	0.00	0.00	0.00		
	134	0.01	23.23	-1.86	0.00	0.00	0.00		
	135	0.03	21.28	-1.62	0.00	0.00	0.00		
	136	0.02	21.51	-1.65	0.00	0.00	0.00		
	137	0.03	23.52	-1.90	0.00	0.00	0.00		
	138	0.02	23.75	-1.93	0.00	0.00	0.00		
	101	-0.02	22.29	1.74	0.00	0.00	0.00		
	102	-0.02	23.14	1.86	0.00	0.00	0.00		
	103	-0.01	6.92	-2.04	0.00	0.00	0.00		
	104	-0.02	10.35	-2.32	0.00	0.00	0.00		
39	105	-0.02	-13.72	-9.77	0.00	0.00	0.00		
	106	-0.03	-15.45	-11.99	0.00	0.00	0.00		
	107	-0.01	-6.64	-6.83	0.00	0.00	0.00		
	108	-0.02	-6.60	-8.31	0.00	0.00	0.00		
	109	-0.02	-26.02	-14.39	0.00	0.00	0.00		
	110	-0.03	-30.83	-17.77	0.00	0.00	0.00		
	111	-0.02	68.96	18.83	0.00	0.00	0.00		
	112	-0.04	87.89	23.77	0.00	0.00	0.00		
	113	-0.03	47.08	10.95	0.00	0.00	0.00		
	114	-0.04	60.55	13.92	0.00	0.00	0.00		
	115	0.00	24.94	2.09	0.00	0.00	0.00		
	116	0.00	24.03	1.97	0.00	0.00	0.00		
39	117	-0.05	23.95	1.97	0.00	0.00	0.00		
	118	-0.05	23.04	1.85	0.00	0.00	0.00		
	119	-0.02	25.63	2.18	0.00	0.00	0.00		
	120	-0.03	25.33	2.14	0.00	0.00	0.00		
	121	-0.02	22.65	1.80	0.00	0.00	0.00		
	122	-0.03	22.35	1.76	0.00	0.00	0.00		
	123	0.00	22.54	1.89	0.00	0.00	0.00		
	124	0.00	21.63	1.78	0.00	0.00	0.00		
	125	-0.05	21.55	1.77	0.00	0.00	0.00		
	126	-0.05	20.64	1.65	0.00	0.00	0.00		
	127	-0.02	23.23	1.98	0.00	0.00	0.00		
	128	-0.03	22.93	1.94	0.00	0.00	0.00		
39	129	-0.02	20.25	1.60	0.00	0.00	0.00		
	130	-0.03	19.95	1.57	0.00	0.00	0.00		
	131	-0.01	23.44	1.89	0.00	0.00	0.00		
	132	-0.01	22.74	1.80	0.00	0.00	0.00		
	133	-0.04	22.70	1.80	0.00	0.00	0.00		
	134	-0.04	21.99	1.71	0.00	0.00	0.00		
	135	-0.02	23.95	1.96	0.00	0.00	0.00		
	136	-0.03	23.72	1.93	0.00	0.00	0.00		
	137	-0.02	21.72	1.67	0.00	0.00	0.00		
	138	-0.03	21.48	1.65	0.00	0.00	0.00		
	101	-0.02	22.09	-1.72	0.00	0.00	0.00		
	102	-0.02	22.98	-1.84	0.00	0.00	0.00		

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE									

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z		
103	-0.02	68.78	-18.81	0.00	0.00	0.00	0.00		
104	-0.04	87.76	-23.75	0.00	0.00	0.00	0.00		
105	-0.03	47.04	-10.95	0.00	0.00	0.00	0.00		
106	-0.04	60.58	-13.92	0.00	0.00	0.00	0.00		
107	-0.01	-6.84	6.85	0.00	0.00	0.00	0.00		
108	-0.02	-6.77	8.33	0.00	0.00	0.00	0.00		
109	-0.02	-26.00	14.39	0.00	0.00	0.00	0.00		
110	-0.03	-30.72	17.75	0.00	0.00	0.00	0.00		
111	-0.01	6.73	2.06	0.00	0.00	0.00	0.00		
112	-0.02	10.20	2.34	0.00	0.00	0.00	0.00		
113	-0.02	-13.84	9.79	0.00	0.00	0.00	0.00		
114	-0.03	-15.51	12.00	0.00	0.00	0.00	0.00		
115	0.00	23.91	-1.96	0.00	0.00	0.00	0.00		
116	0.00	24.82	-2.07	0.00	0.00	0.00	0.00		
117	-0.05	22.94	-1.84	0.00	0.00	0.00	0.00		
118	-0.05	23.85	-1.95	0.00	0.00	0.00	0.00		
119	-0.02	22.53	-1.79	0.00	0.00	0.00	0.00		
120	-0.03	22.24	-1.75	0.00	0.00	0.00	0.00		
121	-0.02	25.52	-2.16	0.00	0.00	0.00	0.00		
122	-0.03	25.23	-2.13	0.00	0.00	0.00	0.00		
123	0.00	21.52	-1.76	0.00	0.00	0.00	0.00		
124	0.00	22.43	-1.88	0.00	0.00	0.00	0.00		
125	-0.05	20.55	-1.64	0.00	0.00	0.00	0.00		
126	-0.05	21.46	-1.76	0.00	0.00	0.00	0.00		
127	-0.02	20.14	-1.59	0.00	0.00	0.00	0.00		
128	-0.03	19.85	-1.55	0.00	0.00	0.00	0.00		
129	-0.02	23.13	-1.97	0.00	0.00	0.00	0.00		
130	-0.03	22.84	-1.93	0.00	0.00	0.00	0.00		
131	-0.01	22.55	-1.78	0.00	0.00	0.00	0.00		
132	-0.01	23.25	-1.87	0.00	0.00	0.00	0.00		
133	-0.04	21.82	-1.69	0.00	0.00	0.00	0.00		
134	-0.04	22.53	-1.78	0.00	0.00	0.00	0.00		
135	-0.02	21.53	-1.65	0.00	0.00	0.00	0.00		
136	-0.03	21.30	-1.62	0.00	0.00	0.00	0.00		
137	-0.02	23.77	-1.93	0.00	0.00	0.00	0.00		
138	-0.03	23.55	-1.90	0.00	0.00	0.00	0.00		
40	101	15.43	1.24	0.00	0.00	0.00	0.00		
102	-0.91	16.23	1.34	0.00	0.00	0.00	0.00		
103	-0.74	3.75	-1.67	0.00	0.00	0.00	0.00		
104	-0.91	6.29	-1.88	0.00	0.00	0.00	0.00		
105	-0.68	-11.74	-7.55	0.00	0.00	0.00	0.00		
106	-0.84	-13.07	-9.22	0.00	0.00	0.00	0.00		
107	-0.72	-6.74	-5.35	0.00	0.00	0.00	0.00		
108	-0.88	-6.83	-6.47	0.00	0.00	0.00	0.00		
109	-0.65	-21.01	-11.06	0.00	0.00	0.00	0.00		
110	-0.80	-24.66	-13.62	0.00	0.00	0.00	0.00		
111	-0.79	51.90	14.39	0.00	0.00	0.00	0.00		
112	-0.98	66.47	18.19	0.00	0.00	0.00	0.00		
52	101	15.32	-1.22	0.00	0.00	0.00	0.00		
102	-0.91	16.15	-1.33	0.00	0.00	0.00	0.00		
103	-0.79	51.80	-14.37	0.00	0.00	0.00	0.00		
104	-0.98	66.41	-18.19	0.00	0.00	0.00	0.00		
105	-0.75	35.30	-8.38	0.00	0.00	0.00	0.00		
106	-0.92	45.78	-10.69	0.00	0.00	0.00	0.00		
107	-0.72	-6.86	5.36	0.00	0.00	0.00	0.00		
108	-0.88	-6.91	6.48	0.00	0.00	0.00	0.00		
109	-0.65	-20.98	11.06	0.00	0.00	0.00	0.00		
110	-0.80	-24.56	13.60	0.00	0.00	0.00	0.00		
111	-0.74	3.63	1.69	0.00	0.00	0.00	0.00		
112	-0.91	6.20	1.89	0.00	0.00	0.00	0.00		
113	-0.68	-11.81	7.56	0.00	0.00	0.00	0.00		
114	-0.84	-13.10	9.23	0.00	0.00	0.00	0.00		
115	-0.87	16.56	-1.39	0.00	0.00	0.00	0.00		
116	-0.87	17.34	-1.49	0.00	0.00	0.00	0.00		
117	-0.94	16.62	-1.40	0.00	0.00	0.00	0.00		
118	-0.94	17.40	-1.50	0.00	0.00	0.00	0.00		
119	-0.90	15.70	-1.28	0.00	0.00	0.00	0.00		
120	-0.92	15.72	-1.28	0.00	0.00	0.00	0.00		
121	-0.90	18.24	-1.60	0.00	0.00	0.00	0.00		
122	-0.92	18.26	-1.60	0.00	0.00	0.00	0.00		